GC9309NA

a-Si TFT LCD Single Chip Driver
240RGBx320 Resolution and 262K color

Datasheet

V1.0

2024-10-16

GENERATION REVISION HISTORY

Version.	Effective Date	Description of Changes	Prepared by
V1.0	2024-10-16	Document Release	Racky

Galaxycore Incorporation

Table of Content

1. Introduction

The GC9309NA is a 262,144-color single-chip SOC driver for a-TFT liquid crystal display with resolution of 240RGBx320dots, comprising a 720-channel source driver, a 320-channel gate driver, 172,800 bytes GRAM for graphic display data of 240RGBx320 dots, and power supply circuit.

The GC9309NA supports parallel 8-/9-/16-/18-bit data bus MCU interface, 6-/16-/18-bit data bus RGB interface and 3-/4-line serial peripheral interface (SPI) and 2 lane SPI and Quad serial peripheral interface data transmission. The moving picture area can be specified in internal GRAM by window address function. The specified window area can be updated selectively, so that moving picture can be displayed simultaneously independent of still picture area.

The GC9309NA can operate with 1.65V ~ 3.5V I/O interface voltage and an incorporated voltage follower circuit to generate voltage levels for driving an LCD. GC9309NA supports full color, 8-color display mode and sleep mode for precise power control by software and these features make the GC9309NA an ideal LCD driver for medium or small size portable products such as digital cellular phones, smart phone, MP3 and PMP where long battery life is a major concern.

2. Features

- No need for external electronic component
- Display resolution: [240xRGB](H) x 320(V)
- Output:
  - 720 source outputs
  - 320 gate outputs
- a-TFT LCD driver with on-chip full display RAM: 172,800 bytes
- System Interface
  - 8-bits, 9-bits, 16-bits, 18-bits interface with 8080-I /8080-II series MCU
  - 6-bits, 16-bits, 18-bits RGB interface with graphic controller
  - 8-bits, 9-bits Serial Peripheral Interface (SPI) and 2 data lane SPI
  - Quad Serial Peripheral Interface
- Display mode:
  - Full color mode (Idle mode OFF): 262K-color (selectable color depth mode by software)
  - Reduce color mode (Idle mode ON): 8-color
- Power saving mode:
  - Sleep mode
  - Deep Standby
- On chip functions:
  - Timing generator
  - Oscillator
  - DC/DC converter
  - Dot/column inversion
- Low -power consumption architecture
  - Low operating power supplies:
    - IOVCC = 1.65V ~ 3.5V (logic)
    - VCI = 2.6V ~ 3.5V (analog)
- LCD Voltage drive:
  - Source power supply voltage

    Source Voltage (Vreg1A to Vreg2A): -4.5V ~6.5V
  - Gate driver output voltage

    VGH - GND = 8V ~ 15.0V

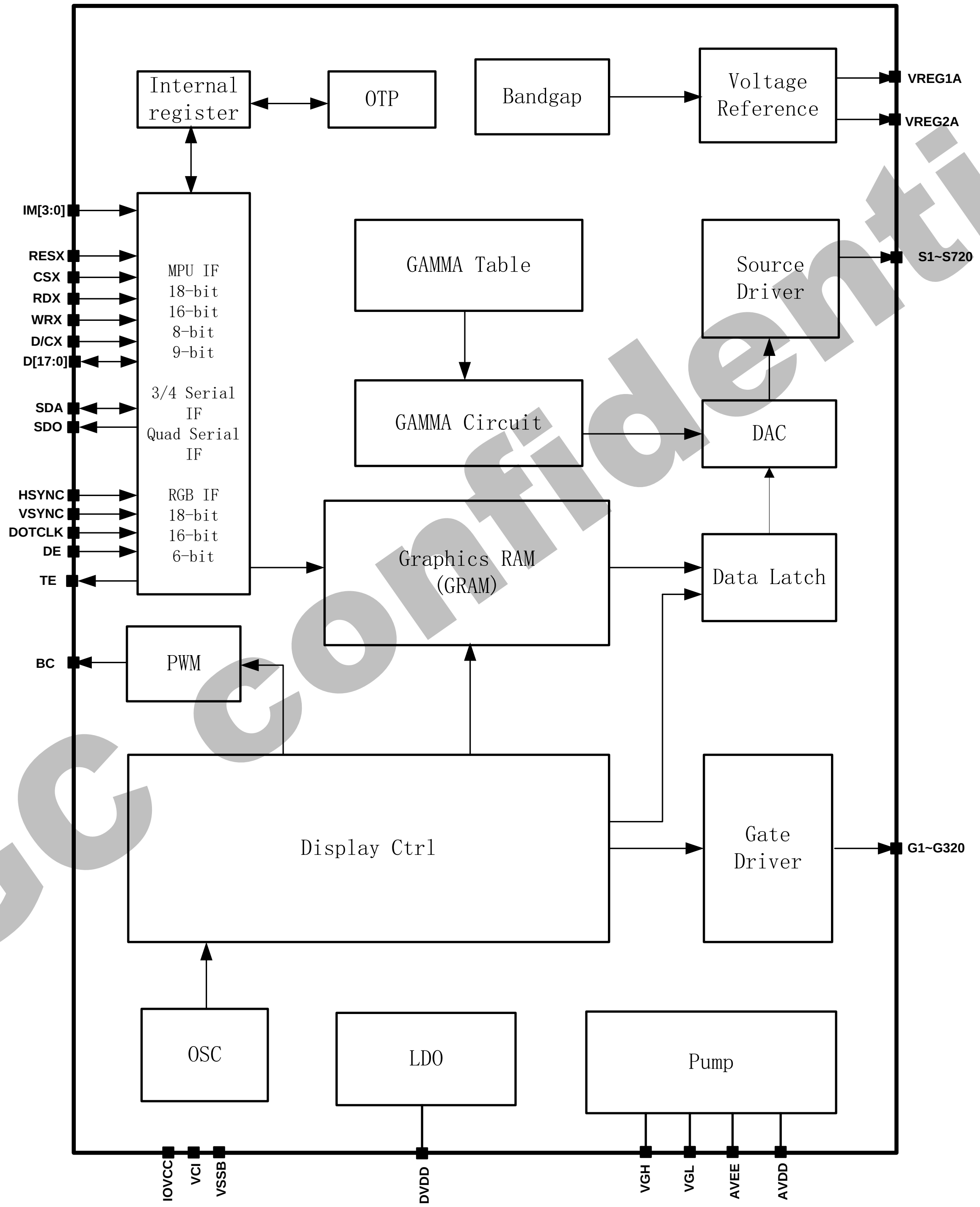
    VGL - GND = -11.5V ~ -7.0V

    VGH - VGL ≦27V
  - VCOM connect to GND
- Operate temperature range: -30℃to 85℃
- a-Si TFT LCD storage capacitor : Cst on Common structure only

3. Block Diagram

3.1. Block diagram

Figure1



3.2. Pin Description

Table 1.

Power Supply Pins			
Pin Name	I/O	Type	Descriptions
VDDI(IOVCC)	I	Digital Power	Low voltage power supply for interface logic circuits(1.65~3.5V)
VDDB(VCI)	I	Analog Power	High voltage power supply for analog circuit blocks(2.6~3.5V)
DVDD	O	Digital Power	Regulated Low voltage level for interface circuits Don't apply any external power to this pad
VSSB/VSSR	I	Ground	System ground level

Table 2

Interface Logic Signals			
Pin Name	I/O	Type	Descriptions
IM[3:0]	I	(VDDI/GND)	-Select the MCU interface mode

IM3	IM2	IM1	IM0	MCU-Interface	Pins in use	
					Register	GRAM
0	0	0	0	8080 MCU 8-bit bus interface I	D[7:0]	D[7:0]
0	0	0	1	8080 MCU16-bit bus interface I	D[7:0]	D[15:0]
0	0	1	0	8080 MCU 9-bit bus interface I	D[7:0]	D[8:0]
0	0	1	1	8080 MCU18-bit bus interface I	D[7:0]	D[17:0]
0	1	0	0	Q_SPI interface I	SDA	SDA, D[0:2]
0	1	0	1	3-wire 9-bit data serial interface I	SDA: In/OUT	
				2 data line serial interface I	SDA: In/OUT WRX:In	
0	1	1	0	4-wire 8-bit data serial interface I	SDA: In/OUT	
0	1	1	1	Q_SPI C0-interface I	SDA	SDA, D[0:2]
1	0	0	0	8080 MCU 16-bit bus interface II	D[8:1]	D[17:10], D[8:1]
1	0	0	1	8080 MCU 8-bit bus interface II	D[17:10]	
1	0	1	0	8080 MCU 18-bit bus interface II	D[8:1]	D[17:0]
1	0	1	1	8080 MCU 9-bit bus interface II	D[17:10]	D[17:9]
1	1	0	0	Q_SPI interface II	SDA:In SDO:Out	SDA, D[0:2]
1	1	0	1	3-wire 9-bit data serial interface II	SDA:In SDO:Out	SDA

<table>
<tr><td></td><td></td><td></td><td><table><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>4-wire 8-bit data serial interface II</td><td>SDA:In
SDO:Out</td><td>SDA</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>Q_SPI C0-interface II</td><td>SDA:In
SDO:Out</td><td>SDA,
D[0:2]</td></tr></table>-MPU Parallel interface bus and serial interface select
-If use RGB Interface must select serial interface.</td></tr>
<tr><td>RESX</td><td>I</td><td>MCU
(VDDI/GND)</td><td>-This signal will reset the device and must be applied to properly initialize the chip.
-Signal is active low.</td></tr>
<tr><td>CSX</td><td>I</td><td>MCU
(VDDI/GND)</td><td>-Chip select input pin("Low" enable).
-This pin can be permanently fixed "Low" in MPU interface mode only.</td></tr>
<tr><td>D/CX
(SCL)</td><td>I</td><td>MCU
(VDDI/ GND)</td><td>-This pin is used to select "Data or Command" in the parallel interface
When DCX='1', data is selected.
When DCX='0', command is selected.
-This pin is used serial interface clock in 3-wire 9-bit / 4-wire 8-bit /QSPI serial interface.
-If not used, please fix this pin at VDDI or GND level.</td></tr>
<tr><td>RDX</td><td>I</td><td>MCU
(VDDI/ GND)</td><td>-8080-I/8080-II system (RDX): Serves as a read signal and MCU read data at the rising edge.
-If not used, please fix this pin at VDDI or GND level.</td></tr>
<tr><td>WRX
(D/CX)</td><td>I</td><td>MCU
(VDDI/ GND)</td><td>-8080-I/8080-II system (WRX): Serves as a write signal and writes data at the rising edge.
-4-line system (D/CX): Serves as command or parameter select.
-2 lane mode serial interface: Serves as the second SDA
-If not used, please fix this pin at VDDI or GND level.</td></tr>
<tr><td>D[17:0]</td><td>I/O</td><td>MCU
(VDDI/ GND)</td><td>-18-bit parallel bi-directional data bus for MCU system and RGB interface mode
-QSPI 4wire mode D[0:2] Server as SDA[1:3]
-If not used, please fix this pin at VDDI or GND level.</td></tr>
<tr><td>SDA</td><td>I/O</td><td>MCU
(VDDI/ GND)</td><td>-When IM[3]:Low, Serial in/out signal in 3-wire 9-bit/4-wire 8-bit/QSPI serial data interface.
-When IM[3]:High, Serial input signal in 3-wire 9-bit/4-wire 8-bit/QSPI serial data interface.
-The data is applied on the rising edge of the SCL signal.
-If not used, please fix this pin at VDDI or GND level.</td></tr>
<tr><td>SDO</td><td>O</td><td>MCU
(VDDI/GND)</td><td>-Serial output signal.
-The data is outputted on the falling edge of the SCL signal.
-If not used, please open this pin</td></tr>
<tr><td>TE</td><td>O</td><td>MCU
(VDDI/ GND)</td><td>-Tearing effect output pin to synchronize MPU to frame writing, activated by S/W command. When this pin is not activated, this pin is low.</td></tr>
</table>

			- If not used, please open this pin.
DOTCLK	I	MCU (VDDI/GND)	-Dot clock signal for RGB interface operation. -If not used, please fix this pin at VDDI or GND level.
VSYNC	I	MCU (VDDI/GND)	-Frame synchronizing signal for RGB interface operation. -If not used, please fix this pin at VDDI or GND level.
HSYNC	I	MCU (VDDI/ GND)	-Line synchronizing signal for RGB interface operation. -If not used, please fix this pin at VDDI or GND level.
DE	I	MCU (VDDI/ GND)	-Data enable signal for RGB interface operation. -If not used, please fix this pin at VDDI or GND level.

Note:

1. If CSX is connected to GND in Parallel interface mode, there will be no abnormal visible effect to the display module. Also there will be no restriction on using the Parallel Read/Write protocols, Power On/Off Sequences or other functions. Furthermore there will be no influence to the Power Consumption of the display module.

2. When CSX='1', there is no influence to the parallel and serial interface.

Table 3

LCD Driver Input/Output Pins			
Pin Name	I/O	Type	Descriptions
S720~S1	O	Source	Source output signals.. Leave the pin to open when not in use.
G320~G1	O	Gate	Gate output signals. Leave the pin to open when not in use.
VCOM	O	GND	Connect to GND.
DVDD	O	Power	Regulated Low voltage level for interface circuits Don’t apply any external power to this pin
AVDD	O	Power	Generated positive power for source driver block.
AVEE	O	Power	Generated negative power for source driver block.
VGH	O	Power	Power supply for the gate driver(Positive).
VGL	O	Power	Power supply for the gate driver(Negative).
VREG1A	O	Ref	VREG1A is the highest positive grayscale reference voltage of source driver
VREG1B	O	Ref	VREG1B is the lowest positive grayscale reference voltage of source driver
VREG2A	O	Ref	VREG2A is the highest negative grayscale reference voltage of source driver
VREG2B	O	Ref	VREG2B is the lowest negative grayscale reference voltage of source driver
BC	O	Dig IO	Output pin for PWM(Pulse width Modulation) signal of LED driving. -If not used,please open this pin.

Table 4

Test Pins			
Pin Name	I/O	Type	Descriptions
OSC_IN	I/O	Open	Test pin
OSC_TEST	I/O	Open	Test pin
VPP_PAD	I/O	Open	Test pin
DUMMY	-	Open	Input pads used only for test purpose at IC-side. During normal operation ,let these pads open.

Liquid crystal power supply specifications Table

Table 5

No.	Item		Description
1	TFT Source Driver		720 pins (240*RGB)
2	TFT Gate Driver		320 pins
3	TFT Display's Capacitor Structure		Cst structure only (Cs on Common)
4	Liquid Crystal Drive Output	S1~S720	V0~V63 grayscales
		G1~G320	VGH-VGL
5	Input Voltage	IOVCC	1.65~3.50V
		VCI	2.60~3.50V
6	Liquid Crystal Drive Voltages	Vreg1A	4~6.5V
		Vreg2A	-4.5V~-2V
		VGH	8.0~15.0V
		VGL	-11.5~-7.0V
		VGH-VGL	Max=27.0V

3.3. PAD coordinates

No.	Pad	X	Y
1	DUMMY	-7292.375	-239
2	DUMMY	-7232.375	-239
3	VCOM	-7172.375	-239
4	VCOM	-7112.375	-239
5	VCOM	-7052.375	-239
6	VCOM	-6992.375	-239
7	VCOM	-6932.375	-239
8	VCOM	-6872.375	-239
9	VGH	-6632.375	-239
10	VGH	-6572.375	-239
11	VGL	-6512.375	-239
12	VGL	-6452.375	-239
13	VREG1A	-6392.375	-239
14	VREG1A	-6332.375	-239
15	DUMMY	-6272.375	-239
16	DUMMY	-6212.375	-239
17	VDDB	-6152.375	-239
18	VDDB	-6092.375	-239
19	VDDB	-6032.375	-239
20	VDDB	-5972.375	-239
21	VDDB	-5912.375	-239
22	DUMMY	-5852.375	-239
23	VGL	-5792.375	-239
24	VGL	-5732.375	-239
25	VGL	-5672.375	-239
26	VGL	-5612.375	-239
27	VGL	-5552.375	-239
28	VGL	-5492.375	-239
29	BVDD	-5432.375	-239
30	BVDD	-5372.375	-239
31	BVDD	-5312.375	-239
32	BVDD	-5252.375	-239
33	BVDD	-5192.375	-239
34	BVDD	-5132.375	-239
35	BVDD	-5072.375	-239
36	VSSB	-5012.375	-239
37	VSSB	-4952.375	-239
38	VSSB	-4892.375	-239
39	VSSB	-4832.375	-239
40	VSSB	-4772.375	-239
41	VSSB	-4712.375	-239
42	VSSB	-4652.375	-239
43	AVDD	-4592.375	-239
44	AVDD	-4532.375	-239
45	AVDD	-4472.375	-239
46	AVDD	-4412.375	-239
47	AVDD	-4352.375	-239
48	AVDD	-4292.375	-239
49	AVDD	-4232.375	-239
50	DUMMY	-4172.375	-239

No.	Pad	X	Y
51	DUMMY	-4172.375	-239
52	DUMMY	-4112.375	-239
53	DUMMY	-4052.375	-239
54	DUMMY	-3992.375	-239
55	DUMMY	-3932.375	-239
56	DUMMY	-3872.375	-239
57	TESTP	-3812.375	-239
58	DUMMY	-3752.375	-239
59	DVDD	-3692.375	-239
60	DVDD	-3632.375	-239
61	DVDD	-3572.375	-239
62	DVDD	-3512.375	-239
63	DVDD	-3452.375	-239
64	DVDD	-3392.375	-239
65	VSSB	-3332.375	-239
66	VSSB	-3272.375	-239
67	VSSB	-3212.375	-239
68	VSSB	-3152.375	-239
69	VSSB	-3092.375	-239
70	VSSB	-3032.375	-239
71	VSSB	-2972.375	-239
72	VDDB	-2912.375	-239
73	VDDB	-2852.375	-239
74	VDDB	-2792.375	-239
75	VDDB	-2732.375	-239
76	VDDB	-2672.375	-239
77	VDDB	-2612.375	-239
78	VDDB	-2552.375	-239
79	VDDB	-2492.375	-239
80	VSSR	-2432.375	-239
81	VSSR	-2372.375	-239
82	VSSR	-2312.375	-239
83	VSSR	-2252.375	-239
84	VSSR	-2192.375	-239
85	VSSR	-2132.375	-239
86	VSSR	-2072.375	-239
87	VSSR	-2012.375	-239
88	VSSR	-1952.375	-239
89	VSSB	-1892.375	-239
90	VSSB	-1832.375	-239
91	VSSB	-1772.375	-239
92	VSSB	-1712.375	-239
93	VSSB	-1652.375	-239
94	VSSB	-1592.375	-239
95	VSSB	-1532.375	-239
96	VSSB	-1472.375	-239
97	VSSB	-1412.375	-239
98	VSSB	-1352.375	-239
99	VSSB	-1292.375	-239
100	VSSB	-1232.375	-239

No.	Pad	X	Y
101	VSSB	-1112.37	-239
102	VSSB	-1052.37	-239
103	DUMMY	-992.375	-239
104	VSSB	-932.375	-239
105	VSSB	-872.375	-239
106	VPP PAD	-812.375	-239
107	IM<3>	-752.375	-239
108	IM<2>	-692.375	-239
109	IM<1>	-632.375	-239
110	IM<0>	-572.375	-239
111	RESX	-512.375	-239
112	CSX	-452.375	-239
113	DCX	-392.375	-239
114	WRX	-332.375	-239
115	RDX	-272.375	-239
116	DUMMY	-212.375	-239
117	VSYNC	-152.375	-239
118	HSYNC	-92.375	-239
119	ENABLE	-32.375	-239
120	DOTCLK	27.625	-239
121	DUMMY	87.625	-239
122	SDA	160.125	-239
123	DB<0>	245.125	-239
124	DB<1>	330.125	-239
125	DB<2>	415.125	-239
126	DB<3>	500.125	-239
127	DUMMY	572.625	-239
128	DB<4>	645.125	-239
129	DB<5>	730.125	-239
130	DB<6>	815.125	-239
131	DB<7>	900.125	-239
132	DUMMY	972.625	-239
133	DB<8>	1045.125	-239
134	DB<9>	1130.125	-239
135	DB<10>	1215.125	-239
136	DB<11>	1300.125	-239
137	DUMMY	1372.625	-239
138	DB<12>	1445.125	-239
139	DB<13>	1530.125	-239
140	DB<14>	1615.125	-239
141	DB<15>	1700.125	-239
142	DUMMY	1772.625	-239
143	DB<16>	1845.125	-239
144	DB<17>	1930.125	-239
145	OSC IN	2002.5	-239
146	TE	2075	-239
147	SDO	2160	-239
148	BC	2245	-239
149	DUMMY	2330	-239
150	DUMMY	2402.625	-239

No.	Pad	X	Y
151	DUMMY	2462.625	-239
152	DBDUM21	2790.125	-239
153	DBDUM22	2875.125	-239
154	DBDUM23	2960.125	-239
155	DUMMY<13>	3032.625	-239
156	VDDI	3092.625	-239
157	VDDI	3152.625	-239
158	VDDI	3212.625	-239
159	VDDI	3272.625	-239
160	VDDI	3332.625	-239
161	VDDI	3392.625	-239
162	VDDI	3452.625	-239
163	DUMMY	3512.625	-239
164	DUMMY	3572.625	-239
165	DUMMY	3632.625	-239
166	DUMMY	3692.625	-239
167	DUMMY	3752.625	-239
168	DUMMY	3812.625	-239
169	DUMMY	3872.625	-239
170	DUMMY	3932.625	-239
171	DUMMY	3992.625	-239
172	DUMMY	4052.625	-239
173	DUMMY	4112.625	-239
174	DUMMY	4172.625	-239
175	DUMMY	4232.625	-239
176	VREGN	4292.625	-239
177	VREGN	4352.625	-239
178	DUMMY	4412.625	-239
179	DUMMY	4472.625	-239
180	DUMMY	4532.625	-239
181	DUMMY	4592.625	-239
182	DUMMY<15>	4652.625	-239
183	DUMMY<16>	4712.625	-239
184	BVEE	4772.625	-239
185	BVEE	4832.625	-239
186	BVEE	4892.625	-239
187	BVEE	4952.625	-239
188	BVEE	5012.625	-239
189	BVEE	5072.625	-239
190	BVEE	5132.625	-239
191	BVEE	5192.625	-239
192	VDDB	5252.625	-239
193	VDDB	5312.625	-239
194	VDDB	5372.625	-239
195	VDDB	5432.625	-239
196	VDDB	5492.625	-239
197	VDDB	5552.625	-239
198	VDDB	5612.625	-239
199	VDDB	5672.625	-239
200	VSSB	5732.625	-239

No.	Pad	X	Y
201	VSSB	5792.625	-239
202	VSSB	5852.625	-239
203	VSSB	5912.625	-239
204	VSSB	5972.625	-239
205	VSSB	6032.625	-239
206	VSSB	6092.625	-239
207	VSSB	6152.625	-239
208	AVEE	6212.625	-239
209	AVEE	6272.625	-239
210	AVEE	6332.625	-239
211	AVEE	6392.625	-239
212	AVEE	6452.625	-239
213	AVEE	6512.625	-239
214	AVEE	6572.625	-239
215	AVEE	6632.625	-239
216	AVEE	6692.625	-239
217	VCOM	6752.625	-239
218	VCOM	6812.625	-239
219	VCOM	6872.625	-239
220	VCOM	6932.625	-239
221	VCOM	6992.625	-239
222	VCOM	7052.625	-239
223	VCOM	7112.625	-239
224	VCOM	7172.625	-239
225	DUMMY<25>	7232.625	-239
226	DUMMY	7292.625	-239
227	MARKR	7480	221.5
228	DUMMY<20>	7399	223
229	DUMMY<19>	7385	124
230	DUMMY<21>	7371	223
231	G<2>	7357	124
232	G<4>	7343	223
233	G<6>	7329	124
234	G<8>	7315	223
235	G<10>	7301	124
236	G<12>	7287	223
237	G<14>	7273	124
238	G<16>	7259	223
239	G<18>	7245	124
240	G<20>	7231	223
241	G<22>	7217	124
242	G<24>	7203	223
243	G<26>	7189	124
244	G<28>	7175	223
245	G<30>	7161	124
246	G<32>	7147	223
247	G<34>	7133	124
248	G<36>	7119	223
249	G<38>	7105	124
250	G<40>	7091	223

No.	Pad	X	Y
251	G<42>	7077	124
252	G<44>	7063	223
253	G<46>	7049	124
254	G<48>	7035	223
255	G<50>	7021	124
256	G<52>	7007	223
257	G<54>	6993	124
258	G<56>	6979	223
259	G<58>	6965	124
260	G<60>	6951	223
261	G<62>	6937	124
262	G<64>	6923	223
263	G<66>	6909	124
264	G<68>	6895	223
265	G<70>	6881	124
266	G<72>	6867	223
267	G<74>	6853	124
268	G<76>	6839	223
269	G<78>	6825	124
270	G<80>	6811	223
271	G<82>	6797	124
272	G<84>	6783	223
273	G<86>	6769	124
274	G<88>	6755	223
275	G<90>	6741	124
276	G<92>	6727	223
277	G<94>	6713	124
278	G<96>	6699	223
279	G<98>	6685	124
280	G<100>	6671	223
281	G<102>	6657	124
282	G<104>	6643	223
283	G<106>	6629	124
284	G<108>	6615	223
285	G<110>	6601	124
286	G<112>	6587	223
287	G<114>	6573	124
288	G<116>	6559	223
289	G<118>	6545	124
290	G<120>	6531	223
291	G<122>	6517	124
292	G<124>	6503	223
293	G<126>	6489	124
294	G<128>	6475	223
295	G<130>	6461	124
296	G<132>	6447	223
297	G<134>	6433	124
298	G<136>	6419	223
299	G<138>	6405	124
300	G<140>	6391	223

No.	Pad	X	Y
301	G<142>	6377	124
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322	G<184>	6083	223
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472	S<639>	3941	223
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614	S<497>	1953	223
615	S<496>	1939	124
616	S<495>	1925	223
617	S<494>	1911	124
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627	S<484>	1771	124
628	S<483>	1757	223
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632	S<479>	1701	223
633	S<478>	1687	124
634	S<477>	1673	223
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638	S<473>	1617	223
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No.	Pad	X	Y
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683	S<428>	987	124
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694	S<417>	833	223
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696	S<415>	805	223
697	S<414>	791	124
698	S<413>	777	223
699	S<412>	763	124
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708	S<403>	637	223
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710	S<401>	609	223
711	S<400>	595	124
712	S<399>	581	223
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714	S<397>	553	223
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716	S<395>	525	223
717	S<394>	511	124
718	S<393>	497	223
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720	S<391>	469	223
721	S<390>	455	124
722	S<389>	441	223
723	S<388>	427	124
724	S<387>	413	223
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729	S<382>	343	124
730	S<381>	329	223
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734	S<377>	273	223
735	S<376>	259	124
736	S<375>	245	223
737	S<374>	231	124
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743	S<368>	147	124
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745	S<366>	119	124
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747	S<364>	91	124
748	S<363>	77	223
749	S<362>	63	124
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No.	Pad	X	Y
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757	S<354>	-133	124
758	S<353>	-147	223
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760	S<351>	-175	223
761	S<350>	-189	124
762	S<349>	-203	223
763	S<348>	-217	124
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767	S<344>	-273	124
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769	S<342>	-301	124
770	S<341>	-315	223
771	S<340>	-329	124
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774	S<337>	-371	223
775	S<336>	-385	124
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780	S<331>	-455	223
781	S<330>	-469	124
782	S<329>	-483	223
783	S<328>	-497	124
784	S<327>	-511	223
785	S<326>	-525	124
786	S<325>	-539	223
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791	S<320>	-609	124
792	S<319>	-623	223
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796	S<315>	-679	223
797	S<314>	-693	124
798	S<313>	-707	223
799	S<312>	-721	124
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No.	Pad	X	Y
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804	S<307>	-791	223
805	S<306>	-805	124
806	S<305>	-819	223
807	S<304>	-833	124
808	S<303>	-847	223
809	S<302>	-861	124
810	S<301>	-875	223
811	S<300>	-889	124
812	S<299>	-903	223
813	S<298>	-917	124
814	S<297>	-931	223
815	S<296>	-945	124
816	S<295>	-959	223
817	S<294>	-973	124
818	S<293>	-987	223
819	S<292>	-1001	124
820	S<291>	-1015	223
821	S<290>	-1029	124
822	S<289>	-1043	223
823	S<288>	-1057	124
824	S<287>	-1071	223
825	S<286>	-1085	124
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833	S<278>	-1197	124
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843	S<268>	-1337	124
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847	S<264>	-1393	124
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No.	Pad	X	Y
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860	S<251>	-1575	223
861	S<250>	-1589	124
862	S<249>	-1603	223
863	S<248>	-1617	124
864	S<247>	-1631	223
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868	S<243>	-1687	223
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870	S<241>	-1715	223
871	S<240>	-1729	124
872	S<239>	-1743	223
873	S<238>	-1757	124
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876	S<235>	-1799	223
877	S<234>	-1813	124
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884	S<227>	-1911	223
885	S<226>	-1925	124
886	S<225>	-1939	223
887	S<224>	-1953	124
888	S<223>	-1967	223
889	S<222>	-1981	124
890	S<221>	-1995	223
891	S<220>	-2009	124
892	S<219>	-2023	223
893	S<218>	-2037	124
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896	S<215>	-2079	223
897	S<214>	-2093	124
898	S<213>	-2107	223
899	S<212>	-2121	124
900	S<211>	-2135	223

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903	S<208>	-2177	124	953	S<158>	-2877	124	1003	S<108>	-3577	124
904	S<207>	-2191	223	954	S<157>	-2891	223	1004	S<107>	-3591	223
905	S<206>	-2205	124	955	S<156>	-2905	124	1005	S<106>	-3605	124
906	S<205>	-2219	223	956	S<155>	-2919	223	1006	S<105>	-3619	223
907	S<204>	-2233	124	957	S<154>	-2933	124	1007	S<104>	-3633	124
908	S<203>	-2247	223	958	S<153>	-2947	223	1008	S<103>	-3647	223
909	S<202>	-2261	124	959	S<152>	-2961	124	1009	S<102>	-3661	124
910	S<201>	-2275	223	960	S<151>	-2975	223	1010	S<101>	-3675	223
911	S<200>	-2289	124	961	S<150>	-2989	124	1011	S<100>	-3689	124
912	S<199>	-2303	223	962	S<149>	-3003	223	1012	S<99>	-3703	223
913	S<198>	-2317	124	963	S<148>	-3017	124	1013	S<98>	-3717	124
914	S<197>	-2331	223	964	S<147>	-3031	223	1014	S<97>	-3731	223
915	S<196>	-2345	124	965	S<146>	-3045	124	1015	S<96>	-3745	124
916	S<195>	-2359	223	966	S<145>	-3059	223	1016	S<95>	-3759	223
917	S<194>	-2373	124	967	S<144>	-3073	124	1017	S<94>	-3773	124
918	S<193>	-2387	223	968	S<143>	-3087	223	1018	S<93>	-3787	223
919	S<192>	-2401	124	969	S<142>	-3101	124	1019	S<92>	-3801	124
920	S<191>	-2415	223	970	S<141>	-3115	223	1020	S<91>	-3815	223
921	S<190>	-2429	124	971	S<140>	-3129	124	1021	S<90>	-3829	124
922	S<189>	-2443	223	972	S<139>	-3143	223	1022	S<89>	-3843	223
923	S<188>	-2457	124	973	S<138>	-3157	124	1023	S<88>	-3857	124
924	S<187>	-2471	223	974	S<137>	-3171	223	1024	S<87>	-3871	223
925	S<186>	-2485	124	975	S<136>	-3185	124	1025	S<86>	-3885	124
926	S<185>	-2499	223	976	S<135>	-3199	223	1026	S<85>	-3899	223
927	S<184>	-2513	124	977	S<134>	-3213	124	1027	S<84>	-3913	124
928	S<183>	-2527	223	978	S<133>	-3227	223	1028	S<83>	-3927	223
929	S<182>	-2541	124	979	S<132>	-3241	124	1029	S<82>	-3941	124
930	S<181>	-2555	223	980	S<131>	-3255	223	1030	S<81>	-3955	223
931	S<180>	-2569	124	981	S<130>	-3269	124	1031	S<80>	-3969	124
932	S<179>	-2583	223	982	S<129>	-3283	223	1032	S<79>	-3983	223
933	S<178>	-2597	124	983	S<128>	-3297	124	1033	S<78>	-3997	124
934	S<177>	-2611	223	984	S<127>	-3311	223	1034	S<77>	-4011	223
935	S<176>	-2625	124	985	S<126>	-3325	124	1035	S<76>	-4025	124
936	S<175>	-2639	223	986	S<125>	-3339	223	1036	S<75>	-4039	223
937	S<174>	-2653	124	987	S<124>	-3353	124	1037	S<74>	-4053	124
938	S<173>	-2667	223	988	S<123>	-3367	223	1038	S<73>	-4067	223
939	S<172>	-2681	124	989	S<122>	-3381	124	1039	S<72>	-4081	124
940	S<171>	-2695	223	990	S<121>	-3395	223	1040	S<71>	-4095	223
941	S<170>	-2709	124	991	S<120>	-3409	124	1041	S<70>	-4109	124
942	S<169>	-2723	223	992	S<119>	-3423	223	1042	S<69>	-4123	223
943	S<168>	-2737	124	993	S<118>	-3437	124	1043	S<68>	-4137	124
944	S<167>	-2751	223	994	S<117>	-3451	223	1044	S<67>	-4151	223
945	S<166>	-2765	124	995	S<116>	-3465	124	1145	S<66>	-4165	124
946	S<165>	-2779	223	996	S<115>	-3479	223	1146	S<65>	-4179	223
947	S<164>	-2793	124	997	S<114>	-3493	124	1147	S<64>	-4193	124
948	S<163>	-2807	223	998	S<113>	-3507	223	1148	S<63>	-4207	223
949	S<162>	-2821	124	999	S<112>	-3521	124	1149	S<62>	-4221	124
950	S<161>	-2835	223	1000	S<111>	-3535	223	1150	S<61>	-4235	223

No.	Pad	X	Y
1051	S<60>	-4249	124
1052	S<59>	-4263	223
1053	S<58>	-4277	124
1054	S<57>	-4291	223
1055	S<56>	-4305	124
1056	S<55>	-4319	223
1057	S<54>	-4333	124
1058	S<53>	-4347	223
1059	S<52>	-4361	124
1060	S<51>	-4375	223
1061	S<50>	-4389	124
1062	S<49>	-4403	223
1063	S<48>	-4417	124
1064	S<47>	-4431	223
1065	S<46>	-4445	124
1066	S<45>	-4459	223
1067	S<44>	-4473	124
1068	S<43>	-4487	223
1069	S<42>	-4501	124
1070	S<41>	-4515	223
1071	S<40>	-4529	124
1072	S<39>	-4543	223
1073	S<38>	-4557	124
1074	S<37>	-4571	223
1075	S<36>	-4585	124
1076	S<35>	-4599	223
1077	S<34>	-4613	124
1078	S<33>	-4627	223
1079	S<32>	-4641	124
1080	S<31>	-4655	223
1081	S<30>	-4669	124
1082	S<29>	-4683	223
1083	S<28>	-4697	124
1084	S<27>	-4711	223
1085	S<26>	-4725	124
1086	S<25>	-4739	223
1087	S<24>	-4753	124
1088	S<23>	-4767	223
1089	S<22>	-4781	124
1090	S<21>	-4795	223
1091	S<20>	-4809	124
1092	S<19>	-4823	223
1093	S<18>	-4837	124
1094	S<17>	-4851	223
1095	S<16>	-4865	124
1096	S<15>	-4879	223
1097	S<14>	-4893	124
1098	S<13>	-4907	223
1099	S<12>	-4921	124
1100	S<11>	-4935	223

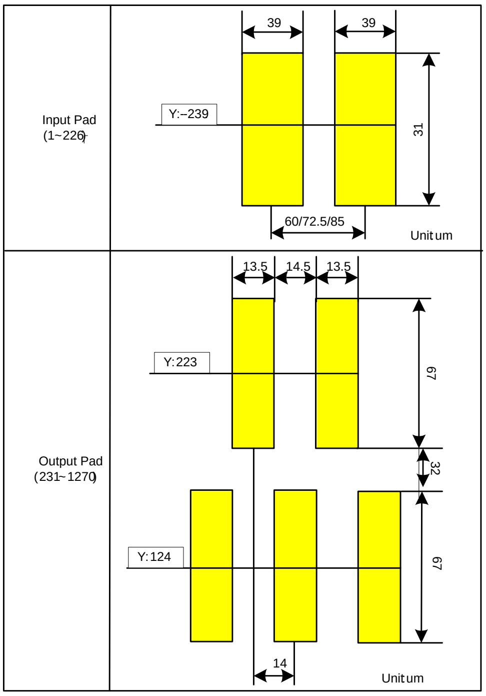
No.	Pad	X	Y
1101	S<10>	-4949	124
1102	S<9>	-4963	223
1103	S<8>	-4977	124
1104	S<7>	-4991	223
1105	S<6>	-5005	124
1106	S<5>	-5019	223
1107	S<4>	-5033	124
1108	S<3>	-5047	223
1109	S<2>	-5061	124
1110	S<1>	-5075	223
1111	G<319>	-5131	124
1112	G<317>	-5145	223
1113	G<315>	-5159	124
1114	G<313>	-5173	223
1115	G<311>	-5187	124
1116	G<309>	-5201	223
1117	G<307>	-5215	124
1118	G<305>	-5229	223
1119	G<303>	-5243	124
1120	G<301>	-5257	223
1121	G<299>	-5271	124
1122	G<297>	-5285	223
1123	G<295>	-5299	124
1124	G<293>	-5313	223
1125	G<291>	-5327	124
1126	G<289>	-5341	223
1127	G<287>	-5355	124
1128	G<285>	-5369	223
1129	G<283>	-5383	124
1130	G<281>	-5397	223
1131	G<279>	-5411	124
1132	G<277>	-5425	223
1133	G<275>	-5439	124
1134	G<273>	-5453	223
1135	G<271>	-5467	124
1136	G<269>	-5481	223
1137	G<267>	-5495	124
1138	G<265>	-5509	223
1139	G<263>	-5523	124
1140	G<261>	-5537	223
1141	G<259>	-5551	124
1142	G<257>	-5565	223
1143	G<255>	-5579	124
1144	G<253>	-5593	223
1145	G<251>	-5607	124
1146	G<249>	-5621	223
1147	G<247>	-5635	124
1148	G<245>	-5649	223
1149	G<243>	-5663	124
1150	G<241>	-5677	223

No.	Pad	X	Y
1151	G<239>	-5691	124
1152	G<237>	-5705	223
1153	G<235>	-5719	124
1154	G<233>	-5733	223
1155	G<231>	-5747	124
1156	G<229>	-5761	223
1157	G<227>	-5775	124
1158	G<225>	-5789	223
1159	G<223>	-5803	124
1160	G<221>	-5817	223
1161	G<219>	-5831	124
1162	G<217>	-5845	223
1163	G<215>	-5859	124
1164	G<213>	-5873	223
1165	G<211>	-5887	124
1166	G<209>	-5901	223
1167	G<207>	-5915	124
1168	G<205>	-5929	223
1169	G<203>	-5943	124
1170	G<201>	-5957	223
1171	G<199>	-5971	124
1172	G<197>	-5985	223
1173	G<195>	-5999	124
1174	G<193>	-6013	223
1175	G<191>	-6027	124
1176	G<189>	-6041	223
1177	G<187>	-6055	124
1178	G<185>	-6069	223
1179	G<183>	-6083	124
1180	G<181>	-6097	223
1181	G<179>	-6111	124
1182	G<177>	-6125	223
1183	G<175>	-6139	124
1184	G<173>	-6153	223
1185	G<171>	-6167	124
1186	G<169>	-6181	223
1187	G<167>	-6195	124
1188	G<165>	-6209	223
1189	G<163>	-6223	124
1190	G<161>	-6237	223
1191	G<159>	-6251	124
1192	G<157>	-6265	223
1193	G<155>	-6279	124
1194	G<153>	-6293	223
1195	G<151>	-6307	124
1196	G<149>	-6321	223
1197	G<147>	-6335	124
1198	G<145>	-6349	223
1199	G<143>	-6363	124
1200	G<141>	-6377	223

No.	Pad	X	Y
1201	G<139>	-6391	124
1202	G<137>	-6405	223
1203	G<135>	-6419	124
1204	G<133>	-6433	223
1205	G<131>	-6447	124
1206	G<129>	-6461	223
1207	G<127>	-6475	124
1208	G<125>	-6489	223
1209	G<123>	-6503	124
1210	G<121>	-6517	223
1211	G<119>	-6531	124
1212	G<117>	-6545	223
1213	G<115>	-6559	124
1214	G<113>	-6573	223
1215	G<111>	-6587	124
1216	G<109>	-6601	223
1217	G<107>	-6615	124
1218	G<105>	-6629	223
1219	G<103>	-6643	124
1220	G<101>	-6657	223
1221	G<99>	-6671	124
1222	G<97>	-6685	223
1223	G<95>	-6699	124
1224	G<93>	-6713	223
1231	G<79>	-6811	124
1232	G<77>	-6825	223
1233	G<75>	-6839	124
1234	G<73>	-6853	223
1235	G<71>	-6867	124
1236	G<69>	-6881	223
1237	G<67>	-6895	124
1238	G<65>	-6909	223
1239	G<63>	-6923	124
1240	G<61>	-6937	223
1241	G<59>	-6951	124
1242	G<57>	-6965	223
1243	G<55>	-6979	124
1244	G<53>	-6993	223
1245	G<51>	-7007	124
1246	G<49>	-7021	223
1247	G<47>	-7035	124
1248	G<45>	-7049	223
1249	G<43>	-7063	124
1250	G<41>	-7077	223

No.	Pad	X	Y
1251	G<39>	-7091	124
1252	G<37>	-7105	223
1253	G<35>	-7119	124
1254	G<33>	-7133	223
1255	G<31>	-7147	124
1256	G<29>	-7161	223
1257	G<27>	-7175	124
1258	G<25>	-7189	223
1259	G<23>	-7203	124
1260	G<21>	-7217	223
1261	G<19>	-7231	124
1262	G<17>	-7245	223
1263	G<15>	-7259	124
1264	G<13>	-7273	223
1265	G<11>	-7287	124
1266	G<9>	-7301	223
1267	G<7>	-7315	124
1268	G<5>	-7329	223
1269	G<3>	-7343	124
1270	G<1>	-7357	223
1271	DUMMY<23>	-7371	124
1272	DUMMY<22>	-7385	223
1273	DUMMY<24>	-7399	124
1274	MARKL	-7480	221.5

BUMP Size



Chip Size:
15304 um x 526 um(No scribe line)

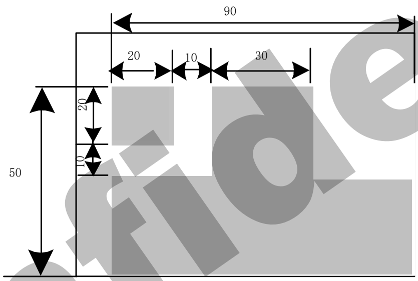
Chip thickness:200um(typ.)

Pad Location:Pad Center

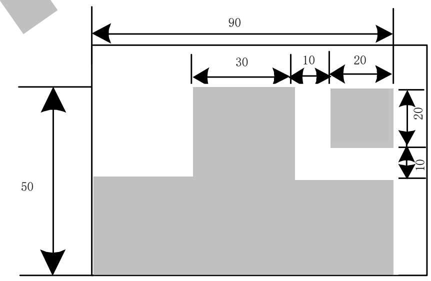
Coordinate Origin:Chip center

Au bump height:9um(typ.)

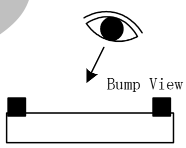
Alignment Marks

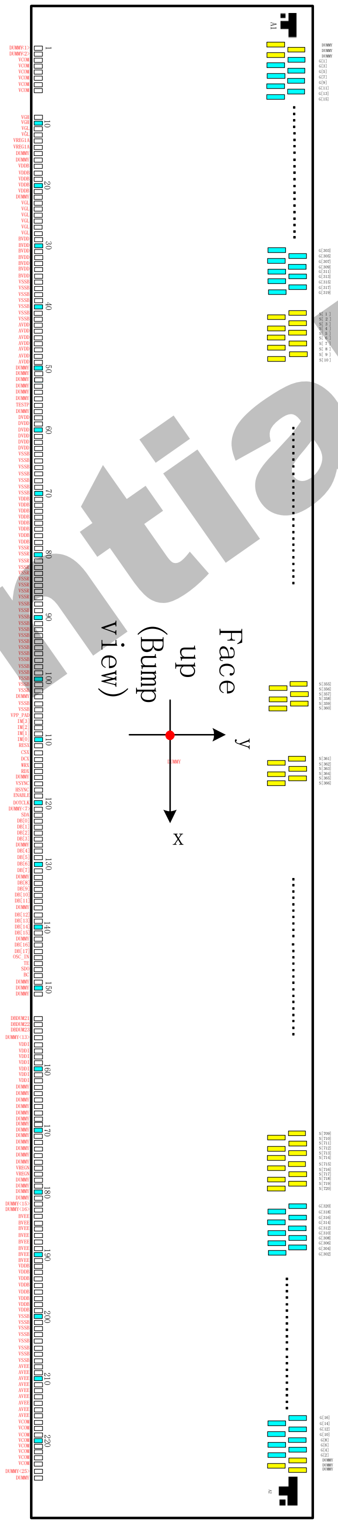


Alignment
Mark : A2



Alignment
Mark:A1





4. Interface setting

4.1. MCU interfaces

GC9309NA provides the 8-/9-/16-/18-bit parallel system interface for 8080-I /8080- II series, and 3-/4-line serial system interface for serial data input. The input system interface is selected by external pins IM [3:0] and the bit formal per pixel color order is selected by DBI [2:0] 3-bits of 3Ah register.

4.1.1. MCU interface selection

The selection of interface is done by setting external pins IM [3:0] as shown in the following table.

Table 6

IM3	IM2	IM1	IM0	MCU-Interface Mode	Pins in use	
					Register/Content	GRAM
0	0	0	0	8080 MCU 8-bit bus interface I	D[7:0]	D[7:0],WRX,RDX,CSX,D/CX
0	0	0	1	8080 MCU 16-bit bus interface I	D[7:0]	D[15:0],WRX,RDX,CSX,D/CX
0	0	1	0	8080 MCU 9-bit bus interface I	D[7:0]	D[8:0],WRX,RDX,CSX,D/CX
0	0	1	1	8080 MCU 18-bit bus interface I	D[7:0]	D[17:0],WRX,RDX,CSX,D/CX
0	1	0	1	3-wire 9-bit data serial interface I	SCL,SDA,CSX	
				2 data lane serial interface I	SCL,SDA,CSX,D/CX	
0	1	1	0	4-wire 8-bit data serial interface I	SCL,SDA,D/CX,CSX	
1	0	0	0	8080 MCU 16-bit bus interface II	D[8:1]	D[17:10],D[8:1],WRX,RDX,CSX,D/CX
1	0	0	1	8080 MCU 8-bit bus interface II	D[17:10]	D[17:10],WRX,RDX,CSX,D/CX
1	0	1	0	8080 MCU 18-bit bus interface II	D[8:1]	D[17:0],WRX,RDX,CSX,D/CX
1	0	1	1	8080 MCU 9-bit bus interface II	D[17:10]	D[17:9],WRX,RDX,CSX,D/CX
1	1	0	1	3-wire 9-bit data serial interface II	SCL,SDI,SDO,CSX	
1	1	1	0	4-wire 8-bit data serial interface II	SCL,SDI,SDO,D/CX,CSX	

4.1.2. 8080-I Series Parallel Interface

GC9309NA can be accessed via 8-/9-/16-/18-bit MCU 8080-I series parallel interface. The chip select CSX (active low) is used to enable or disable GC9309NA chip. The RESX (active low) is an external reset signal. WRX is the parallel data write strobe, RDX is the parallel data read strobe and D[17:0] is parallel data bus.

GC9309NA latches the input data at the rising edge of WRX signal. The D/CX is the signal of data/command selection. When D/CX='1', D [17:0] bits are display RAM data or command's parameters. When D/CX='0', D[17:0] bits are commands.

The 8080-I series bi-directional interface can be used for communication between the MCU controller and LCD driver chip. The 8080-I Interface selection is done when IM3 pin is low state (VSSC level). Interface bus width can be selected by IM [2:0] bits.
The selection of 8080-I series parallel interface is shown as the table in the following.

Table 7

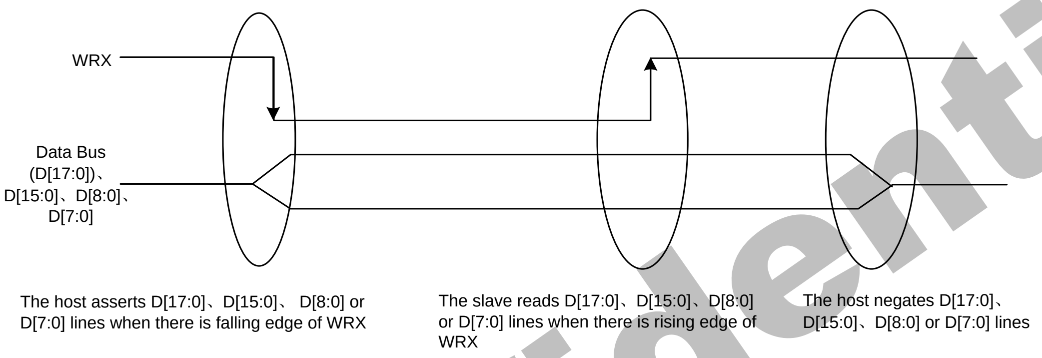
IM 3	IM 2	IM 1	IM 0	MCU-Interface	CSX	WRX	RDX	D/CX	Function
0	0	0	0	8080 MCU 8-bit bus interface I	"L"	↑	"H"	"L"	Write command code.
					"L"	"H"	↑	"H"	Read internal status.
					"L"	↑	"H"	"H"	Write parameter or display data.
					"L"	"H"	↑	"H"	Reads parameter or display data.
0	0	0	1	8080 MCU 16-bit bus interface I	"L"	↑	"H"	"L"	Write command code.
					"L"	"H"	↑	"H"	Read internal status.
					"L"	↑	"H"	"H"	Write parameter or display data.
					"L"	"H"	↑	"H"	Reads parameter or display data.
0	0	1	0	8080 MCU 9-bit bus interface I	"L"	↑	"H"	"L"	Write command code.
					"L"	"H"	↑	"H"	Read internal status.
					"L"	↑	"H"	"H"	Write parameter or display data.
					"L"	"H"	↑	"H"	Reads parameter or display data.
0	0	1	1	8080 MCU 18-bit bus interface I	"L"	↑	"H"	"L"	Write command code.
					"L"	"H"	↑	"H"	Read internal status.
					"L"	↑	"H"	"H"	Write parameter or display data.
					"L"	"H"	↑	"H"	Reads parameter or display data.

4.1.3. Write Cycle Sequence

The WRX signal is driven from high to low and then be pulled back to high during the write cycle. The host processor provides information during the write cycle when the display module captures the information from host processor on the rising edge of WRX. When the D/CX signal is driven to low level, then input data on the interface is interpreted as command information. The D/CX signal also can be pulled high level when the data on the interface is SRAM data or command's parameter.

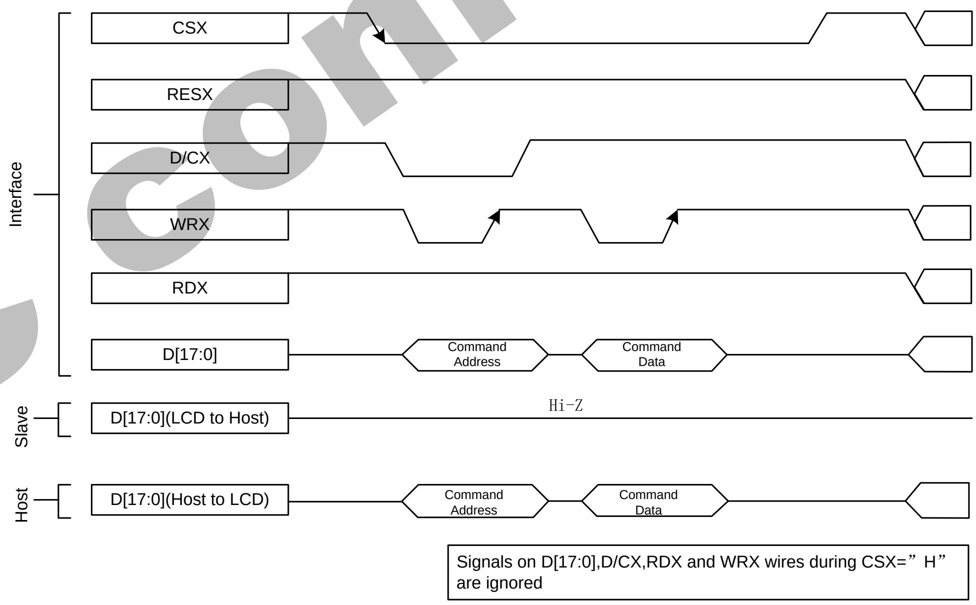
The following figure shows a write cycle for the 8080-I MCU interface.

Figure 2.



Note: WRX is an unsynchronized signal (It can be stopped)

Figure 3.

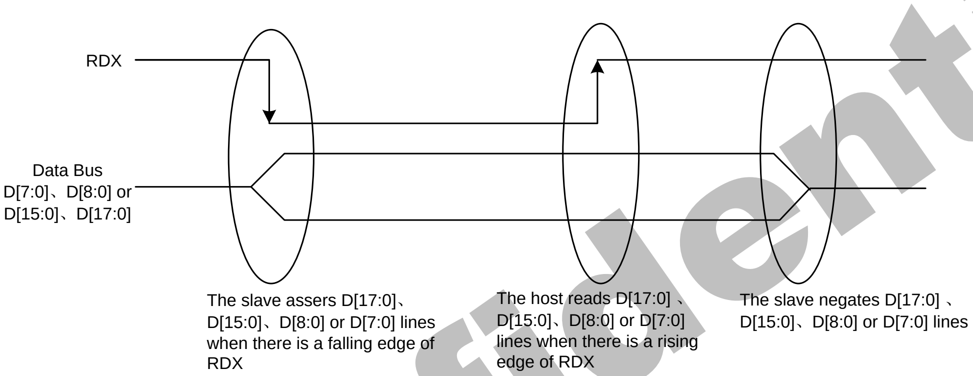


4.1.4. Read Cycle Sequence

The RDX signal is driven from high to low and then allowed to be pulled back to high during the read cycle. The display module provides information to the host processor during the read cycle, while the host processor reads the display module information on the rising edge of RDX signal. When the D/CX signal is driven to low level, then input data on the interface is interpreted as command. The D/CX signal also can be pulled high level when the data on the interface is RAM data or command parameter.

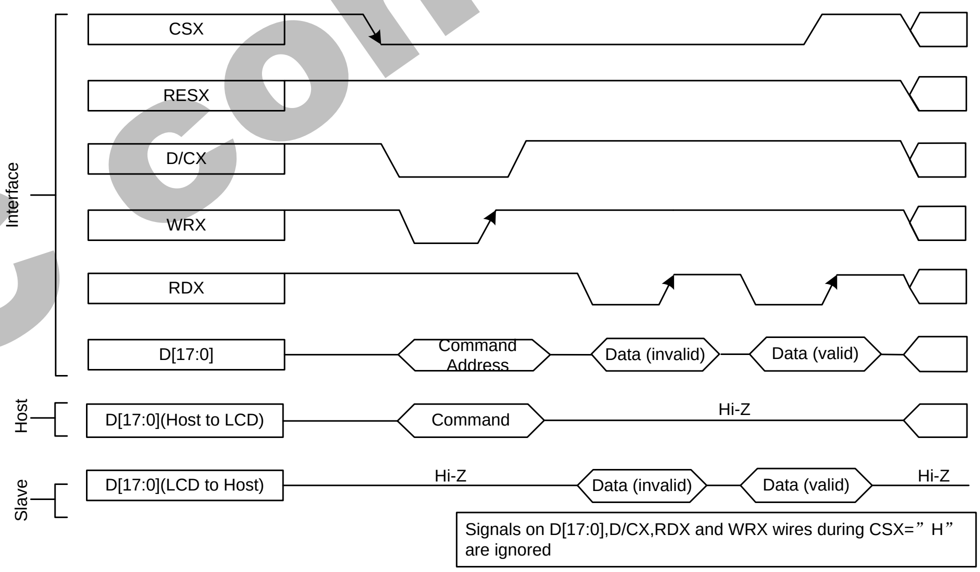
The following figure shows the read cycle for the 8080-I MCU interface.

Figure 4.



Note: RDX is an unsynchronized signal (It can be stopped).

Figure 5.



Note: Read data is only valid when the D/CX input is pulled high. If D/CX is driven low during read then the display information outputs will be High-Z.

4.1.5. 8080-Ⅱ Series Parallel Interface

GC9309NA can be accessed via 8-/9-/16-/18-bit MCU 8080-Ⅱ series parallel interface. The chip select CSX (active low) is used to enable or disable GC9309NA chip. The RESX (active low) is an external reset signal. WRX is the parallel data write strobe, RDX is the parallel data read strobe and D[17:0] is parallel data bus.

GC9309NA latches the input data at the rising edge of WRX signal. The D/CX is the signal of data/command selection. When D/CX='1', D [17:0] bits are display RAM data or command's parameters. When D/CX='0', D[17:0] bits are commands.

The 8080-II series bi-directional interface can be used for communication between the MCU controller and LCD driver chip. The 8080-II Interface selection is done when IM3 pin is high state (IOVCC level). Interface bus width can be selected by IM [2:0] bits.

The selection of 8080-II series parallel interface is shown as the table in the following.

Table 8

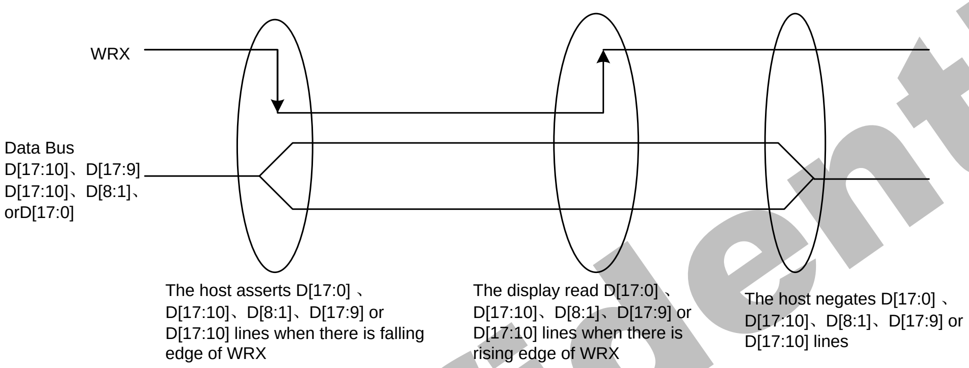
IM3	IM2	IM1	IM0	MCU-Interface	CSX	WRX	RDX	D/CX	Function
1	0	0	0	8080 MCU 16-bit bus interface II	"L"	↑	"H"	"L"	Write command code.
					"L"	"H"	↑	"H"	Read internal status.
					"L"	↑	"H"	"H"	Write parameter or display data.
					"L"	"H"	↑	"H"	Reads parameter or display data.
1	0	0	1	8080 MCU 8-bit bus interface II	"L"	↑	"H"	"L"	Write command code.
					"L"	"H"	↑	"H"	Read internal status.
					"L"	↑	"H"	"H"	Write parameter or display data.
					"L"	"H"	↑	"H"	Reads parameter or display data.
1	0	1	0	8080 MCU 18-bit bus interface II	"L"	↑	"H"	"L"	Write command code.
					"L"	"H"	↑	"H"	Read internal status.
					"L"	↑	"H"	"H"	Write parameter or display data.
					"L"	"H"	↑	"H"	Reads parameter or display data.
1	0	1	1	8080 MCU 9-bit bus interface II	"L"	↑	"H"	"L"	Write command code.
					"L"	"H"	↑	"H"	Read internal status.
					"L"	↑	"H"	"H"	Write parameter or display data.
					"L"	"H"	↑	"H"	Reads parameter or display data.

4.1.6. Write Cycle Sequence

The WRX signal is driven from high to low and then be pulled back to high during the write cycle. The host processor provides information during the write cycle when the display module captures the information from host processor on the rising edge of WRX. When the D/CX signal is driven to low level, then input data on the interface is interpreted as command information. The D/CX signal also can be pulled high level when the data on the interface is RAM data or command's parameter.

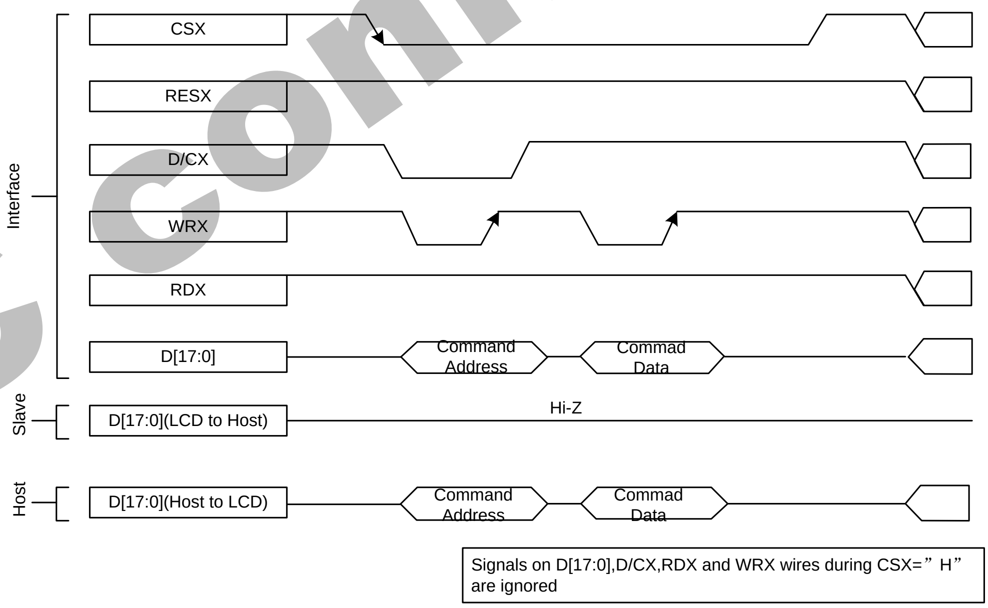
The following figure shows a write cycle for the 8080-II MCU interface.

Figure 6.



Note: WRX is an unsynchronized signal (It can be stopped)

Figure 7.

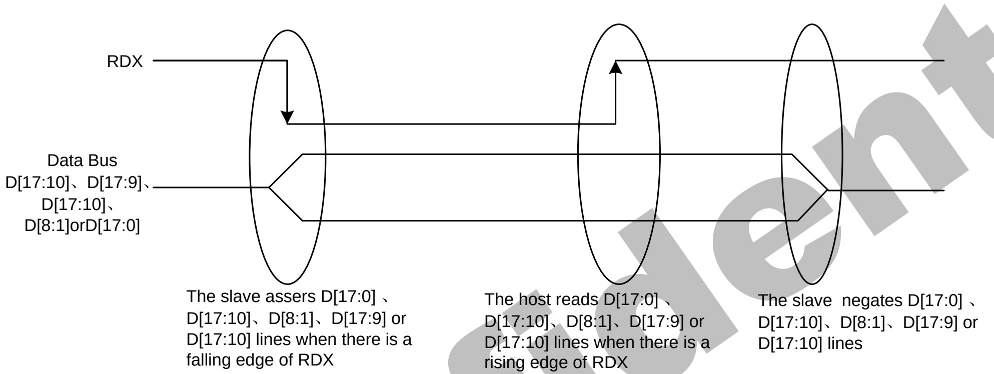


4.1.7. Read Cycle Sequence

The RDX signal is driven from high to low and then allowed to be pulled back to high during the read cycle. The display module provides information to the host processor during the read cycle while the host processor reads the display module information on the rising edge of RDX signal. When the D/CX signal is driven to low level, then input data on the interface is interpreted as command. The D/CX signal also can be pulled high level when the data on the interface is RAM data or command parameter.

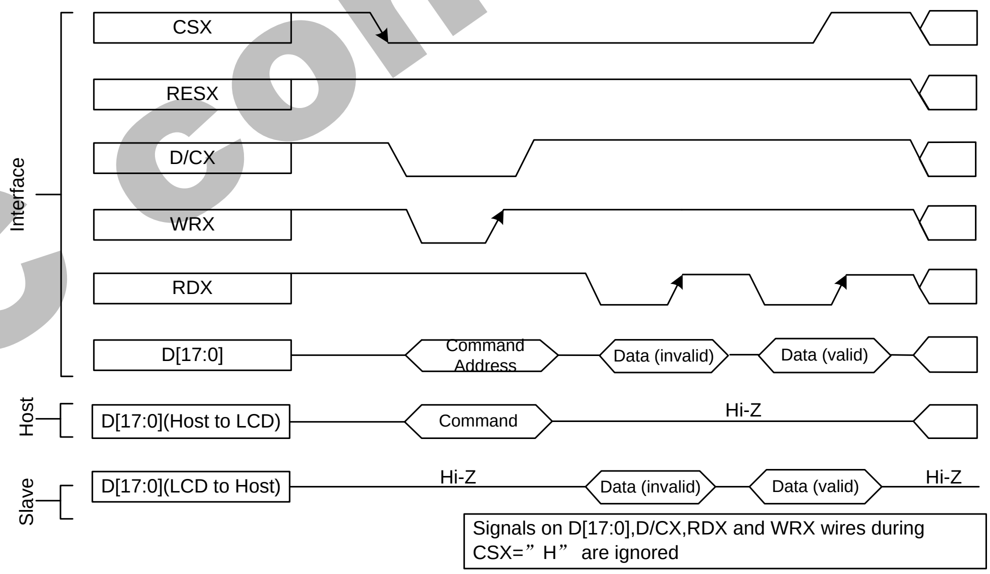
The following figure shows the read cycle for the 8080-II MCU interface.

Figure 8.



Note: RDX is an unsynchronized signal (It can be stopped).

Figure 9.



Note: Read data is only valid when the D/CX input is pulled high. If D/CX is driven low during read then the display information outputs will be High-Z.

4.1.8. Serial Interface

The selection of interface is done by IM [3:0] bits. Please refer to the Table in the following.

Table 8.

IM 3	IM 2	IM 1	IM 0	MCU-Interface Mode	CSX	D/CX	SCL	Function
0	1	0	1	3-line serial interface	“L”	-	↑	Read/Write command, parameter or display data.
0	1	1	0	4-line serial interface	“L”	"H/L"	↑	Read/Write command, parameter or display data.
1	1	0	1	3-line serial interface	“L”	-	↑	Read/Write command, parameter or display data.
1	1	1	0	4-line serial interface	“L”	"H/L"	↑	Read/Write command, parameter or display data.

GC9309NA supplies 3-lines/ 9-bit and 4-line/8-bit bi-directional serial interfaces for communication between host and GC9309NA. The 3-line serial mode consists of the chip enable input (CSX), the serial clock input (SCL) and serial data Input/Output (SDA or SDI/SDO). The 4-line serial mode consists of the Data/ Command selection input (D/CX), chip enable input (CSX), the serial clock input (SCL) and serial data Input/Output (SDA or SDI/SDO) for data transmission. The data bus (D [17:0]), which are not used, must be connected to GND. Serial clock (SCL) is used for interface with MCU only, so it can be stopped when no communication is necessary.

4.1.9. Write Cycle Sequence

The write mode of the interface means that host writes commands or data to GC9309NA. The 3-lines serial data packet contains a data/command select bit (D/CX) and a transmission byte. If the D/CX bit is “low”, the transmission byte is interpreted as a command byte. If the D/CX bit is “high”, the transmission byte is stored as the display data RAM(Memory write command),or command register as parameter.

Any instruction can be sent in any order to GC9309NA and the MSB is transmitted first. The serial interface is initialized when CSX is high status. In this state, SCL clock pulse and SDA data are no effect. A falling edge on CSX enables the serial interface and indicates the start of data transmission. See the detailed data format for 3-/4-line serial interface.

Figure 10.

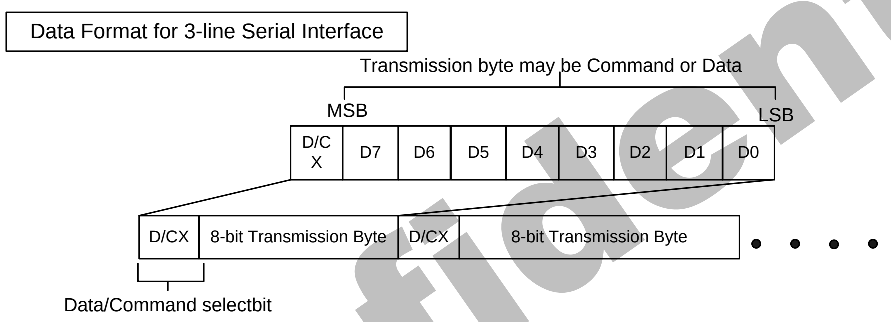
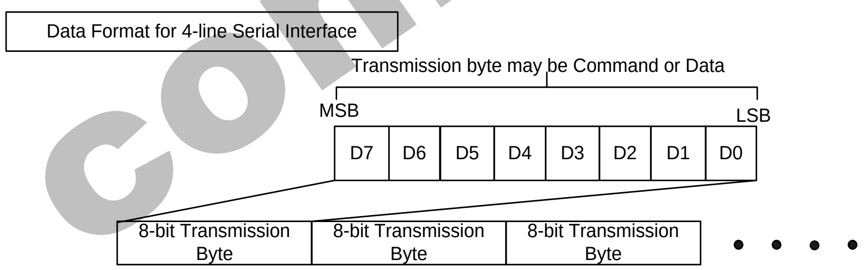


Figure11.



Host processor drives the CSX pin to low and starts by setting the D/CX bit on SDA. The bit is read by GC9309NA on the first rising edge of SCL signal. On the next falling edge of SCL, the MSB data bit (D7) is set on SDA by the host. On the next falling edge of SCL, the next bit (D6) is set on SDA. If the optional D/CX signal is used, a byte is eight read cycle width. The 3/4-line serial interface writes sequence described in the figure as below.

Figure 12.

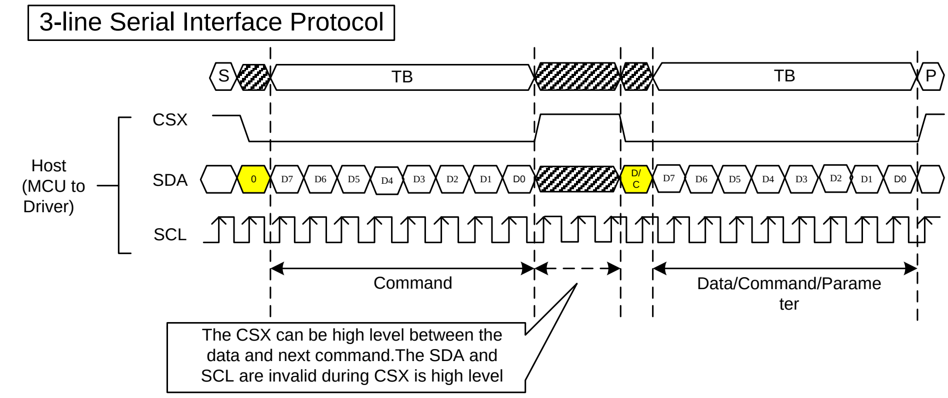
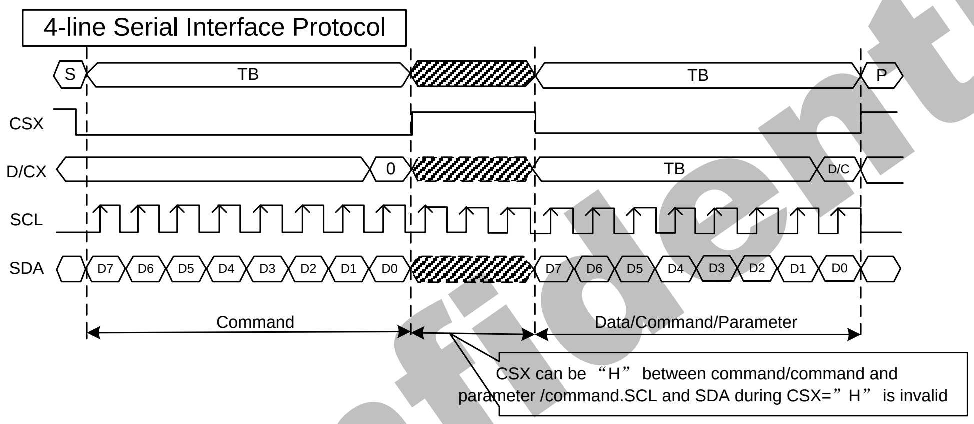


Figure 13.



4.1.10. Read Cycle Sequence

The read mode of interface means that the host reads register's parameter from GC9309NA. The host has to send a command (Read ID or register command) and then the following byte is transmitted in the opposite direction. GC9309NA latches the SDA (input data) at the rising edges of SCL (serial clock), and then shifts SDA (output data) at falling edges of SCL (serial clock). After the read status command has been sent, the SDA line must be set to tri-state and no later than at the falling edge of SCL of the last bit. The read mode has three types of transmitted command data (8-/24-/32-bit) according command code.

Figure 14.

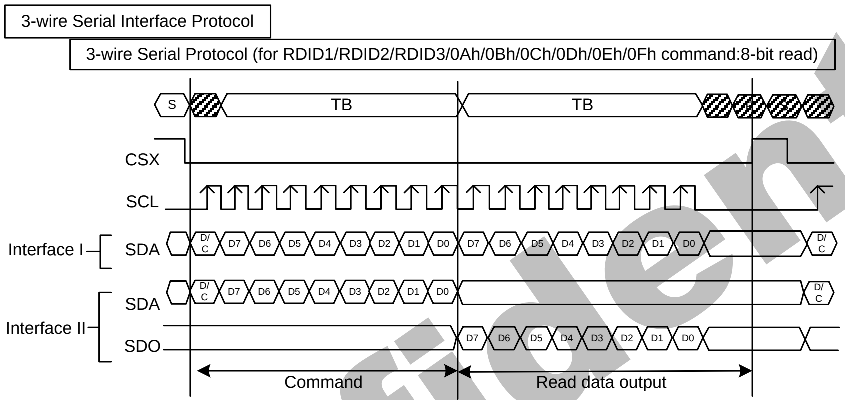


Figure 15.

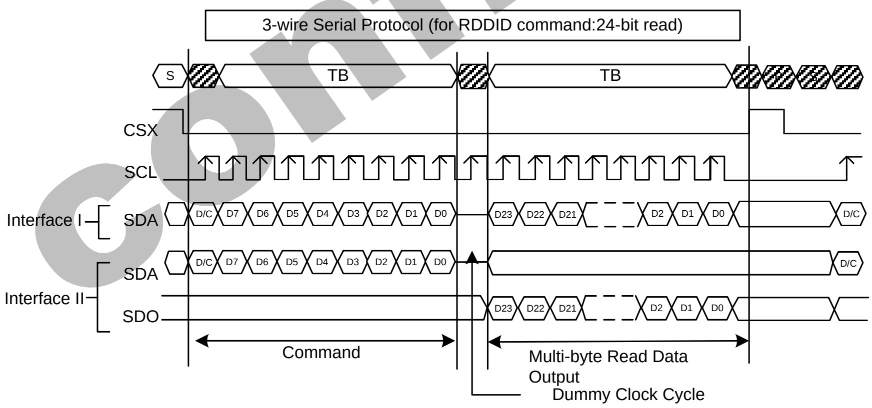


Figure 16.

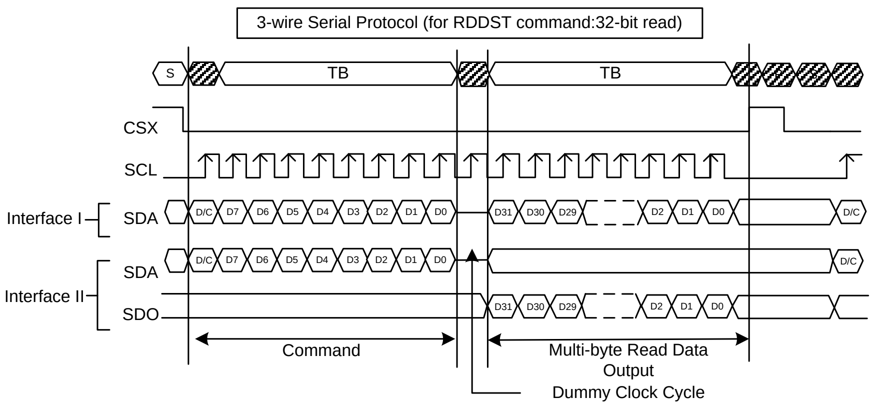


Figure 17.

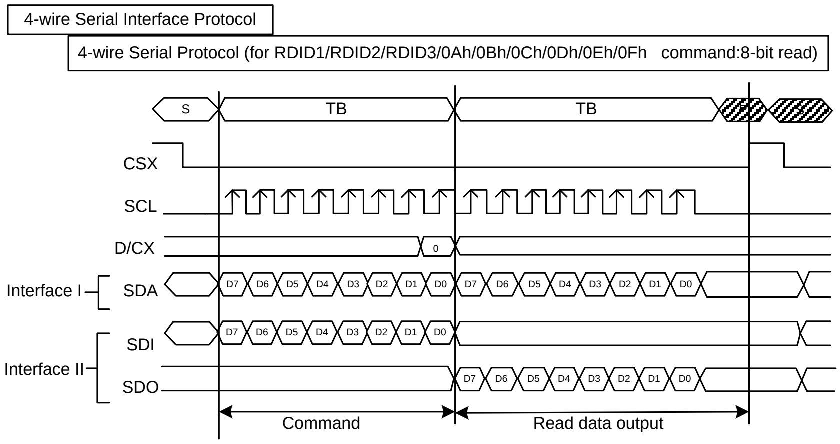


Figure 18.

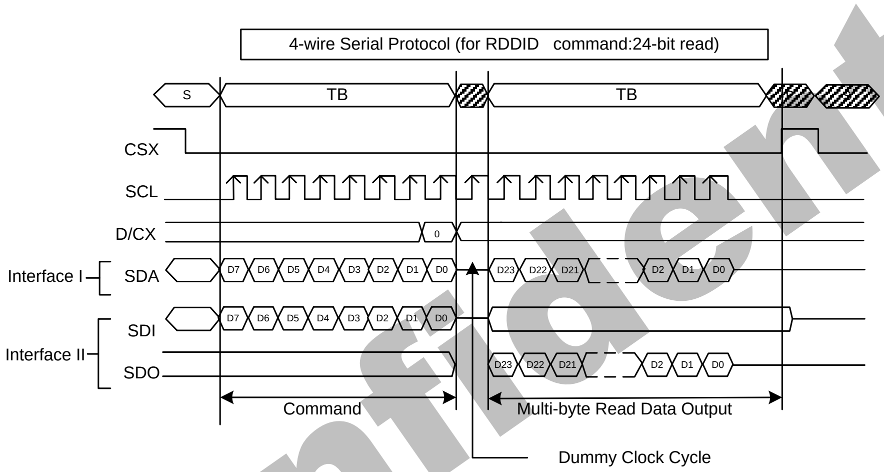
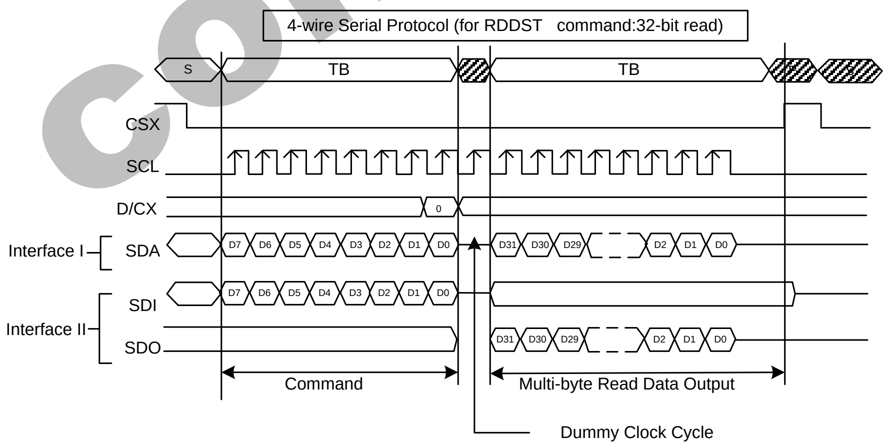


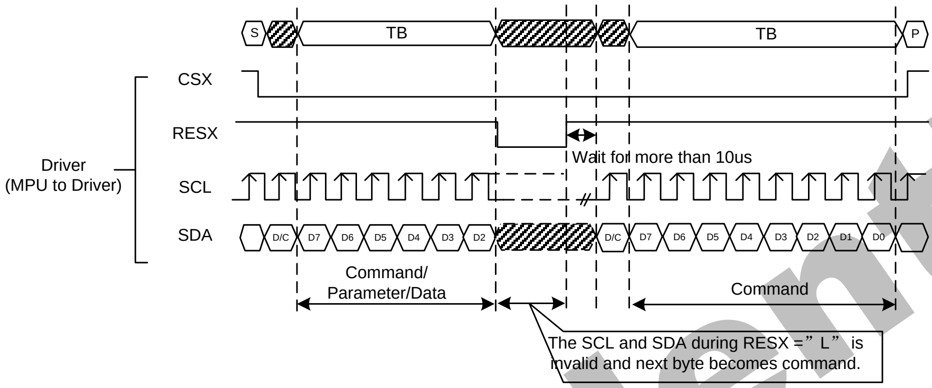
Figure 19.



4.1.11. Data Transfer Break and Recovery

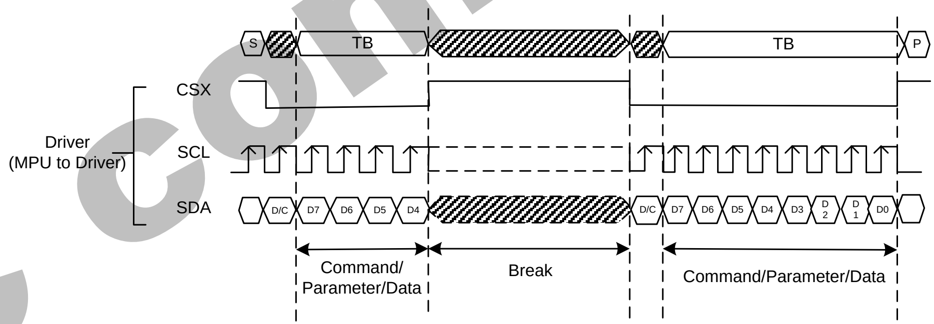
If there is a break in data transmission by RESX pulse, while transferring a command or multiple parameter command data, before Bit D0 of the byte has been completed, then the driver will reject the previous bits and have reset the interface such that it will be ready to receive command data again when the chip select pin (CSX) is activated after RESX have been high state.

Figure 20.



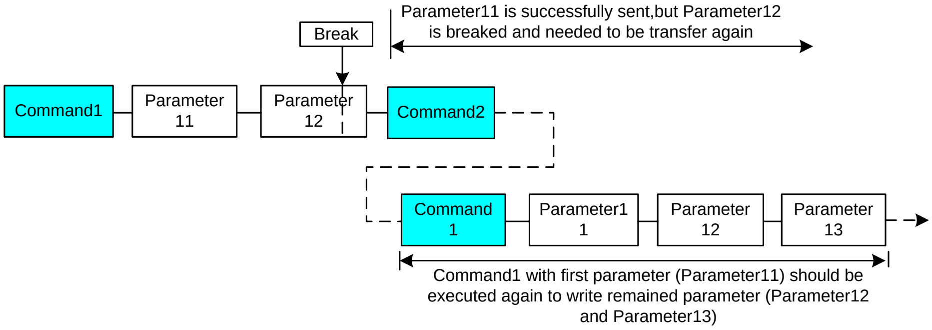
If there is a break in data transmission by CSX pulse, while transferring a command or frame memory data or multiple parameter command data, before Bit D0 of the byte has been completed, then the driver will reject the previous bits and have reset the interface such that it will be ready to receive the same byte re-transmitted when the chip select pin (CSX) is next activated.

Figure 21.



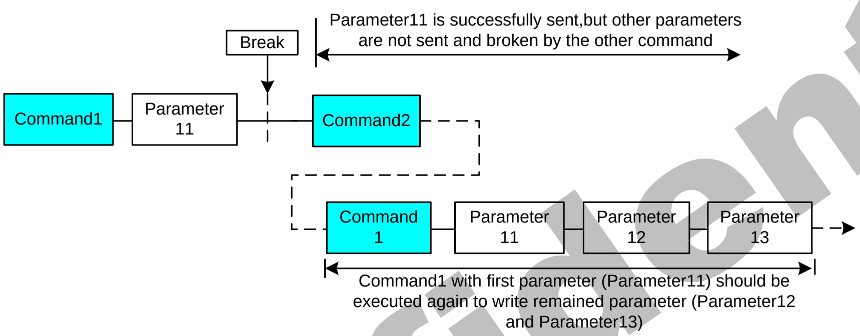
If a two or more parameter command is being sent and a break occurs while sending any parameter before the last one and if the host then sends a new command rather than continue to send the remained parameters that was interrupted, then the parameters which had been successfully sent are stored and the parameter where the break occurred is rejected. The interface is ready to receive next byte as shown below.

Figure 22.



If a two or more parameter command is being sent and a break occurs by the other command before the last one is sent, then the parameters which had been successfully sent are stored and the other parameter of that command remains previous value.

Figure 23.

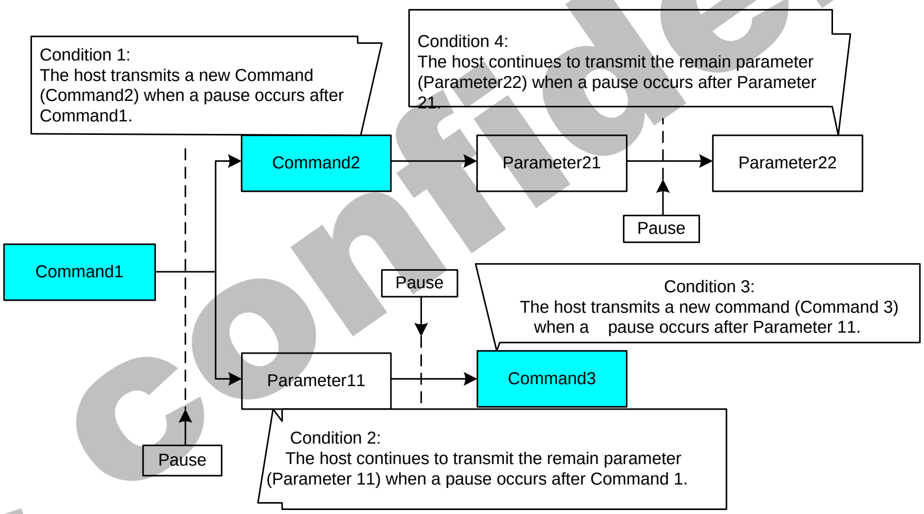


4.1.12. Data Transfer Pause

It will be possible when transferring a command, frame memory data or multiple parameter data to invoke a pause in the data transmission. If the chip select pin (CSX) is released to high state after a whole byte of a frame memory data or multiple parameter data has been completed, then GC9309NA will wait and continue the frame memory data or parameter data transmission from the point where it was paused. If the chip select pin is released after a whole byte of a command has been completed, then the display module will receive either the command's parameters(if appropriate) or a new command when the chip select pin is next enabled as shown below.

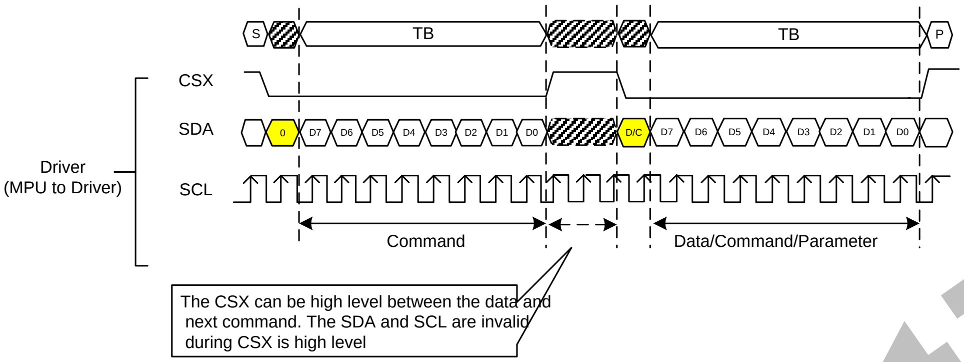
This applies to the following 4 conditions:
1) Command-Pause-Command
2) Command-Pause-Parameter
3) Parameter-Pause-Command
4) Parameter-Pause-Parameter

Figure 24.



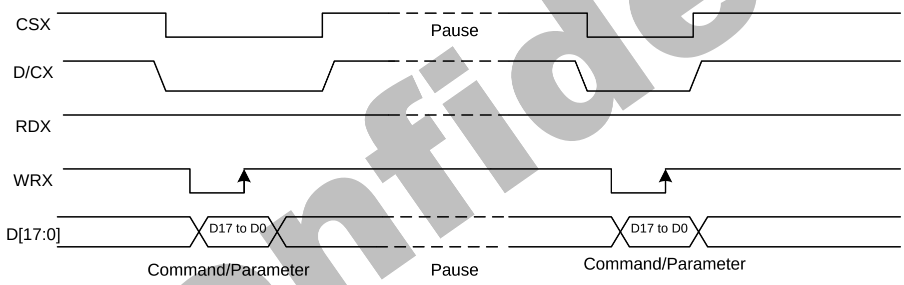
4.1.13. Serial Interface Pause (3_wire)

Figure 25.



4.1.14. Parallel Interface Pause

Figure 26.



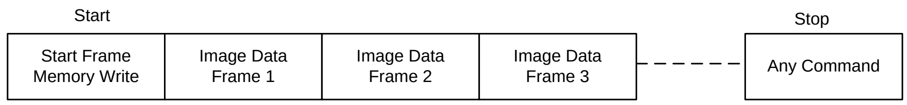
4.1.15. Data Transfer Mode

GC9309NA can provide two different kinds of color depth (16-bit/pixel and 18-bit/pixel) display data to the graphic RAM. The data format is described for each interface. Data can be downloaded to the frame memory by 2 methods.

4.1.16. Data Transfer Method 1

The image data is sent to the frame memory in the successive frame writing, each time the frame memory is filled by image data, the frame memory pointer is reset to the start point and the next frame is written.

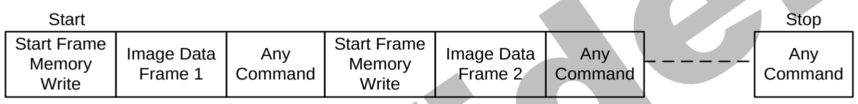
Figure 27.



4.1.17. Data Transfer Method 2

Image data is sent and at the end of each frame memory download, a command is sent to stop frame memory writing. Then start memory write command is sent, and a new frame is downloaded.

Figure 28.

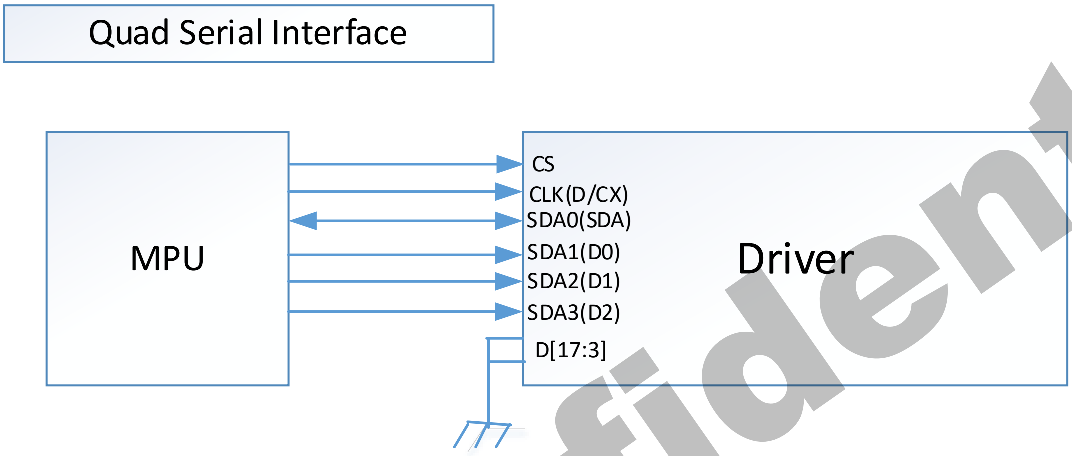


Note 1: These methods are applied to all data transfer color modes on both serial and parallel interfaces.

Note 2: The frame memory can contain both odd and even number of pixels for both methods. Only complete pixel data will be stored in the frame memory.

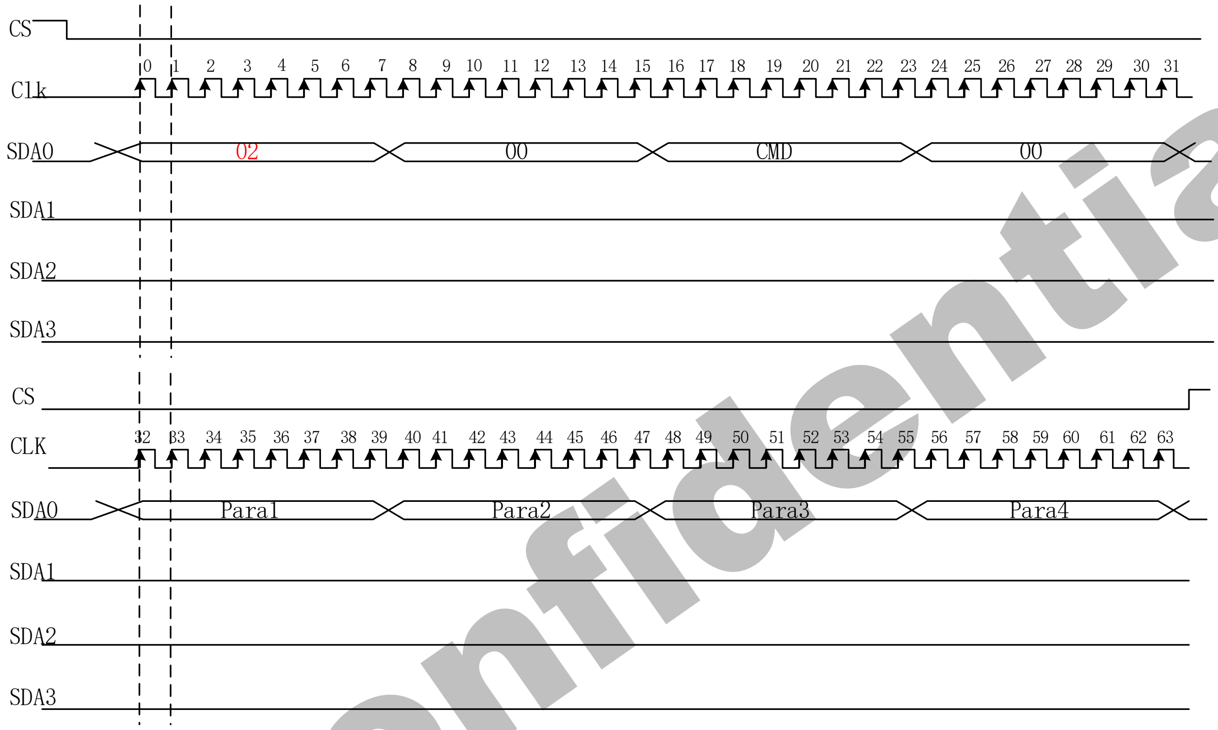
4.1.18. Quad Serial Peripheral Interface

The Quad Serial Peripheral Interface of GC9309NA can be selected by setting hardware pin IM [3:0] to “0100 or 0111”.
The following shown figure is the example of interface with Quad Serial Peripheral Interface.

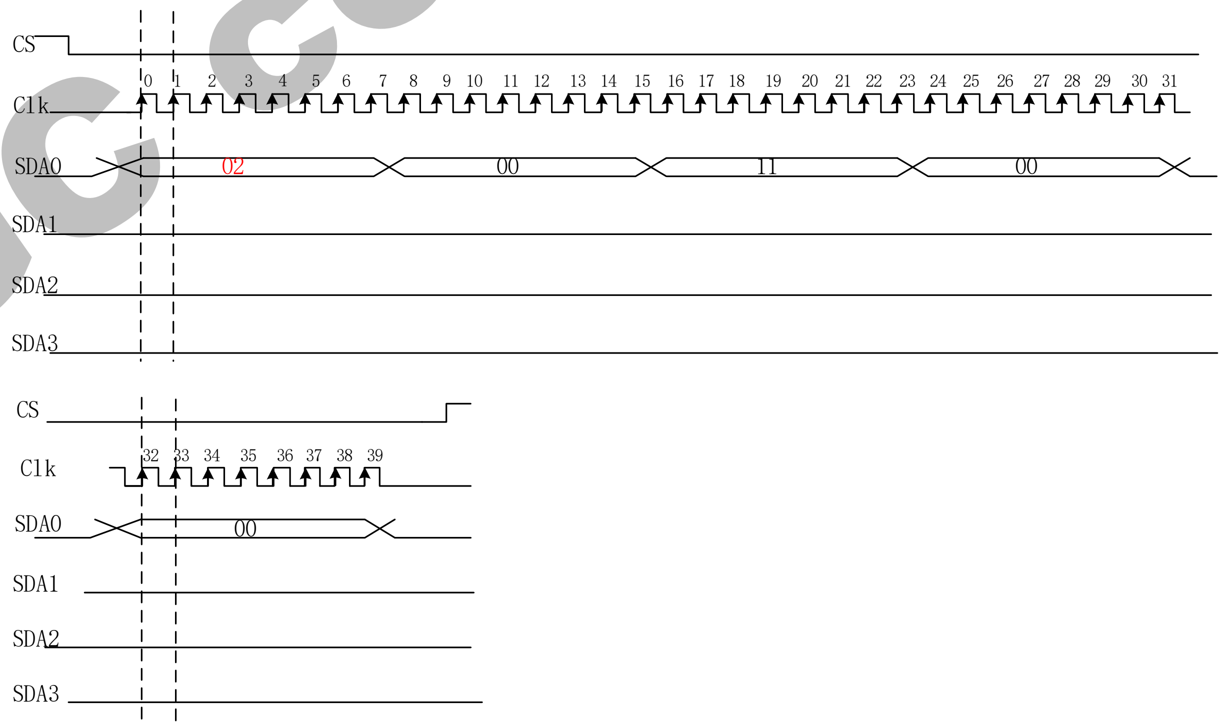


4.1.19. Write Cycle Sequence

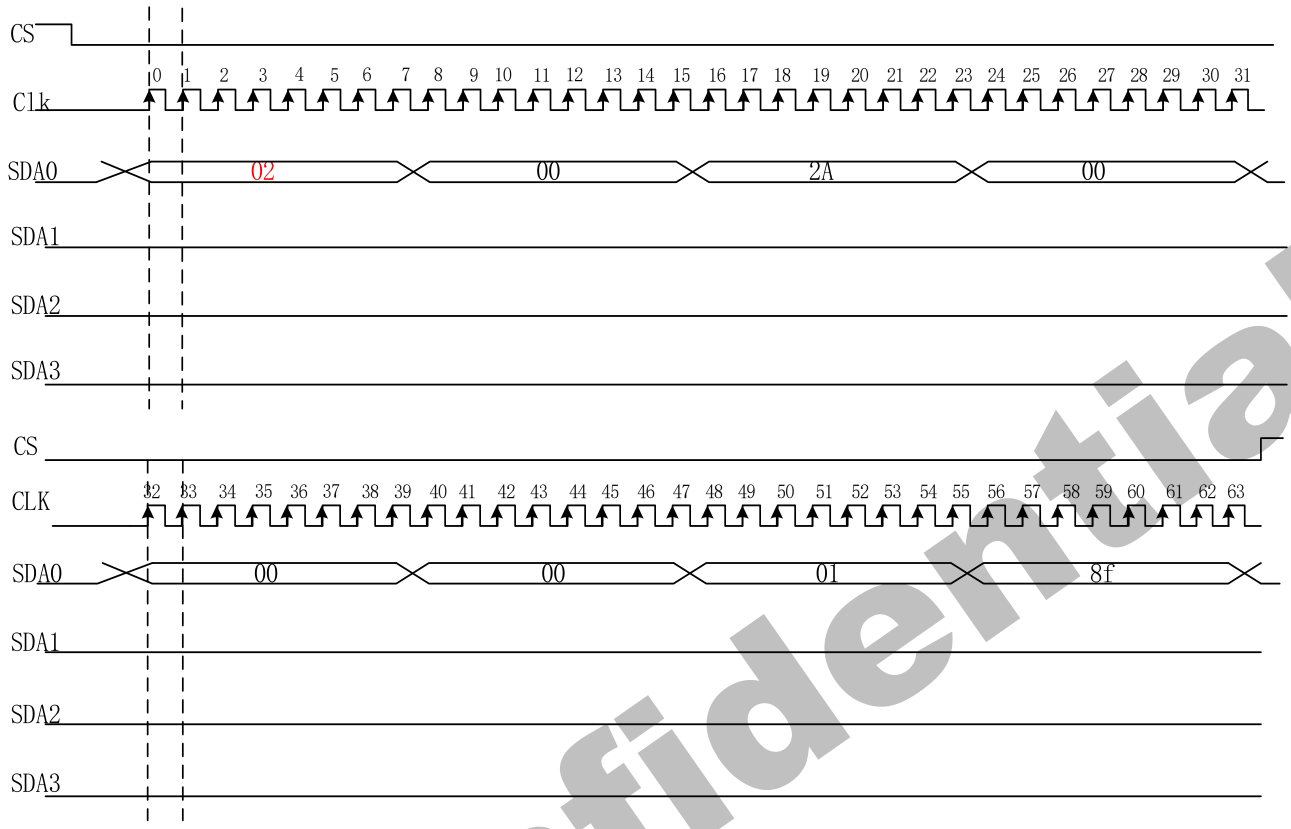
The GC9309NA reads the data at the rising edge of SCL signal.
The timing of Write Cycle Sequence is shown as below
CMD_WR only use SDA0, first byte=0x02



Eg:11h



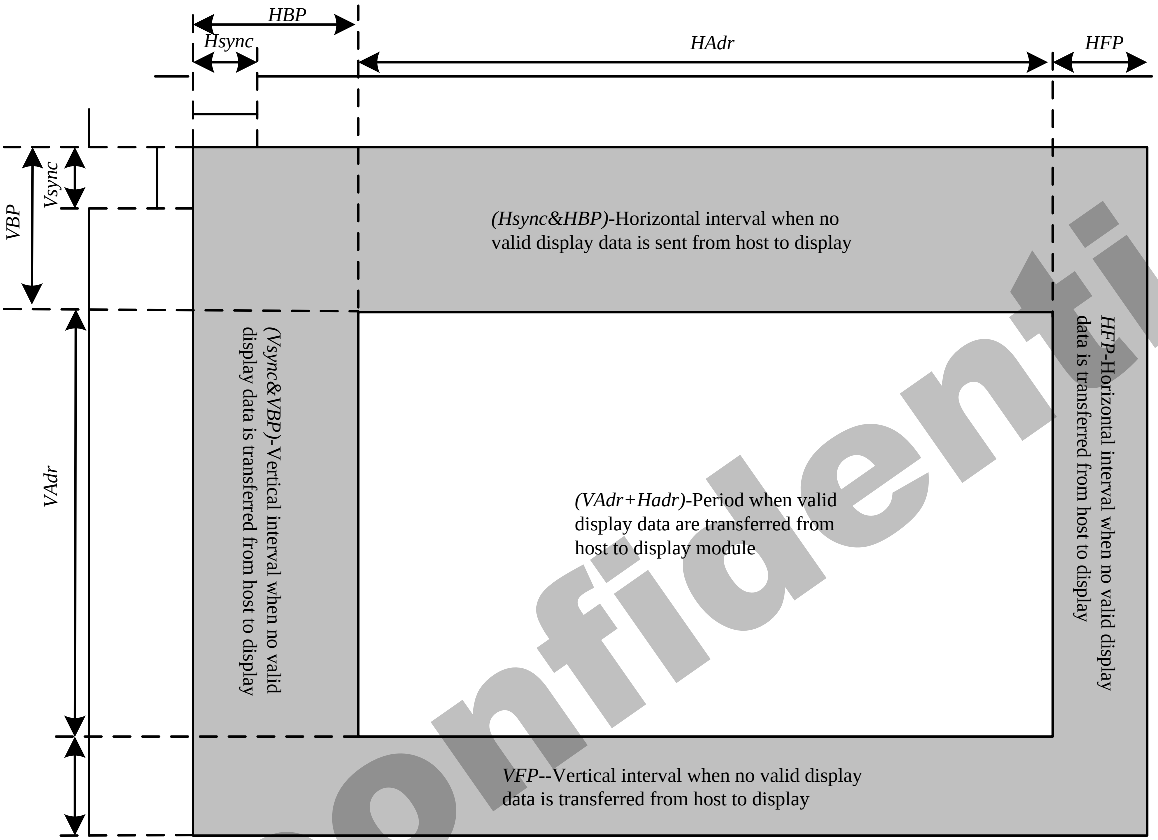
Eg:2A=00,00,01,8f



4.1.20. Read Cycle Sequence

The timing of Read Cycle Sequence is shown as below

CMD_RD: Only Use SDA0, First Byte=0x03



4.2. RGB Interface

4.2.1. RGB Interface Selection

GC9309NA has two kinds of RGB interface and these interfaces can be selected by RCM [1:0] bits. When RCM [1:0] bits are set to “10”, the DE mode is selected which utilizes VSYNC, HSYNC, DOTCLK, DE, D [17:0] pins; when RCM [1:0] bits are set to “11”, the SYNC mode is selected which utilizes which utilizes VSYNC, HSYNC,DOTCLK, D [17:0] pins. Using RGB interface must selection serial interface.
GC9309NA supports several pixel formats that can be selected by RIM bit of BAh command. The selection of a given interfaces is done by setting RCM [1:0] as show in the following table.

Table 9

RCM[1:0]		RIM	DPI[1:0]			RGB interface Mode	RGB Mode	Used Pins
1	0	0	1	1	0	18-bit RGB interface (262K colors)	DE Mode Valid data is determined by the DE signal	VSYNC,HSYNC,DE,DOTCLK,D[17:0]
1	0	0	1	0	1	16-bit RGB interface (65K colors)		VSYNC,HSYNC,DE,DOTCLK,D[17:13] & D[11:1]
1	0	1	-			6-bit RGB interface (262K colors)		VSYNC,HSYNC,DE,DOTCLK,D[5:0]
1	1	0	1	1	0	18-bit RGB interface (262K colors)	SYNC Mode In SYNC mode, DE signal is ignored;blanking porch is determined by B5h command	VSYNC,HSYNC,DOTCLK, D[17:0]
1	1	0	1	0	1	16-bit RGB interface (65K colors)		VSYNC,HSYNC,DOTCLK, D[17:13] & D[11:1]
1	1	1	-			6-bit RGB interface (262K colors)		VSYNC,HSYNC,DOTCLK, D[5:0]

18-bit data bus interface (D[17:0] is used) , RIM=0

Figure 29.

	D17	D16	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
18bpp Frame Memory Write	R[5]	R[4]	R[3]	R[2]	R[1]	R[0]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	B[5]	B[4]	B[3]	B[2]	B[1]	B[0]

16-bit data bus interface (D[17:13] & D[11:1] is used) , DPI[2:0] = 101, and RIM=0

Figure 30.

	D17	D16	D15	D14	D13	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1
16bpp Frame Memory Write	R[4]	R[3]	R[2]	R[1]	R[0]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	B[4]	B[3]	B[2]	B[1]	B[0]

The LSB data of red/blue color are same as MSB data.

6-bit data bus interface (D[5:0] is used) , RIM=1

Figure 31.

	D5	D4	D3	D2	D1	D0	D5	D4	D3	D2	D1	D0	D5	D4	D3	D2	D1	D0
18bpp Frame Memory Write	R[5]	R[4]	R[3]	R[2]	R[1]	R[0]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	B[5]	B[4]	B[3]	B[2]	B[1]	B[0]

Pixel clock (DOTCLK) is running all the time without stopping and used to enter VSYNC, HSYNC, DE and D[17:0] states when there is a rising edge of the DOTCLK. Vertical synchronization (VSYNC) is used to tell when there is received a new frame of the display. This is low enable and its state is read to the display module by a rising edge of the DOTCLK signal.

Horizontal synchronization (HSYNC) is used to tell when there is received a new line of the frame. This is low enable and its state is read to the display module by a rising edge of the DOTCLK signal. In DE mode, Data Enable (DE) is used to tell when there is received RGB information that should be transferred on the display. This is a high enable and its state is read to the display module by a rising edge of the DOTCLK signal. D [17:0] are used to tell what is the information of the image that is transferred on the display (When DE= '0' (low) and there is a rising edge of DOTCLK). D [17:0] can be '0' (low) or '1' (high). These lines are read by a rising edge of the DOTCLK signal. In SYNC mode, the valid display data in inputted in pixel unit via D [17:0] according to HFP/HBP settings of HSYNC signal and VFP/VBP setting of VSYNC. In both RGB interface modes, the input display data is written to GRAM first then outputs corresponding source voltage according the gray data from GRAM.

Figure32.

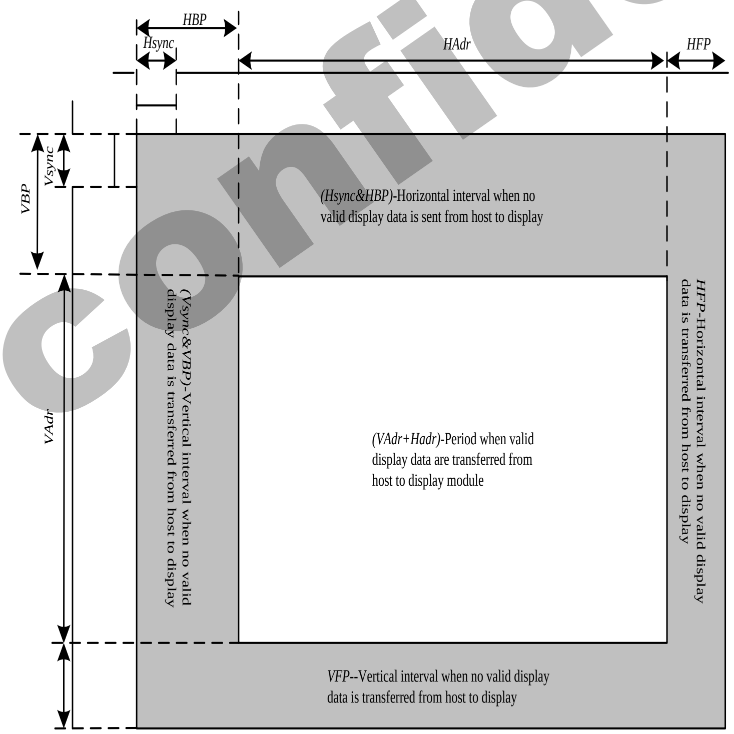


Table 10.

Parameters	Symbols	Condition	Min.	Typ.	Max.	Units
Horizontal Synchronization	Hsync		2	10	16	DOTCLK
Horizontal Back Porch	HBP		2	20	24	DOTCLK
Horizontal Address	HAdr		-	240	-	DOTCLK
Horizontal Front Porch	HFP		2	10	16	DOTCLK

Vertical Synchronization	Vsync		1	2	4	Line
Vertical Back Porch	VBP		1	2	-	Line
Vertical Address	VAdr		-	320	-	Line
Vertical Front Porch	VFP		3	4	-	Line

Notes:

1. Vertical period (one frame) shall be equal to the sum of VBP + VAdr + VFP.

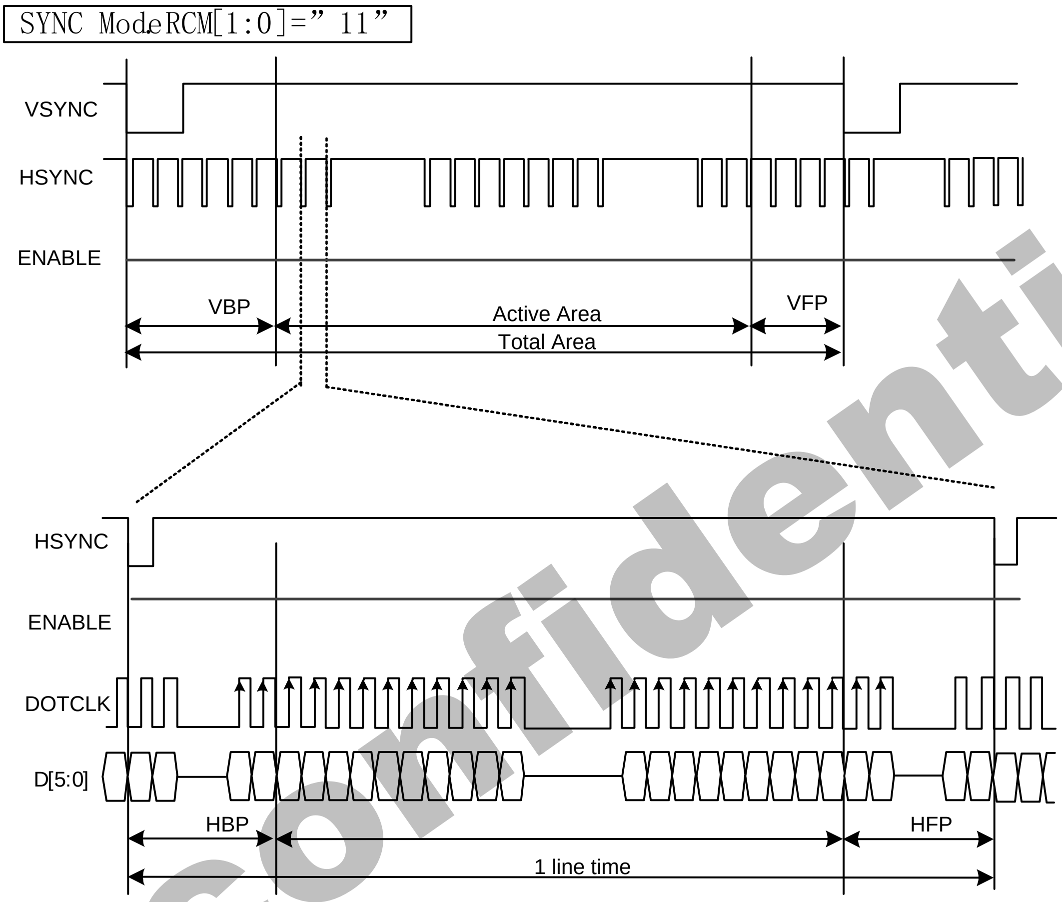
2. Horizontal period (one line) shall be equal to the sum of HBP + HAdr + HFP.

3. Control signals Hsync shall be transmitted as specified at all times while valid pixels are transferred between the host processor and the display module.

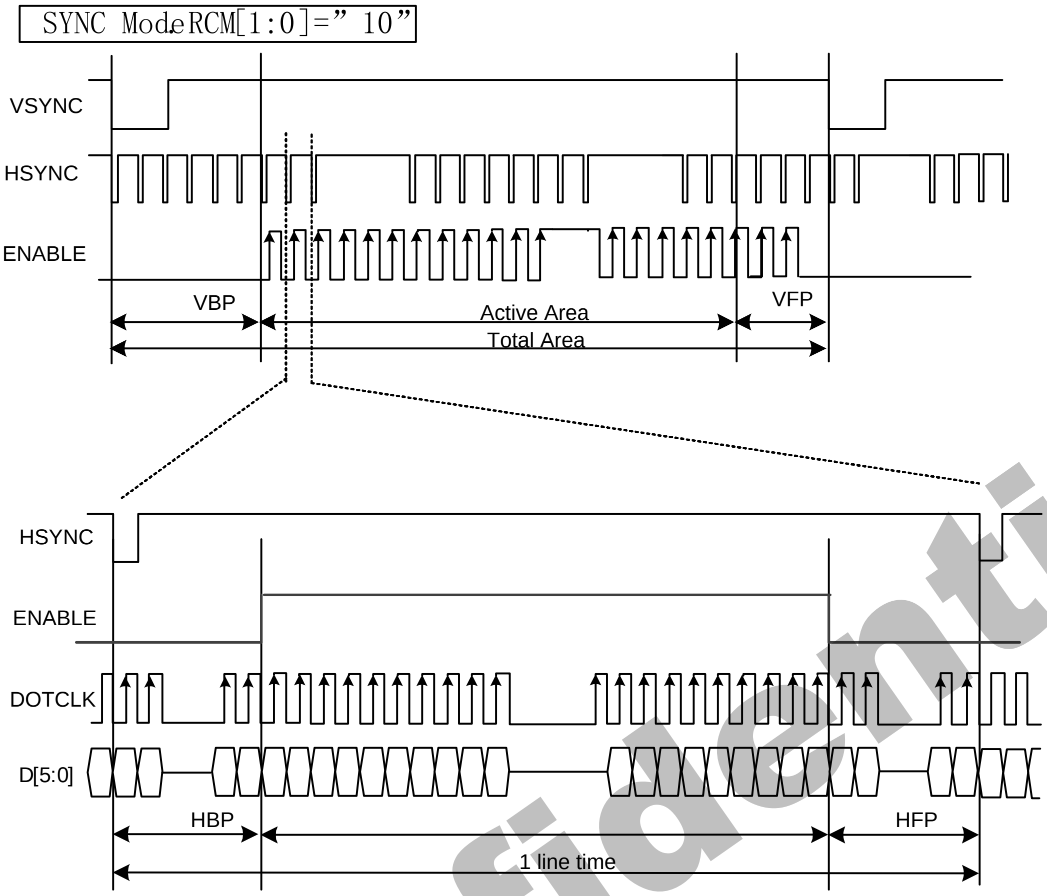
4.2.2. RGB Interface Timing

The timing chart of 18/16-bit RGB interface mode1 and mode 2 is shown as below.

Figure33.



Mode1 RGB SYNC mode



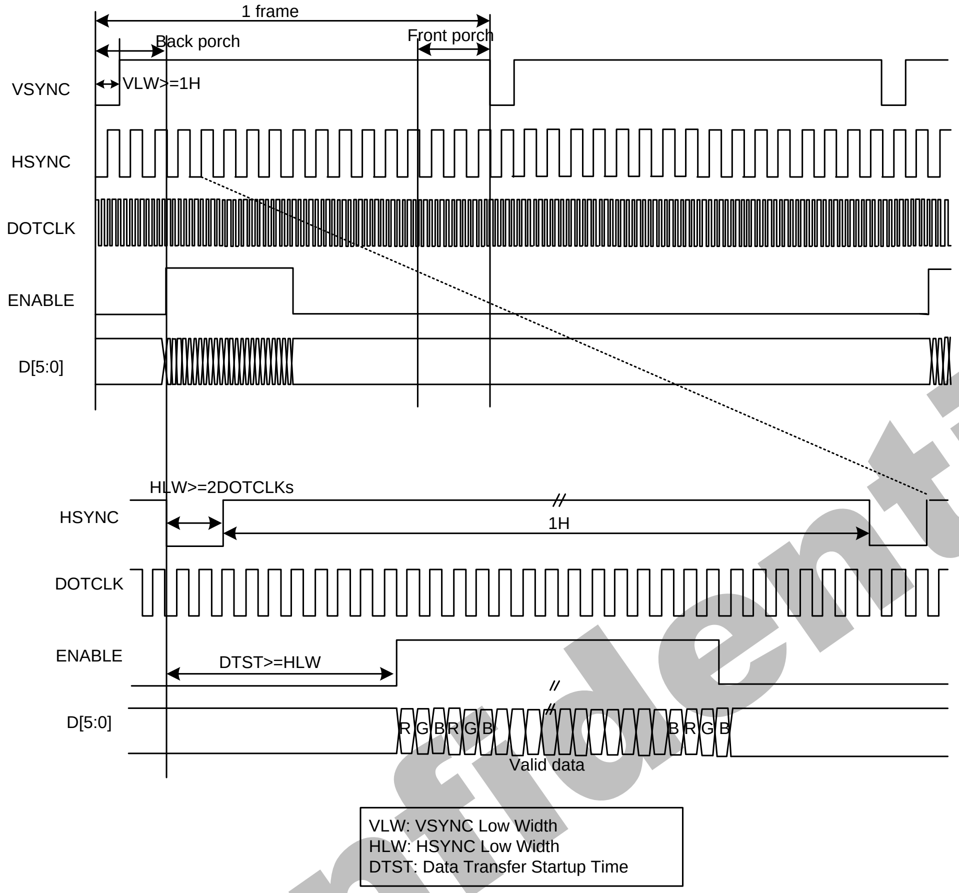
Mode2 RGB SYNC+DE mode

Note 1: The DE signal is not needed when RGB interface SYNC mode is selected.

Note 2: VSPL='0', HSPL='0', DPL='0' and EPL='0' of "Interface Mode Control (B0h)" command.

The timing chart of 6-bit RGB interface mode is shown as below:

Figure34.



Note 1: 6-bit RGB interface mode only used in the DE interface.

Note 2: VSPL='0', HSPL='0', DPL='0' and EPL='0' of "Interface Mode Control (B0h)" command.

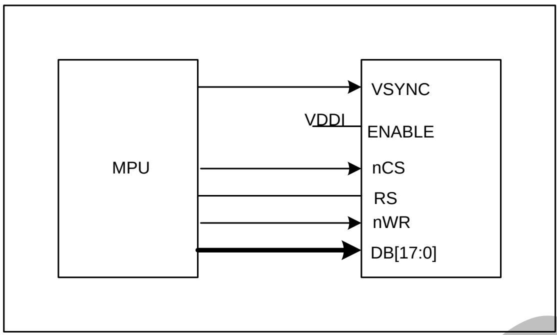
Note 3: In 6-bit RGB interface mode, each dot of one pixel (R, G and B) is transferred in synchronization with DOTCLK.

Note 4: In 6-bit RGB interface mode, set the cycles of VSYNC, HSYNC and DE to 3 multiples of DOTCLK.

4.3. VSYNC Interface

GC9309NA supports the VSYNC interface in synchronization with the frame-synchronizing signal VSYNC to display the moving picture with the 8080-Ⅰ/8080-Ⅱ system interface. When the VSYNC interface is selected to display a moving picture, the minimum GRAM update speed is limited and the VSYNC interface is enabled by setting DM[1:0] = "10" and RM = "0".

Figure35.



Note 1:In the VSYNC mode,the pin ENABLE should connect to IOVCC.

In the VSYNC mode, the display operation is synchronized with the internal clock and VSYNC input and the frame rate is determined by the pulse rate of VSYNC signal. All display data are stored in GRAM to minimize total data transfer required for moving picture display.

Figure36.

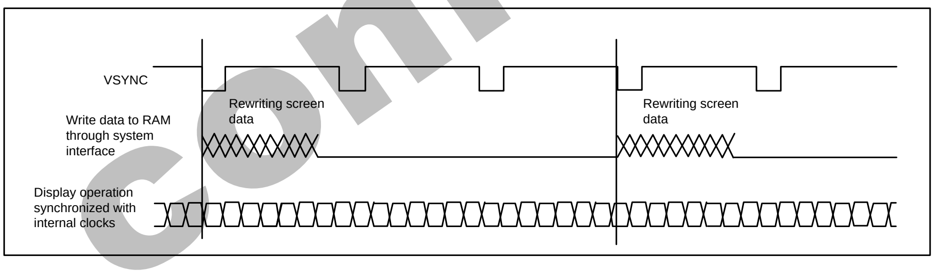
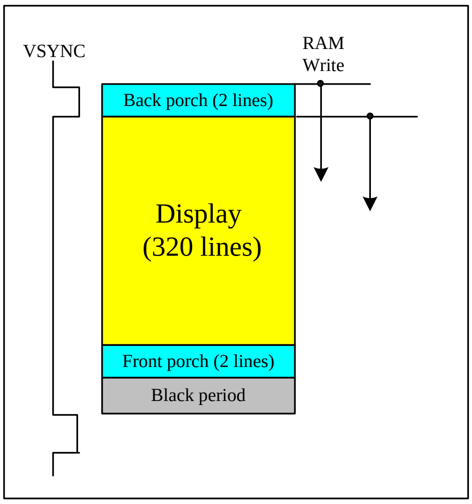


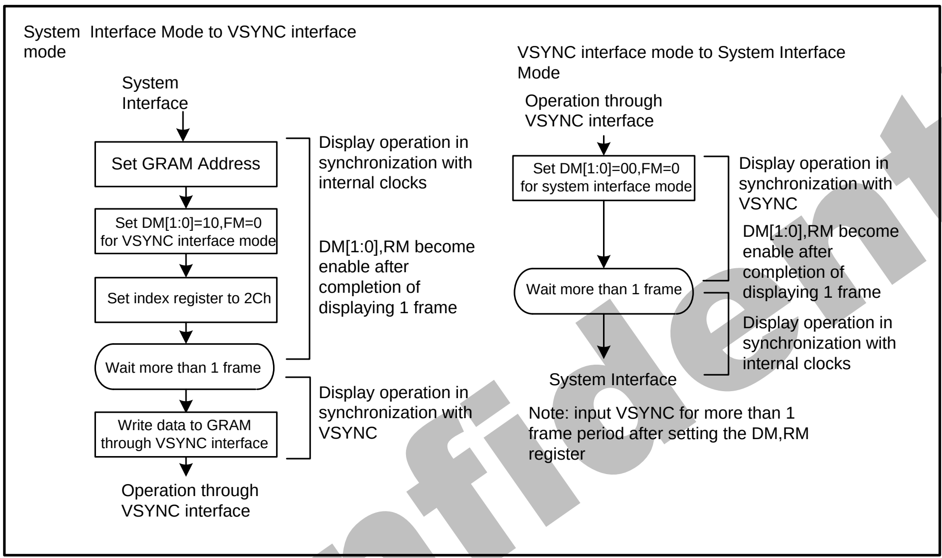
Figure37.



Notes in using the VSYNC interface

1. The minimum GRAM write speed must be satisfied and the frequency variation must be taken into consideration.
2. The display frame rate is determined by the VSYNC signal and the period of VSYNC must be longer than the scan period of an entire display.
3. When switching from the internal clock operation mode (DM[1:0] = "00") to the VSYNC interface mode or inversely, the switching starts from the next VSYNC cycle, i.e. after completing the display of the frame.
4. The partial display, vertical scroll, and interlaced scan functions are not available in VSYNC interface mode.

Figure38.



4.4. Display Data RAM (DDRAM)

GC9309NA has an integrated 240x320x18-bit graphic type static RAM. This 172,800-byte memory allows storing a 240xRGBx320 image with an 18-bit resolution (262K-color). There is no abnormal visible effect on the display when there are simultaneous panel display read and interface read/write to the same location of the frame memory.

4.5. Display Data Format

GC9309NA supplies 18-/16-/9-/8-bit parallel MCU interface with 8080-Ⅰ/8080-Ⅱ series, 3-/4-line serial interface and 6-/16-18-bit parallel RGB interface. The parallel MCU interface and serial interface mode can be selected by external pins IM [3:0] and RGB interface mode can be selected by software command parameters RCM[1:0].

4.5.1. 3-line Serial Interface

The 3-line/9-bit serial bus interface of GC9309NA can be used by setting external pin as IM [3:0] to "0101" for serial interface I or IM [3:0] to "1101" for serial interface II. The shown figure is the example of 3-line SPI interface.

Figure39.

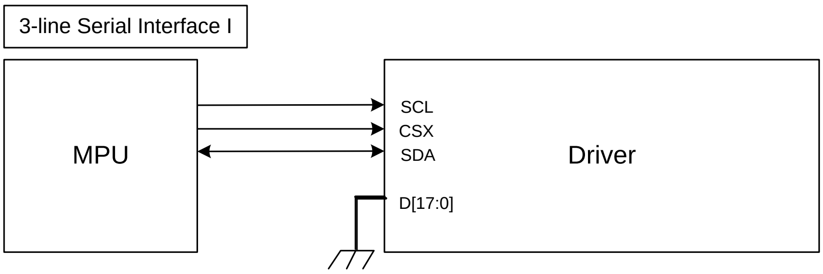
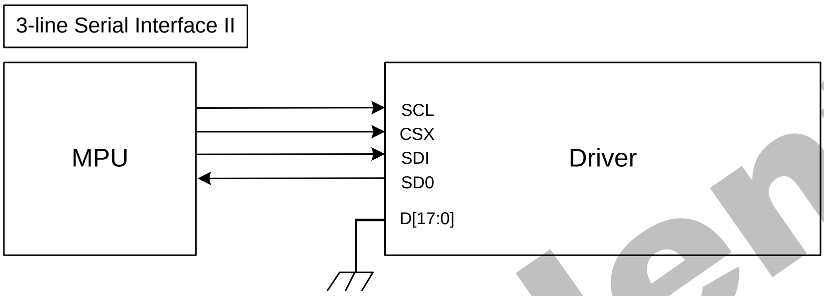


Figure40.



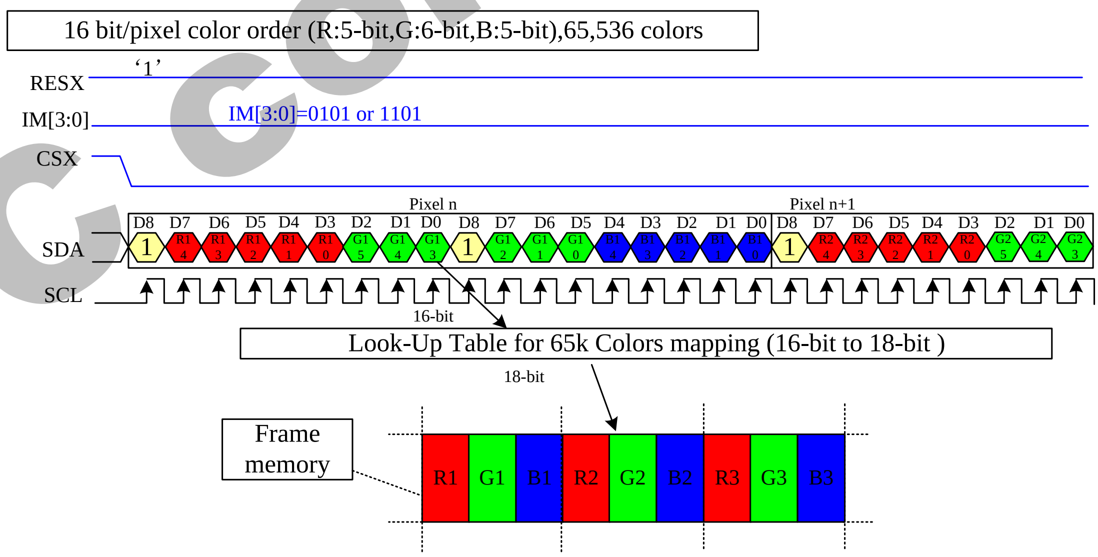
In 3-line serial interface, different display data format is available for two color depths supported by the LCM listed below.

-65k colors, RGB 5, 6, 5 -bits input

-262k colors, RGB 6, 6, 6 -bits input.

1)65K-Colors:16-bit/pixel(RGB 5, 6, 5 -bits input).

Figure41.



Note 1: The pixel data with 16-bit color depth information.

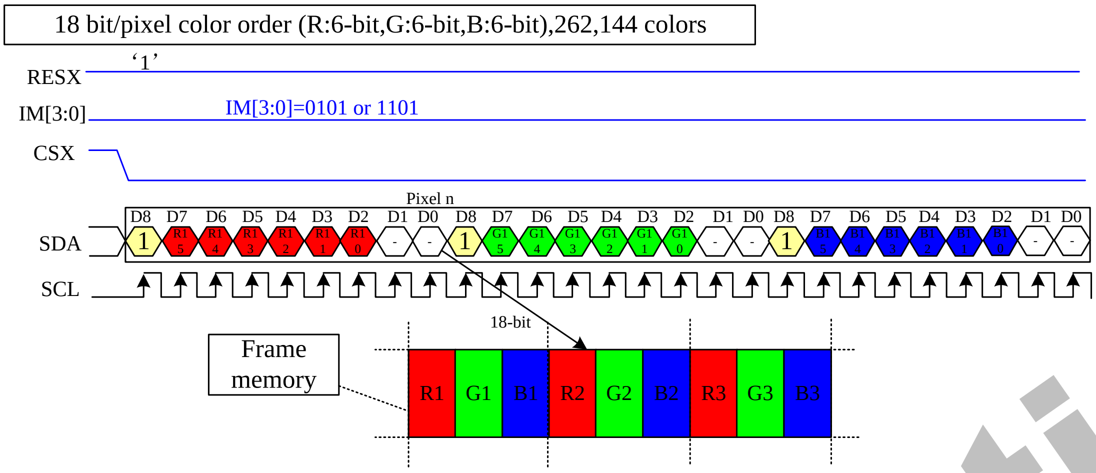
Note 2: The most significant bits are: Rx4, Gx5 and Bx4.

Note 3: The least significant bits are: Rx0, Gx0 and Bx0.

Note 4: '-'= Don't care –Can be set "0" or "1".

2)262K-Colors:18-bit/pixel(RGB 6, 6, 6 -bits input).

Figure42.



Note 1: The pixel data with 18-bit color depth information.

Note 2: The most significant bits are: Rx5, Gx5 and Bx5.

Note 3: The least significant bits are : Rx0, Gx0 and Bx0.

Note 4: '-'= Don't care - Can be set "0" or "1".

4.5.2. 4-line Serial Interface

The 4-line/8-bit serial bus interface of GC9309NA can be used by setting external pin as IM [3:0] to "0110" for serial interface I or IM [3:0] to "1110" for serial interface II. The shown figure is the example of 4-line SPI interface.

Figure43.

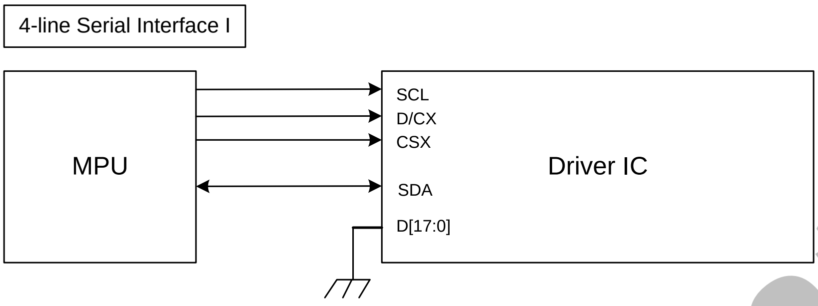
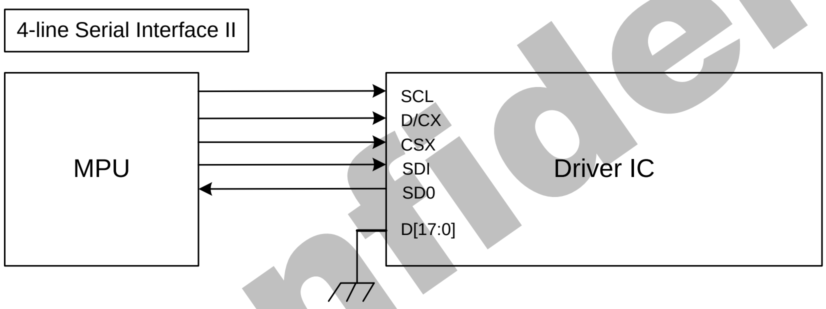


Figure44.

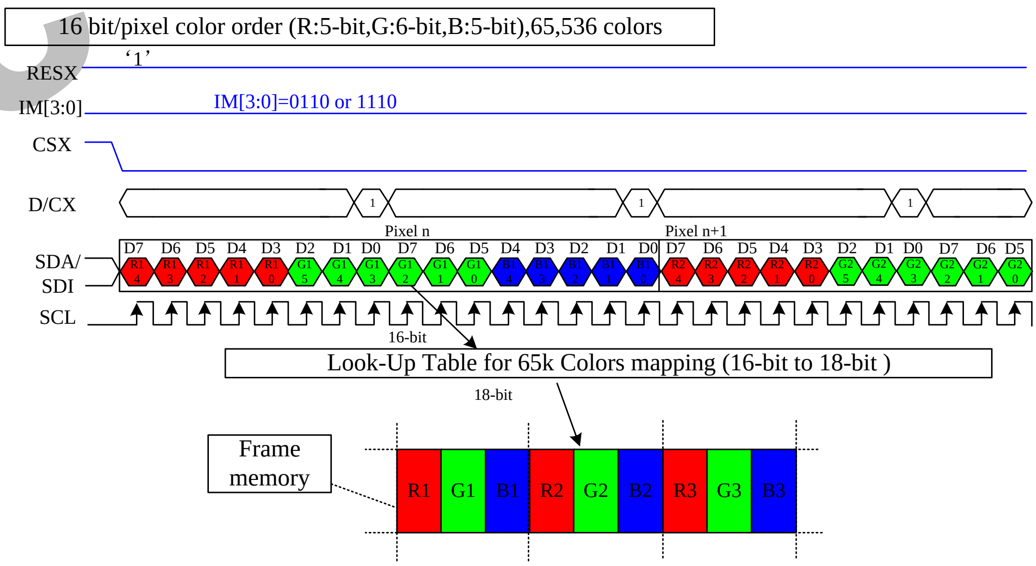


In 4-line serial interface, different display data format is available for two color depths supported by the LCM listed below.

-65k colors, RGB 5, 6, 5 -bits input.

-262k colors, RGB 6, 6, 6 -bits input.

Figure45.



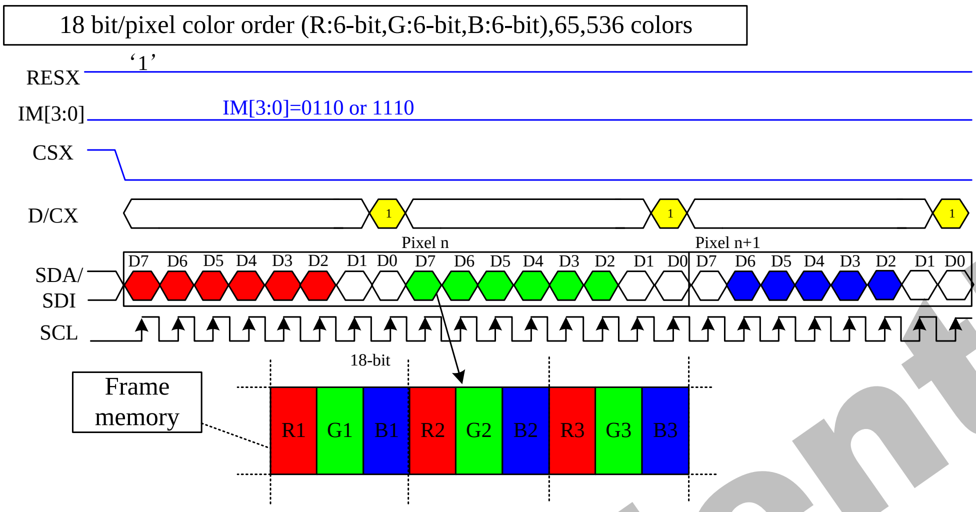
Note 1: The pixel data with 16-bit color depth information.

Note 2: The most significant bits are: Rx4, Gx5 and Bx4.

Note 3: The least significant bits are: Rx0, Gx0 and Bx0.

Note 4: '-'= Don't care –Can be set "0" or "1".

Figure46.



Note 1: The pixel data with 18-bit color depth information.

Note 2: The most significant bits are: Rx5, Gx5 and Bx5.

Note 3: The least significant bits are: Rx0, Gx0 and Bx0.

Note 4: '-'= Don't care –Can be set "0" or "1".

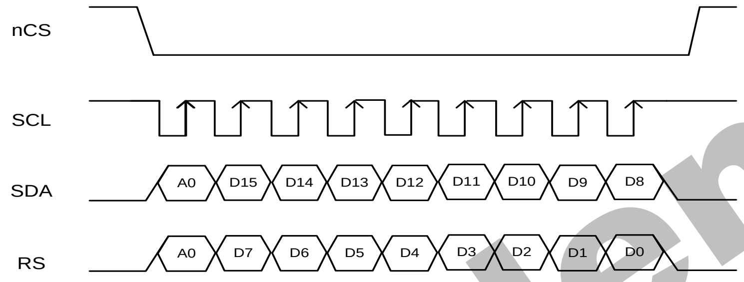
4.5.3. 2-data-line mode

This mode is active when 2data_en (B4h[3]) set to “1” in 3-wire. Only frame pixle data write transitions are sent in 2-data-line mode, register write/read is still sent in 3-wire.

The chip-select nCS (active low) enables and disables the serial interface. SCL is the serial data clock. SDA and DCX are serial data lines.

Serial data must be input to SDA in the sequence A0, D15 to D10 and DCX in the sequence A0, D7 to D0. The GC9309NA reads the data at the rising edge of SCL signal. The first bit of serial data A0 is data/command flag. It must be set to "1", D15 to D0 bits are display RAM data.

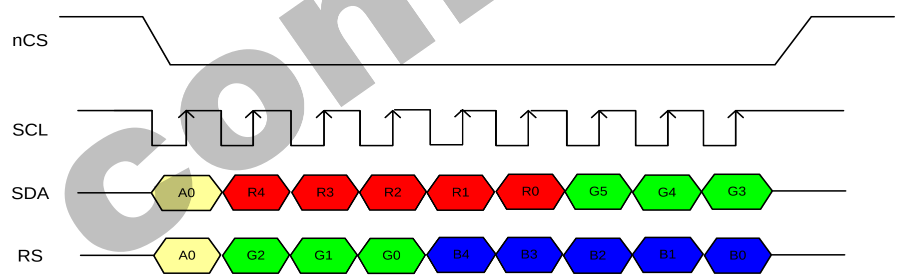
Figure47.



Five data formats are supported in 2-data-line mode, which is indicated by 2data_mdt (E9h[2:0]) .

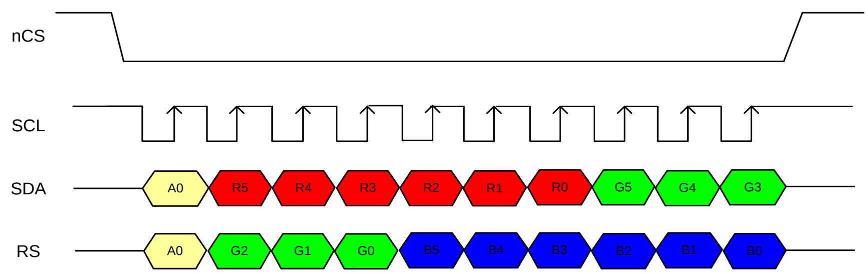
1)RGB565 1pixel/transition(65K color,2data_mdt[2:0]=’000’)

Figure48.



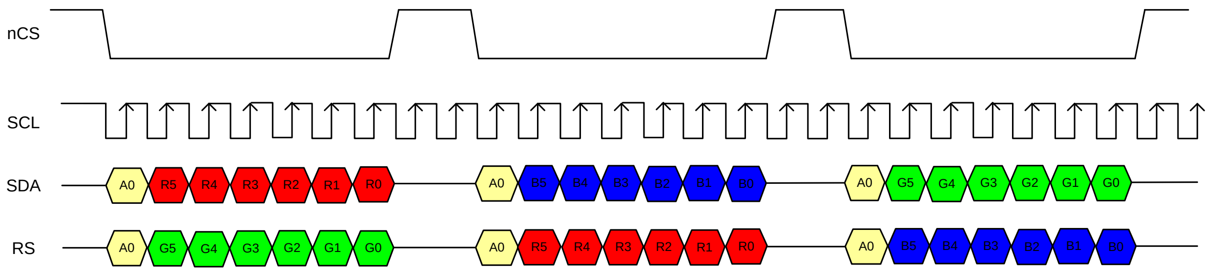
2)RGB666 1pixel/transition(262K color,2data_mdt[2:0]=’001’)

Figure49.



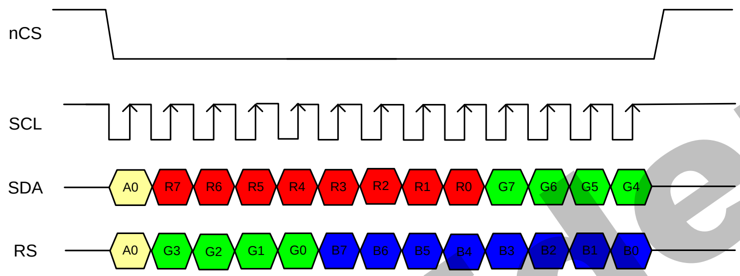
3)RGB666 2/3pixel/transition(262K color,2data_mdt[2:0]=’010’)

Figure50.



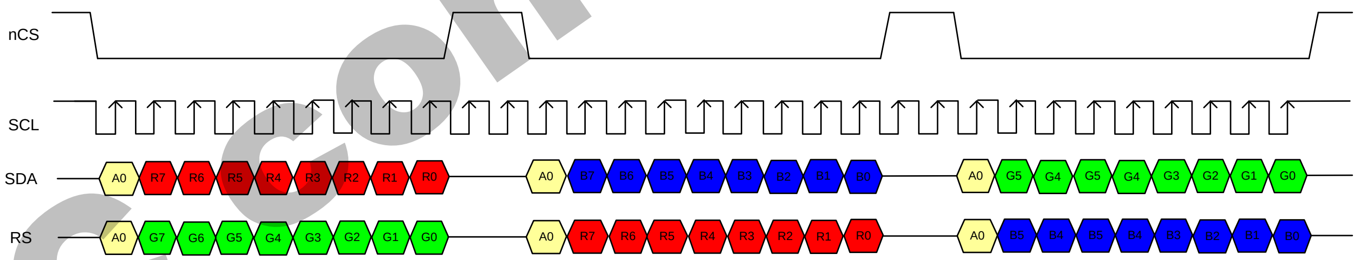
4)RGB888 1pixel/transition(4M color,2data_mdt[2:0]='100')

Figure51.



5)RGB888 2/3pixel/transition(4M color,2data_mdt[2:0]='110')

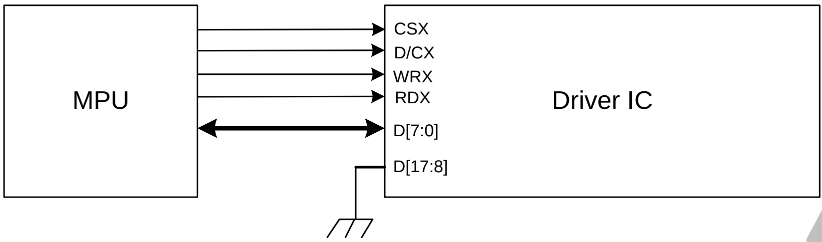
Figure52.



4.5.4. 8-bit Parallel MCU Interface

The 8080- Ⅰ system 8-bit parallel bus interface of GC9309NA can be used by setting external pin as IM [3:0] to"0000".The following shown figure is the example of interface with 8080- Ⅰ MCU system interface.

Figure53.



Different display data formats are available for two color depths supported by listed below.

- 65K-Colors, RGB 5, 6, 5 -bits input data.
- 262K-Colors, RGB 6, 6, 6 -bits input data.

1) 65K-Colors:16-bit/pixel(RGB 5, 6, 5 -bits input).

One pixel (3 sub-pixels) display data is sent by 2 byte transfers when DBI [2:0] bits of 3Ah register are set to "101".

Table 11.

Count	0	1	2	3	4	…	477	478	479	480
D/CX	0	1	1	1	1	…	1	1	1	1
D7	C7	0R4	0G2	1R4	1G2	…	238R4	238G2	239R4	239G2
D6	C6	0R3	0G1	1R3	1G1	…	238R3	238G1	239R3	239G1
D5	C5	0R2	0G0	1R2	1G0	…	238R2	238G0	239R2	239G0
D4	C4	0R1	0B4	1R1	1B4	…	238R1	238B4	239R1	239B4
D3	C3	0R0	0B3	1R0	1B3	…	238R0	238B3	239R0	239B3
D2	C2	0G5	0B2	1G5	1B2	…	238G5	238B2	239G5	239B2
D1	C1	0G4	0B1	1G4	1B1	…	238G4	238B1	239G4	239B1
D0	C0	0G3	0B0	1G3	1B0	…	238G3	238B0	239G3	239B0

2) 262K-Colors:18-bit/pixel(RGB 6, 6, 6 -bits input).

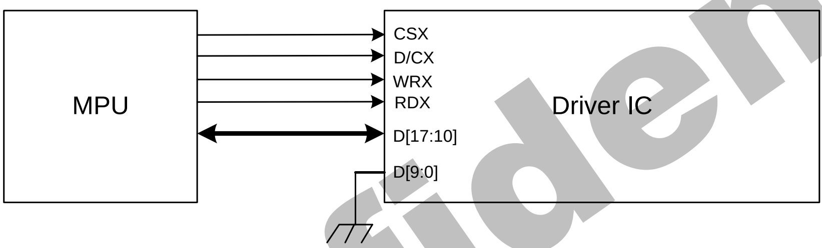
One pixel (3 sub-pixels) display data is sent by 3 bytes transfer when DBI [2:0] bits of 3Ah register are set to "110".

Table12.

Count	0	1	2	3	…	718	719	720
D/CX	0	1	1	1	…	1	1	1
D7	C7	0R5	0G5	0B5	…	239R5	239G5	239B5
D6	C6	0R4	0G4	0B4	…	239R4	239G4	239B4
D5	C5	0R3	0G3	0B3	…	239R3	239G3	239B3
D4	C4	0R2	0G2	0B2	…	239R2	239G2	239B2
D3	C3	0R1	0G1	0B1	…	239R1	239G1	239B1
D2	C2	0R0	0G0	0B0	…	239R0	239G0	239B0
D1	C1				…			
D0	C0				…			

The 8080-II system 8-bit parallel bus interface of GC9309NA can be used by settings as IM [3:0] ="1001". The following shown figure is the example of interface with 8080-Ⅱ MCU system interface.

Figure54.



Different display data formats are available for two color depths supported by listed below.
- 65K-Colors, RGB 5, 6, 5 -bits input data.
- 262K-Colors, RGB 6, 6, 6 -bits input data.

1) 65K-Colors:16-bit/pixel(RGB 5, 6, 5 -bits input).

One pixel (3 sub-pixels) display data is sent by 2 byte transfers when DBI [2:0] bits of 3Ah register are set to "101".

Table13.

Count	0	1	2	3	4	…	477	478	479	480
D/CX	0	1	1	1	1	…	1	1	1	1
D17	C7	0R4	0G2	1R4	1G2	…	238R4	238G2	239R4	239G2
D16	C6	0R3	0G1	1R3	1G1	…	238R3	238G1	239R3	239G1
D15	C5	0R2	0G0	1R2	1G0	…	238R2	238G0	239R2	239G0
D14	C4	0R1	0B4	1R1	1B4	…	238R1	238B4	239R1	239B4
D13	C3	0R0	0B3	1R0	1B3	…	238R0	238B3	239R0	239B3
D12	C2	0G5	0B2	1G5	1B2	…	238G5	238B2	239G5	239B2
D11	C1	0G4	0B1	1G4	1B1	…	238G4	238B1	239G4	239B1
D10	C0	0G3	0B0	1G3	1B0	…	238G3	238B0	239G3	239B0

2) 262K-Colors:18-bit/pixel(RGB 6, 6, 6 -bits input).

One pixel (3 sub-pixels) display data is sent by 3 bytes transfer when DBI [2:0] bits of 3Ah register are set to"110".

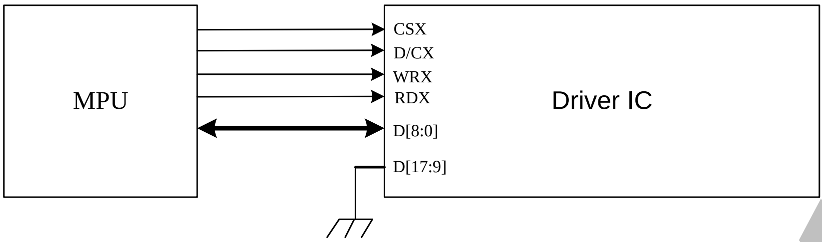
Table14.

Count	0	1	2	3	…	718	719	720
D/CX	0	1	1	1	…	1	1	1
D17	C7	0R5	0G5	0B5	…	239R5	239G5	239B5
D16	C6	0R4	0G4	0B4	…	239R4	239G4	239B4
D15	C5	0R3	0G3	0B3	…	239R3	239G3	239B3
D14	C4	0R2	0G2	0B2	…	239R2	239G2	239B2
D13	C3	0R1	0G1	0B1	…	239R1	239G1	239B1
D12	C2	0R0	0G0	0B0	…	239R0	239G0	239B0
D11	C1				…			
D10	C0				…			

4.5.5. 9-bit Parallel MCU Interface

The 8080-I system 9-bit parallel bus interface of GC9309NA can be selected by setting hardware pin IM [3:0] to “0010”. The following shown figure is the example of interface with 8080-Ⅰ MCU system interface.

Figure55.



1)262K-Colors,:18-bit/pixel(RGB 6, 6, 6 -bits input).

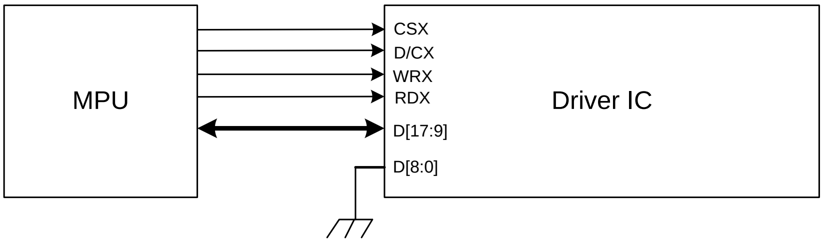
There are 2 pixels (6 sub-pixels) display data is sent by 4 transfers, when DBI [2:0] bits of 3Ah register are set to “110”.

Table15.

Count	0	1	2	3	4	…	477	478	479	480
D/CX	0	1	1	1	1	…	1	1	1	1
D8		0R5	0G2	1R5	1G2	…	238R5	238G2	239R5	239G2
D7	C7	0R4	0G1	1R4	1G1	…	238R4	238G1	239R4	239G1
D6	C6	0R3	0G0	1R3	1G0	…	238R3	238G0	239R3	239G0
D5	C5	0R2	0B5	1R2	1B5	…	238R2	238B5	239R2	239B5
D4	C4	0R1	0B4	1R1	1B4	…	238R1	238B4	239R1	239B4
D3	C3	0R0	0B3	1R0	1B3	…	238R0	238B3	239R0	239B3
D2	C2	0G5	0B2	1G5	1B2	…	238G5	238B2	239G5	239B2
D1	C1	0G4	0B1	1G4	1B1	…	238G4	238B1	239G4	239B1
D0	C0	0G3	0B0	1G3	1B0	…	238G3	238B0	239G3	239B0

The 8080- Ⅱ system 9-bit parallel bus interface of GC9309NA can be selected by setting hardware pin IM [3:0] to “1011”. The following shown figure is the example of interface with 8080- MCU system interface.

Figure56.



1)262K-Colors,:18-bit/pixel(RGB 6, 6, 6 -bits input).

There are 2 pixels (6 sub-pixels) display data is sent by 4 transfers, when DBI [2:0] bits of 3Ah register are set to “110”.

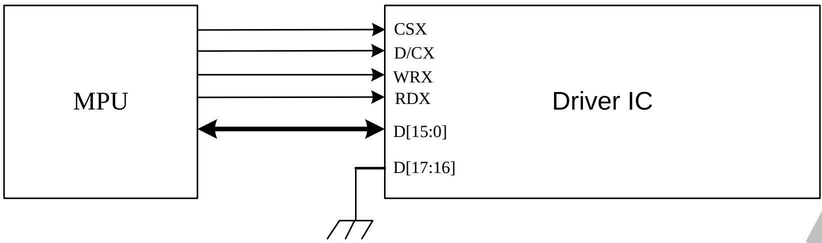
Table16.

Count	0	1	2	3	4	…	477	478	479	480
D/CX	0	1	1	1	1	…	1	1	1	1
D17	C7	0R5	0G2	1R5	1G2	…	238R5	238G2	239R5	239G2
D16	C6	0R4	0G1	1R4	1G1	…	238R4	238G1	239R4	239G1
D15	C5	0R3	0G0	1R3	1G0	…	238R3	238G0	239R3	239G0
D14	C4	0R2	0B5	1R2	1B5	…	238R2	238B5	239R2	239B5
D13	C3	0R1	0B4	1R1	1B4	…	238R1	238B4	239R1	239B4
D12	C2	0R0	0B3	1R0	1B3	…	238R0	238B3	239R0	239B3
D11	C1	0G5	0B2	1G5	1B2	…	238G5	238B2	239G5	239B2
D10	C0	0G4	0B1	1G4	1B1	…	238G4	238B1	239G4	239B1
D9		0G3	0B0	1G3	1B0	…	238G3	238B0	239G3	239B0

4.5.6. 16-bit Parallel MCU Interface

The 8080-Ⅰ system 16-bit parallel bus interface of GC9309NA can be selected by setting hardware pin IM[3:0] to “0001”.The following shown figure is the example of interface with 8080-Ⅰ MCU system interface.

Figure57.



Different display data format is available for two colors depth supported by listed below.
- 65K-Colors, RGB 5, 6, 5 -bits input data.
- 262K-Colors, RGB 6, 6, 6 -bits input data.

1)65K-Colors:16-bit/pixel(RGB 5, 6, 5 -bits input).

One pixel (3 sub-pixels) display data is sent by 1 transfer when DBI [2:0] bits of 3Ah register are set to “101”.

Table17.

Count	0	1	2	3	…	238	239	240
D/CX	0	1	1	1	…	1	1	1
D15		0R4	1R4	2R4	…	237R4	238R4	239R4
D14		0R3	1R3	2R3	…	237R3	238R3	239R3
D13		0R2	1R2	2R2	…	237R2	238R2	239R2
D12		0R1	1R1	2R1	…	237R1	238R1	239R1
D11		0R0	1R0	2R0	…	237R0	238R0	239R0
D10		0G5	1G5	2G5	…	237G5	238G5	239G5
D9		0G4	1G4	2G4	…	237G4	238G4	239G4
D8		0G3	1G3	2G3	…	237G3	238G3	239G3
D7	C7	0G2	1G2	2G2	…	237G2	238G2	239G2
D6	C6	0G1	1G1	2G1	…	237G1	238G1	239G1
D5	C5	0G0	1G0	2G0	…	237G0	238G0	239G0
D4	C4	0B4	1B4	2B4	…	237B4	238B4	239B4
D3	C3	0B3	1B3	2B3	…	237B3	238B3	239B3
D2	C2	0B2	1B2	2B2	…	237B2	238B2	239B2
D1	C1	0B1	1B1	2B1	…	237B1	238B1	239B1
D0	C0	0B0	1B0	2B0	…	237B0	238B0	239B0

2)262K-Colors:18-bit/pixel(RGB 6, 6, 6 -bits input).

One pixel (3 sub-pixels) display data is sent by 2 transfers when DBI [2:0] bits of 3Ah register are set to “110”.

1)MDT[1:0]=“00”

Table18.

Count	0	1	2	3	…	238	239	240
D/CX	0	1	1	1	…	1	1	1
D15		0R5	0B5	1G5	…	238R5	238B5	239G5
D14		0R4	0B4	1G4	…	238R4	238B4	239G4
D13		0R3	0B3	1G3	…	238R3	238B3	239G3
D12		0R2	0B2	1G2	…	238R2	238B2	239G2
D11		0R1	0B1	1G1	…	238R1	238B1	239G1
D10		0R0	0B0	1G0	…	238R0	238B0	239G0
D9								
D8								
D7	C7	0G5	1R5	1B5	…	238G5	239R5	239B5
D6	C6	0G4	1R4	1B4	…	238G4	239R4	239B4
D5	C5	0G3	1R3	1B3	…	238G3	239R3	239B3
D4	C4	0G2	1R2	1B2	…	238G2	239R2	239B2
D3	C3	0G1	1R1	1B1	…	238G1	239R1	239B1
D2	C2	0G0	1R0	1B0	…	238G0	239R0	239B0
D1	C1							
D0	C0							

2)MDT[1:0]="01"

Table19.

Count	0	1	2	3		…	357	358	479	480
D/CX	0	1	1	1		…	1	1	1	1
D15		0R5	0B5	1R5	1B5	…	238R5	238B5	239R5	239B5
D14		0R4	0B4	1R4	1B4	…	238R4	238B4	239R4	239B4
D13		0R3	0B3	1R3	1B3	…	238R3	238B3	239R3	239B3
D12		0R2	0B2	1R2	1B2	…	238R2	238B2	239R2	239B2
D11		0R1	0B1	1R1	1B1	…	238R1	238B1	239R1	239B1
D10		0R0	0B0	1R0	1B0	…	238R0	238B0	239R0	239B0
D9						…				
D8						…				
D7	C7	0G5		1G5		…	238G5		239G5	
D6	C6	0G4		1G4		…	238G4		239G4	
D5	C5	0G3		1G3		…	238G3		239G3	
D4	C4	0G2		1G2		…	238G2		239G2	
D3	C3	0G1		1G1		…	238G1		239G1	
D2	C2	0G0		1G0		…	238G0		239G0	
D1	C1					…				
D0	C0					…				

3)MDT[1:0]="10"

Table20.

Count	0	1	2	3		…	357	358	479	480
D/CX	0	1	1	1		…	1	1	1	1
D15		0R5	0B1	1R5	1B1	…	238R5	238B1	239R5	239B1
D14		0R4	0B0	1R4	1B0	…	238R4	238B0	239R4	239B0
D13		0R3		1R3		…	238R3		239R3	
D12		0R2		1R2		…	238R2		239R2	
D11		0R1		1R1		…	238R1		239R1	
D10		0R0		1R0		…	238R0		239R0	
D9		0G5		1G5		…	238G5		239G5	
D8		0G4		1G4		…	238G4		239G4	
D7	C7	0G3		1G3		…	238G3		239G3	
D6	C6	0G2		1G2		…	238G2		239G2	
D5	C5	0G1		1G1		…	238G1		239G1	
D4	C4	0G0		1G0		…	238G0		239G0	
D3	C3	0B5		1B5		…	238B5		239B5	
D2	C2	0B4		1B4		…	238B4		239B4	
D1	C1	0B3		1B3		…	238B3		239B3	
D0	C0	0B2		1B2		…	238B2		239B2	

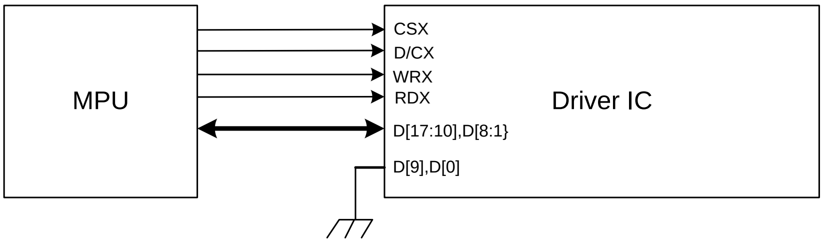
4)MDT[1:0]="11"

Table21.

Count	0	1	2	3		…	357	358	479	480
D/CX	0	1	1	1		…	1	1	1	1
D15			0R3		1R3	…		238R3		239R3
D14			0R2		1R2	…		238R2		239R2
D13			0R1		1R1	…		238R1		239R1
D12			0R0		1R0	…		238R0		239R0
D11			0G5		1G5	…		238G5		239G5
D10			0G4		1G4	…		238G4		239G4
D9			0G3		1G3	…		238G3		239G3
D8			0G2		1G2	…		238G2		239G2
D7	C7		0G1		1G1	…		238G1		239G1
D6	C6		0G0		1G0	…		238G0		239G0
D5	C5		0B5		1B5	…		238B5		239B5
D4	C4		0B4		1B4	…		238B4		239B4
D3	C3		0B3		1B3	…		238B3		239B3
D2	C2		0B2		1B2	…		238B2		239B2
D1	C1	0R5	0B1	1R5	1B1	…	238R5	238B1	239R5	239B1
D0	C0	0R4	0B0	1R4	1B0	…	238R4	238B0	239R4	239B0

The 8080-II system 16-bit parallel bus interface of GC9309NA can be selected by settings IM [3:0] ="1000". The following shown figure is the example of interface with 8080- MCU system interface.

Figure58.



Different display data format is available for two colors depth supported by listed below.
- 65K-Colors, RGB 5, 6, 5 -bits input data.
- 262K-Colors, RGB 6, 6, 6 -bits input data.

1) 65K-Colors:16-bit/pixel(RGB 5, 6, 5 -bits input).

One pixel (3 sub-pixels) display data is sent by 1 transfer when DBI [2:0] bits of 3Ah register are set to "101".

Table22.

Count	0	1	2	3	…	238	239	240
D/CX	0	1	1	1	…	1	1	1
D17		0R4	1R4	2R4	…	237R4	238R4	239R4
D16		0R3	1R3	2R3	…	237R3	238R3	239R3
D15		0R2	1R2	2R2	…	237R2	238R2	239R2
D14		0R1	1R1	2R1	…	237R1	238R1	239R1
D13		0R0	1R0	2R0	…	237R0	238R0	239R0
D12		0G5	1G5	2G5	…	237G5	238G5	239G5
D11		0G4	1G4	2G4	…	237G4	238G4	239G4
D10		0G3	1G3	2G3	…	237G3	238G3	239G3
D8	C7	0G2	1G2	2G2	…	237G2	238G2	239G2
D7	C6	0G1	1G1	2G1	…	237G1	238G1	239G1
D6	C5	0G0	1G0	2G0	…	237G0	238G0	239G0
D5	C4	0B4	1B4	2B4	…	237B4	238B4	239B4
D4	C3	0B3	1B3	2B3	…	237B3	238B3	239B3
D3	C2	0B2	1B2	2B2	…	237B2	238B2	239B2
D2	C1	0B1	1B1	2B1	…	237B1	238B1	239B1
D1	C0	0B0	1B0	2B0	…	237B0	238B0	239B0

2)262K-Colors:18-bit/pixel(RGB 6, 6, 6 -bits input).

One pixel (3 sub-pixels) display data is sent by 2 transfers when DBI [2:0] bits of 3Ah register are set to "110".

1)MDT[1:0]=00

Table23.

Count	0	1	2	3	…	238	239	240
D/CX	0	1	1	1	…	1	1	1
D17		0R5	0B5	1G5	…	238R5	238B5	239G5
D16		0R4	0B4	1G4	…	238R4	238B4	239G4

D15		0R3	0B3	1G3	…	238R3	238B3	239G3
D14		0R2	0B2	1G2	…	238R2	238B2	239G2
D13		0R1	0B1	1G1	…	238R1	238B1	239G1
D12		0R0	0B0	1G0	…	238R0	238B0	239G0
D11								
D10								
D8	C7	0G5	1R5	1B5	…	238G5	239R5	239B5
D7	C6	0G4	1R4	1B4	…	238G4	239R4	239B4
D6	C5	0G3	1R3	1B3	…	238G3	239R3	239B3
D5	C4	0G2	1R2	1B2	…	238G2	239R2	239B2
D4	C3	0G1	1R1	1B1	…	238G1	239R1	239B1
D3	C2	0G0	1R0	1B0	…	238G0	239R0	239B0
D2	C1							
D1	C0							

2)MDT[1:0]=01

Table24.

Count	0	1	2	3		…	357	358	479	480
D/CX	0	1	1	1		…	1	1	1	1
D17		0R5	0B5	1R5	1B5	…	238R5	238B5	239R5	239B5
D16		0R4	0B4	1R4	1B4	…	238R4	238B4	239R4	239B4
D15		0R3	0B3	1R3	1B3	…	238R3	238B3	239R3	239B3
D14		0R2	0B2	1R2	1B2	…	238R2	238B2	239R2	239B2
D13		0R1	0B1	1R1	1B1	…	238R1	238B1	239R1	239B1
D12		0R0	0B0	1R0	1B0	…	238R0	238B0	239R0	239B0
D11						…				
D10						…				
D8	C7	0G5		1G5		…	238G5		239G5	
D7	C6	0G4		1G4		…	238G4		239G4	
D6	C5	0G3		1G3		…	238G3		239G3	
D5	C4	0G2		1G2		…	238G2		239G2	
D4	C3	0G1		1G1		…	238G1		239G1	
D3	C2	0G0		1G0		…	238G0		239G0	
D2	C1					…				
D1	C0					…				

3)MDT[1:0]=10

Table25.

Count	0	1	2	3		…	357	358	479	480
D/CX	0	1	1	1		…	1	1	1	1
D17		0R5	0B1	1R5	1B1	…	238R5	238B1	239R5	239B1
D16		0R4	0B0	1R4	1B0	…	238R4	238B0	239R4	239B0
D15		0R3		1R3		…	238R3		239R3	
D14		0R2		1R2		…	238R2		239R2	
D13		0R1		1R1		…	238R1		239R1	
D12		0R0		1R0		…	238R0		239R0	
D11		0G5		1G5		…	238G5		239G5	
D10		0G4		1G4		…	238G4		239G4	
D8	C7	0G3		1G3		…	238G3		239G3	
D7	C6	0G2		1G2		…	238G2		239G2	
D6	C5	0G1		1G1		…	238G1		239G1	
D5	C4	0G0		1G0		…	238G0		239G0	
D4	C3	0B5		1B5		…	238B5		239B5	
D3	C2	0B4		1B4		…	238B4		239B4	
D2	C1	0B3		1B3		…	238B3		239B3	
D1	C0	0B2		1B2		…	238B2		239B2	

4)MDT[1:0]=11

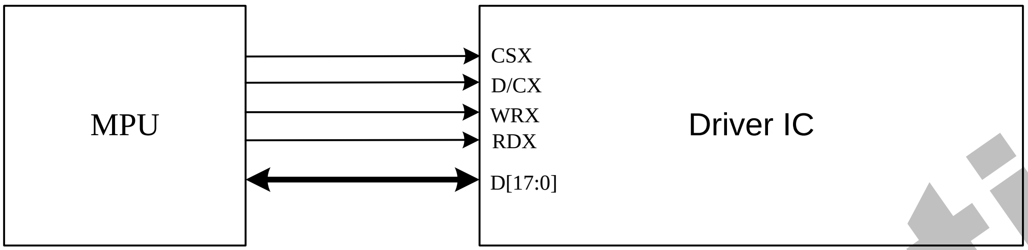
Table26.

Count	0	1	2	3		…	357	358	479	480
D/CX	0	1	1	1		…	1	1	1	1
D17			0R3		1R3	…		238R3		239R3
D16			0R2		1R2	…		238R2		239R2
D15			0R1		1R1	…		238R1		239R1
D14			0R0		1R0	…		238R0		239R0
D13			0G5		1G5	…		238G5		239G5
D12			0G4		1G4	…		238G4		239G4
D11			0G3		1G3	…		238G3		239G3
D10			0G2		1G2	…		238G2		239G2
D8	C7		0G1		1G1	…		238G1		239G1
D7	C6		0G0		1G0	…		238G0		239G0
D6	C5		0B5		1B5	…		238B5		239B5
D5	C4		0B4		1B4	…		238B4		239B4
D4	C3		0B3		1B3	…		238B3		239B3
D3	C2		0B2		1B2	…		238B2		239B2
D2	C1	0R5	0B1	1R5	1B1	…	238R5	238B1	239R5	239B1
D1	C0	0R4	0B0	1R4	1B0	…	238R4	238B0	239R4	239B0

4.5.7. 18-bit Parallel MCU Interface

The 8080-I system 18-bit parallel bus interface of GC9309NA can be selected by setting hardware pin IM[3:0] to “0011”.The following shown figure is the example of interface with 8080-I MCU system interface.

Figure58.



Different display data format is available for one color depth only supported by listed below.

- 65K-Colors, RGB 5, 6, 5 -bits input data.
- 262K-Colors, RGB 6, 6, 6 -bits input data.

1) 65K-Colors:16-bit/pixel(RGB 5, 6, 5 -bits input).

One pixel (3 sub-pixels) display data is sent by 1 transfer when DBI [2:0] bits of 3Ah register are set to “101”.

Table27.

Count	0	1	2	3	…	238	239	240
D/CX	0	1	1	1	…	1	1	1
D17								
D16								
D15		0R4	1R4	2R4	…	237R4	238R4	239R4
D14		0R3	1R3	2R3	…	237R3	238R3	239R3
D13		0R2	1R2	2R2	…	237R2	238R2	239R2
D12		0R1	1R1	2R1	…	237R1	238R1	239R1
D11		0R0	1R0	2R0	…	237R0	238R0	239R0
D10		0G5	1G5	2G5	…	237G5	238G5	239G5
D9		0G4	1G4	2G4	…	237G4	238G4	239G4
D8		0G3	1G3	2G3	…	237G3	238G3	239G3
D7	C7	0G2	1G2	2G2	…	237G2	238G2	239G2
D6	C6	0G1	1G1	2G1	…	237G1	238G1	239G1
D5	C5	0G0	1G0	2G0	…	237G0	238G0	239G0
D4	C4	0B4	1B4	2B4	…	237B4	238B4	239B4
D3	C3	0B3	1B3	2B3	…	237B3	238B3	239B3
D2	C2	0B2	1B2	2B2	…	237B2	238B2	239B2
D1	C1	0B1	1B1	2B1	…	237B1	238B1	239B1
D0	C0	0B0	1B0	2B0	…	237B0	238B0	239B0

2)262K-Colors:18-bit/pixel(RGB 6, 6, 6 -bits input).

One pixel (3 sub-pixels) display data is sent by 1 transfer when DBI [2:0] bits of 3Ah register are set to "110".

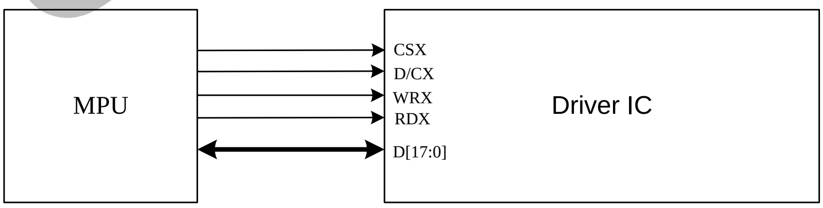
Table28.

Count	0	1	2	3	…	238	239	240
D/CX	0	1	1	1	…	1	1	1
D17		0R5	1R5	2R5	…	237R5	238R5	239R5
D16		0R4	1R4	2R4	…	237R4	238R4	239R4
D15		0R3	1R3	2R3	…	237R3	238R3	239R3
D14		0R2	1R2	2R2	…	237R2	238R2	239R2
D13		0R1	1R1	2R1	…	237R1	238R1	239R1
D12		0R0	1R0	2R0	…	237R0	238R0	239R0
D11		0G5	1G5	2G5	…	237G5	238G5	239G5
D10		0G4	1G4	2G4	…	237G4	238G4	239G4
D9		0G3	1G3	2G3	…	237G3	238G3	239G3
D8		0G2	1G2	2G2	…	237G2	238G2	239G2
D7	C7	0G1	1G1	2G1	…	237G1	238G1	239G1
D6	C6	0G0	1G0	2G0	…	237G0	238G0	239G0
D5	C5	0B5	1B5	2B5	…	237B5	238B5	239B5
D4	C4	0B4	1B4	2B4	…	237B4	238B4	239B4
D3	C3	0B3	1B3	2B3	…	237B3	238B3	239B3
D2	C2	0B2	1B2	2B2	…	237B2	238B2	239B2
D1	C1	0B1	1B1	2B1	…	237B1	238B1	239B1
D0	C0	0B0	1B0	2B0	…	237B0	238B0	239B0

The 8080-II system 18-bit parallel bus interface mode can be selected by settings IM [3:0] ="1010".

The following shown figure is the example of interface with 8080- MCU system interface.

Figure59.



Different display data format is available for one color depth only supported by listed below.

- 65K-Colors, RGB 5, 6, 5 -bits input data.
- 262K-Colors, RGB 6, 6, 6 -bits input data.

1)65K-Colors:16-bit/pixel(RGB 5, 6, 5 -bits input).

One pixel (3 sub-pixels) display data is sent by 1 transfer when DBI [2:0] bits of 3Ah register are set to "101".

Table29.

Count	0	1	2	3	…	238	239	240
D/CX	0	1	1	1	…	1	1	1
D17								
D16								
D15		0R4	1R4	2R4	…	237R4	238R4	239R4
D14		0R3	1R3	2R3	…	237R3	238R3	239R3
D13		0R2	1R2	2R2	…	237R2	238R2	239R2
D12		0R1	1R1	2R1	…	237R1	238R1	239R1
D11		0R0	1R0	2R0	…	237R0	238R0	239R0
D10		0G5	1G5	2G5	…	237G5	238G5	239G5
D9		0G4	1G4	2G4	…	237G4	238G4	239G4
D8	C7	0G3	1G3	2G3	…	237G3	238G3	239G3
D7	C6	0G2	1G2	2G2	…	237G2	238G2	239G2
D6	C5	0G1	1G1	2G1	…	237G1	238G1	239G1
D5	C4	0G0	1G0	2G0	…	237G0	238G0	239G0
D4	C3	0B4	1B4	2B4	…	237B4	238B4	239B4
D3	C2	0B3	1B3	2B3	…	237B3	238B3	239B3
D2	C1	0B2	1B2	2B2	…	237B2	238B2	239B2
D1	C0	0B1	1B1	2B1	…	237B1	238B1	239B1
D0		0B0	1B0	2B0	…	237B0	238B0	239B0

2)262K-Colors:18-bit/pixel(RGB 6, 6, 6 -bits input).

One pixel (3 sub-pixels) display data is sent by 1 transfer when DBI [2:0] bits of 3Ah register are set to "110".

Table30.

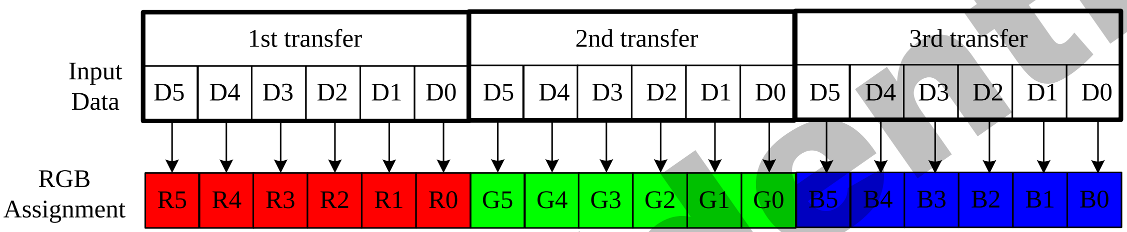
Count	0	1	2	3	…	238	239	240
D/CX	0	1	1	1	…	1	1	1
D17		0R5	1R5	2R5	…	237R5	238R5	239R5
D16		0R4	1R4	2R4	…	237R4	238R4	239R4
D15		0R3	1R3	2R3	…	237R3	238R3	239R3
D14		0R2	1R2	2R2	…	237R2	238R2	239R2
D13		0R1	1R1	2R1	…	237R1	238R1	239R1
D12		0R0	1R0	2R0	…	237R0	238R0	239R0
D11		0G5	1G5	2G5	…	237G5	238G5	239G5
D10		0G4	1G4	2G4	…	237G4	238G4	239G4
D9		0G3	1G3	2G3	…	237G3	238G3	239G3
D8	C7	0G2	1G2	2G2	…	237G2	238G2	239G2
D7	C6	0G1	1G1	2G1	…	237G1	238G1	239G1
D6	C5	0G0	1G0	2G0	…	237G0	238G0	239G0
D5	C4	0B5	1B5	2B5	…	237B5	238B5	239B5
D4	C3	0B4	1B4	2B4	…	237B4	238B4	239B4

D3	C2	0B3	1B3	2B3	…	237B3	238B3	239B3
D2	C1	0B2	1B2	2B2	…	237B2	238B2	239B2
D1	C0	0B1	1B1	2B1	…	237B1	238B1	239B1
D0		0B0	1B0	2B0	…	237B0	238B0	239B0

4.5.8. 6-bit Parallel RGB Interface

The 6-bit RGB interface is selected by setting the RIM bit to “1”. When RCM [1:0] are set to “10” and DE mode is selected, the display operation is synchronized with VSYNC, HSYNC and DOTCLK signals. The display data are transferred to the internal GRAM in synchronization with the display operation via 6-bit RGB data bus (D [5:0]) according to the data enable signal (DE) when RCM [1:0] are set to “10”. the valid display data is inputted in pixel unit via D [5:0] according to the VFP/VBP and HFP/HBP settings. Unused pins must be connected to GND to ensure normally operation. Registers can be set by the SPI system interface.

1)262K-Colors:18-bit/pixel(RGB 6, 6, 6 -bits input).
Figure60.



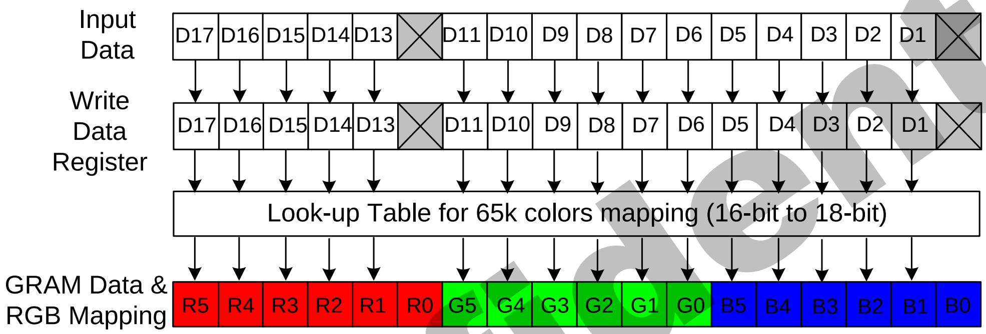
GC9309NA has data transfer counters to count the first, second, third data transfer in 6-bit RGB interface mode. The transfer counter is always reset to the state of first data transfer on the falling edge of VSYNC. If a mismatch arises in the number of each data transfer, the counter is reset to the state of first data transfer at the start of the frame (i.e. on the falling edge of VSYNC) to restart data transfer in the correct order from the next frame. This function is expedient for moving picture display, which requires consecutive data transfer in light of minimizing effects from failed data transfer and enabling the system to return to a normal state.

Note that internal display operation is performed in units of pixels (RGB: taking 3 inputs of DOTCLK).Accordingly, the number of DOTCLK inputs in one frame period must be a multiple of 3 to complete data transfer correctly. Otherwise it will affect the display of that frame as well as the next frame.

4.5.9. 16-bit Parallel RGB Interface

The 16-bit RGB interface is selected by setting the DPI [2:0] bits to “101”. When RCM [1:0] are set to “10” and DE mode is selected, the display operation is synchronized with VSYNC, HSYNC and DOTCLK signals. The display data is transferred to the internal GRAM in synchronization with the display operation via 16-bit RGB data bus (D[17:13] & D[11:0]) according to the data enable signal (DE). The RGB interface SYNC mode is selected by setting the RCM [1:0] to “11”, the valid display data is inputted in pixel unit via D[17:13] & D[11:0] according to the VFP/VBP and HFP/HBP settings. The unused D12 and D0 pins must be connected to GND for ensure normally operation. Registers can be set by the SPI system interface.

Figure62.



4.5.10. 18-bit Parallel RGB Interface

The 18-bit RGB interface is selected by setting the DPI [2:0] bits to “110”. When RCM [1:0] are set to “10” and DE mode is selected, the display operation is synchronized with VSYNC, HSYNC and DOTCLK signals. The display data are transferred to the internal GRAM in synchronization with the display operation via 18-bit RGB data bus (D [17:0]) according to the data enable signal (DE) when RCM [1:0] are set to “10”. The RGB interface SYNC mode is selected by setting the RCM [1:0] to “11”, the valid display data is inputted in pixel unit via D[17:0] according to the VFP/VBP and HFP/HBP settings. Registers can be set by the SPI system interface.

Figure63.

Input Data	D17	D16	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
RGB Assignment	R5	R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1	B0

4.5.11. Quad Serial Peripheral Interface

In quad serial Peripheral interface, different display data format is available for three color depths supported by the LCM listed below.

-262k colors, RGB 6, 6, 6 -bits input.

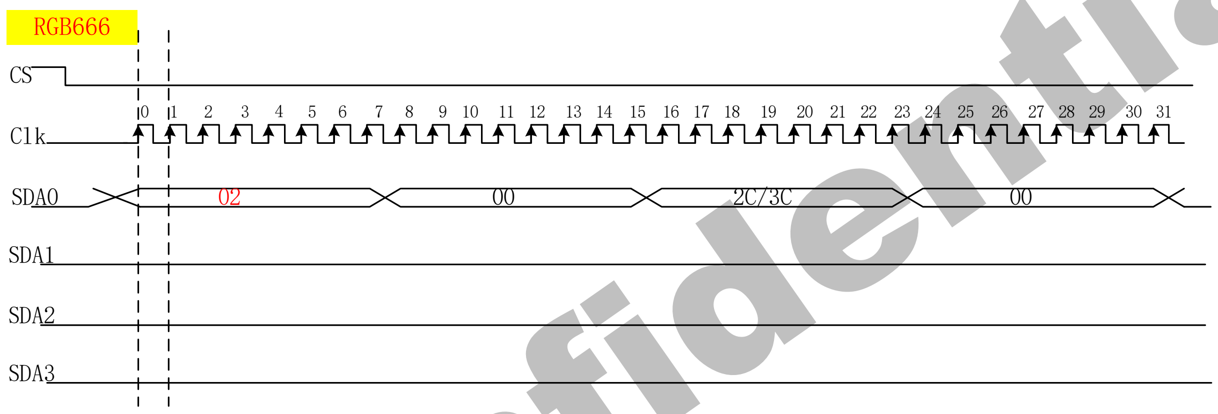
-65k colors, RGB 5, 6, 5 -bits input

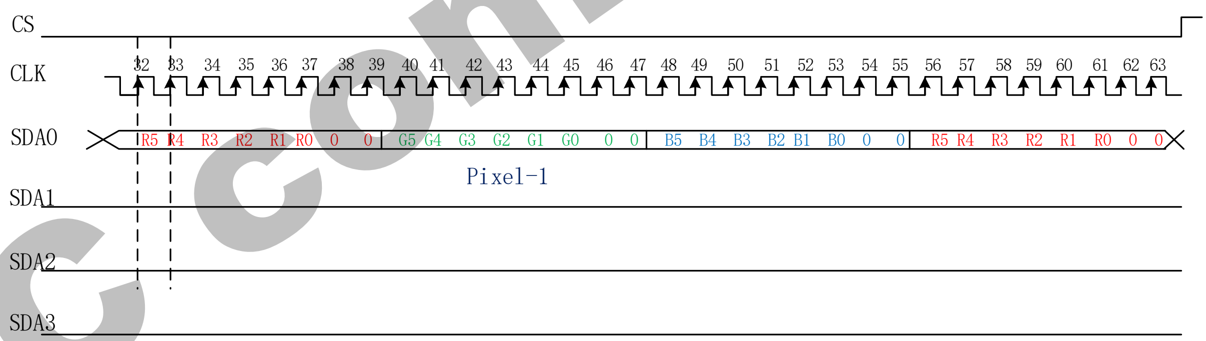
-256 colors, RGB 3, 3, 2 -bits input

-8 colors，RGB 1, 1, 1-bits input

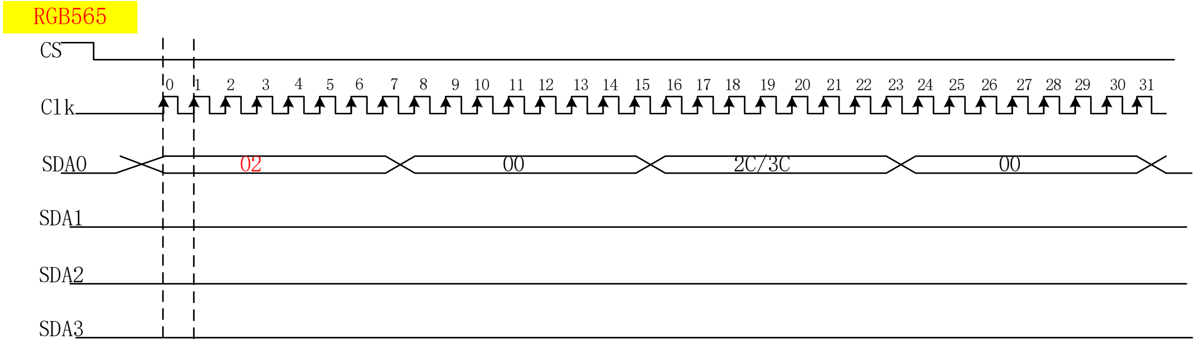
1wire data: only use SDA0, first byte=0x02

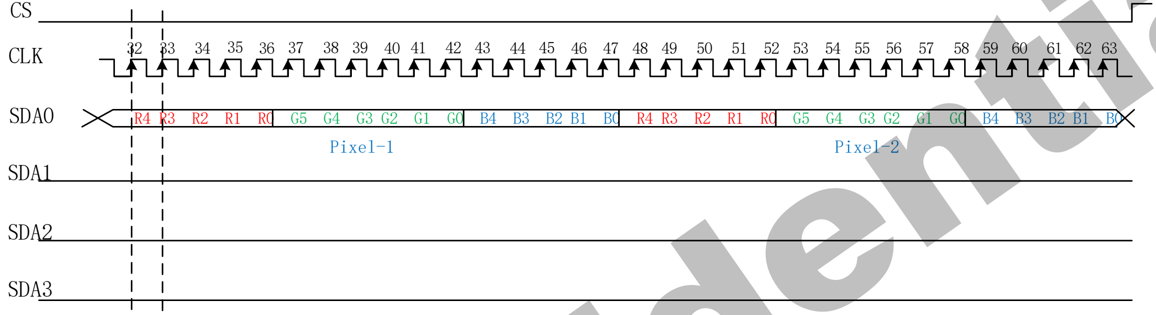
1） 262k-Colors:18-bit/pixel (RGB 6, 6, 6-bits input).



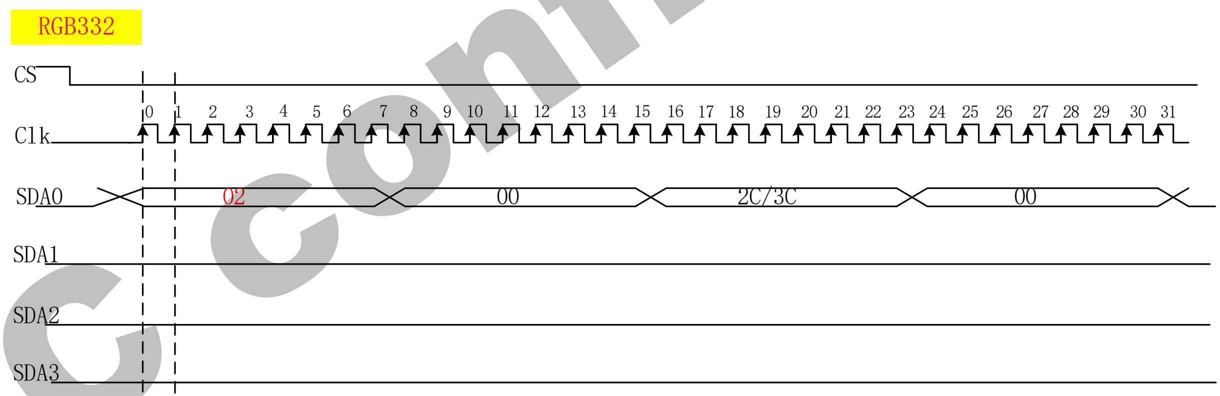


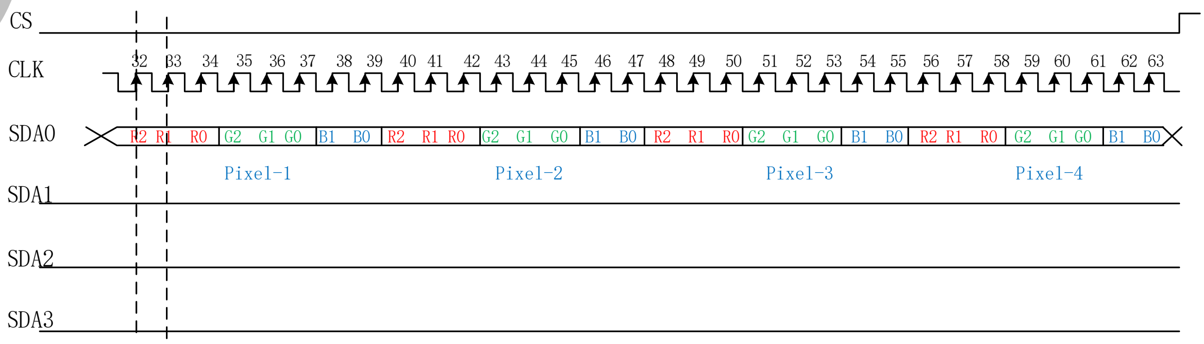
2） 65k-Colors:16-bit/pixel (RGB 5, 6,5 -bits input).



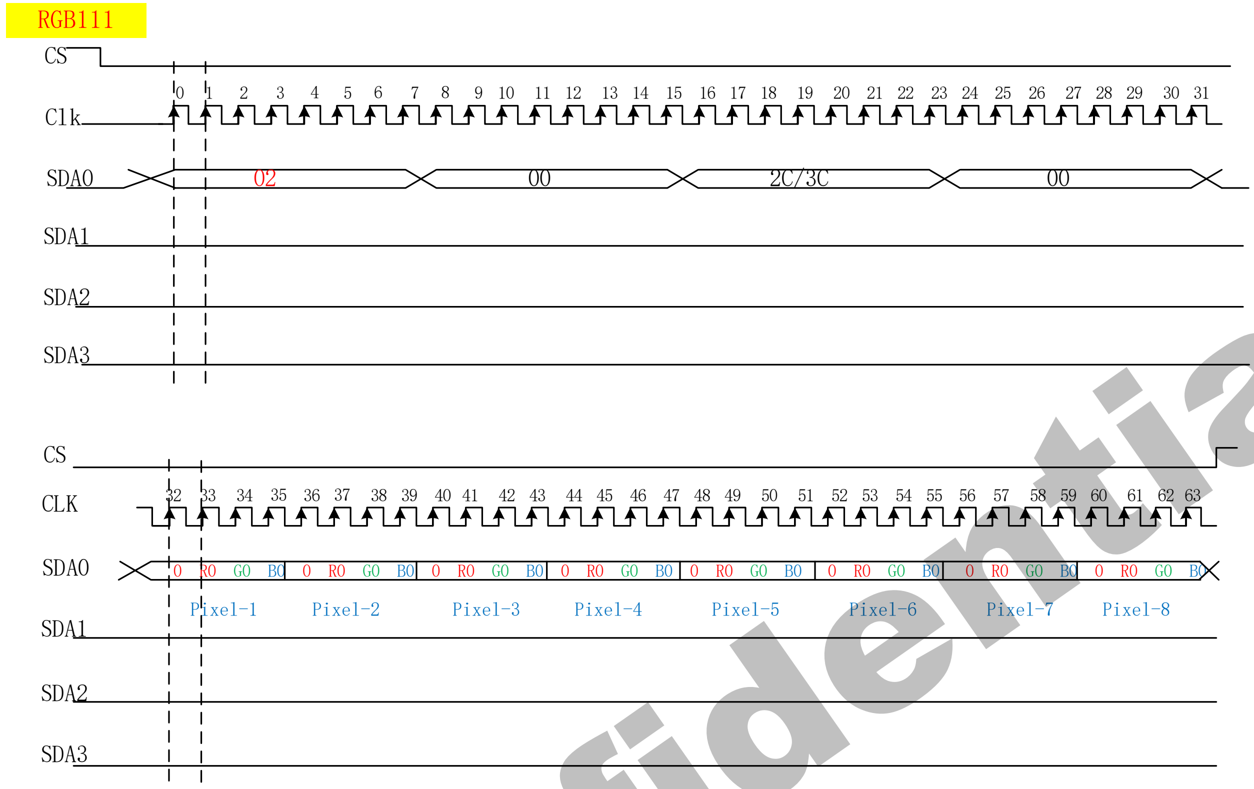


3） 256-Colors:8-bit/pixel (RGB 3, 3, 2 -bits input).



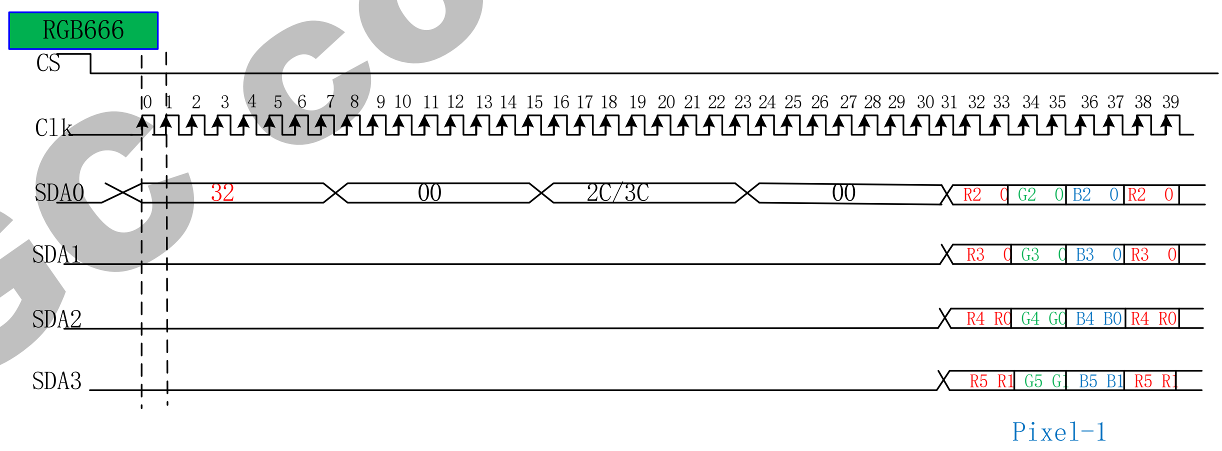


4） **8-Colors:3-bit/pixel (RGB 1, 1, 1 -bits input).**

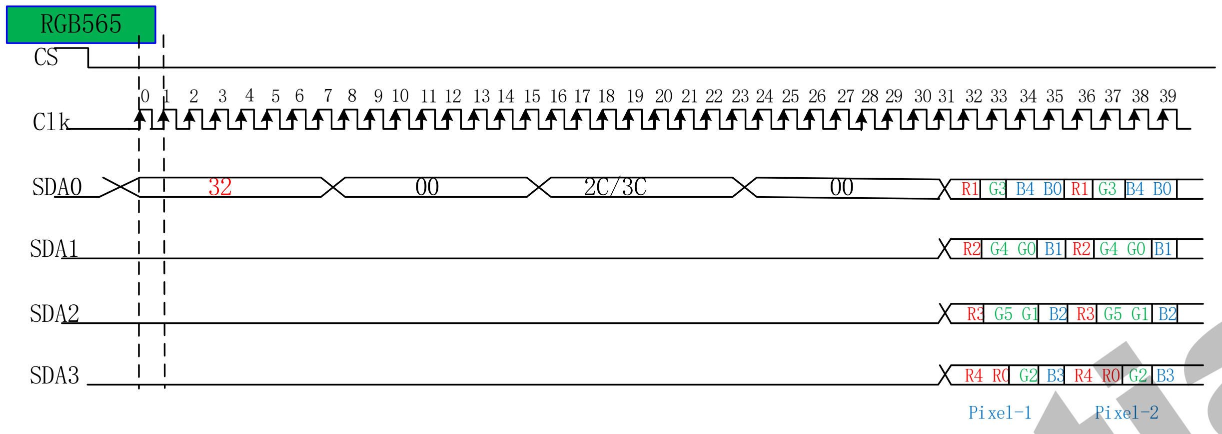


4wire data 1wire Addr: first byte=0x32

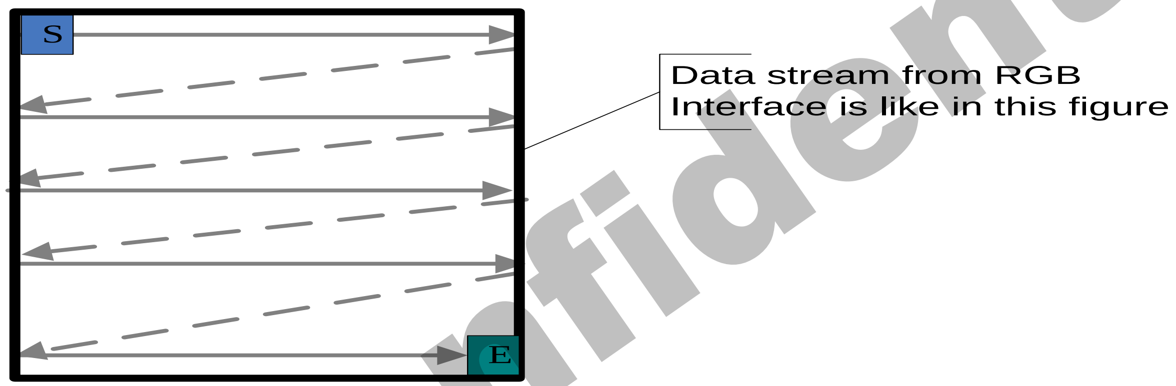
1） **262k-Colors:18-bit/pixel (RGB 6, 6, 6 -bits input).**



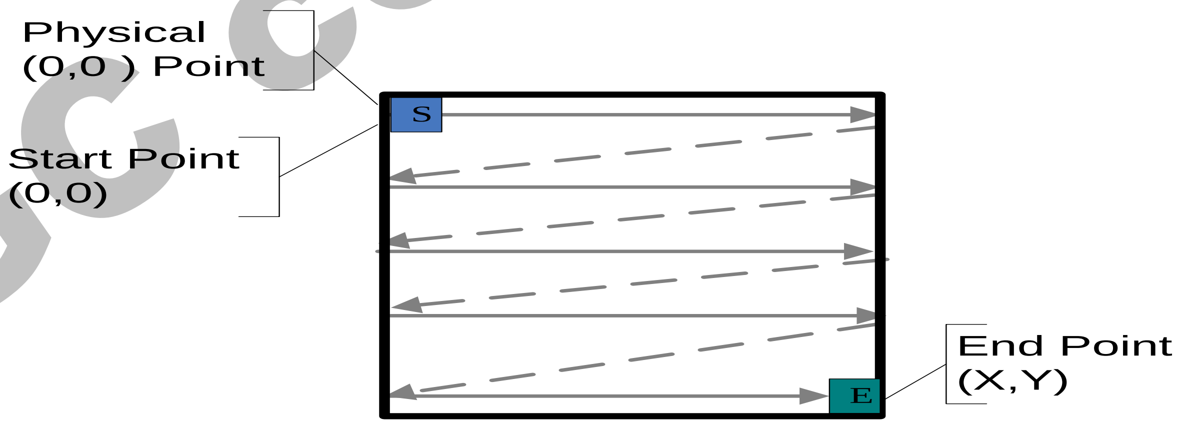
2） 65k-Colors:16-bit/pixel (RGB 5, 6,5 -bits input).



3） 256-Colors:8-bit/pixel (RGB 3, 3, 2 -bits input).

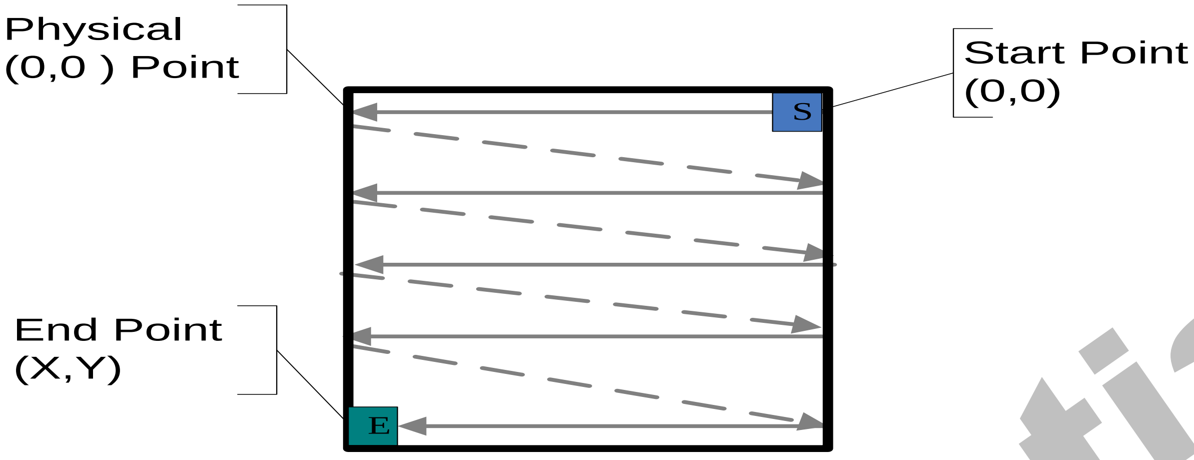


4） 8-Colors:3-bit/pixel (RGB 1, 1, 1 -bits input).

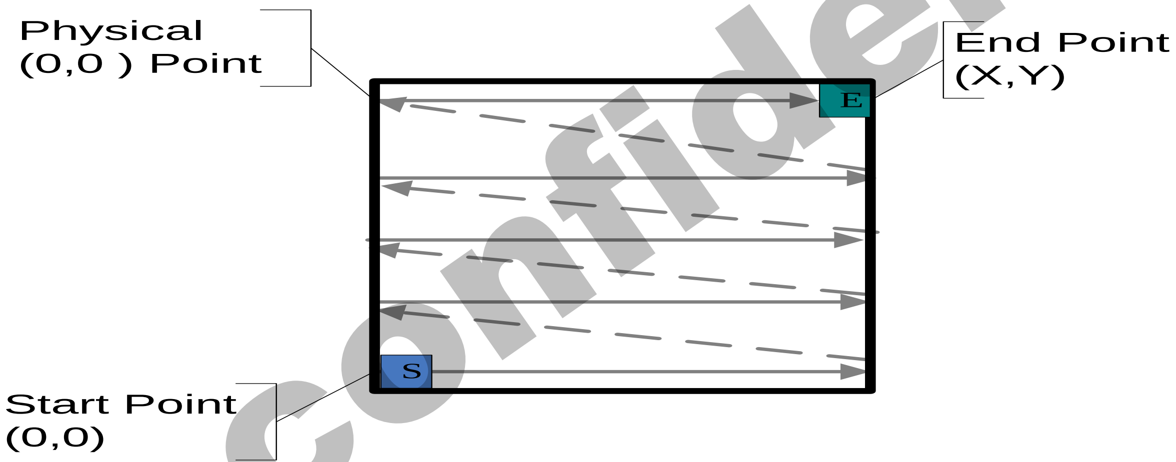


4wire data 4wire Addr: first byte=0x12

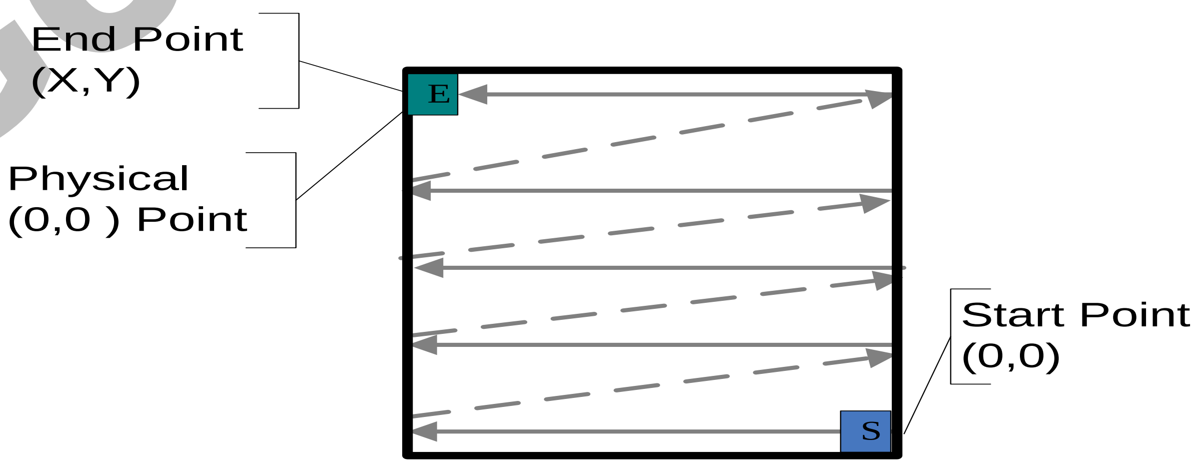
1）262k-Colors:18-bit/pixel (RGB 6, 6, 6-bits input).



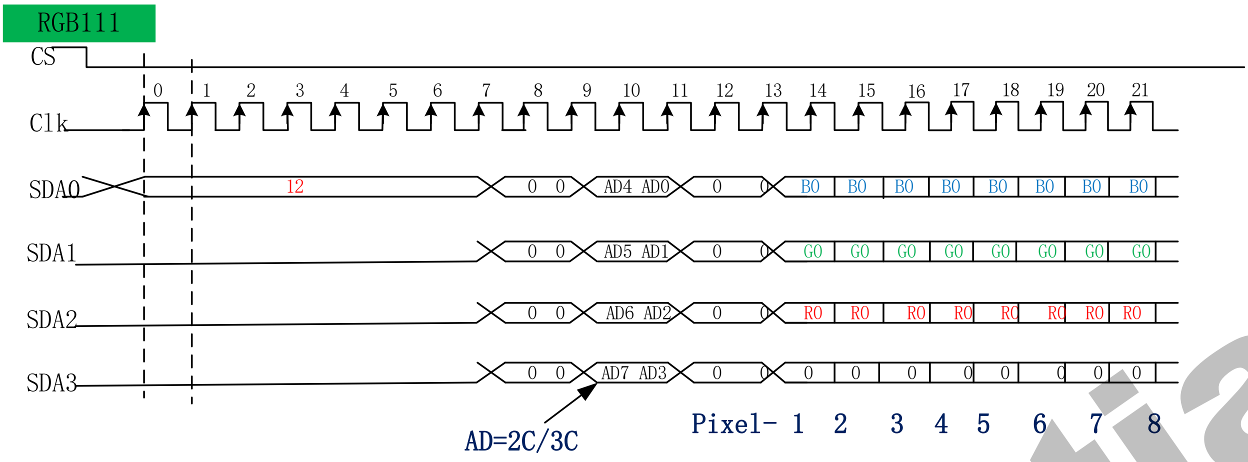
3）65k-Colors:16-bit/pixel (RGB 5, 6,5 -bits input).



4）256-Colors:8-bit/pixel (RGB 3, 3, 2 -bits input).

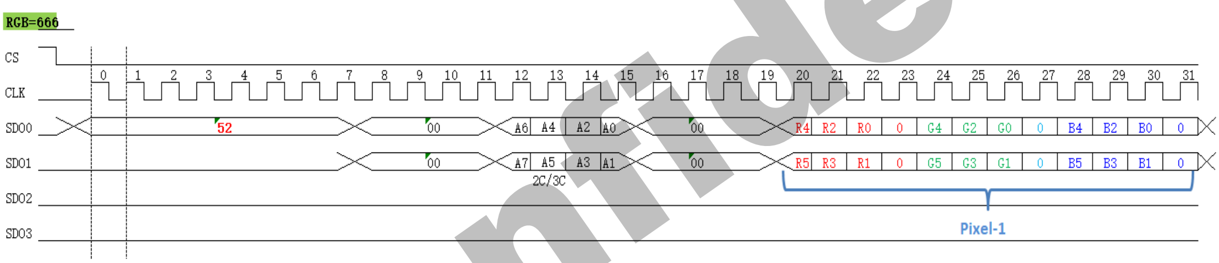


5）8-Colors:3-bit/pixel (RGB 1, 1, 1 -bits input).

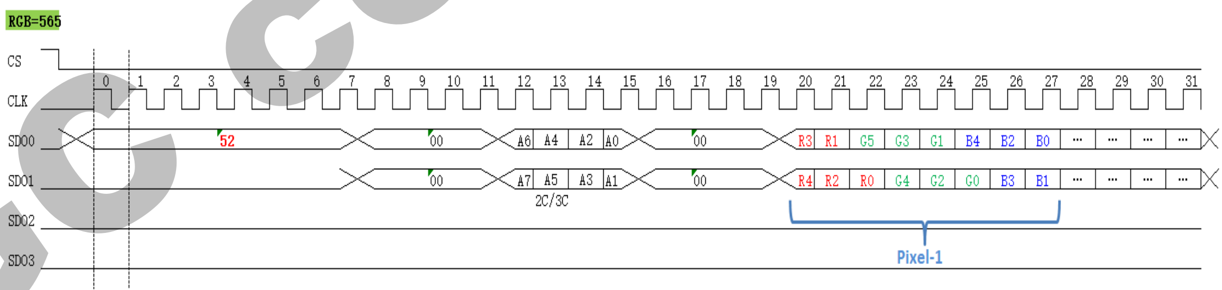


2wire data: only use SDA0, first byte=0x52

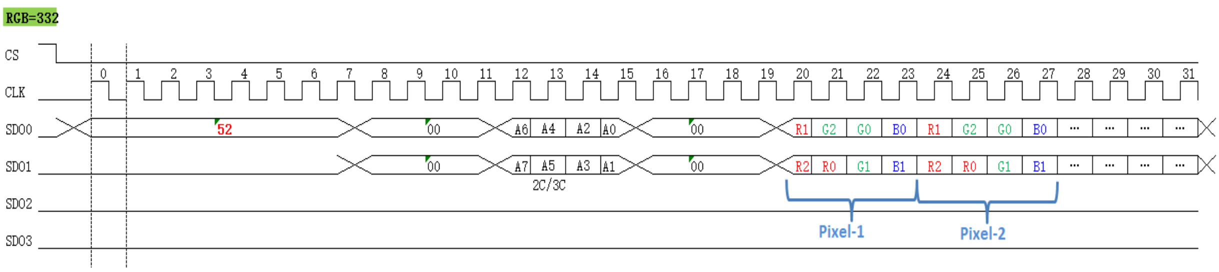
1) 262k-Colors:18-bit/pixel (RGB 6, 6, 6-bits input).



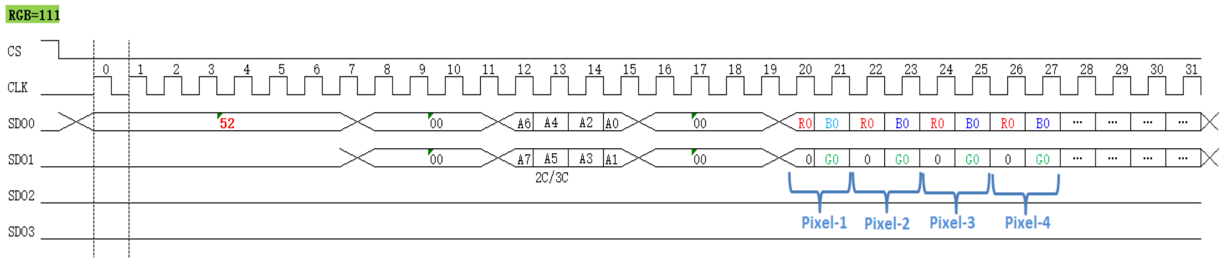
2) 65k-Colors:16-bit/pixel (RGB 5, 6,5 -bits input).



3) 256-Colors:8-bit/pixel (RGB 3, 3, 2 -bits input).

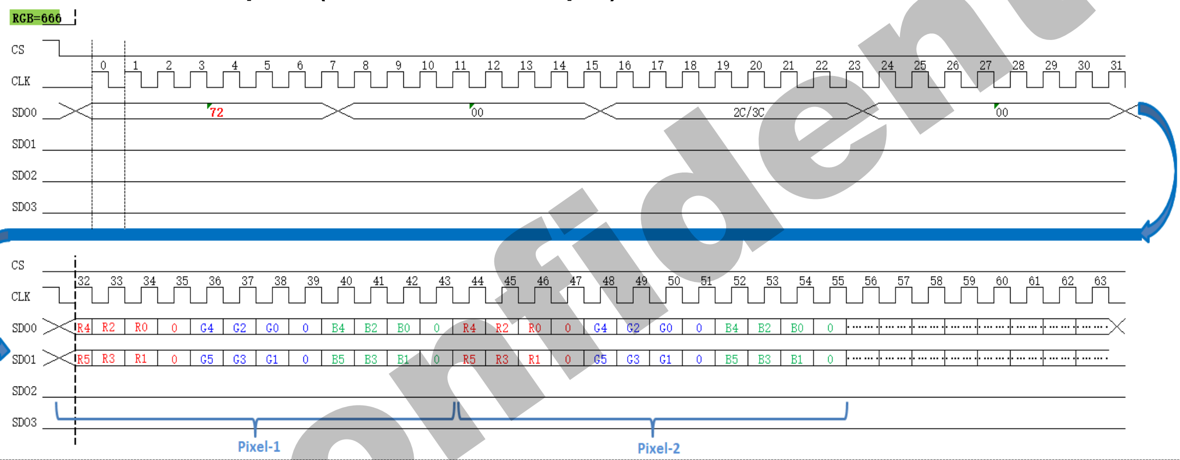


4) 8-Colors:3-bit/pixel (RGB 1, 1, 1 -bits input).

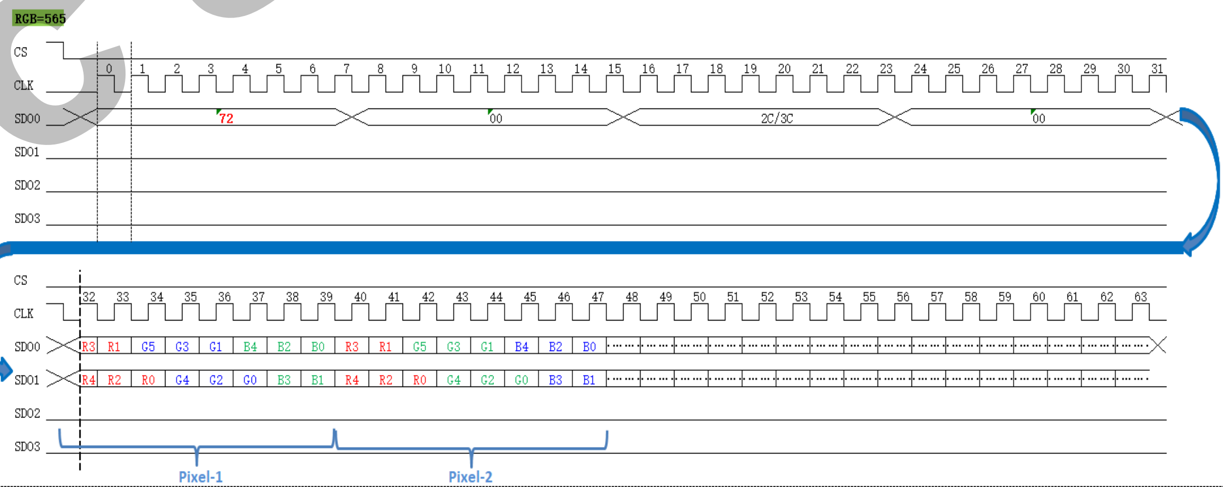


2wire data: only use SDA0, first byte=0x72

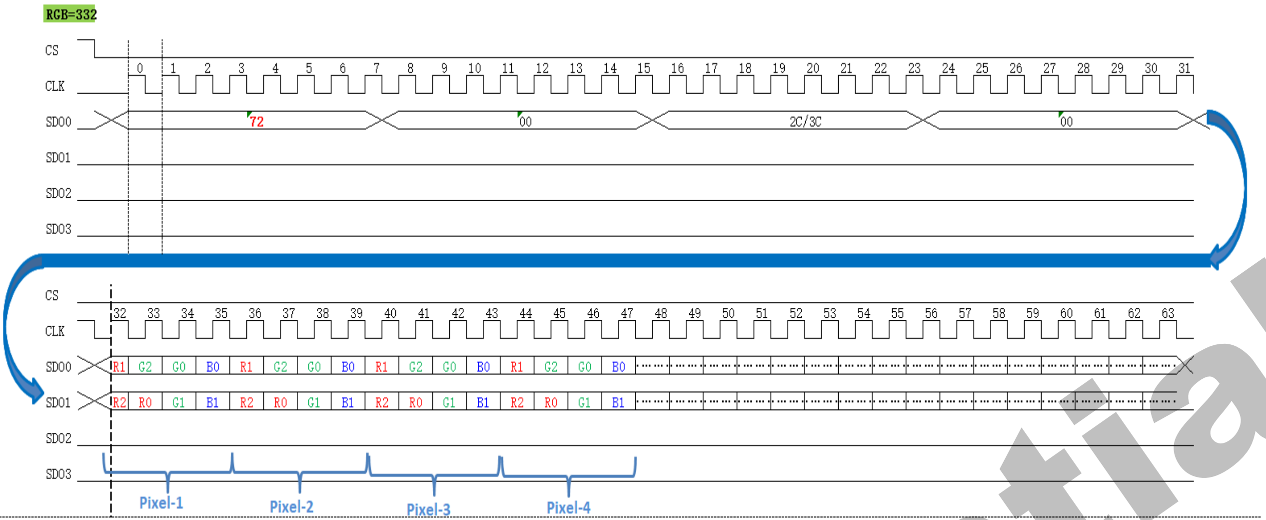
1) 262k-Colors:18-bit/pixel (RGB 6, 6, 6-bits input).



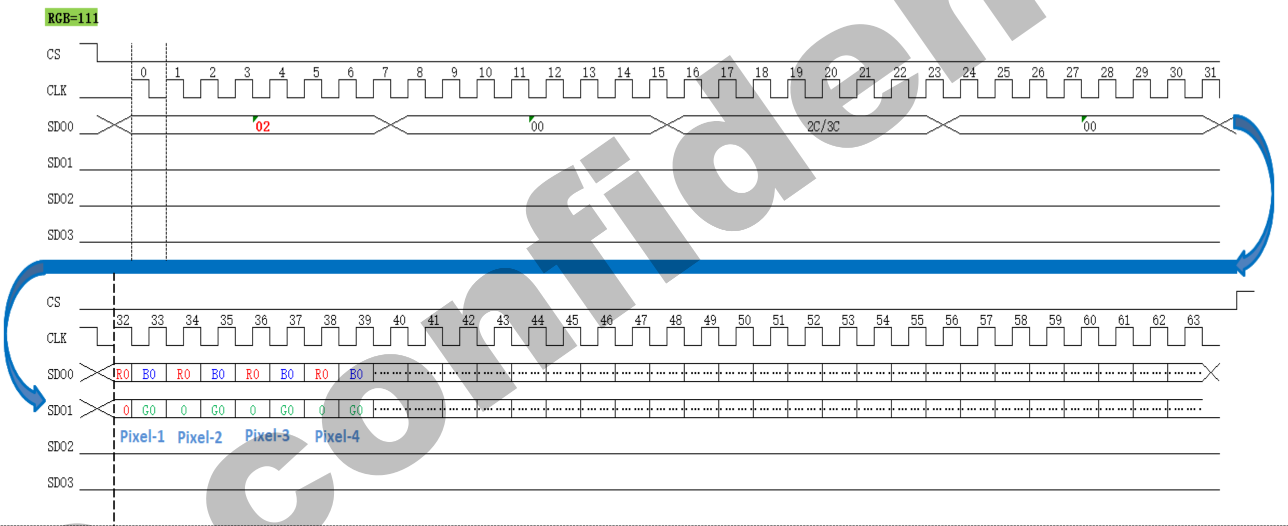
2) 65k-Colors:16-bit/pixel (RGB 5, 6,5 -bits input).



3) 256-Colors:8-bit/pixel (RGB 3, 3, 2 -bits input).



4) 8-Colors:3-bit/pixel (RGB 1, 1, 1 -bits input).



5. Function Description

5.1. Display data GRAM mapping

The display data RAM stores display dots and consists of 1,382,400 bits (240x18x320 bits). There is no restriction on access to the RAM even when the display data on the same address is loaded to DAC. There will be no abnormal visible effect on the display when there is a simultaneous Panel Read and Interface Read or Write to the same location of the Frame Memory.

Every pixel (18-bit) data in GRAM is located by a (Page, Column) address (Y, X). By specifying the arbitrary window address **SC, EC** bits and **SP, EP** bits, it is possible to access the GRAM by setting RAMWR or RAMRD commands from start positions of the window address.

GRAM address for display panel position as shown in the following table

Table31.

(00,00)h	(00,01)h	…… …	(00,ED)h	(00,EE)h	(00,EF)h
(01,00)h	(01,01)h	…… …	(01,ED)h	(01,EE)h	(01,EF)h
(02,00)h	(02,01)h	…… …	(02,ED)h	(02,EE)h	(02,EF)h
(03,00)h	(03,01)h	…… …	(03,ED)h	(03,EE)h	(03,EF)h
. . .	. . .	. . .	. . .	. . .	. . .
(13D,00)h	(13D,01)h	…… …	(13D,ED)h	(13D,EE)h	(13D,EF)h
(13E,00)h	(13E,01)h	…… …	(13E,ED)h	(13E,EE)h	(13E,EF)h
(13F,00)h	(13F,01)h	…… …	(13F,ED)h	(13F,EE)h	(13F,EF)h

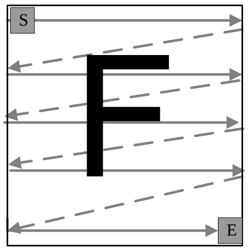
5.2. Address Counter (AC) of GRAM

The GC9309NA contains an address counter (AC) which assigns address for writing/reading pixel data to/from GRAM. The address pointers set the position of GRAM. Every time when a pixel data is written into the GRAM, the X address or Y address of AC will be automatically increased by 1 (or decreased by 1), which is decided by the register (**MV, MX** and **MY** bits) setting.

To simplify the address control of GRAM access, the window address function allows for writing data only to a window area of GRAM specified by registers. After data being written to the GRAM, the AC will be increased or decreased within setting window address-range which is specified by the (start: **SC**, end: **EC**) and the (start: **SP**, end: **EP**). Therefore, the data can be written consecutively without thinking a data wrap by those bit function.

Image data sending order from host and data stream update as shown in the following figure.

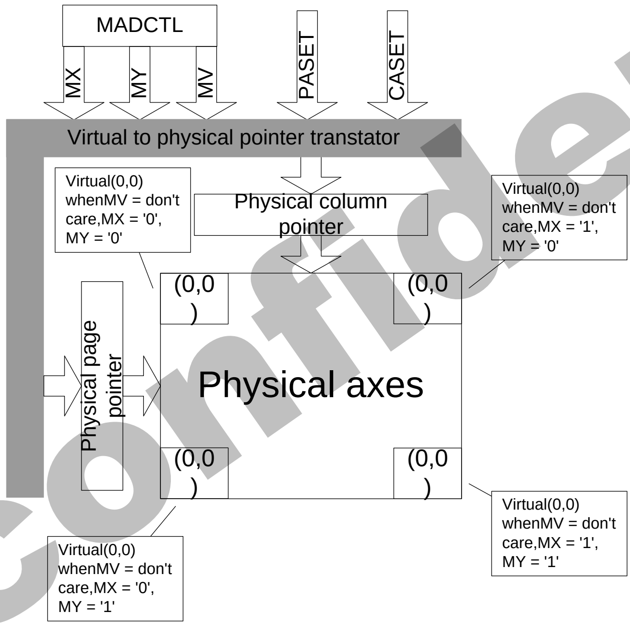
Figure64.



The data is written in the order illustrated above. The counter which dictates where in the physical memory the data is to be written is controlled by **MV, MX** and **MY** bits setting

Image data writing control:

Figure65.



CASET and PASET control for physical column/page pointers:

Table32.

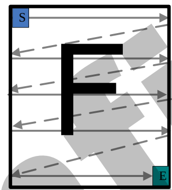
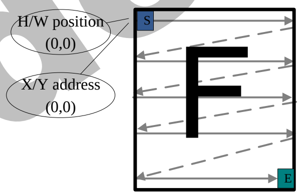
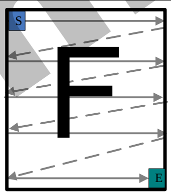
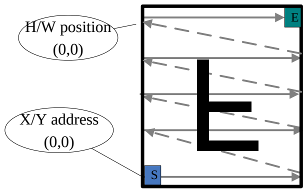
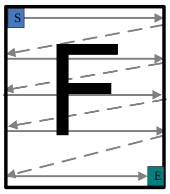
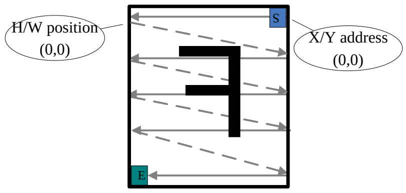
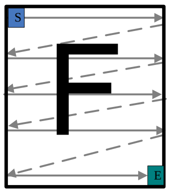
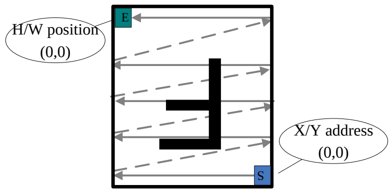
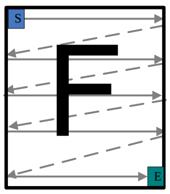
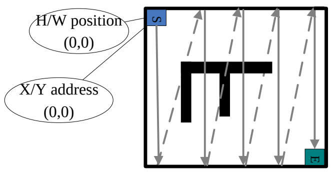
MV	MX	MY	CASET	PASET
0	0	0	Direct to Physical Column Pointer	Direct to Physical Page Pointer
0	0	1	Direct to Physical Column Pointer	Direct to (319 - Physical Page Pointer)
0	1	0	Direct to (239 - Physical Column Pointer)	Direct to Physical Page Pointer
0	1	1	Direct to (239 - Physical Column Pointer)	Direct to (319 - Physical Page Pointer)
0	0	0	Direct to Physical Page Pointer	Direct to Physical Column Pointer
0	0	1	Direct to (319 - Physical Page Pointer)	Direct to Physical Column Pointer
0	1	0	Direct to Physical Page Pointer	Direct to (239 - Physical Column

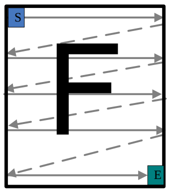
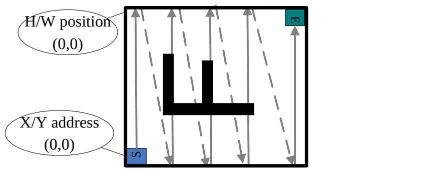
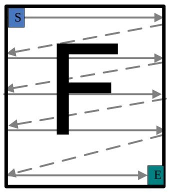
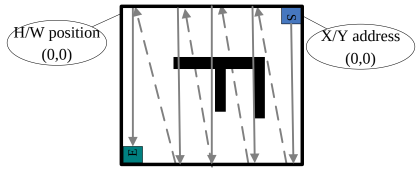
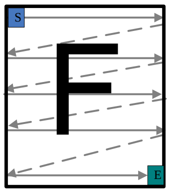
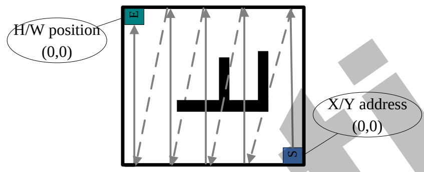
				Pointer)
0	1	1	Direct to (319 - Physical Page Pointer)	Direct to (239 - Physical Column Pointer)

condition	Column Counter	Page Counter
When RAMWR/RAMRD command is accepted	Return to “Start Column”	Return to “Start Page”
Complete Pixel Pair Write/Read action	Increment by 1	No change
The Column counter value is larger than “End column.”	Return to “Start Column”	Increment by 1
The Page counter value is larger than “End page”.	Return to “Start column”	Return to “Start Page”

The following figure depicts the GRAM address update method with MV, MX and MY bit setting.

Table33.

Display data direction	MV	MX	MY	Image in the Host	Image in the Driver (GRAM)
normal	0	0	0		
Y-invert	0	0	1		
X-invert	0	1	0		
Y-invert X-invert	0	1	1		
X-Y exchange	1	0	0		

X-Y exchange Y-invert	1	0	1		
X-Y exchange X-invert	1	1	0		
X-Y exchange Y-invert X-invert	1	1	1		

5.3. GRAM to display address mapping

By setting the **SS**, the relation between the source output channel and the GRAM address can be changed as reverse display. By setting the **GS** , the relation between the gate output channel and the GRAM address can be changed as reverse display. By setting the **BGR**, the relation between the source output channel and the <R>, <G>, <B> dot allocation can be reversed for different LCD color filter arrangement.

The following Tables show relations among the GRAM data allocation, the source output channel, and the R, G, B dot allocation.

GRAM X address and display panel position:

Table34.

BGR="0"														
Source Output	**SS="0"**	S1	S2	S3	S4	S5	S6	------	S715	S716	S717	S718	S719	S720
	SS="1"	S718	S719	S720	S715	S716	S717	------	S4	S5	S6	S1	S2	S3
GRAM X address		"00"h			"01"h			------	"EE"h			"EF"h		
RGB data		R	G	B	R	G	B	------	R	G	B	R	G	B
Pixel		Pixel1			Pixel2			------	Pixel239			Pixel240		
BGR="1"														
Source Output	**SS="0"**	S3	S2	S1	S6	S5	S4	------	S717	S716	S715	S720	S719	S718
	SS="1"	S720	S719	S718	S717	S716	S715	------	S6	S5	S4	S3	S2	S1

GRAM X address	"00"h			"01"h			------	"EE"h			"EF"h		
RGB data	R	G	B	R	G	B	------	R	G	B	R	G	B
Pixel	Pixel1			Pixel2			------	Pixel239			Pixel240		

GRAM address and display panel position (GS_Panel ='0'):

Table35.

S/G pins	S1	S2	S3	S4	S5	S6	S7	S8	S9	----	S712	S713	S714	S715	S716	S717	S718	S719	S720
G1	0000h			0001h			0002h			----	00EDh			00EEh			00EFh		
G2	0100h			0101h			0102h			----	01EDh			01EEh			01EFh		
G3	0200h			0201h			0202h			----	02EDh			02EEh			02EFh		
G4	0300h			0301h			0302h			----	03EDh			03EEh			03EFh		
G5	0400h			0401h			0402h			----	04EDh			04EEh			04EFh		
G6	0500h			0501h			0502h			----	05EDh			05EEh			05EFh		
⋮	⋮			⋮			⋮			⋮	⋮			⋮			⋮		
G315	13A00h			13A01h			13A02h			----	13AEDh			13AEEh			13AEFh		
G316	13B00h			13B01h			13B02h			----	13BEDh			13BEEh			13BEFh		
G317	13C00h			13C01h			13C02h			----	13CEDh			13CEEh			13CEFh		
G318	13D00h			13D01h			13D02h			----	13DEDh			13DEEh			13DEFh		
G319	13E00h			13E01h			13E02h			----	13EEDh			13EEEh			13EEFh		
G320	13F00h			13F01h			13F02h			----	13FEDh			13FEEh			13FEFh		

GRAM address and display panel position (GS_Panel ='1'):

Table36.

S/G pins	S1	S2	S3	S4	S5	S6	S7	S8	S9	----	S712	S713	S714	S715	S716	S717	S718	S719	S720
G320	0000h			0001h			0002h			----	00EDh			00EEh			00EFh		
G319	0100h			0101h			0102h			----	01EDh			01EEh			01EFh		
G318	0200h			0201h			0202h			----	02EDh			02EEh			02EFh		
G317	0300h			0301h			0302h			----	03EDh			03EEh			03EFh		
G316	0400h			0401h			0402h			----	04EDh			04EEh			04EFh		
G315	0500h			0501h			0502h			----	05EDh			05EEh			05EFh		

┆	┆	┆	┆	┆	┆	┆	┆
G6	13A00h	13A01h	13A02h	----	13AEDh	13AEEh	13AEFh
G5	13B00h	13B01h	13B02h	----	13BEDh	13BEEh	13BEFh
G4	13C00h	13C01h	13C02h	----	13CEDh	13CEEh	13CEFh
G3	13D00h	13D01h	13D02h	----	13DEDh	13DEEh	13DEFh
G2	13E00h	13E01h	13E02h	----	13EEDh	13EEEh	13EEFh
G1	13F00h	13F01h	13F02h	----	13FEDh	13FEEh	13FEFh

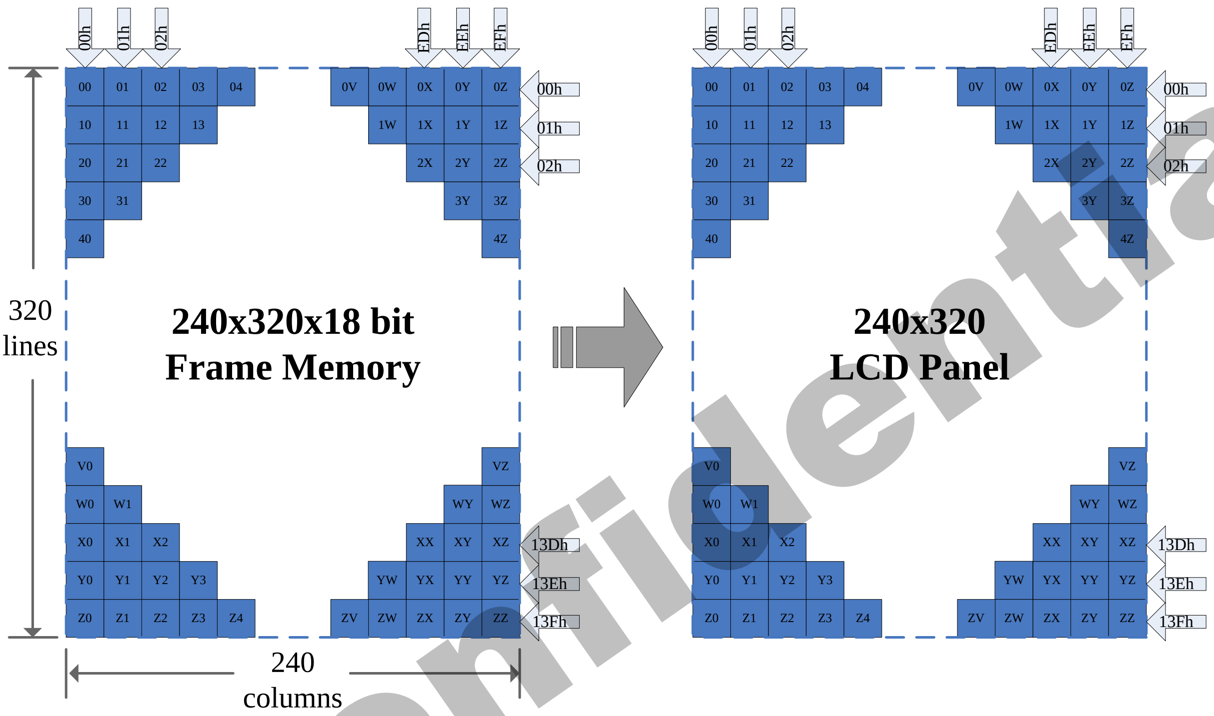
GC9309NA supports three kinds of display mode: one is Normal Display Mode, the other is Partial Display Mode, and Scrolling Display Mode.

5.3.1. Normal display on or partial mode on, vertical scroll off

In this mode, content of the frame memory within an area where column pointer is 0000h to 00EFh and page pointer is 0000h to 013Fh is displayed.

To display a dot on leftmost top corner, store the dot data at (column pointer, page pointer) = (0,0)

Figure66.

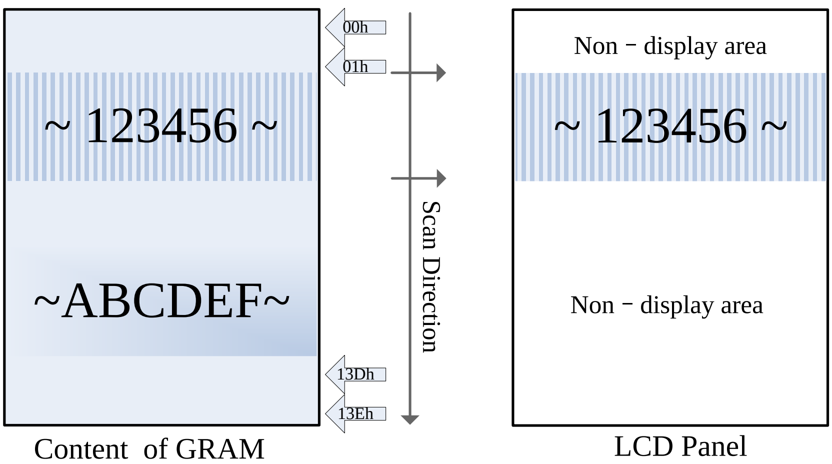


Example1:

(1) partial mode on (setting 12h)

(2) SR [15:0] =50DEC, ER [15:0] =150DEC, MADCTL's **B4(ML)='0'** (GS='0').

Figure67.

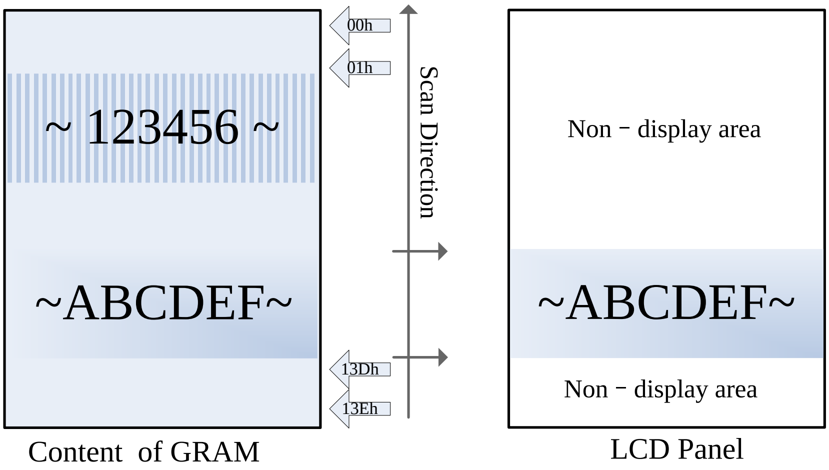


Example2:

(1) partial mode on (setting 12h)

(2) SR [15:0] =50DEC, ER [15:0] =150DEC, MADCTL's **B4(ML)='1'** (GS='0').

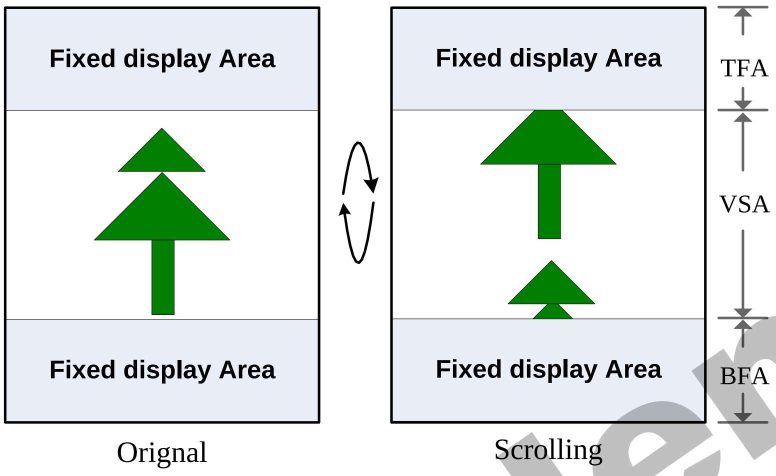
Figure68.


00h
01h
~ 123456 ~
~ABCDEF~
13Dh
13Eh
Scan Direction
Content of GRAM
Non－display area
~ABCDEF~
Non－display area
LCD Panel

5.3.2. Vertical scroll display mode

When setting R37h, the scrolling display mode is active, and the vertical scrolling display is specified by **TFA, VSA ,BFA** bits (R33h) and **VSP** bits (R37h).

Figure69.

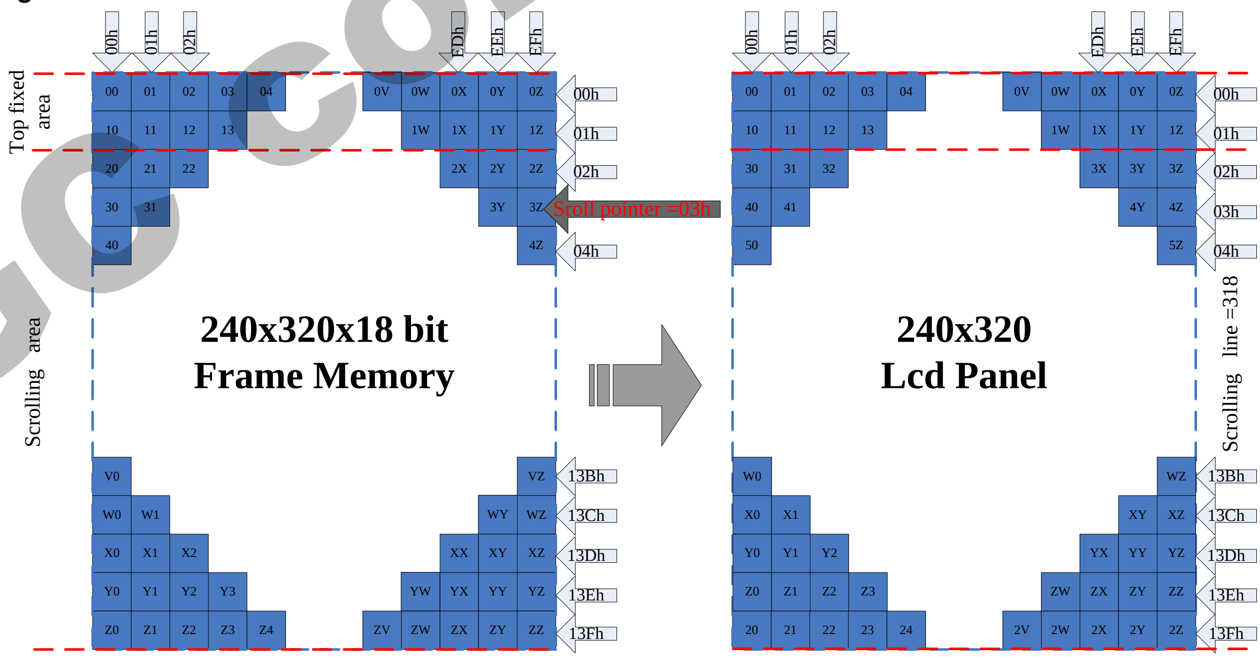


When Vertical Scrolling Definition Parameters (TFA+VSA+BFA) =320. In this case, scrolling is applied as shown below.

Example 1 .TFA='2d', VSA='318d', BFA='0d', VSP='3d' (SS='0', GS='0')

Memory map of vertical scrolling 1:

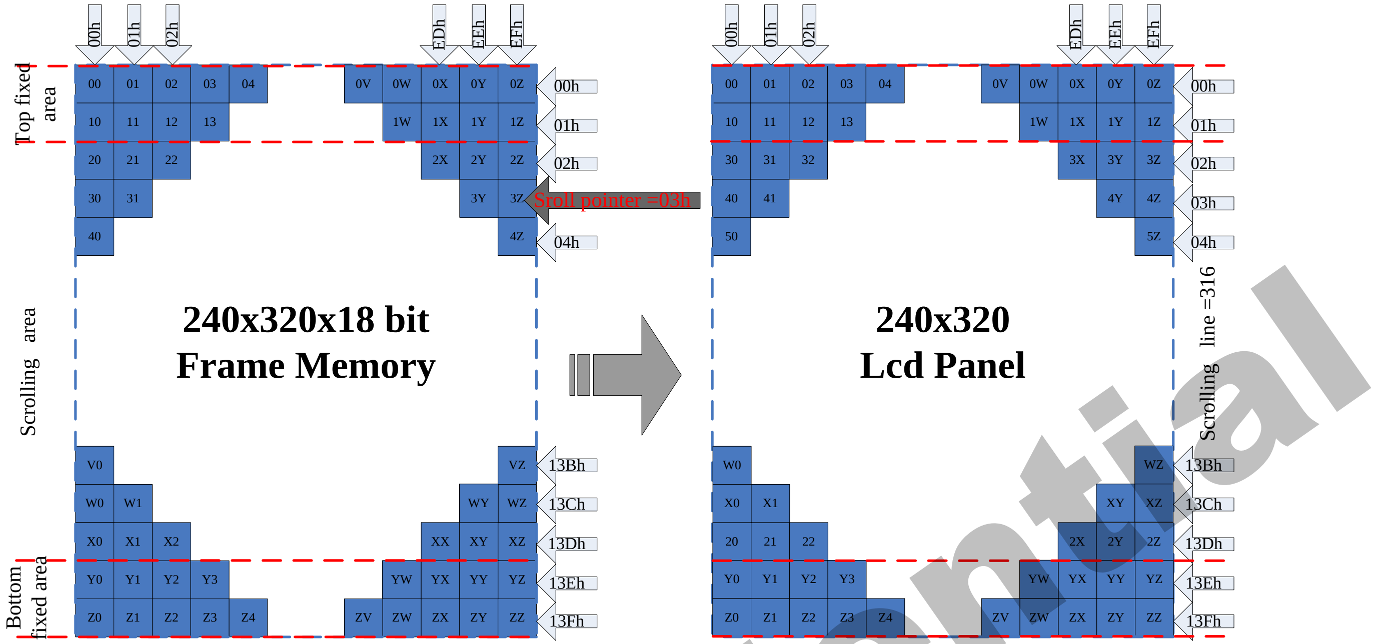
Figure70.



Example 2 .TFA='2d', VSA='316d', BFA='2d', VSP='3d' (SS='0', GS='0')

Memory map of vertical scrolling 2:

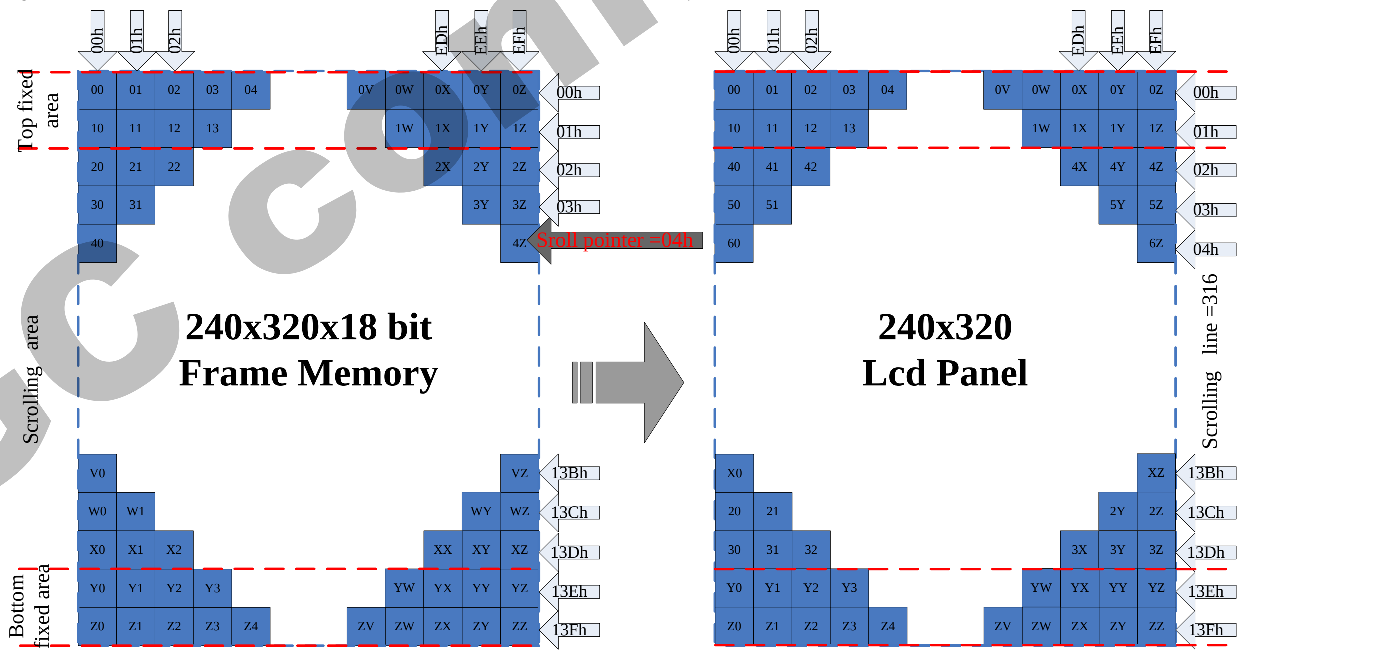
Figure71.



Example 3 .TFA='2d', VSA='316d', BFA='2d', VSP='4d' (SS='0', GS='0')

Memory map of vertical scrolling 3:

Figure72.



Vertical scroll example

There are 2 types of vertical scrolling, which are determined by the **TFA, VSA, BFA** bits and **VSP** bits

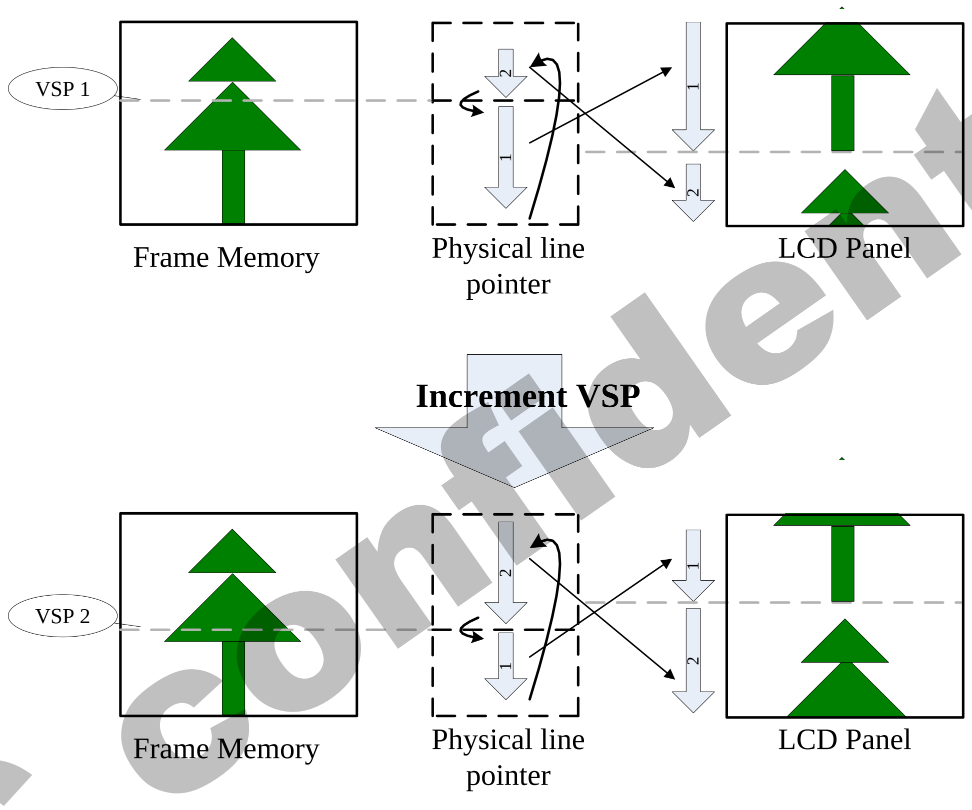
Case 1: TFA + VSA + BFA ≠ ‘320d’

N/A. Do not set TFA + VSA + BFA ≠ ‘320d’. In that case, unexpected picture will be shown.

Case 2: TFA + VSA + BFA = ‘320d’ (Scrolling)

Example (1) When TFA=’0d’, VSA=’320d’, BFA=’0d’ and VSP1=’40d’ & VSP2=’140d’ (SS =’0’,GS=’0’)

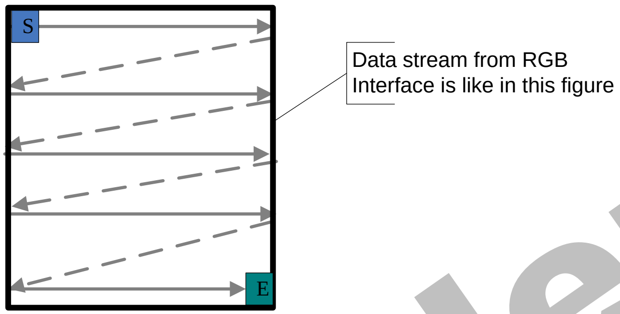
Figure73.



5.3.3. Updating order on display active area in RGB interface mode

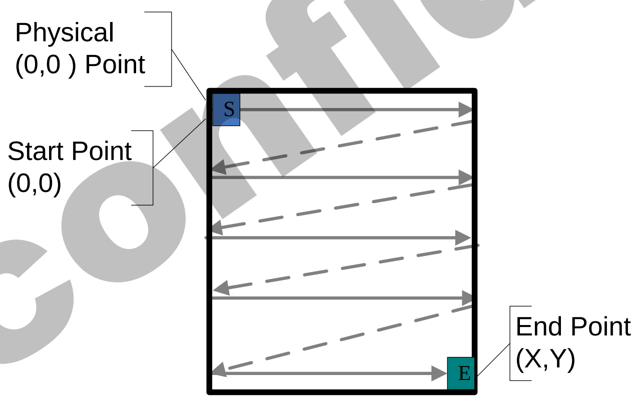
There is defined different kind of updating orders for display in RGB interface mode
(**RCM [1:0]** ='1x').
These updating are controlled by **MY** and **MX** bits. Data streaming direction from the host to the display is described in the following figure.

Figure74.



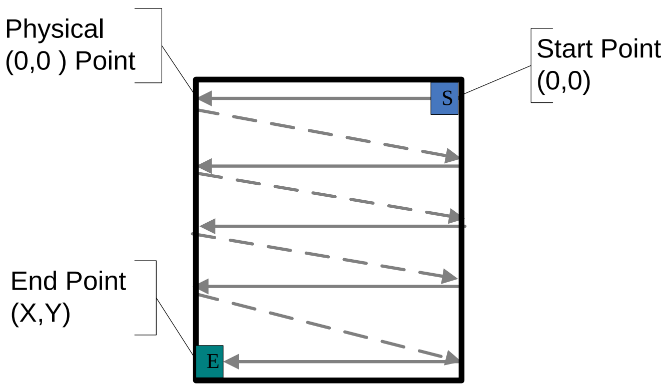
Updating order when MY = '0' and MX = '0'

Figure75.



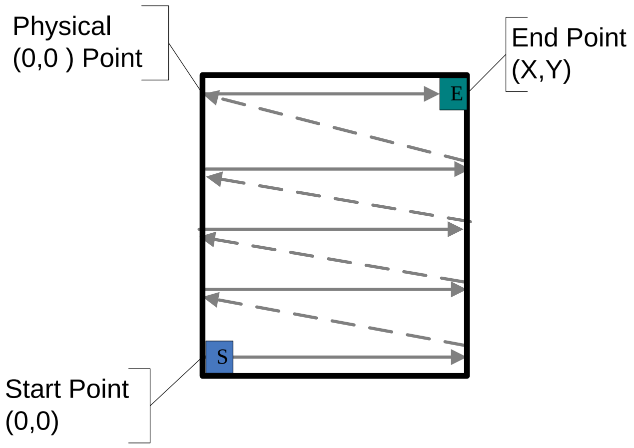
Updating order when MY = '0' and MX = '1'

Figure76.



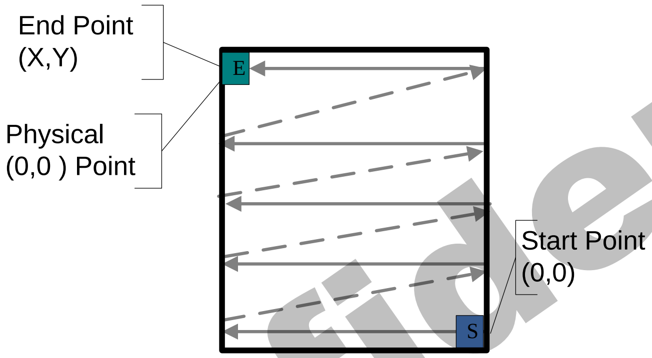
Updating order when MY = '1' and MX = '0'

Figure77.



Updating order when MY = ‘1’ and MX = ‘1’

Figure78.



Rules for updating order on display active area in RGB interface display mode:

Table37.

Condition	Horizontal Counter	Vertical Counter
An active VS signal is received	Return to 0	Return to 0
Single Pixel information of the active area is received	Increment by 1	No change
An active HS signal between two active area lines	Return to 0	Increment by 1
The Horizontal counter value is larger than X and the Vertical counter value is larger than Y	Return to 0 “Start Column”	Return to “Start Page”

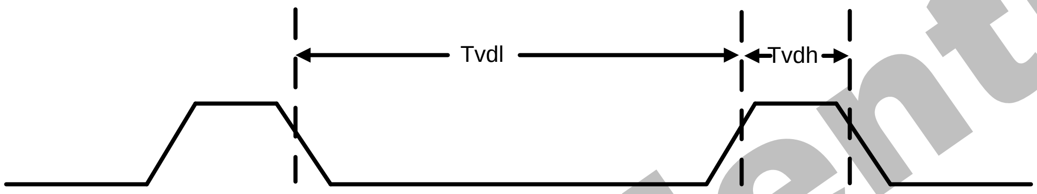
Note: Pixel order is RGB on the display.

5.4. Tearing effect output line

The Tearing Effect output line supplies to the MPU a Panel synchronization signal. This signal can be enabled or disabled by the Tearing Effect Line Off & On commands. The mode of the Tearing Effect signal is defined by the parameter of the Tearing Effect Line On command. The signal can be used by the MPU to synchronize Frame Memory Writing when displaying video images.

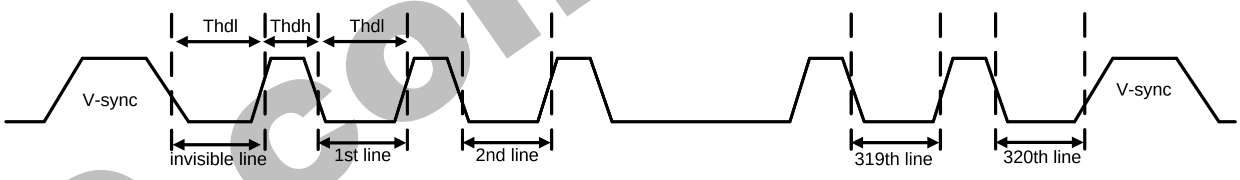
5.4.1. Tearing effect line modes

Mode 1, The Tearing Effect Output signal consists of V-Blanking Information only:
Figure79.



tVdh= The LCD display is not updated from the Frame Memory
tvdl = The LCD display is updated from the Frame Memory (except Invisible Line – see below)

Mode 2, The Tearing Effect Output signal consists of V-Blanking and H-Blanking Information, there is one V-sync and 320 H-sync pulses per field.
Figure80.

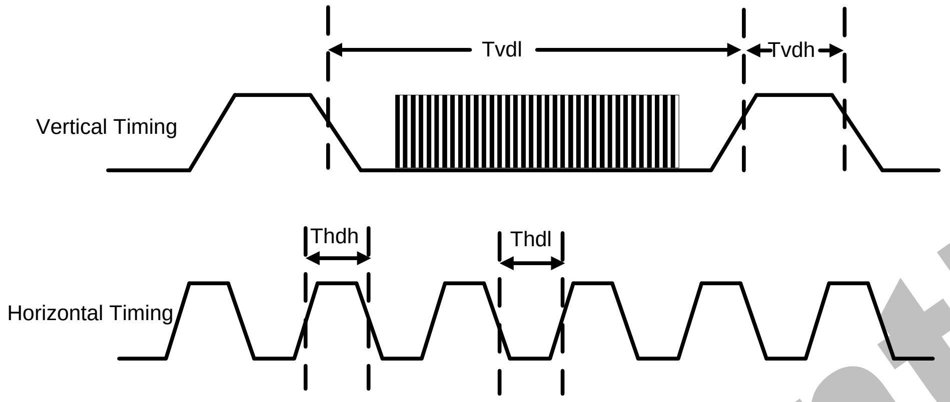


thdh= The LCD display is not updated from the Frame Memory
thdl= The LCD display is updated from the Frame Memory (except Invisible Line – see above)

5.4.2. Tearing effect line timing

The Tearing Effect signal is described below.

Figure81.



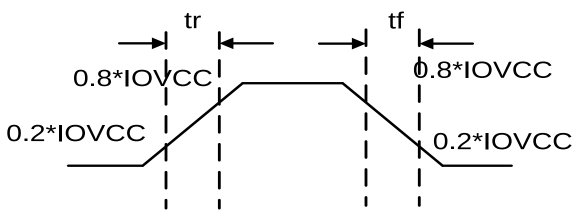
Idle Mode Off (Frame Rate = 60 Hz)

Table38.

Symbol	Parameter	Spec.			Description
		Min.	Max.	Unit	
tvdl	Vertical Timing Low Duration	TBD	-	ms	-
tvdh	Vertical Timing High Duration	1000	-	us	-
thdl	Horizontal Timing Low Duration	TBD	-	us	-
thdh	Horizontal Timing High Duration	TBD	500	us	-

Note: Idle Mode Off (Frame Rate = 60 Hz) ,The signal's rise and fall times (tf, tr) are stipulated to be equal to or less than 15ns.

Figure82.



The Tearing Effect Output Line is fed back to the MCU and should be used to avoid Tearing Effect.

5.5. Source driver

The GC9309NA contains a 720 channels of source driver (S1~S720) which is used for driving the source line of TFT LCD panel. The source driver converts the digital data from GRAM into the analog voltage for 720 channels and generates corresponding
gray scale voltage output, which can realize a 262K colors display simultaneously. Since the output circuit of this source driver incorporates an operational amplifier, a positive and a negative voltage can be alternately outputted from each channel.

5.6. Gate driver

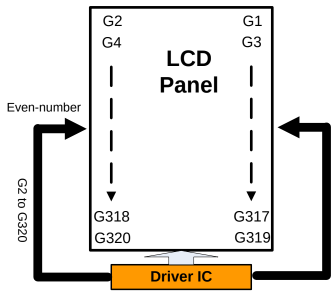
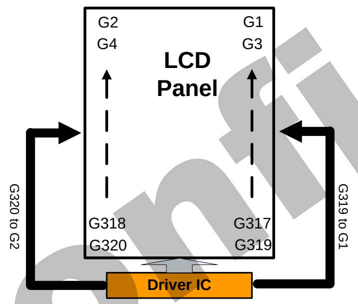
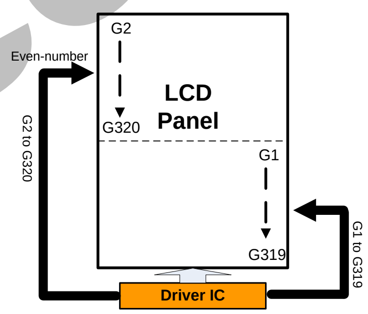
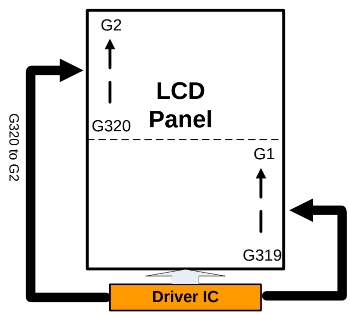
The GC9309NA contains a 320 gate channels of gate driver (G1~G320) which is usedfor driving the gate. The gate driver level is VGH when scan some line, VGL the other lines.

5.7. Scan mode setting

GS: Sets the direction of scan by the gate driver, The scan direction determined by GS = 0 can be reversed by setting GS = 1.

SM: Sets the gate driver pin arrangement in combination with the GS bit to select the optimal scan mode for the module.

Table39.

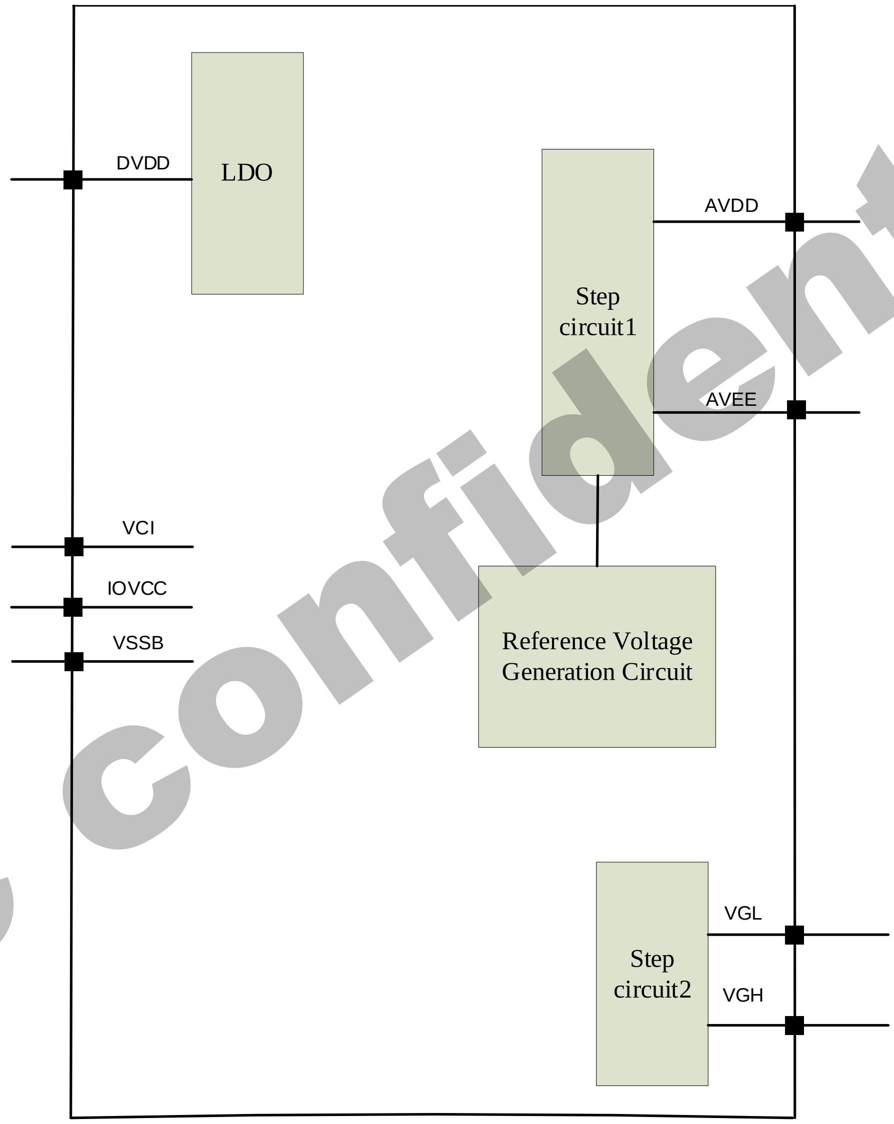
SM	GS	Scan Direction	Gate Output Sequence
0	0		G1 G2 G3 G4 ---> G317 G318 G319 G320
0	1		G320 G319 G 318 G317 ---> G4 G3 G2 G1
1	0		G1 G3 ---> G317 G319 --> G2 G4 ---> G318 G320
1	1		G320 G318 ---> G4 G2 --> G319 G317 ---> G3 G1

5.8. LCD power generation circuit

5.8.1. Power supply circuit

The power circuit of GC9309NA is used to generate supply voltages for LCD panel driving.

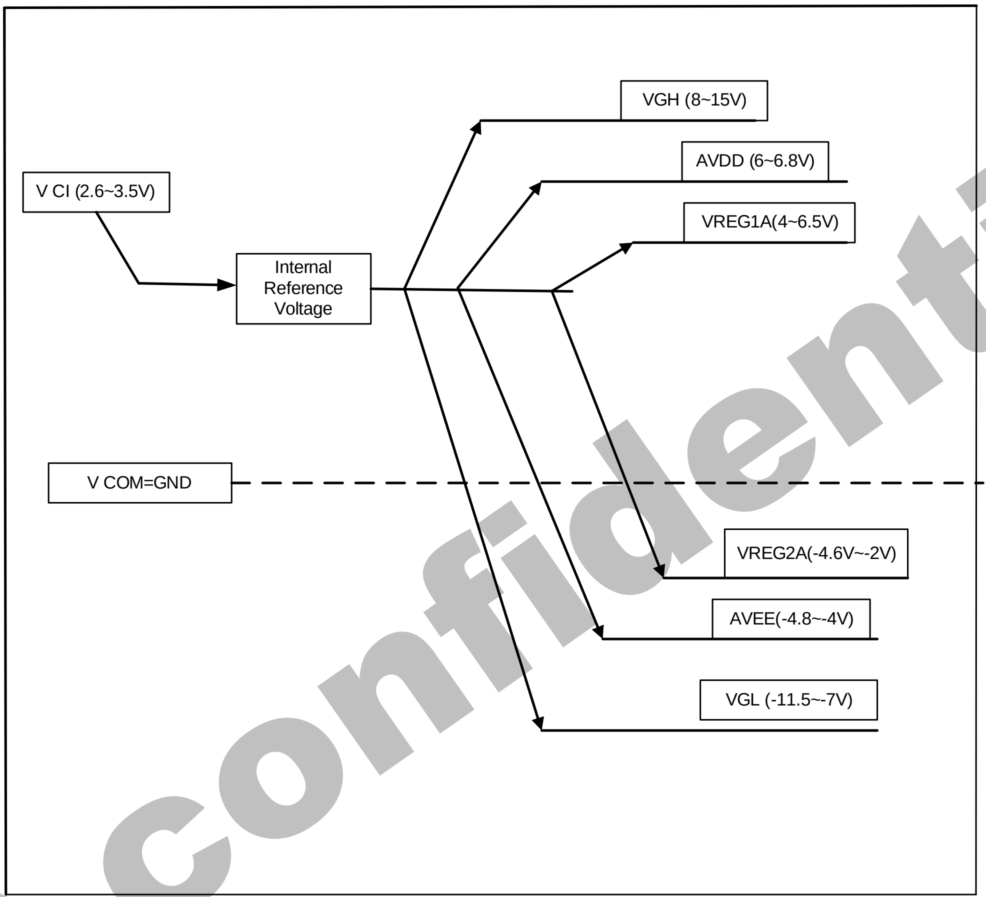
Figure83.



5.8.2. LCD power generation scheme

The boost voltage generated is shown as below.

Figure84.



LCD power generation scheme

5.9. Gamma Correction

GC9309NA incorporates the γ-correction function to display 262,144 colors for the LCD panel. The γ-correction is performed with 2 groups of registers determining eight reference grayscale levels, which are gradient adjustment, amplitude adjustment and fine-adjustment registers for positive and negative polarities, to make GC9309NA available with liquid crystal panels of various characteristics.

Figure85.

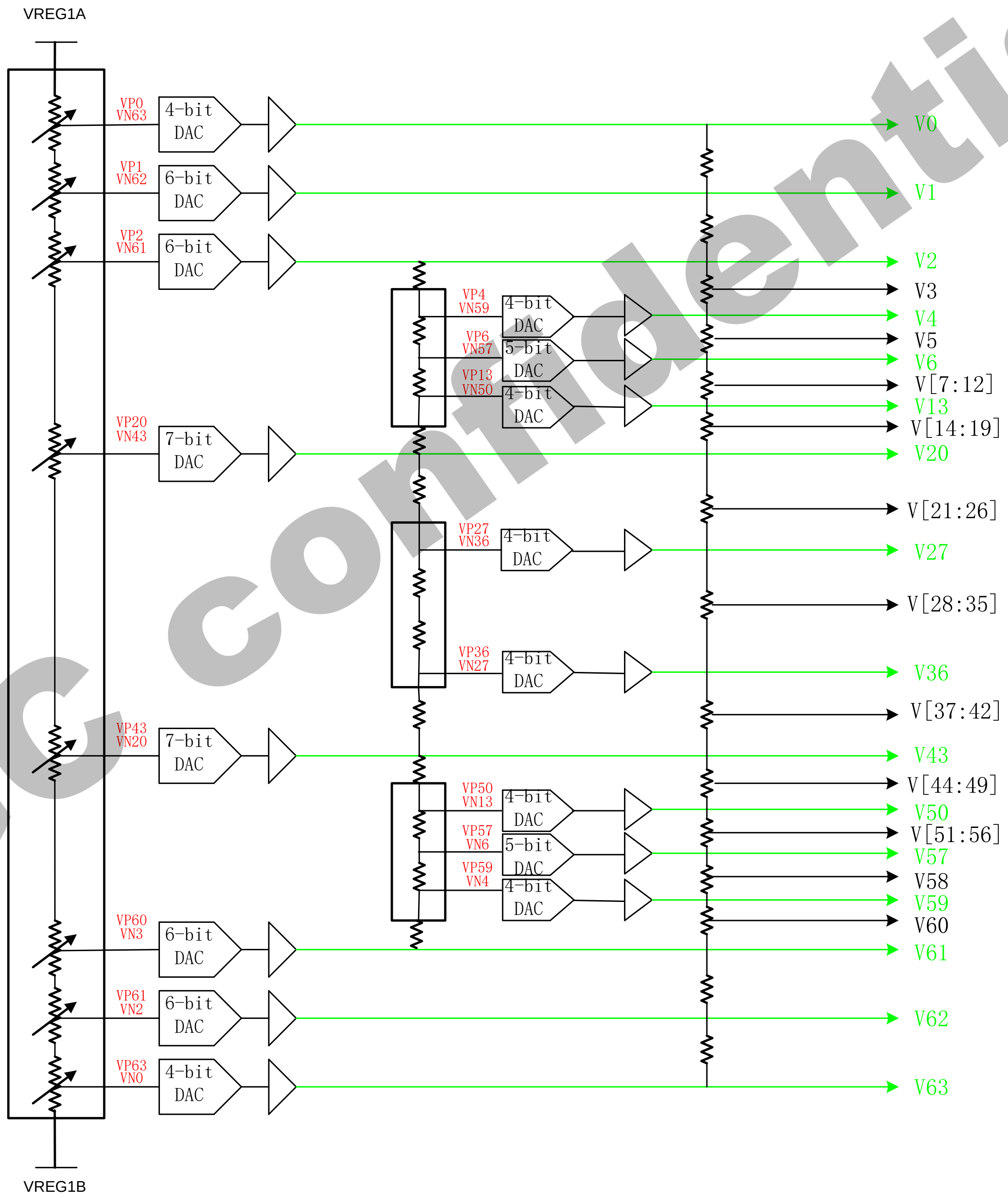
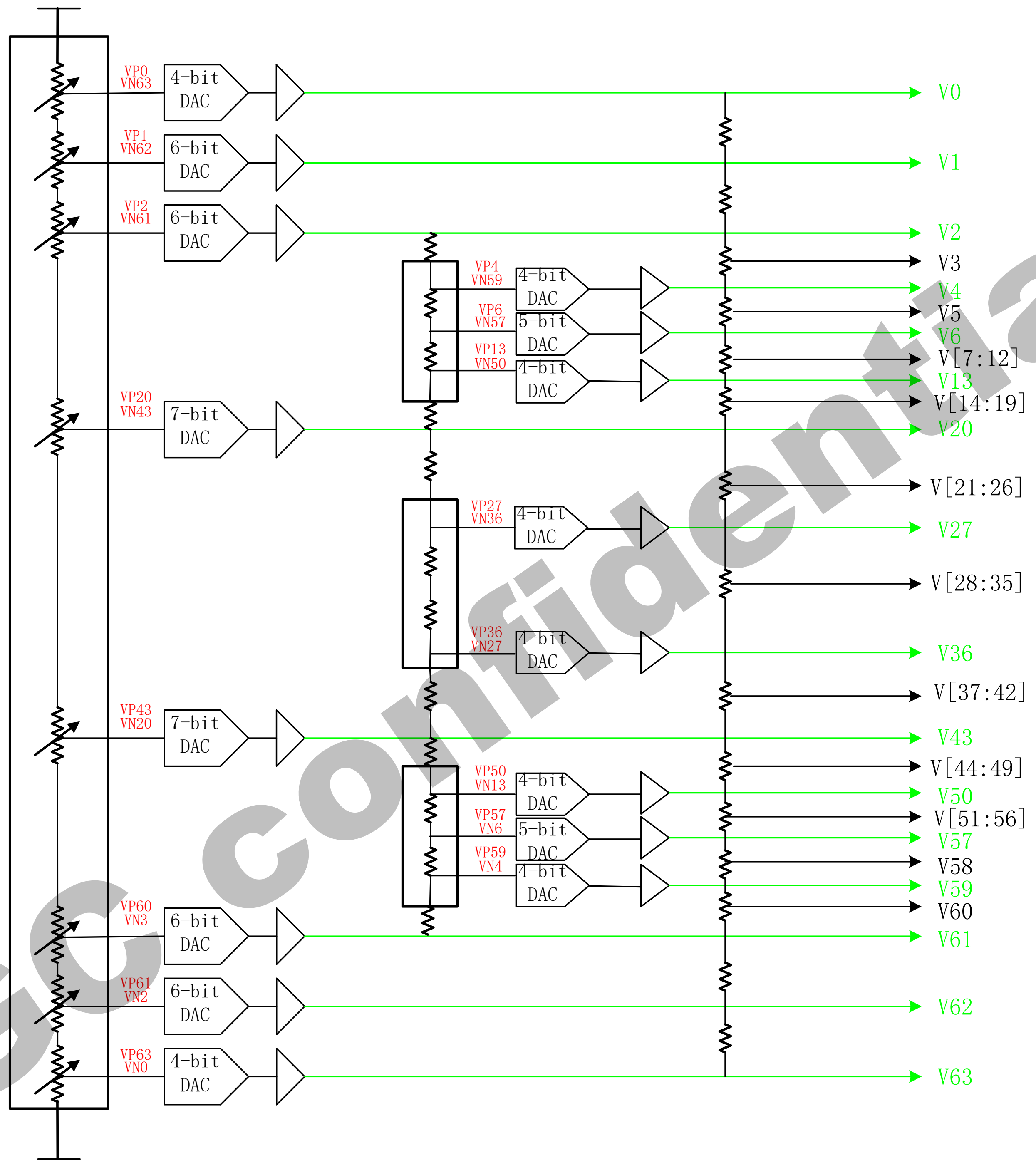
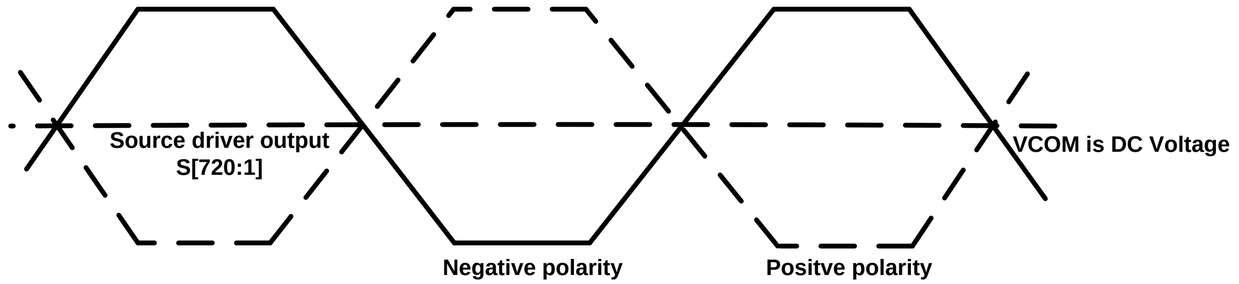


Figure86.



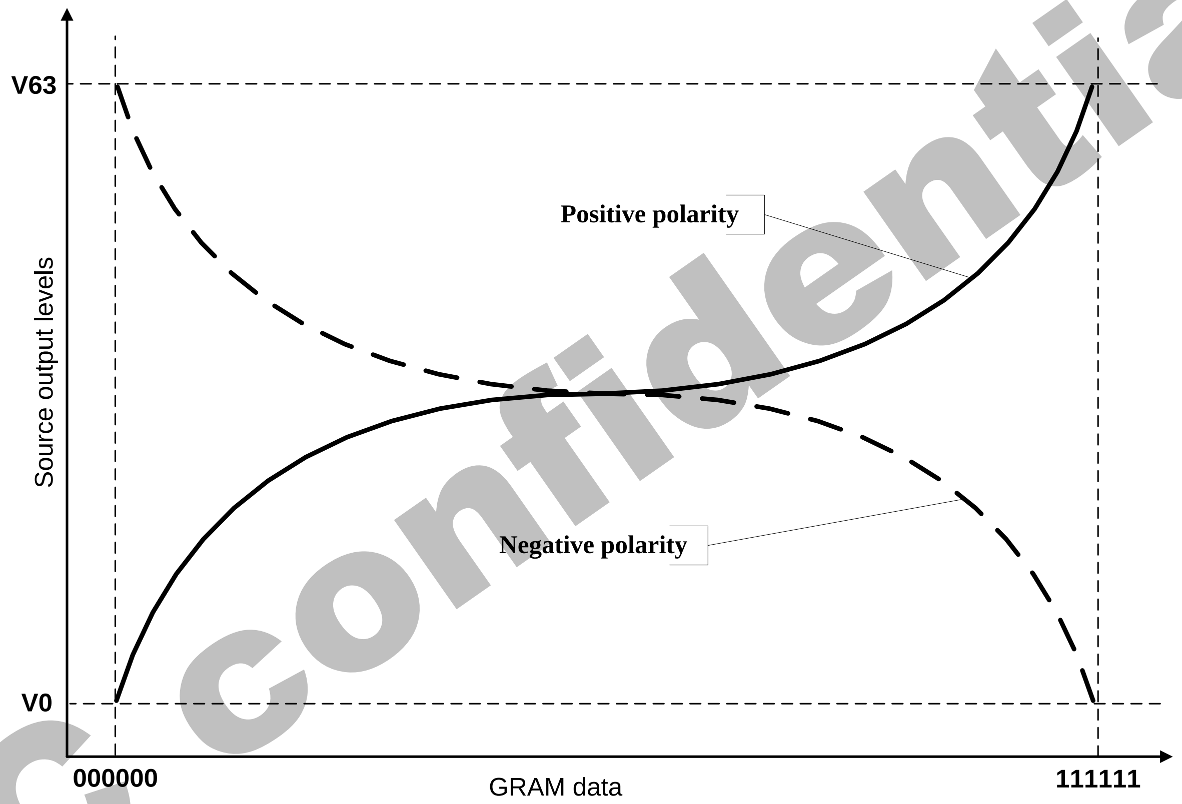
Grayscale Voltage Generation

Figure87.Dot inversion



Relationship between Source Output and VCOM

Figure88.



5.10. Power Level Definition

5.10.1. Power Levels

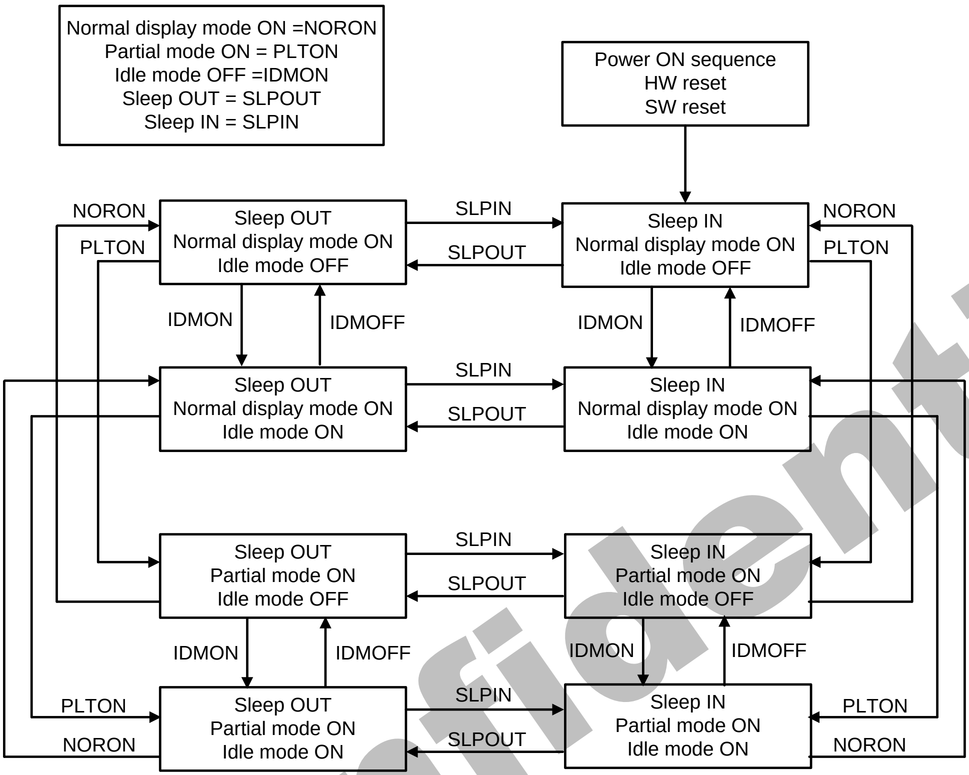
6 level modes are defined they are in order of Maximum Power consumption to Minimum Power Consumption:

1. Normal Mode On (full display), Idle Mode Off, Sleep Out.
   In this mode, the display is able to show maximum 262,144 colors.
2. Partial Mode On, Idle Mode Off, Sleep Out.
   In this mode part of the display is used with maximum 262,144 colors.
3. Normal Mode On (full display), Idle Mode On, Sleep Out.
   In this mode, the full display area is used but with 8 colors.
4. Partial Mode On, Idle Mode On, Sleep Out.
   In this mode, part of the display is used but with 8 colors.
5. Sleep In Mode.
   In this mode, the DC : DC converter, Internal oscillator and panel driver circuit are stopped. Only the MCU interface and memory works with IOVCC power supply. Contents of the memory are safe.
6. Power Off Mode.
   In this mode, both VCI and IOVCC are removed.

Note1: Transition between modes 1-5 is controllable by MCU commands. Mode 6 is entered only when both Power supplies are removed.

5.10.2. Power Flow Chart

Figure89.

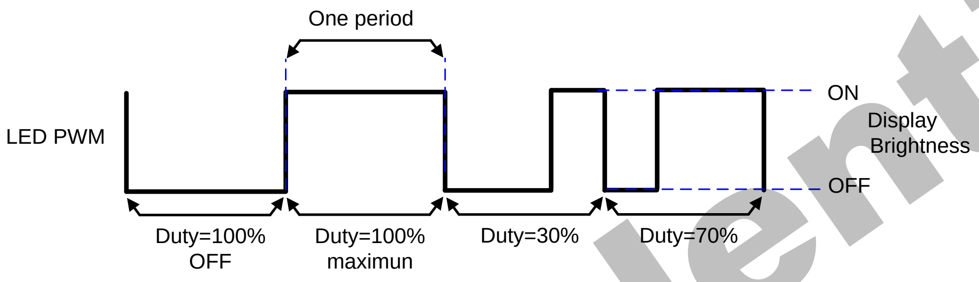


Note 1: There is not any abnormal visual effect when there is changing from one power mode to another power mode.

Note 2: There is not any limitation, which is not specified by User, when there is changing from one power mode to another power mode.

5.10.3.Brightness control block

There is an external output signal from brightness block, LEDPWM to control the LED driver IC in order to control display brightness.
There are resister bits, DBV[7:0] of R51h, for display brightness of manual brightness setting. The LEDPWM duty is calculated as DBV[7:0]/255 x period (affected by OSC frequency).
For example: LEDPWM period = 3ms, and DBV[7:0] = ‘200DEC’. Then LEDPWM duty = 200 / 255=78.1%. Correspond to the LEDPWM period = 3 ms, the high-level of LEDPWM (high effective) = 2.344ms, and the low-level of LEDPWM = 0.656ms.
Figure90.



LEDPWM output duty

5.11. Input/output pin state

5.11.1. Output pins

Table40.

Output or Bi-directional pins	After Power On	After Hardware Reset
DB17 to DB0 (Output driver)	High-Z (Inactive)	High-Z (Inactive)
SDA	High-Z (Inactive)	High-Z (Inactive)
SDO	High-Z (Inactive)	High-Z (Inactive)
TE	Low	Low
LEDPWM	Low	Low

Characteristics of output pins

5.11.2. Input pins

Table41.

Input pins	During Power On Process	After Power On	After Hardware Reset	During Power Off Process
RESX	Input valid	Input valid	Input valid	Input valid
CSX	Input invalid	Input valid	Input valid	Input invalid
WRX	Input invalid	Input valid	Input valid	Input invalid
RDX	Input invalid	Input valid	Input valid	Input invalid
D/CX	Input invalid	Input valid	Input valid	Input invalid
SDA	Input invalid	Input valid	Input valid	Input invalid
VSYNC	Input invalid	Input valid	Input valid	Input invalid
HSYNC	Input invalid	Input valid	Input valid	Input invalid
DE	Input invalid	Input valid	Input valid	Input invalid
DOTCLK	Input invalid	Input valid	Input valid	Input invalid
D[17:0]	Input invalid	Input valid	Input valid	Input invalid
IM[3:0]	Input invalid	Input valid	Input valid	Input invalid

Characteristics of input pins

6. Command

6.1. Command List

Regulative Command Set													
Command Function	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Read Display Identification Information 2	0	1	↑	XX	0	0	0	0	0	1	0	0	04h
	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XX	ID_1[7:0]								00
	1	↑	1	XX	ID_2[7:0]								93
	1	↑	1	XX	ID_3[7:0]								09
Read Display Status	0	1	↑	XX	0	0	0	0	1	0	0	1	09h
	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XX	D[31:25]							X	00
	1	↑	1	XX	X	D[22:20]			D[19:16]				61
	1	↑	1	XX	X	X	X	X	X	D[10:8]			00
	1	↑	1	XX	D[7:5]			X	X	X	X	X	00
Enter Sleep Mode	0	1	↑	XX	0	0	0	1	0	0	0	0	10h
Sleep OUT	0	1	↑	XX	0	0	0	1	0	0	0	1	11h
Partial Mode ON	0	1	↑	XX	0	0	0	1	0	0	1	0	12h
Normal Display Mode ON	0	1	↑	XX	0	0	0	1	0	0	1	1	13h
Display Inversion OFF	0	1	↑	XX	0	0	1	0	0	0	0	0	20h
Display Inversion ON	0	1	↑	XX	0	0	1	0	0	0	0	1	21h
Display OFF	0	1	↑	XX	0	0	1	0	1	0	0	0	28h
Display ON	0	1	↑	XX	0	0	1	0	1	0	0	1	29h
Column Address Set	0	1	↑	XX	0	0	1	0	1	0	1	0	2Ah
	1	1	↑	XX	SC[15:8]								00
	1	1	↑	XX	SC[7:0]								00
	1	1	↑	XX	EC[15:8]								00
	1	1	↑	XX	EC[7:0]								EFh
Page Address Set	0	1	↑	XX	0	0	1	0	1	0	1	1	2Bh
	1	1	↑	XX	SP[15:8]								00
	1	1	↑	XX	SP[7:0]								00

	1	1	↑	XX	EP[15:8]								01h
	1	1	↑	XX	EP[7:0]								3Fh
Memory Write	0	1	↑	XX	0	0	1	0	1	1	0	0	2Ch
	1	1	↑	D[17:0]									XX
Partial Area	0	1	↑	XX	0	0	1	1	0	0	0	0	30h
	1	1	↑	XX	SR[15:8]								00
	1	1	↑	XX	SR[7:0]								00
	1	1	↑	XX	ER[15:8]								01
	1	1	↑	XX	ER[7:0]								3F
Vertical Scrolling Definition	0	1	↑	XX	0	0	1	1	0	0	1	1	33h
	1	1	↑	XX	TFA[15:8]								00
	1	1	↑	XX	TFA[7:0]								00
	1	1	↑	XX	VSA[15:8]								01
	1	1	↑	XX	VSA[7:0]								40
Tearing Effect Line OFF	0	1	↑	XX	0	0	1	1	0	1	0	0	34h
Tearing Effect Line ON	0	1	↑	XX	0	0	1	1	0	1	0	1	35h
	1	1	↑	XX	X	X	X	X	X	X	X	M	00
Memory Access Control	0	1	↑	XX	0	0	1	1	0	1	1	0	36h
	1	1	↑	XX	MY	MX	MV	ML	BGR	MH	X	X	00
Vertical Scrolling Start Address	0	1	↑	XX	0	0	1	1	0	1	1	1	37h
	1	1	↑	XX	VSP[15:8]								00
	1	1	↑	XX	VSP[7:0]								00
Idle Mode OFF	0	1	↑	XX	0	0	1	1	1	0	0	0	38h
Idle Mode ON	0	1	↑	XX	0	0	1	1	1	0	0	1	39h
Pixel Format Set	0	1	↑	XX	0	0	1	1	1	0	1	0	3Ah
	1	1	↑	XX	X	DPI[2:0]			X	DBI[2:0]			66
Write Memory Continue	0	1	↑	XX	0	0	1	1	1	1	0	0	3Ch
	1	1	↑	D[17:0]									XX
Set Tear Scanline	0	1	↑	XX	0	1	0	0	0	1	0	0	44h
	1	1	↑	XX	X	X	X	X	X	X	X	STS[8]	00
	1	1	↑	XX	STS[7:0]								00
Get Scanline	0	1	↑	XX	0	1	0	0	0	1	0	1	45h
	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XX	X	X	X	X	X	X	X	GTS [8]	00

	1	↑	1	XX	GTS[7:0]								00
Write Display Brightness	0	1	↑	XX	0	1	0	1	0	0	0	1	51h
	1	↑	1	XX	DBV[7:0]								00
Write CTRL Display	0	1	↑	XX	0	1	0	1	0	0	1	1	53h
	1	1	↑	XX	X	X	BCTRL	X	DD	BL	X	X	00
Read ID1	0	1	↑	XX	1	1	0	1	1	0	1	0	DAh
	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XX	LCD Module / Driver ID [7:0]								00
Read ID2	0	1	↑	XX	1	1	0	1	1	0	1	1	DBh
	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XX	LCD Module / Driver ID [7:0]								93
Read ID3	0	1	↑	XX	1	1	0	1	1	1	0	0	DCh
	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XX	LCD Module / Driver ID [7:0]								09

Inter Command Set													
Command Function	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RGB Interface Signal Control	0	1	↑	XX	1	0	1	1	0	0	0	0	B9h
	1	1	↑	XX	X	RCM[1:0]		X	VS PL	HSPL	DPL	EPL	01
Blanking Porch Control	0	1	↑	XX	1	0	1	1	0	1	0	1	B2h
	1	1	↑	XX	VFP [7:0]								08
	1	1	↑	XX	VBP [7:0]								08
	1	1	↑	XX	HX_HBP [7:0]								14
	1	1	↑	XX	HX_VBP [7:0]								08
Display Function Control	0	1	1	XX	1	0	1	1	0	1	1	0	AE
	1	1	1	XX	X	X	X	X	X	X	GS	SS	00
Interface Control	0	1	↑	XX	1	1	1	1	0	1	1	0	BAh
	1	1	↑	XX	1	1	0	0	DM[1:0]		RM	RIM	C0
Frame Rate	0	1	↑	XX	1	0	1	1	0	1	1	0	B6h
	1	1	↑	XX	RTN[7:0]								5C
	1	1	↑	XX	RTN_IDLE[7:0]								40
	1	1	↑	XX	RTN_DUMMY[7:0]								40
SPI 2DATA control	0	1	↑	XX	1	0	1	1	0	1	0	0	B4h
	1	1	↑	XX	1	1	0	0	2data_en	2data_mdt[2:0]			00
DINV	0	1	↑	XX	1	0	1	1	0	1	0	1	B5h
	1	1	↑	XX	Dinv_idle[1:0]		dinv[1:0]		0				50
Power Control 1	0	1	↑	XX	0	1	1	0	0	1	1	1	67h
	1	1	↑	XX	X	vbp_d[6:0]							27
Power Control 2	0	1	↑	XX	0	1	1	0	1	0	0	0	68h
	1	1	↑	XX	X	vbn_d[6:0]							27
Power Control 3	0	1	↑	XX	0	1	1	0	0	1	1	0	66h
	1	1	↑	XX	X	vrh[5:0]							18
Power Control 4	0	1	↑	XX	1	1	0	0	1	0	1	0	CAh
	1	1	↑	XX	X	0	0	0	VGH_AD[4:0]				09
Power Control 5	0	1	↑	XX	1	1	0	0	1	0	1	1	CBh
	1	1	↑	XX	X	0	0	0	0	0	VGL_AD[2:0]		04
Inter Register Enable1	0	1	↑	XX	1	1	1	1	1	1	1	0	FEh
Inter Register Enable2	0	1	↑	XX	1	1	1	0	1	1	1	1	EFh

Inter Command Set												
Command Function	D/CX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SET_GAMMA1	0	1	↑	1	1	1	1	0	0	0	0	F0h
	1	1	↑	1	0	VR1_N[5:0]						86
	1	1	↑	1	0	VR2_N[5:0]						85
	1	1	↑	0	0	0	VR4_N[4:0]					05
	1	1	↑	0	0	0	VR6_N[4:0]					05
	1	1	↑	VR0_N[3:0]				VR13_N[3:0]				84
	1	1	↑		VR20_N[6:0]							07
	1	1	↑		VR43_N[6:0]							07
	1	1	↑	VR27_N[2:0]			VR57_N[4:0]					65
	1	1	↑	VR36_N[2:0]			VR59_N[4:0]					65
	1	1	↑			VR61_N[5:0]						06
	1	1	↑			VR62_N[5:0]						06
	1	1	↑	VR50_N[3:0]				VR63_N[3:0]				44
SET_GAMMA2	0	1	↑	1	1	1	1	0	0	0	1	F1h
	1	1	↑	1	0	VR1_P[5:0]						86
	1	1	↑	1	0	VR2_P[5:0]						85
	1	1	↑	0	0	0	VR4_P[4:0]					05
	1	1	↑	0	0	0	VR6_P[4:0]					05
	1	1	↑	VR0_P[3:0]				VR13_P[3:0]				84
	1	1	↑		VR20_P[6:0]							07
	1	1	↑		VR43_P[6:0]							07
	1	1	↑	VR27_P[2:0]			VR57_P[4:0]					65
	1	1	↑	VR36_P[2:0]			VR59_P[4:0]					65
	1	1	↑			VR61_P[5:0]						06
	1	1	↑			VR62_P[5:0]						06
	1	1	↑	VR50_P[3:0]				VR63_P[3:0]				44

6.2. Description of Level 1 Command

6.2.1. Read display identification information (04h)

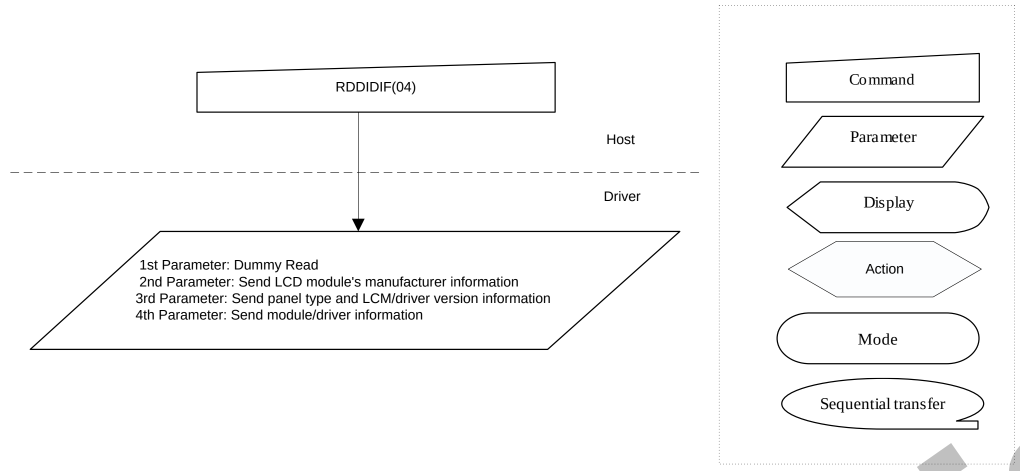
04h	Read display identification information 2												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	0	1	0	0	04h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	X
2nd Parameter	1	↑	1	XX	ID_1[7:0]								00
3rd Parameter	1	↑	1	XX	ID_2[7:0]								93
4th Parameter	1	↑	1	XX	ID_3[7:0]								09
Description	This read byte returns 24 bits display identification information. The 1st parameter is dummy data. The 2nd parameter (ID2_1 [7:0]): LCD module's manufacturer ID. The 3rd parameter (ID2_2 [7:0]): LCD module/driver version ID. The 4th parameter (ID2_3 [7:0]): LCD module/driver ID.												
Restriction													
Register Availability	(see table below)												
Default	(see table below)												
Flow Chart													

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	24'h009309
SW Reset	24'h009309
HW Reset	24'h009309


RDDIDIF(04)
Host
Driver
1st Parameter: Dummy Read
2nd Parameter: Send LCD module's manufacturer information
3rd Parameter: Send panel type and LCM/driver version information
4th Parameter: Send module/driver information
Command
Parameter
Display
Action
Mode
Sequential transfer

6.2.2. Read Display Status (09h)

09h	Read Display Status												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	1	0	0	1	09h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	X
2nd Parameter	1	↑	1	XX	D[31:25]							X	00
3rd Parameter	1	↑	1	XX	0	D[22:20]			D[19:16]				61
4th Parameter	1	↑	1	XX	0	0	0	0	0	D[10:8]			00
5th Parameter	1	↑	1	XX	D[7:5]			0	0	0	0	0	00

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Value	Status
D31	Booster voltage status	O	Booster OFF
		1	Booster ON
D30	Row address order	O	Top to Bottom (When MADCTL B7='0')
		1	Bottom to Top (When MADCTL B7='1')
D29	Column address order	O	Left to Right (When MADCTL B6='0').
		1	Right to Left (When MADCTL B6='1').
D28	Row/column exchange	O	Normal Mode (When MADCTL B5='0').
		1	Reverse Mode (When MADCTL B5='1').
D27	Vertical refresh	O	LCD Refresh Top to BoUom (When MADCTL B4='0')
		1	LCD Refresh BoUom to Top (When MADCTL B4='1').
D26	RGB/BGR order	O	RGB (When MADCTL B3='0')
		1	BGR (When MADCTL B3='1')
D25	Horizontal refresh order	O	LCD Refresh Left to Right (When MADCTL B2='0')
		1	LCD Refresh Right to Left (When

			MADCTL B2='1')
D24	Not used	O	-
D23	Not used	O	-
D22 D21 D20	Interface color pixel format definition	101	16-bit/pixel
		110	18-bit/pixel
D19	Idle mode ON/OFF	O	Idle Mode OFF
		1	Idle Mode ON
D18	Partial mode ON/OFF	O	Partial Mode OFF
		1	Partial Mode ON
D17	Sleep IN/OUT	O	Sleep IN Mode
		1	Sleep OUT Mode
D16	Display normal mode ON/OFF	O	Display Normal Mode OFF.
		1	Display Normal Mode ON.
D15	Vertical scrolling status	O	Scroll OFF
D14	Not used	O	-
D13	Inversion status	O	Not defined
D12	All pixel ON	O	Not defined
D11	All pixel OFF	O	Not defined
D10	Display ON/OFF	O	
		1	Display is ON
D9	Tearing effect line ON/OFF	O	Tearing Effect Line OFF
		1	Tearing Effect ON
D5	Tearing effect line mode	0	Mode 1, V-Blanking only
		1	Mode 2, both H-Blanking and V-Blanking
D4	Not used	O	-
D3	Not used	O	-
D2	Not used	O	-
D1	Not used	O	-
D0	Not used	O	-

Restriction

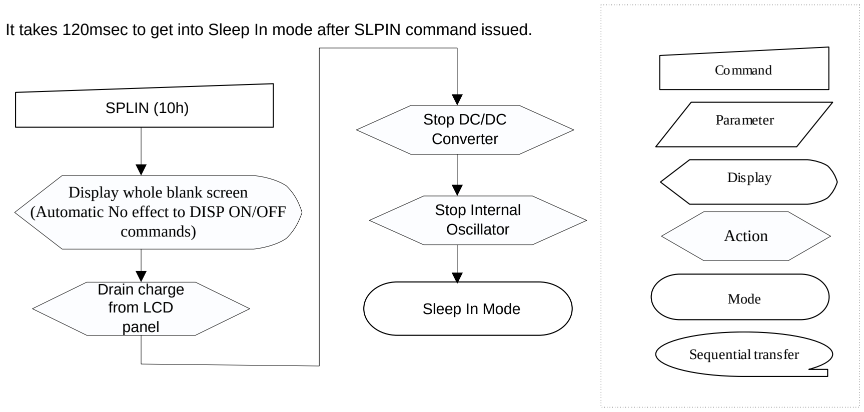
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

6.2.3. Enter Sleep Mode (10h)

<table>
<tr><td>10h</td><td colspan="12">Enter Sleep Mode</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D17-8</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>10h</td></tr>
<tr><td>Parameter</td><td colspan="13">No Parameter</td></tr>
<tr><td>Description</td><td colspan="13">This command causes the LCD module to enter the minimum power consumption mode. In this mode e.g. the DC/DC converter is stopped, Internal oscillator is stopped, and panel scanning is stopped


MCU interface and memory are still working and the memory keeps its contents.
X = Don't care</td></tr>
<tr><td>Restriction</td><td colspan="13">This command has no effect when module is already in sleep in mode. Sleep In Mode can only be left by the Sleep Out Command (11h). It will be necessary to wait 5msec before sending next to command, this is to allow time for the supply voltages and clock circuits to stabilize. It will be necessary to wait 120msec after sending Sleep Out command (when in Sleep In Mode) before Sleep In command can be sent.</td></tr>
<tr><td>Register Availability</td><td colspan="13"><table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="13"><table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Sleep IN Mode</td></tr>
<tr><td>SW Reset</td><td>Sleep IN Mode</td></tr>
<tr><td>HW Reset</td><td>Sleep IN Mode</td></tr>
</table></td></tr>
</table>

Flow Chart	

It takes 120msec to get into Sleep In mode after SLPIN command issued.
SPLIN (10h)
Display whole blank screen (Automatic No effect to DISP ON/OFF commands)
Drain charge from LCD panel
Stop DC/DC Converter
Stop Internal Oscillator
Sleep In Mode
Command
Parameter
Display
Action
Mode
Sequential transfer

6.2.4. Sleep Out Mode (11h)

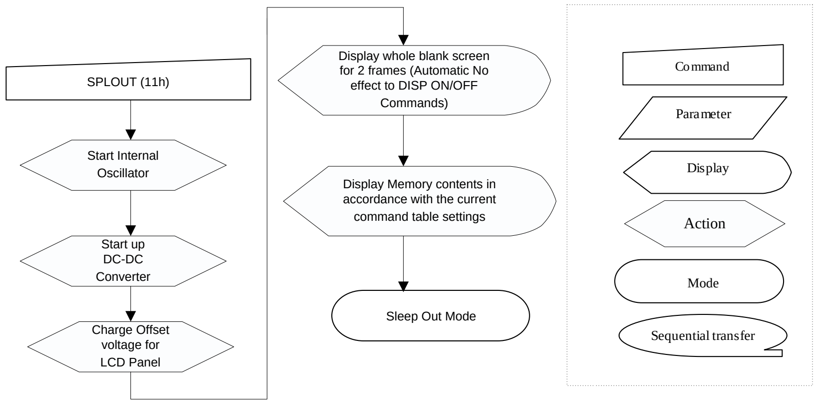
11h	Sleep Out Mode											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	1	0	0	0	1	11h
Parameter	No Parameter												
Description	This command turns off sleep mode. the DC/DC converter is enabled, Internal oscillator is started, and panel scanning is started. X = Don't care												
Restriction	This command has no effect when module is already in sleep out mode. Sleep Out Mode can only be left by the Sleep In Command (10h). It will be necessary to wait 5msec before sending next command, this is to allow time for the supply voltages and clock circuits stabilize. The display module loads all display supplier's factory default values to the registers during this 5msec and there cannot be any abnormal visual effect on the display image if factory default and register values are same when this load is done and when the display module is already Sleep Out –mode. The display module is doing self-diagnostic functions during this 5msec. It will be necessary to wait 120msec after sending Sleep In command (when in Sleep Out mode) before Sleep Out command can be sent.												
Register Availability	(see table below)												
Default	(see table below)												

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

Status	Default Value
Power On Sequence	Sleep IN Mode
SW Reset	Sleep IN Mode
HW Reset	Sleep IN Mode

Flow Chart	

SPLOUT (11h)
Start Internal Oscillator
Start up DC-DC Converter
Charge Offset voltage for LCD Panel
Display whole blank screen for 2 frames (Automatic No effect to DISP ON/OFF Commands)
Display Memory contents in accordance with the current command table settings
Sleep Out Mode
Command
Parameter
Display
Action
Mode
Sequential transfer

6.2.5. Partial Mode ON (12h)

12h	Partial Mode ON											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	1	0	0	1	0	12h
Parameter	No Parameter												
Description	This command turns on partial mode The partial mode window is described by the Partial Area command (30H). To leave Partial mode, the Normal Display Mode On command (13H) should be written. X = Don't care												
Restriction	This command has no effect when Partial mode is active.												
Register Availability	see table below												
Default	see table below												
Flow Chart	See Partial Area (30h)												

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

Status	Default Value
Power On Sequence	Normal Display Mode ON
SW Reset	Normal Display Mode
HW Reset	Normal Display Mode ON

6.2.6. Normal Display Mode ON (13h)

13h	Normal Display Mode ON											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	1	0	0	1	1	13h
Parameter	No Parameter												
Description	This command returns the display to normal mode. Normal display mode on means Partial mode off. Exit from NORON by the Partial mode On command (12h) X = Don't care												
Restriction	This command has no effect when Normal Display mode is active.												
Register Availability	(see table below)												
Default	(see table below)												
Flow Chart	See Partial Area (30h)												

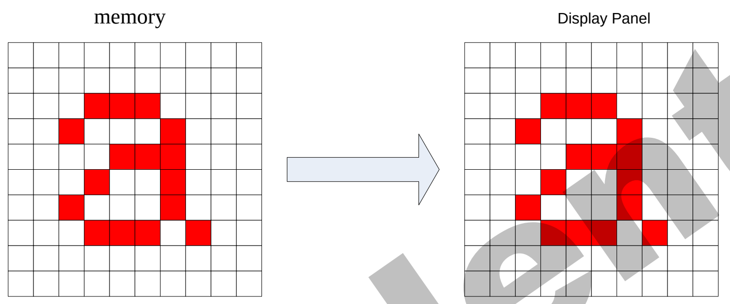
Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

Status	Default Value
Power On Sequence	Normal Display Mode ON
SW Reset	Normal Display Mode
HW Reset	Normal Display Mode ON

6.2.7. Display Inversion OFF (20h)

20h	Display Inversion OFF												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	0	0	0	0	20h
Parameter	No Parameter												
Description	This command is used to recover from display inversion mode. This command makes no change of the content of frame memory. This command doesn't change any other status.  X = Don't care												
Restriction	This command has no effect when module already is inversion OFF mode.												

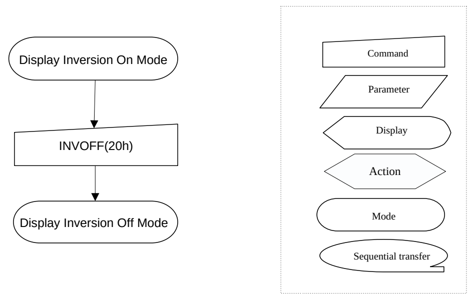
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

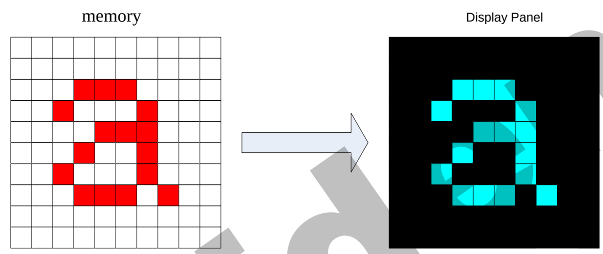
Default

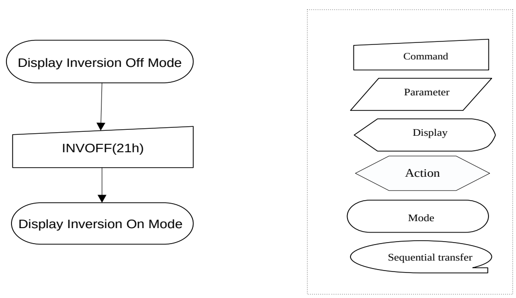
Status	Default Value
Power On Sequence	Display Inversion OFF
SW Reset	Display Inversion OFF
HW Reset	Display Inversion OFF

Flow Chart

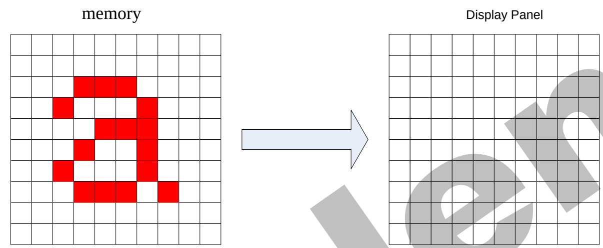


6.2.8. Display Inversion ON (21h)

<table>
<tr><td>21h</td><td colspan="13">Display Inversion ON</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D17-8</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>21h</td></tr>
<tr><td>Parameter</td><td colspan="13">No Parameter</td></tr>
<tr><td>Description</td><td colspan="13">This command is used to enter into display inversion mode.
This command makes no change of the content of frame memory. Every bit is inverted from the frame memory to the display.
This command doesn’t change any other status.
To exit Display inversion mode, the Display inversion OFF command (20h) should be written..


X = Don’t care</td></tr>
<tr><td>Restriction</td><td colspan="13">This command has no effect when module already is inversion ON mode.</td></tr>
<tr><td>Register Availability</td><td colspan="13"><table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="13"><table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Display Inversion OFF</td></tr>
<tr><td>SW Reset</td><td>Display Inversion OFF</td></tr>
<tr><td>HW Reset</td><td>Display Inversion OFF</td></tr>
</table></td></tr>
<tr><td>Flow Chart</td><td colspan="13">
</td></tr>
</table>

6.2.9. Display OFF (28h)

28h	Display OFF											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	1	0	0	0	28h
Parameter	No Parameter												
Description	This command is used to enter into DISPLAY OFF mode. In this mode, the output from Frame Memory is disabled and blank page inserted. This command makes no change of contents of frame memory. This command does not change any other status. There will be no abnormal visible effect on the display.  X = Don't care												
Restriction	This command has no effect when module is already in display off mode.												

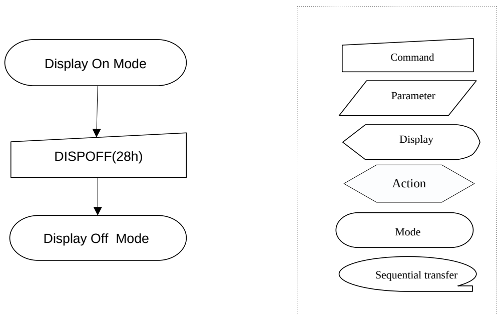
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

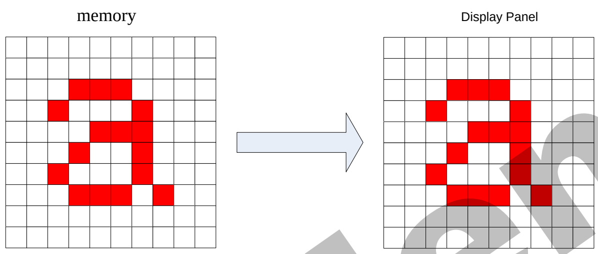
Default

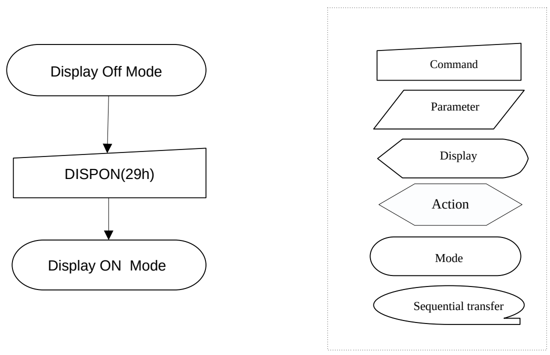
Status	Default Value
Power On Sequence	Display OFF
SW Reset	Display OFF
HW Reset	Display OFF

Flow Chart



6.2.10. Display ON (29h)

<table>
<tr><td>29h</td><td colspan="13">Display ON</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D17-8</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>29h</td></tr>
<tr><td>Parameter</td><td colspan="13">No Parameter</td></tr>
<tr><td>Description</td><td colspan="13">This command is used to recover from DISPLAY OFF mode. Output from the Frame Memory is enabled.
This command makes no change of contents of frame memory.
This command does not change any other status.


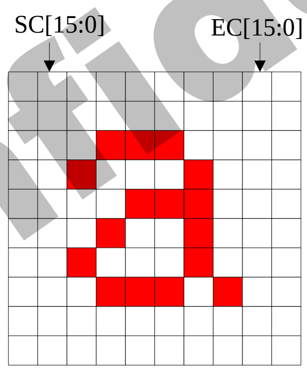
X = Don't care</td></tr>
<tr><td>Restriction</td><td colspan="13">This command has no effect when module is already in display on mode.</td></tr>
<tr><td>Register Availability</td><td colspan="13">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="13">
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Display OFF</td></tr>
<tr><td>SW Reset</td><td>Display OFF</td></tr>
<tr><td>HW Reset</td><td>Display OFF</td></tr>
</table></td></tr>
<tr><td>Flow Chart</td><td colspan="13">
</td></tr>
</table>

6.2.11. Column Address Set (2Ah)

2Ah	Column Address Set											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	1	0	1	0	2Ah
1st Parameter	1	1	↑	XX	SC15	SC14	SC13	SC12	SC11	SC10	SC9	SC8	Note1
2nd Parameter	1	1	↑	XX	SC7	SC6	SC5	SC4	SC3	SC2	SC1	SC0	
3rd Parameter	1	1	↑	XX	EC15	EC14	EC13	EC12	EC11	EC10	EC9	EC8	Note1
4th Parameter	1	1	↑	XX	EC7	EC6	EC5	EC4	EC3	EC2	EC1	EC0	

Description

This command is used to define area of frame memory where MCU can access. This command makes no change on the other driver status. The values of SC [15:0] and EC [15:0] are referred when RAMWR command comes. Each value represents one column line in the Frame Memory..



X = Don't care

Restriction

SC [15:0] always must be equal to or less than EC [15:0].
Note 1: When SC [15:0] or EC [15:0] is greater than 00EFh (When MADCTL's B5 = 0) or 013Fh (When MADCTL's B5 = 1), data of out of range will be ignored

Register Availability

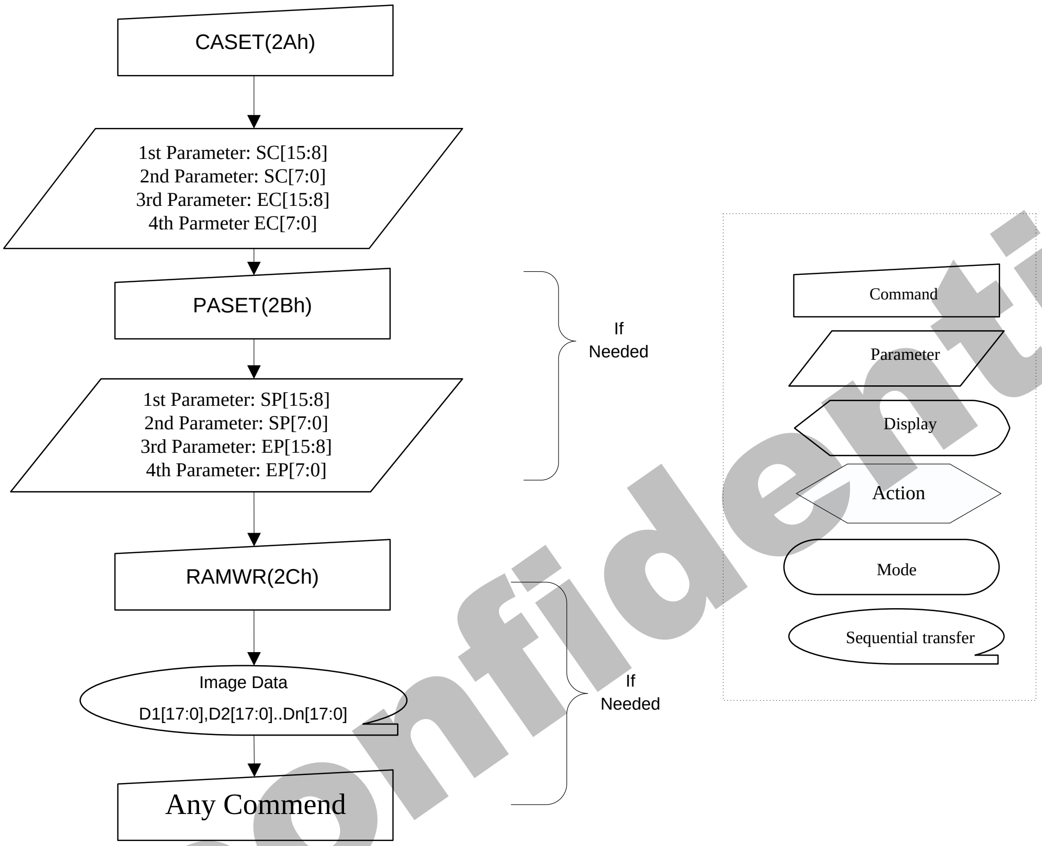
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

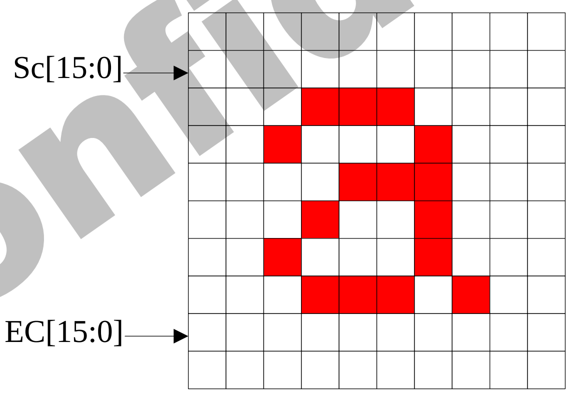
Status	Default Value	
Power On Sequence	SC [15:0]=0000h	EC [15:0]=00EFh

SW Reset	SC [15:0]=0000h	If MADCTL's B5 = 0: EC [15:0]=00EFh
		If MADCTL's B5 = 1: EC [15:0]=013Fh
HW Reset	SC [15:0]=0000h	EC [15:0]=00EFh

Flow Chart



6.2.12. Row Address Set (2Bh)

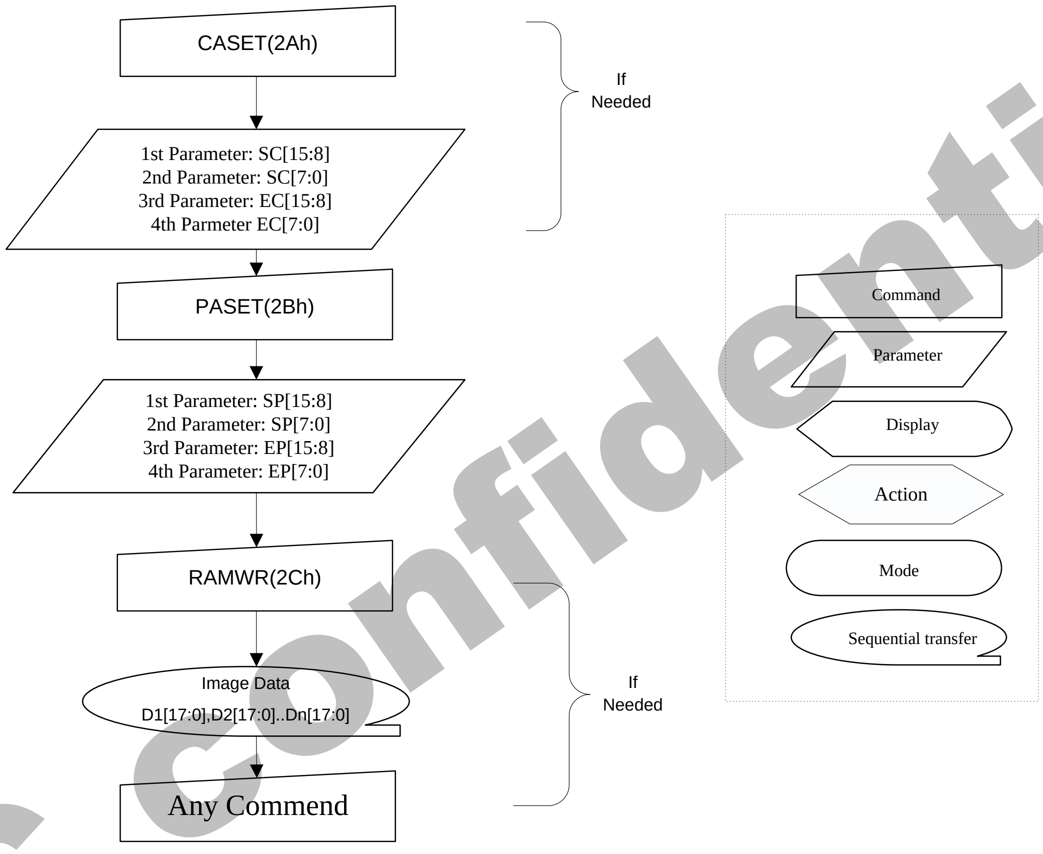
2Bh	Row Address Set												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	1	0	1	1	2Bh
1st Parameter	1	1	↑	XX	SP15	SP14	SP13	SP12	SP11	SP10	SP9	SP8	Note1
2nd Parameter	1	1	↑	XX	SP7	SP6	SP5	SP4	SP3	SP2	SP1	SP0	
3rd Parameter	1	1	↑	XX	EP15	EP14	EP13	EP12	EP11	EP10	EP9	EP8	Note1
4th Parameter	1	1	↑	XX	EP7	EP6	EP5	EP4	EP3	EP2	EP1	EP0	
Description	This command is used to define area of frame memory where MCU can access. This command makes no change on the other driver status. The values of SP [15:0] and EP [15:0] are referred when RAMWR command comes. Each value represents one Page line in the Frame Memory.  X = Don't care												
Restriction	SP [15:0] always must be equal to or less than EP [15:0] Note 1: When SP [15:0] or EP [15:0] is greater than 013Fh (When MADCTL's B5 = 0) or 00EFh (When MADCTL's B5 = 1), data of out of range will be ignored.												
Register Availability	see table below												
Default													

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Status	Default Value	
Power On Sequence	SP [15:0]=0000h	EP [15:0]=013Fh
SW Reset	SP [15:0]=0000h	If MADCTL’s B5 = 0: EP [15:0]=013Fh
		If MADCTL’s B5 = 1: EP [15:0]=0EFh
HW Reset	SP [15:0]=0000h	EP [15:0]=013Fh

Flow Chart



6.2.13. Memory Write (2Ch)

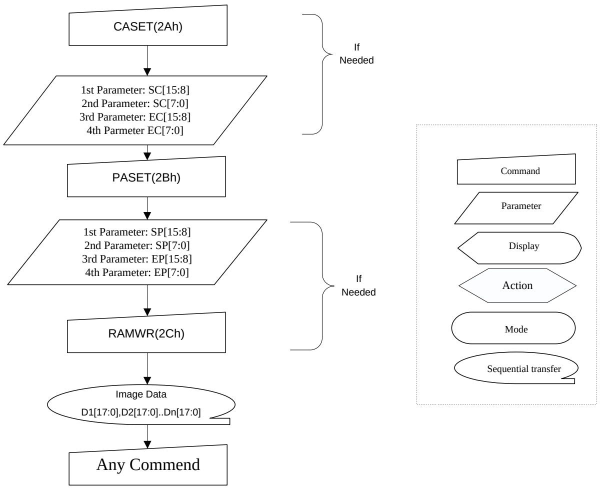
2Ch	Memory Write											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	1	1	0	0	2Ch
1st Parameter	1	1	↑	D1 [17:0]									XX
:	1	1	↑	Dx [17:0]									XX
Nth Parameter	1	1	↑	Dn [17:0]									XX
Description	This command is used to transfer data from MCU to frame memory. This command makes no change to the other driver status. When this command is accepted, the column register and the page register are reset to the Start Column/Start Page positions. The Start Column/Start Page positions are different in accordance with MADCTL setting.) Then D [17:0] isstored in frame memory and the column register and the page register incremented. Sending any other command can stop frame Write. X = Don't care.												
Restriction	In all color modes, there is no restriction on length of parameters.												

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

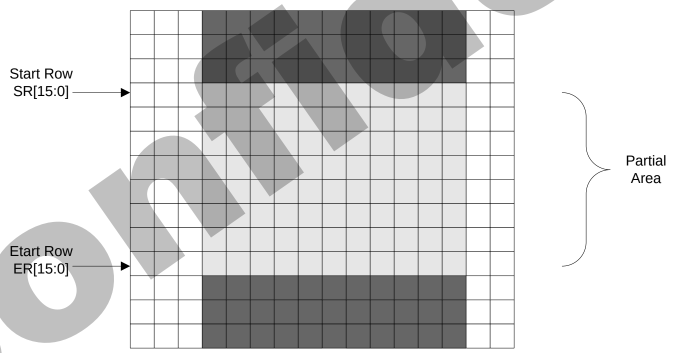
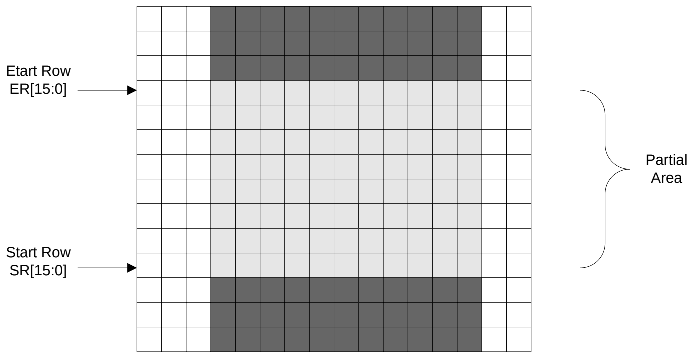
Default

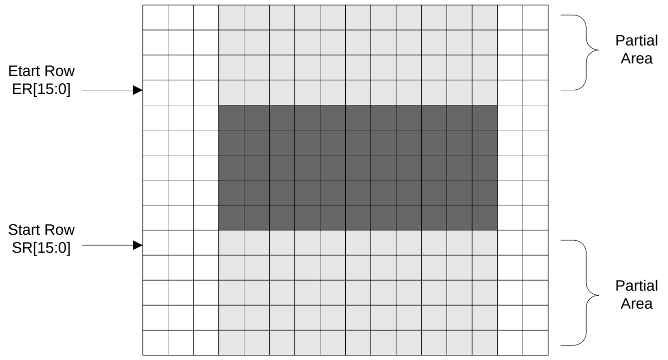
Status	Default Value
Power On Sequence	Contents of memory is set randomly
SW Reset	Contents of memory is not cleared
HW Reset	Contents of memory is not cleared

Flow Chart	

CASET(2Ah)
1st Parameter: SC[15:8]
2nd Parameter: SC[7:0]
3rd Parameter: EC[15:8]
4th Parmeter EC[7:0]
If Needed
PASET(2Bh)
1st Parameter: SP[15:8]
2nd Parameter: SP[7:0]
3rd Parameter: EP[15:8]
4th Parameter: EP[7:0]
If Needed
RAMWR(2Ch)
Image Data
D1[17:0],D2[17:0]..Dn[17:0]
Any Commend
Command
Parameter
Display
Action
Mode
Sequential transfer

6.2.14. Partial Area (30h)

30h	Partial Area											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	0	0	0	0	30h
1st Parameter	1	1	↑	XX	SR15	SR14	SR13	SR12	SR11	SR10	SR9	SR8	00
2nd Parameter	1	1	↑	XX	SR7	SR6	SR5	SR4	SR3	SR2	SR1	SR0	00
3rd Parameter	1	1	↑	XX	ER15	ER14	ER13	ER12	ER11	ER10	ER9	ER8	01
4th Parameter	1	1	↑	XX	ER7	ER6	ER5	ER4	ER3	ER2	ER1	ER0	3F
Description	This command defines the partial mode's display area. There are 2 parameters associated with this command, the first defines the Start Row (SR) and the second the End Row (ER), as illustrated in the figures below. SR and ER refer to the Frame Memory Line Pointer. If End Row>Start Row when MADCTL B4=0:-  If End Row>Start Row when MADCTL B4=1:-  If End Row<Start Row when MADCTL B4=0:-												



If End Row = Start Row then the Partial Area will be one row deep.

X = Don't care.

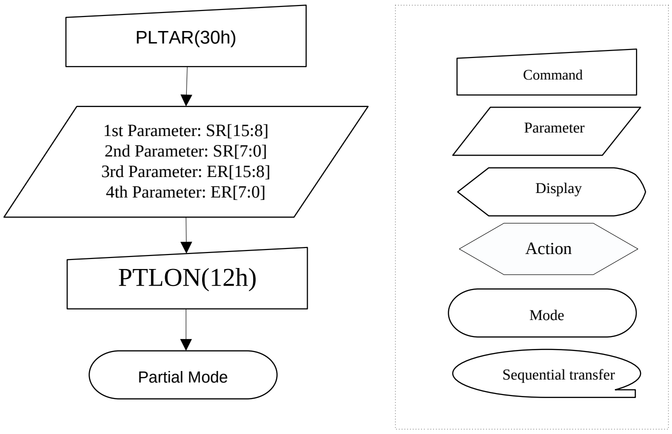
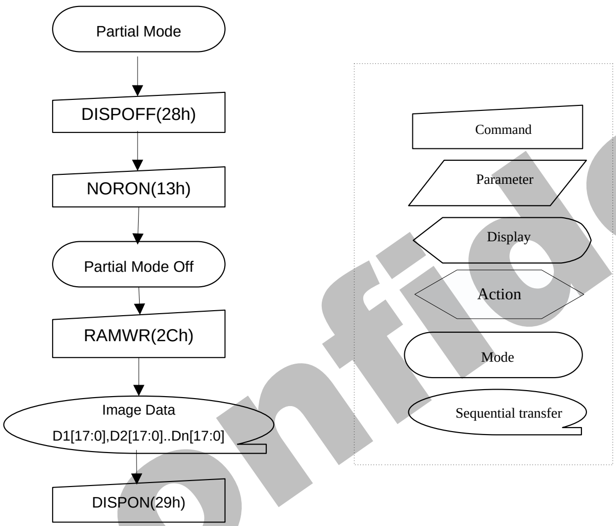
Restriction	SR [15…0] and ER [15…0] cannot be 0000h nor exceed 013Fh.

Register Availability

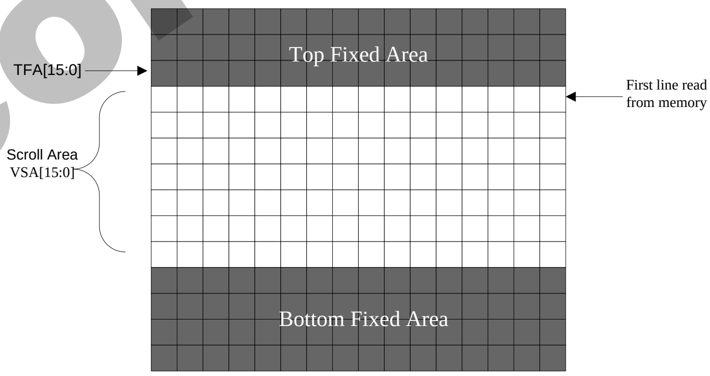
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

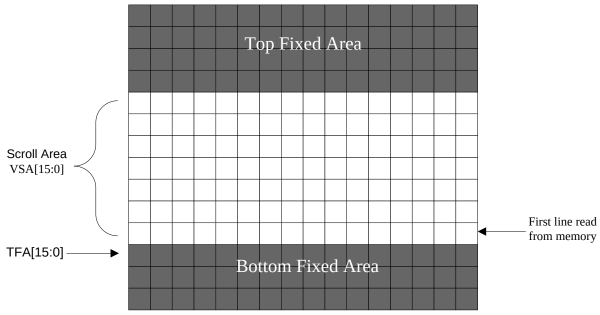
Default

Status	Default Value	
	SR [15:0]	ER [15:0]
Power On Sequence	16'h0000h	16'h013Fh
SW Reset	16'h0000h	16'h013Fh
HW Reset	16'h0000h	16'h013Fh

Flow Chart	1. To Enter Partial Mode  2. To Leave Partial Mode 

6.2.15. Vertical Scrolling Definition (33h)

33h	Vertical Scrolling Definition												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	0	0	1	1	33h
1st Parameter	1	1	↑	XX	TFA [15:8]								00
2nd Parameter	1	1	↑	XX	TFA [7:0]								00
3rd Parameter	1	1	↑	XX	VSA [15:8]								01
4th Parameter	1	1	↑	XX	VSA [7:0]								40
Description	This command defines the Vertical Scrolling Area of the display. When MADCTL B4=0 The 1st & 2nd parameter TFA [15...0] describes the Top Fixed Area (in No. of lines from Top of the Frame Memory and Display). The 3rd & 4th parameter VSA [15...0] describes the height of the Vertical Scrolling Area (in No. of lines of the Frame Memory [not the display] from the Vertical Scrolling Start Address). The first line read from Frame Memory appears immediately after the bottom most line of the Top Fixed Area.  When MADCTL B4=1 The 1st & 2nd parameter TFA [15...0] describes the Top Fixed Area (in No. of lines from Bottom of the Frame Memory and Display). The 3rd & 4th parameter VSA [15...0] describes the height of the Vertical Scrolling Area (in No. of lines of the Frame Memory [not the display] from the Vertical Scrolling Start Address). The first line read from Frame Memory appears immediately after the top most line of the Top Fixed Area.													



X = Don't care.

Restriction	

Register Availability

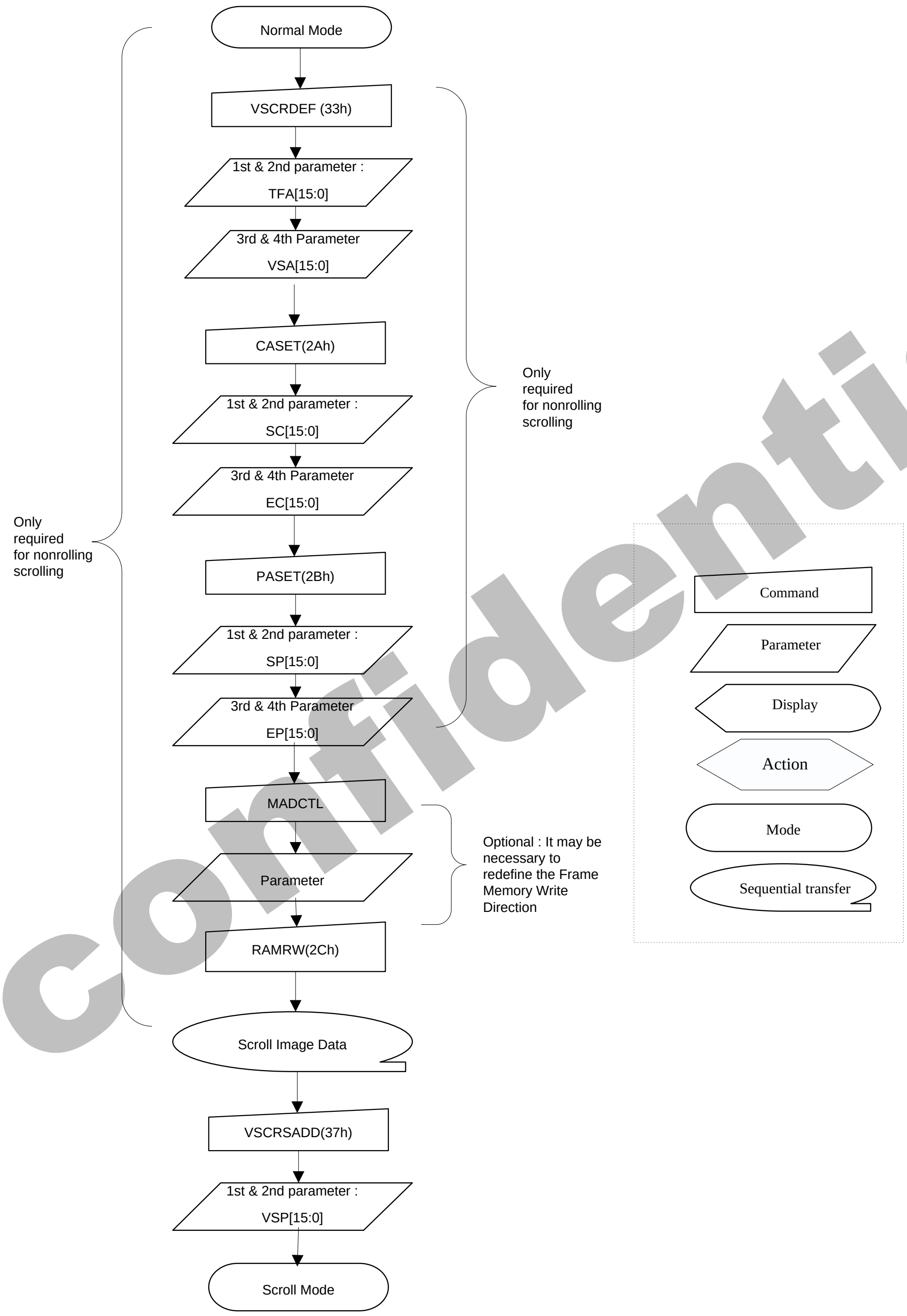
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
	TFA [15:0]	VSA [15:0]
Power On Sequence	16'h0000h	16'h0140h
SW Reset	16'h0000h	16'h0140h
HW Reset	16'h0000h	16'h0140h

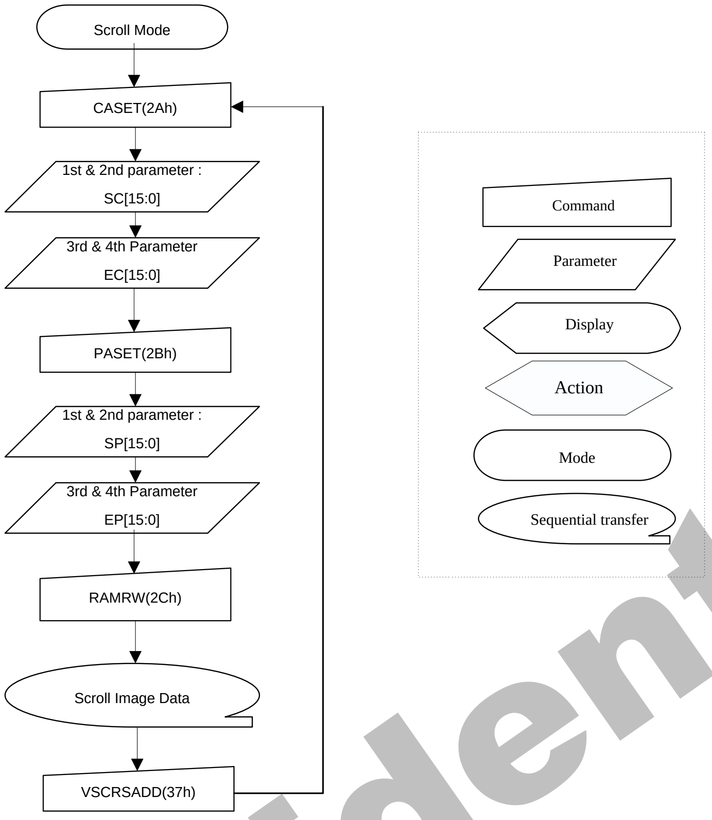
Flow Chart

1. To enter Vertical Scroll Mode :

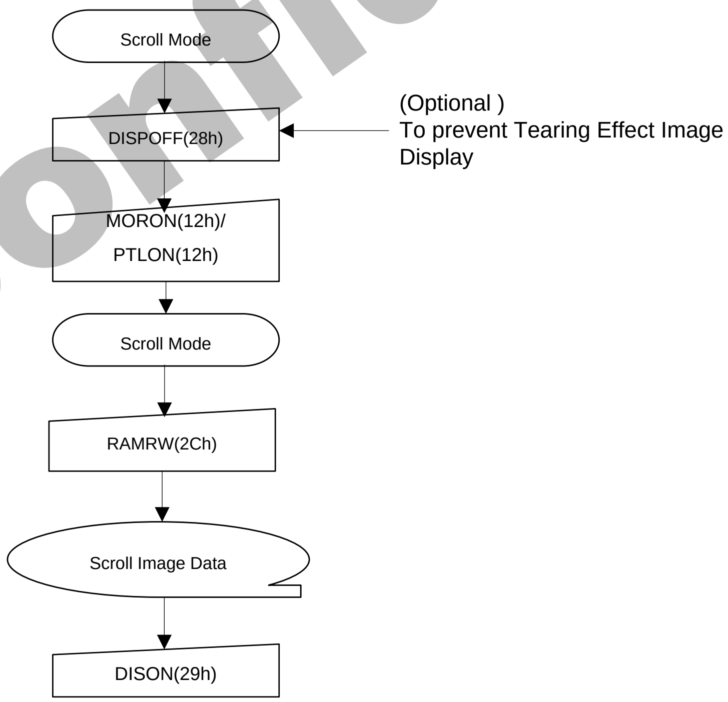


Note : The Frame Memory Window size ,must be defined correctly otherwise undesirable image will be displayed.

2.Continuous Scroll :

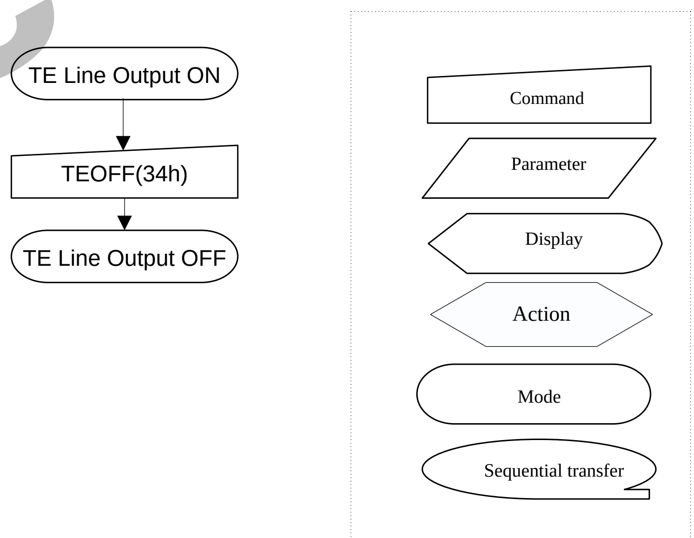


3.To Leave Vertical Scroll Mode:



Note: Scroll Mode can be left by both the Normal Display Mode ON (13h) and Partial Mode ON (12h) commands.

6.2.16. Tearing Effect Line OFF (34h)

<table>
<tr><td>34h</td><td colspan="13">Tearing Effect Line OFF</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D17-8</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>34h</td></tr>
<tr><td>Parameter</td><td colspan="13">No Parameter</td></tr>
<tr><td>Description</td><td colspan="13">This command is used to turn OFF (Active Low) the Tearing Effect output signal from the TE signal line.
X = Don’t care.</td></tr>
<tr><td>Restriction</td><td colspan="13">This command has no effect when Tearing Effect output is already OFF.</td></tr>
<tr><td>Register Availability</td><td colspan="13"><table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="13"><table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>OFF</td></tr>
<tr><td>SW Reset</td><td>OFF</td></tr>
<tr><td>HW Reset</td><td>OFF</td></tr>
</table></td></tr>
<tr><td>Flow Chart</td><td colspan="13">
</td></tr>
</table>

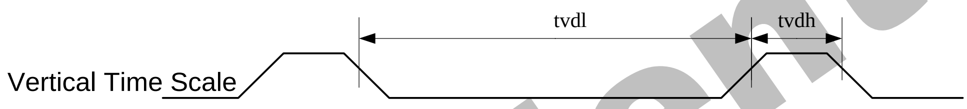
6.2.17. Tearing Effect Line ON (35h)

35h	Tearing Effect Line ON											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	0	1	0	1	35h
Parameter	1	1	↑	XX	0	0	0	0	0	0	0	M	00

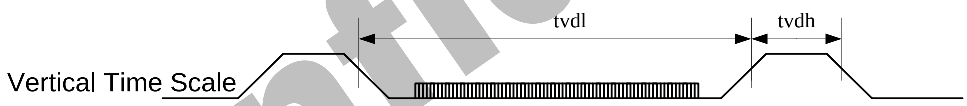
Description

This command is used to turn ON the Tearing Effect output signal from the TE signal line. This output is not affected by changing MADCTL bit B4. The Tearing Effect Line On has one parameter which describes the mode of the Tearing Effect Output Line.

When **M=0**:
The Tearing Effect Output line consists of V-Blanking information only:



When **M=1**:
The Tearing Effect Output Line consists of both V-Blanking and H-Blanking information:



Note: During Sleep In Mode with Tearing Effect Line On, Tearing Effect Output pin will be active Low.
X = Don’t care.

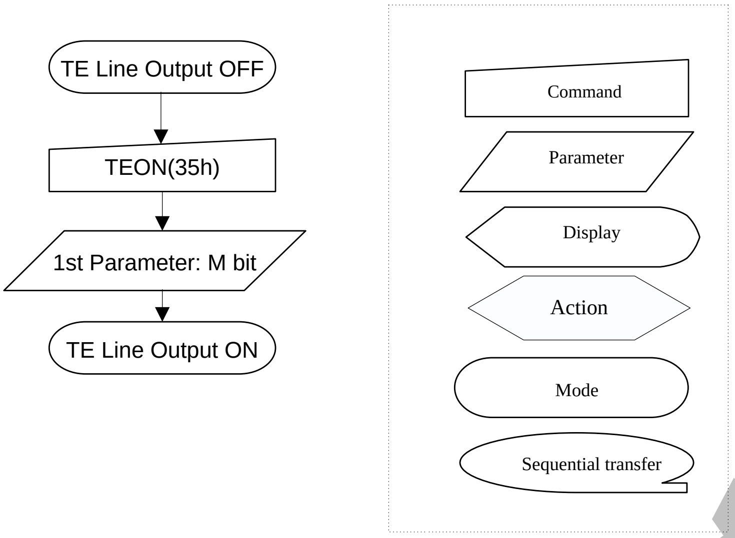
Restriction	This command has no effect when Tearing Effect output is already ON

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	OFF
SW Reset	OFF
HW Reset	OFF

Flow Chart	 TE Line Output OFF TEON(35h) 1st Parameter: M bit TE Line Output ON Command Parameter Display Action Mode Sequential transfer

6.2.18. Memory Access Control(36h)

36h	Tearing Effect Line ON											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	0	1	1	0	36h
Parameter	1	1	↑	XX	MY	MX	MV	ML	BGR	MH	0	0	00

Description

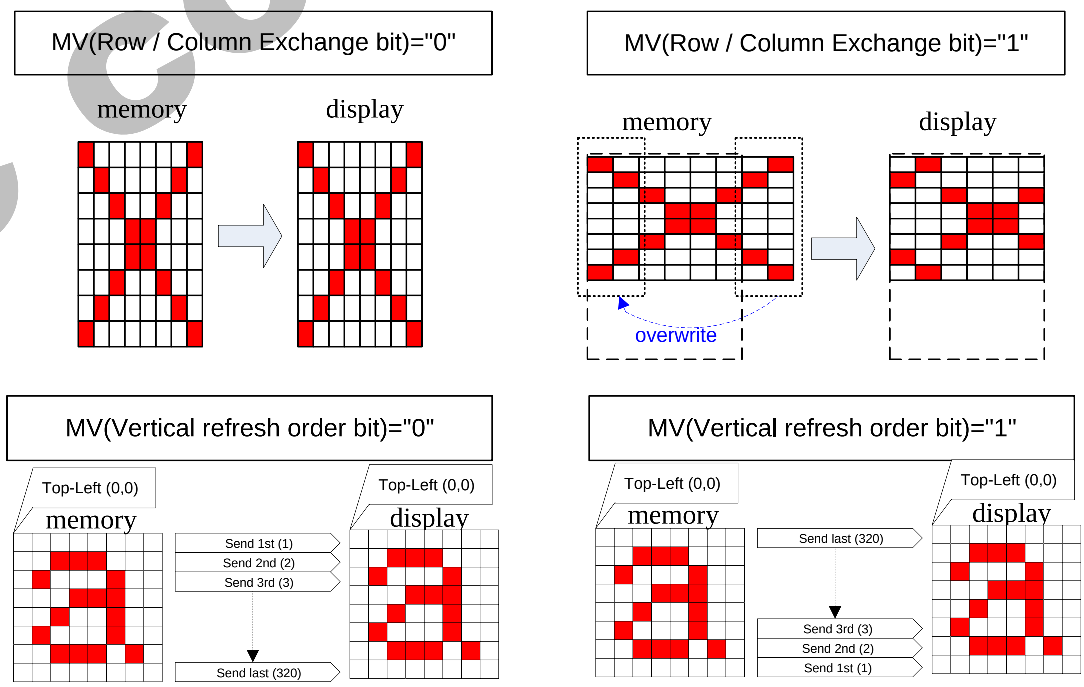
This command defines read/write scanning direction of frame memory.

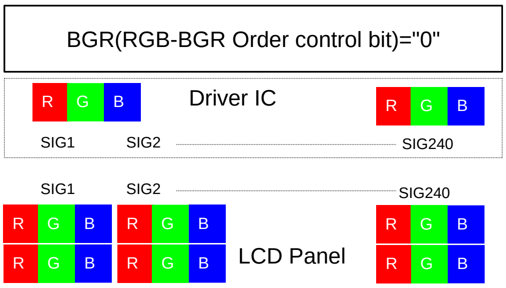
This command makes no change on the other driver status.

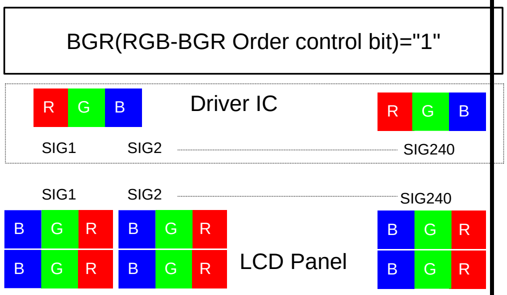
Bit	Name	Description
MY	Row Address Order	These 3 bits control MCU to memory write/read direction.
MX	Column Address Order	
MV	Row / Column Exchange	
ML	Vertical Refresh Order	LCD vertical refresh direction control.
BGR	RGB-BGR Order	Color selector switch control (0=RGB color filter panel, 1=BGR color filter panel)
MH	Horizontal Refresh ORDER	LCD horizontal refreshing direction control.

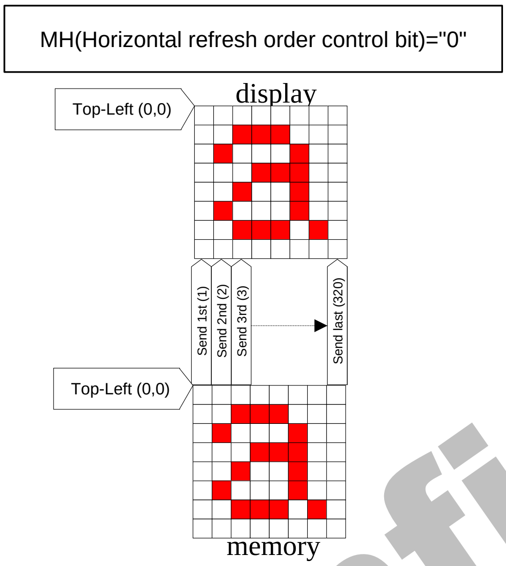
Note: When BGR bit is changed, the new setting is active immediately without update the content in Frame Memory again.

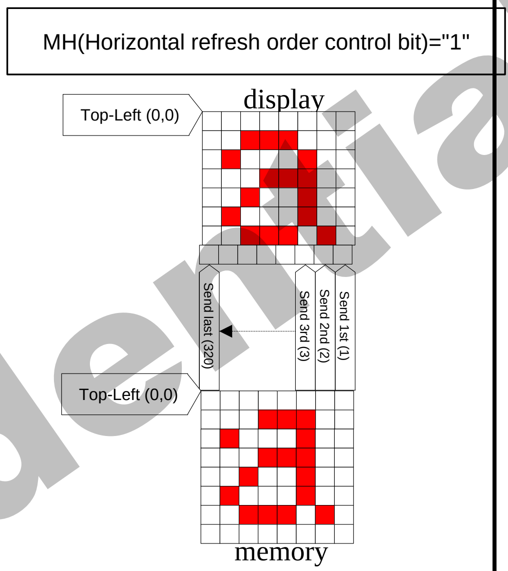
X = Don't care.









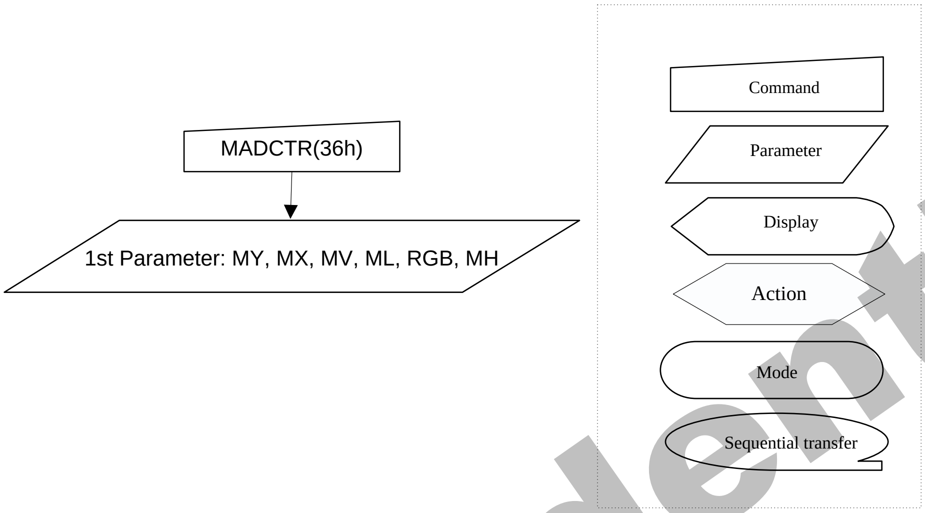


Note: Top-Left (0,0) means a physical memory location.

Restriction	This command has no effect when Tearing Effect output is already ON

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

<table>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>8’h00h</td></tr>
<tr><td>SW Reset</td><td>No change</td></tr>
<tr><td>HW Reset</td><td>8’h00h</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td>


</td></tr>
</table>

6.2.19. Vertical Scrolling Start Address (37h)

37h	VSCRSADD (Vertical Scrolling Start Address)												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	0	1	1	1	37h
1st Parameter	1	1	↑	XX	VSP [15:8]								00
2nd Parameter	1	1	↑	XX	VSP [7:0]								00

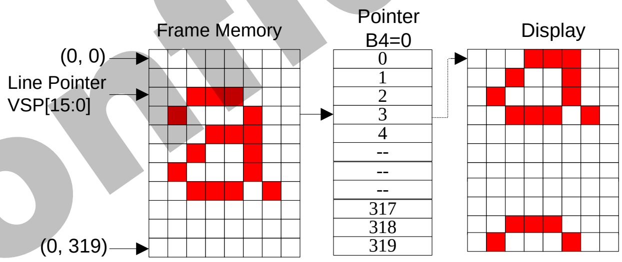
Description

This command is used together with Vertical Scrolling Definition (33h). These two commands describe the scrolling area
and the scrolling mode. The Vertical Scrolling Start Address command has one parameter which describes the address of the line in the Frame Memory that will be written as the first line after the last line of the Top Fixed Area on the display as illustrated below:-

When MADCTL B4=0

Example:

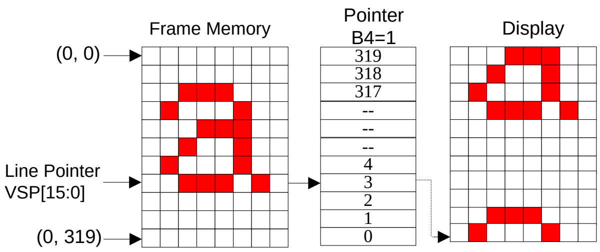
When Top Fixed Area = Bottom Fixed Area = 00, Vertical Scrolling Area = 320 and VSP='3'.



When MADCTL B4=1

Example:

When Top Fixed Area = Bottom Fixed Area = 00, Vertical Scrolling Area = 320 and VSP='3'.



Note: (1) When new Pointer position and Picture Data are sent, the result on the display will happen at the next Panel Scan to avoid tearing effect. VSP refers to the Frame Memory line Pointer.

(2) This command is ignored when the GC9309NA enters Partial mode.

<table>
<tr><td></td><td>X = Don’t care</td></tr>
<tr><td>Restriction</td><td>This command has no effect when Tearing Effect output is already ON</td></tr>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>No</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>No</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th rowspan="2">Status</th><th>Default Value</th></tr>
<tr><th>VSP [15:0]</th></tr>
<tr><td>Power On Sequence</td><td>16’h0000h</td></tr>
<tr><td>SW Reset</td><td>16’h0000h</td></tr>
<tr><td>HW Reset</td><td>16’h0000h</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td>See Vertical Scrolling Definition (33h) description.</td></tr>
</table>

6.2.20. Idle Mode OFF (38h)

38h	Idle Mode OFF												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	1	0	0	0	38h
Parameter	No Parameter												
Description	This command is used to recover from Idle mode on. In the idle off mode, LCD can display maximum 262,144 colors. X = Don’t care.												
Restriction	This command has no effect when module is already in idle off mode.												

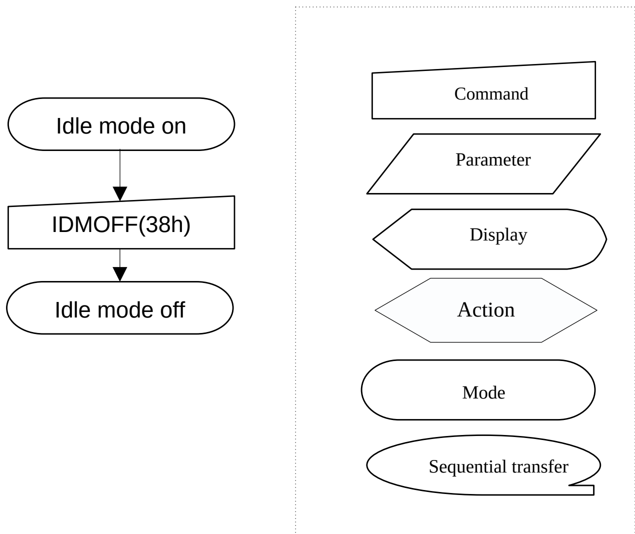
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

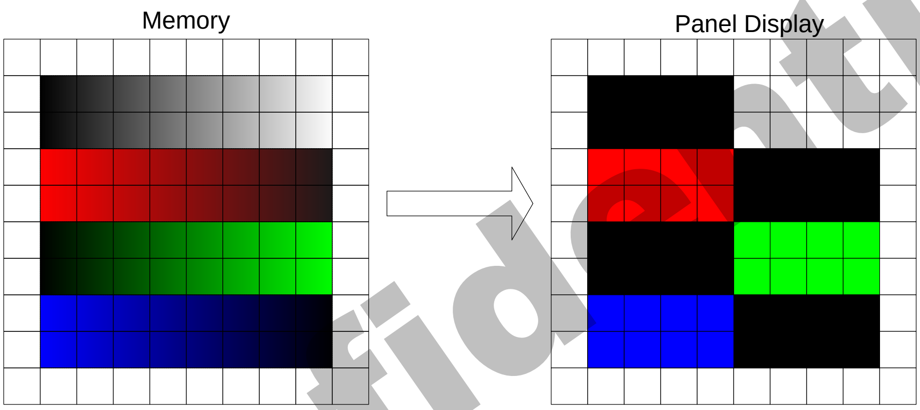
Status	Default Value
Power On Sequence	Idle mode OFF
SW Reset	Idle mode OFF
HW Reset	Idle mode OFF

Flow Chart



6.2.21. Idle Mode ON (39h)

39h	Idle Mode ON											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	1	0	0	1	39h
Parameter	No Parameter												
Description	This command is used to enter into Idle mode on. In the idle on mode, color expression is reduced. The primary and the secondary colors using MSB of each R, G and B in the Frame Memory, 8 color depth data is displayed.												



	Memory Contents vs. Display Color		
	R5 R4 R3 R2 R1 R0	G5 G4 G3 G2 G1 G0	B5 B4 B3 B2 B1 B0
Black	0XXXXX	0XXXXX	0XXXXX
Blue	0XXXXX	0XXXXX	1XXXXX
Red	1XXXXX	0XXXXX	0XXXXX
Magenta	1XXXXX	0XXXXX	1XXXXX
Green	0XXXXX	1XXXXX	0XXXXX
Cyan	0XXXXX	1XXXXX	1XXXXX
Yellow	1XXXXX	1XXXXX	0XXXXX
White	1XXXXX	1XXXXX	1XXXXX

X = Don’t care.

Restriction	This command has no effect when module is already in idle off mode.

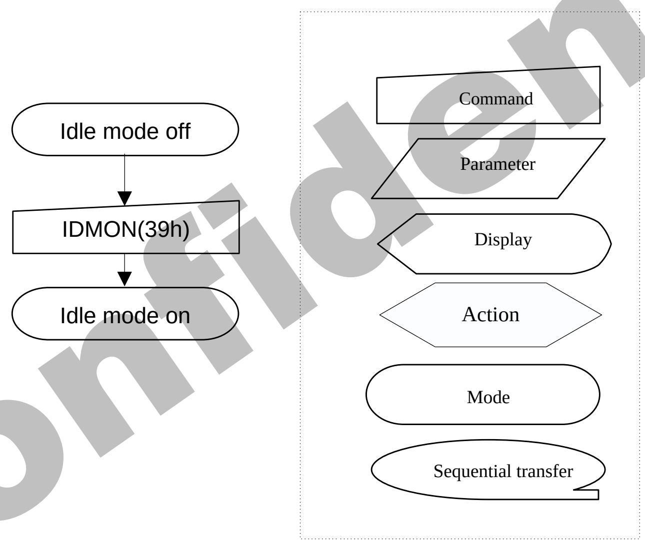
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Idle mode OFF
SW Reset	Idle mode OFF
HW Reset	Idle mode OFF

Flow Chart



6.2.22. COLMOD: Pixel Format Set (3Ah)

3Ah	Pixel Format Set											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	1	0	1	0	3Ah
Parameter	1	1	↑	XX	0	DPI [2:0]			0	DBI [2:0]			66

Description

This command sets the pixel format for the RGB image data used by the interface. DPI [2:0] is the pixel format select of RGB interface and DBI [2:0] is the pixel format of MCU interface. If a particular interface, either RGB interface or MCU interface, is not used then the corresponding bits in the parameter are ignored. The pixel format is shown in the table below.

DPI [2:0]			RGB Interface Format
0	0	0	Reserved
0	0	1	Reserved
0	1	0	Reserved
0	1	1	Reserved
1	0	0	Reserved
1	0	1	16 bits / pixel
1	1	0	18 bits / pixel
1	1	1	Reserved

DBI [2:0]			MCU Interface Format
0	0	0	Reserved
0	0	1	Reserved
0	1	0	Reserved
0	1	1	Reserved
1	0	0	Reserved
1	0	1	16 bits / pixel
1	1	0	18 bits / pixel
1	1	1	Reserved

If using RGB Interface must selection serial interface.
X = Don't care.

Restriction

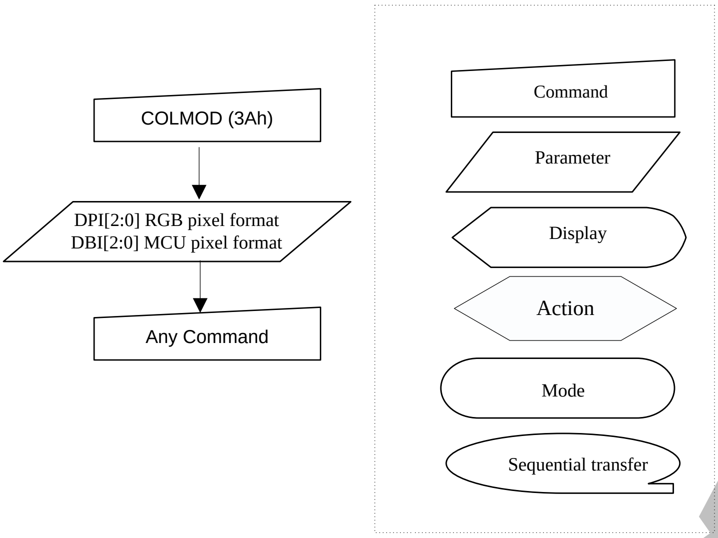
This command has no effect when module is already in idle off mode.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

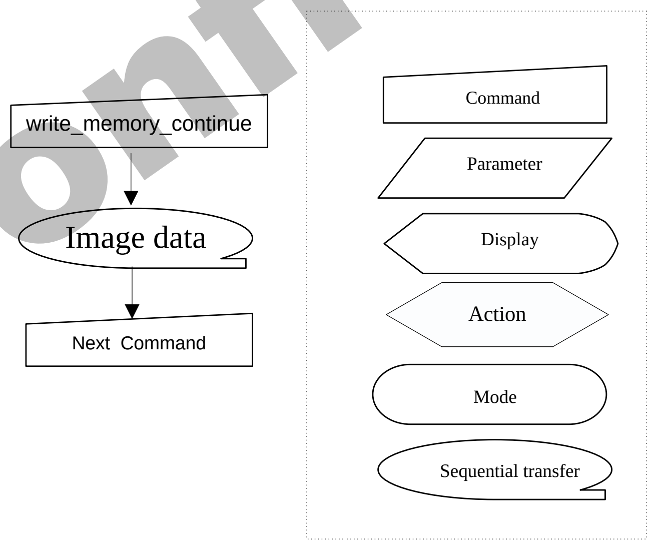
Status	Default Value	
	DPI [2:0]	DBI [2:0]
Power On Sequence	3'b110	3'b110
SW Reset	No Change	No Change
HW Reset	3'b110	3'b110

Flow Chart	

COLMOD (3Ah)
DPI[2:0] RGB pixel format
DBI[2:0] MCU pixel format
Any Command
Command
Parameter
Display
Action
Mode
Sequential transfer

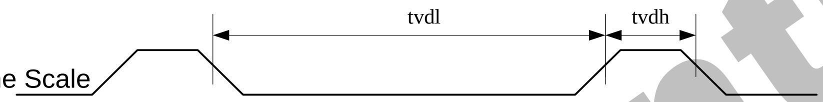
6.2.23. Write Memory Contiue (3Ch)

3Ch	write_memory_continue											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	D1[17..8]	0	0	1	1	1	1	0	0	3Ch
1st Parameter	1	1	↑	Dx[17..8]	D1[7]	D1[6]	D1[5]	D1[4]	D1[3]	D1[2]	D1[1]	D1[0]	0003FF
Xth Parameter	1	1	↑	D1[17..8]	Dx[7]	Dx[6]	Dx[5]	Dx[4]	Dx[3]	Dx[2]	Dx[1]	Dx[0]	0003FF
Nth Parameter	1	1	↑	Dn[17..8]	Dn[7]	Dn[6]	Dn[5]	Dn[4]	Dn[3]	Dn[2]	Dn[1]	Dn[0]	0003FF
iption	This command transfers image data from the host processor to the display module's frame memory continuing from the pixel location following the previous write_memory_continue or write_memory_start command. **If set_address_mode B5 = 0:** Data is written continuing from the pixel location after the write range of the previous write_memory_start or write_memory_continue. The column register is then incremented and pixels are written to the frame memory until the column register equals the End Column (EC) value. The column register is then reset to SC and the page register is incremented. Pixels are written to the frame memory until the page register equals the End Page (EP) value and the column register equals the EC value, or the host processor sends another command. If the number of pixels exceeds (EC –SC + 1) * (EP – SP + 1) the extra pixels are ignored. **If set_address_mode B5 = 1:** Data is written continuing from the pixel location after the write range of the previous write_memory_start or write_memory_continue. The page register is then incremented and pixels are written to the frame memory until the page register equals the End Page (EP) value. The page register is then reset to SP and the column register is incremented. Pixels are written to the frame memory until the column register equals the End column (EC) value and the page register equals the EP value, or the host processor sends another command. If the number of pixels exceeds (EC – SC + 1) * (EP –SP + 1) the extra pixels are ignored. Sending any other command can stop frame Write. Frame Memory Access and Interface setting (B3h), WEMODE=0 When the transfer number of data exceeds (EC-SC+1)*(EP-SP+1), the exceeding data will be ignored. Frame Memory Access and Interface setting (B3h), WEMODE=1 When the transfer number of data exceeds (EC-SC+1)*(EP-SP+1), the column and page											

<table>
<tr><td></td><td>number will be reset, and the
exceeding data will be written into the following column and page.</td></tr>
<tr><td>Restriction</td><td>A write_memory_start should follow a set_column_address, set_page_address or set_address_mode to define the write address. Otherwise, data written with write_memory_continue is written to undefined addresses.</td></tr>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Random value</td></tr>
<tr><td>SW Reset</td><td>No change</td></tr>
<tr><td>HW Reset</td><td>No change</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td>


</td></tr>
</table>

6.2.24. Set_Tear_Scanline (44h)

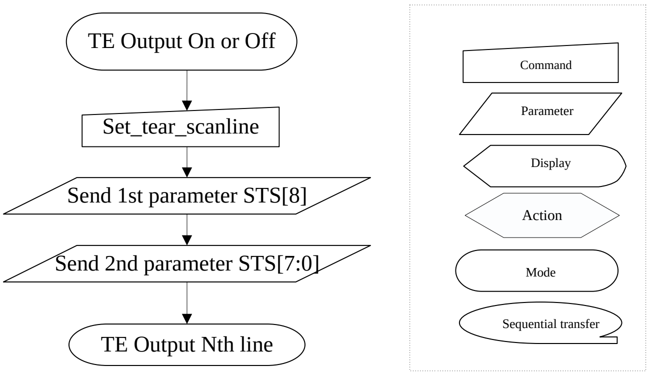
44h	Set_Tear_Scanline												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	0	0	1	0	0	44h
1st Parameter	1	1	↑	XX	0	0	0	0	0	0	0	STS [8]	00
2nd Parameter	1	1	↑	XX	STS [7]	STS [6]	STS [5]	STS [4]	STS [3]	STS [2]	STS [1]	STS [0]	00
Description	This command turns on the display Tearing Effect output signal on the TE signal line when the display reaches line equal the value of STS[8:0]  Note:that set_tear_scanline with STS is equivalent to set_tear_on with 8+GateN(N=1、2、3...320) eg:when the STS[8:0]=8,the TE will output at the position of Gate1. when the STS[8:0]=9,the TE will output at the position of Gate2. when the STS[8:0]=10,the TE will output at the position of Gate3. ………. The Tearing Effect Output line shall be active low when the display module is in Sleep mode.												
Restriction													
Register Availability	(see table below)												
Default	(see table below)												

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

Status	Default Value
Power On Sequence	STS [8:0]=0000h
SW Reset	STS [8:0]=0000h
HW Reset	STS [8:0]=0000h

Flow Chart	

TE Output On or Off
Set_tear_scanline
Send 1st parameter STS[8]
Send 2nd parameter STS[7:0]
TE Output Nth line
Command
Parameter
Display
Action
Mode
Sequential transfer

6.2.25. Get_Scanline (45h)

45h	Get_Scanline												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	0	0	1	0	1	45h
1st Parameter	1	↑	1	XX	0	0	0	0	0	0	0	GTS [8]	00
2nd Parameter	1	↑	1	XX	GTS [7]	GTS [6]	GTS [5]	GTS [4]	GTS [3]	GTS [2]	GTS [1]	GTS [0]	00
Description	This command returns the setting value of STS[8:0] . When in Sleep Mode, the value returned by get_scanline is undefined.												
Restriction	None												

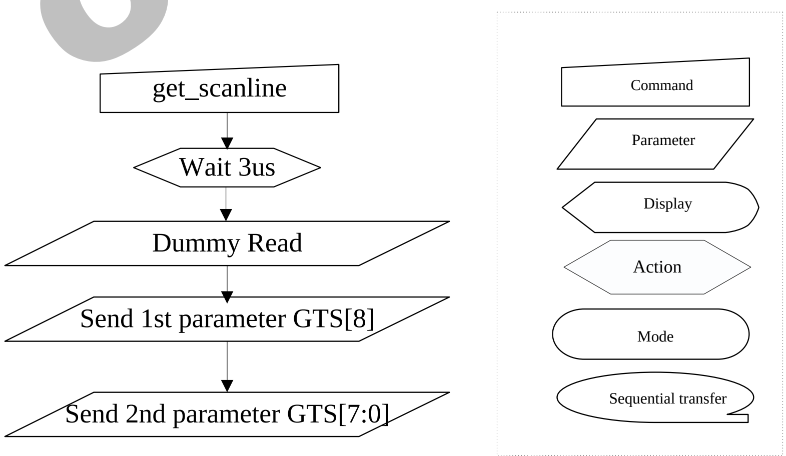
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	GTS [9:0]=0000h
SW Reset	GTS [9:0]=0000h
HW Reset	GTS [9:0]=0000h

Flow Chart



6.2.26. Write Display Brightness (51h)

51h	Write Display Brightness												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	1	0	0	0	1	51h
1st Parameter	1	1	↑	XX	DBV [7]	DBV [6]	DBV [5]	DBV [4]	DBV [3]	DBV [6]	DBV [5]	DBV [4]	00

Description	This command is used to adjust the brightness value of the display. It should be checked what is the relationship between this written value and output brightness of the display. This relationship is defined on the display module specification. In principle relationship is that 00h value means the lowest brightness and FFh value means the highest brightness.
Restriction	None

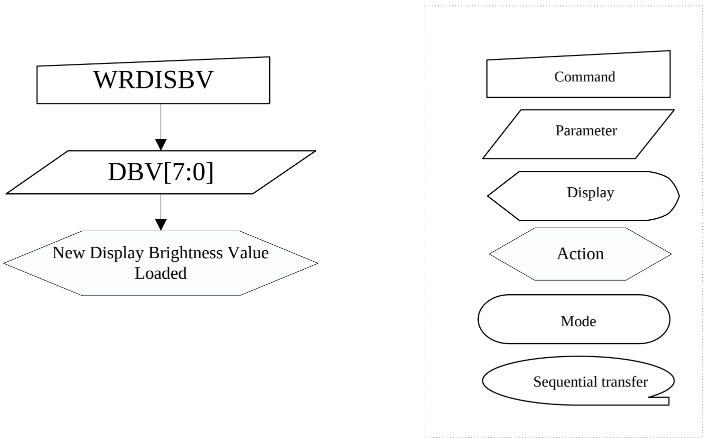
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	DBV [7:0]= 8'h00
SW Reset	DBV [7:0]= 8'h00
HW Reset	DBV [7:0]= 8'h00

Flow Chart



6.2.27. Write CTRL Display (53h)

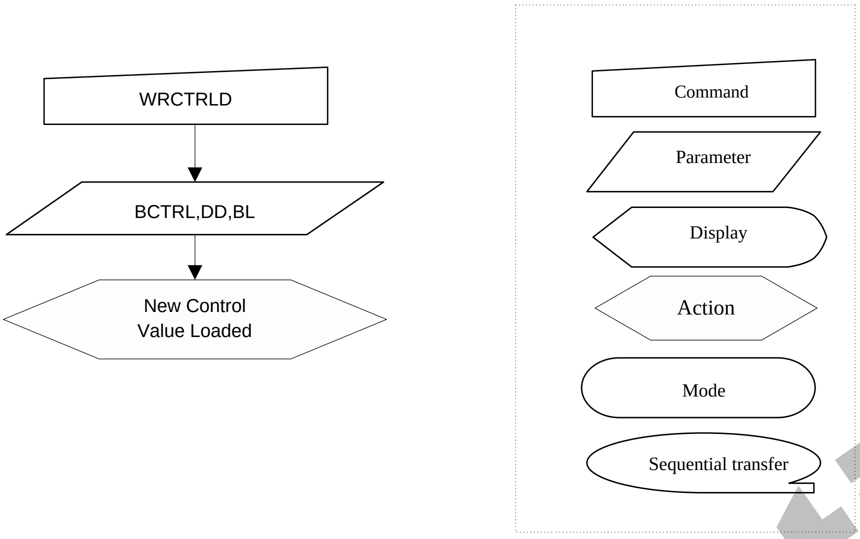
53h	Write CTRL Display											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	1	0	0	1	1	53h
1st Parameter	1	1	↑	XX	0	0	BCTRL	0	DD	BL	0	0	00
Description	This command is used to return brightness setting. **BCTRL**: Brightness Control Block On/Off, ‘0’ = Off (Brightness registers are 00h) ‘1’ = On (Brightness registers are active, according to the DBV[7..0] parameters.) **DD**: Display Dimming ‘0’ = Display Dimming is off ‘1’ = Display Dimming is on **BL**: Backlight On/Off ‘0’ = Off (Completely turn off backlight circuit. Control lines must be low.) ‘1’ = On												
Restriction	The display module is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on DBI. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).												
Register Availability	(see table below)												
Default	(see table below)												

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

Status	Default Value		
	BCTRL	DD	BL
Power On Sequence	1’b0	1’b0	1’b0
SW Reset	1’b0	1’b0	1’b0
HW Reset	1’b0	1’b0	1’b0

Flow Chart	 WRCTRLD BCTRL,DD,BL New Control Value Loaded Command Parameter Display Action Mode Sequential transfer

6.2.28. Read ID1 (DAh)

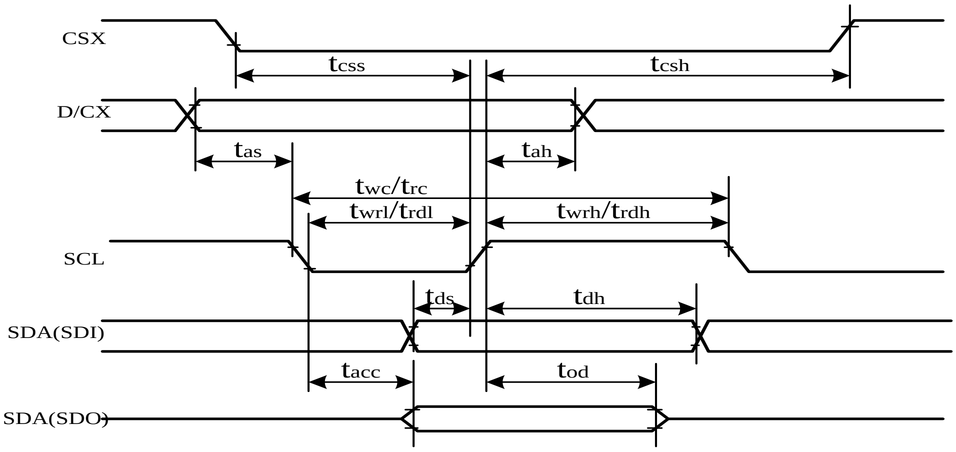
DCh	Read ID2												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	1	1	0	1	0	DAh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	X
2nd Parameter	1	↑	1	XX	ID3 [7:0]								Program value
Description	This read byte is used to track the LCD module/driver version. It is defined by display supplier (with User's agreement) and changes each time a revision is made to the display, material or construction specifications. The 1st parameter is dummy data. The 2nd parameter is LCD module/driver version ID The ID3 can be programmed by MTP function. X = Don't care												
Restriction	None												
Register Availability	(see table below)												
Default	(see table below)												

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

Status	Default Value (After MTP program)
Power On Sequence	8'h00
SW Reset	8'h00
HW Reset	8'h00

Flow Chart	

CSX
tcss
tcsh
D/CX
tas
tah
twc/trc
twrl/trdl
twrh/trdh
SCL
tds
tdh
SDA(SDI)
tacc
tod
SDA(SDO)

6.2.29. Read ID2 (DBh)

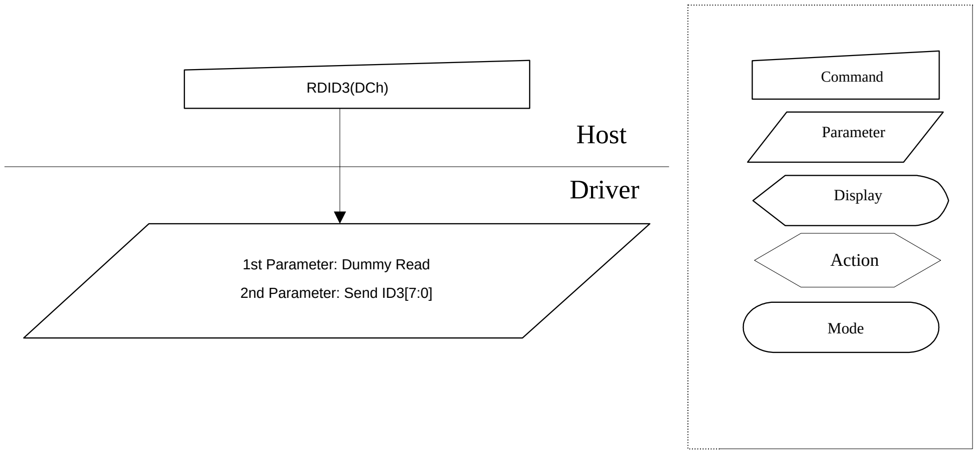
DCh	Read ID2												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	1	1	0	1	1	DBh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	X
2nd Parameter	1	↑	1	XX	ID3 [7:0]								Program value
Description	This read byte is used to track the LCD module/driver version. It is defined by display supplier (with User's agreement) and changes each time a revision is made to the display, material or construction specifications. The 1st parameter is dummy data. The 2nd parameter is LCD module/driver version ID The ID3 can be programmed by MTP function. X = Don't care												
Restriction	None												

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value (After MTP program)
Power On Sequence	8'h93
SW Reset	8'h93
HW Reset	8'h93

Flow Chart	

RDID3(DCh)
Host
Driver
1st Parameter: Dummy Read
2nd Parameter: Send ID3[7:0]
Command
Parameter
Display
Action
Mode

6.2.30. Read ID3 (DCh)

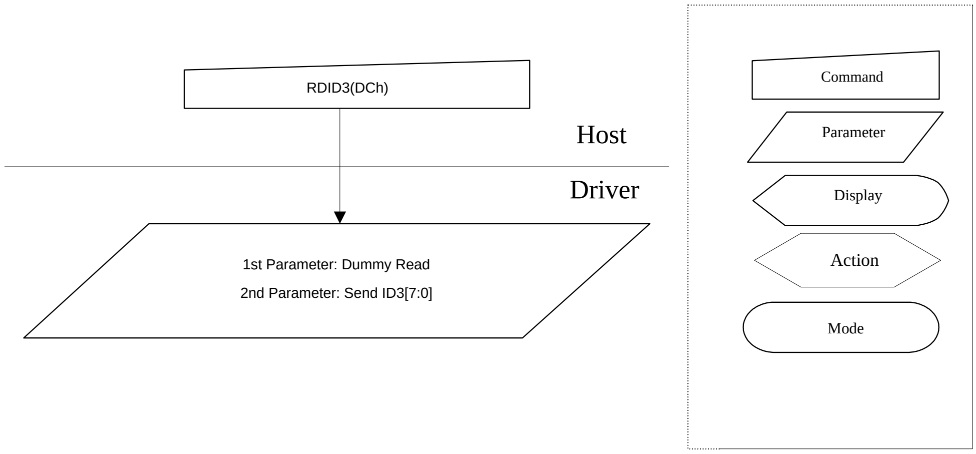
DCh	Read ID2												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	1	1	1	0	0	DCh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	X
2nd Parameter	1	↑	1	XX	ID3 [7:0]								Program value
Description	This read byte is used to track the LCD module/driver version. It is defined by display supplier (with User's agreement) and changes each time a revision is made to the display, material or construction specifications. The 1st parameter is dummy data. The 2nd parameter is LCD module/driver version ID The ID3 can be programmed by MTP function. X = Don't care												
Restriction	None												

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value (After MTP program)
Power On Sequence	8'h09
SW Reset	8'h09
HW Reset	8'h09

Flow Chart	RDID3(DCh) Host Driver 1st Parameter: Dummy Read 2nd Parameter: Send ID3[7:0] Command Parameter Display Action Mode

6.3. Description of Level 2 Command

6.3.1. RGB Interface Signal Control (B9h)

B9h	RGB Interface Signal Control												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	1	0	0	1	B9h
1st Parameter	1	1	↑	XX	0	RCM [1]	RCM [0]	0	VSPL	HSPL	DPL	EPL	01

Description:

Sets the operation status of the display interface. The setting becomes effective as soon as the command is received.

EPL: DE polarity (“0”= High enable for RGB interface, “1”= Low enable for RGB interface)

DPL: DOTCLK polarity set (“0”= data fetched at the rising time, “1”= data fetched at the falling time)

HSPL: HSYNC polarity (“0”= Low level sync clock, “1”= High level sync clock)

VSPL: VSYNC polarity (“0”= Low level sync clock, “1”= High level sync clock)

RCM [1:0]: RGB interface selection (refer to the RGB interface section).

RCM [1:0]		RIM	DPI[1:0]			RGB interface Mode	RGB Mode	Used Pins
1	0	0	1	1	0	18-bit RGB interface (262K colors)	DE Mode Valid data is determined by the DE signal	VSYNC,HSYNC,DE, DOTCLK,D[17:0]
1	0	0	1	0	1	16-bit RGB interface (65K colors)		VSYNC,HSYNC,DE, DOTCLK,D[17:13] & D[11:1]
1	0	1	-			6-bit RGB interface (262K colors)		VSYNC,HSYNC,DE, DOTCLK,D[5:0]
1	1	0	1	1	0	18-bit RGB interface (262K colors)	SYNC Mode In SYNC mode, DE signal is ignored; blanking porch is determined by B2h command	VSYNC,HSYNC, DOTCLK,D[17:0]
1	1	0	1	0	1	16-bit RGB interface (65K colors)		VSYNC,HSYNC,DOTCLK, D[17:13] & D[11:1]
1	1	1	-			6-bit RGB interface (262K colors)		VSYNC,HSYNC, DOTCLK,D[5:0]

Restriction	None

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value				
	RCM[1:0]	VSPL	HSPL	DPL	EPL
Power On Sequence	2’b00	1’b0	1’b0	1’b0	1’b1
SW Reset	2’b00	1’b0	1’b0	1’b0	1’b1
HW Reset	2’b00	1’b0	1’b0	1’b0	1’b1

6.3.2. Blanking Porch Control (B2h)

B2h	Blanking Porch Control												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	0	0	1	0	B2h
1st Parameter	1	1	↑	XX	VFP [7:0]								08
2nd Parameter	1	1	↑	XX	VBP [7:0]								08
3rd Parameter	1	1	↑	XX	HX_HBP [7:0]								14
4th Parameter	1	1	↑	XX	HX_VBP [7:0]								08

Description

Note: The Third parameter must write,but it is not valid.

VFP [7:0] / VBP [7:0]: The VFP [7:0] and VBP [7:0] bits specify the line number of vertical front and back porch period respectively.

VFP [7:0] VBP [7:0]	Number of HSYNC of front/back porch	VFP [7:0] VBP [7:0]	Number of HSYNC of front/back porch
0000000	Setting inhibited	10000000	128
0000001	Setting inhibited	10000001	129
0000010	2	10000010	130
0000011	3	10000011	131
0000100	4	10000100	132
0000101	5	10000101	133
: :	: :	: :	: :
01111101	125	11111101	253
01111110	126	11111110	254
01111111	127	11111111	255

HX_HBP [7:0]: HBP [7:0] bits specify the line number of horizontal back porch period respectively for RGB SYNC mode.

HX_VBP [7:0]: VBP [7:0] bits specify the line number of vertical back porch period respectively for RGB SYNC mode.

HBP [7:0]	Number of HSYNC of front/back porch
000000	Setting inhibited
000001	Setting inhibited
000010	2
000011	3
000100	4
000101	5
: :	: :
11111101	253
11111110	254
11111111	255

Restriction

None

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

<table>
<tr><td rowspan="5">Default</td><td>
<table>
<tr><th rowspan="2">Status</th><th colspan="4">Default Value</th></tr>
<tr><th>VFP [7:0]</th><th>VBP [7:0]</th><th>HX_HBP [7:0]</th><th>HX_VBP [7:0]</th></tr>
<tr><td>Power On Sequence</td><td>8’h08</td><td>8’h08</td><td>8’h14</td><td>8’h08</td></tr>
<tr><td>SW Reset</td><td>8’h08</td><td>8’h08</td><td>8’h14</td><td>8’h08</td></tr>
<tr><td>HW Reset</td><td>8’h08</td><td>8’h08</td><td>8’h14</td><td>8’h08</td></tr>
</table>
</td></tr>
</table>

6.3.3. Display Function Control (AEh)

AEh	Display Function Control											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	0	1	1	1	0	AEh
1st Parameter	1	1	↑	XX	X	X	X	X	X	X	GS	SS	00

Description

SS: Select the shift direction of outputs from the source driver.

SS	Sourc Output Scan Direction
0	S1 → S720
1	S720 → S1

In addition to the shift direction, the settings for both SS and BGR bits are required to change the assignment of R, G, and B dots to the source driver pins.
To assign R, G, B dots to the source driver pins from S1 to S720, set SS = 0.
To assign R, G, B dots to the source driver pins from S720 to S1, set SS = 1.
GS: Sets the direction of scan by the gate driver in the range determined by SCN [4:0] and NL [4:0]. The scan direction determined by GS = 0 can be reversed by setting GS = 1.

GS	Gate Output Scan Direction
0	G1→G320
1	G320→G1

Restriction

None

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
	GS	SS
Power On Sequence	1’b0	1’b0
HW Reset	1’b0	1’b0

6.3.4. Interface Control (BAh)

BAh	Interface Control												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	1	0	1	0	BAh
1st Parameter	1	1	1	XX	1	1	0	0	DM [1:0]		RM	RIM	C0

Description

DM [1:0]: Select the display operation mode.

DM[1]	DM[0]	Display Operation Mode
0	0	Internal clock operation
0	1	RGB Interface Mode
1	0	VSYNC interface Mode
1	1	Setting disabled

RM: Select the interface to access the GRAM.
Set RM to “1” when writing display data by the RGB interface.

RM	Interface for RAM Access
0	System interface/VSYNC interface
1	RGB interface

RIM: Specify the RGB interface mode when the RGB interface is used. These bits should be set before display operation through the RGB interface and should not be set during operation.

RIM	COLMOD [6:4]	RGB Interface Mode
0	110 (262K color)	18- bit RGB interface (1 transfer/pixel)
	101 (65K color)	16- bit RGB interface (1 transfer/pixel)
1	(262K color)	6- bit RGB interface (3 transfer/pixel)

Restriction: None

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

<table>
<tr><td>Default</td><td>
<table>
<tr><th rowspan="2">Status</th><th colspan="4">Default Value</th></tr>
<tr><th>MDT[1:0]</th><th>DM [1:0]</th><th>RM</th><th>RIM</th></tr>
<tr><td>Power On Sequence</td><td>2’b00</td><td>2’b00</td><td>1’b0</td><td>1’b0</td></tr>
<tr><td>SW Reset</td><td>2’b00</td><td>2’b00</td><td>1’b0</td><td>1’b0</td></tr>
<tr><td>HW Reset</td><td>2’b00</td><td>2’b00</td><td>1’b0</td><td>1’b0</td></tr>
</table>
</td></tr>
</table>

6.4. Description of Level 3 Command

6.4.1. Frame Rate (B6h)

B6h	Frame Rate											
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	0	1	1	0	B6h
1st Parameter	1	1	↑	XX	RTN[7:0]								5C
2nd Parameter	1	1	↑	XX	RTN_IDLE[7:0]								40
3rd Parameter	1	1	↑	XX	RTN_DUMMY[7:0]								40

Description	**RTN[7:0] :**Set the frame rate when the internal resistor is used for oscillator circuit for normal mode; **RTN_IDLE[7:0] :**Set the frame rate when the internal resistor is used for oscillator circuit for idle mode; **RTN_DUMMY[7:0]:**Set the frame rate when the internal resistor is used for oscillator circuit for interrupt mode; Frame Rate = 7.44KHz/(32*(RTN[7:5]+1)+RTN[4:0])） **note: set rtn[7:5] =2** (see table 1) **note: set rtn[7:5] =1** (see table 2)
Restriction	Inter_command should be set high to enable this command
Register Availability	(see table 3)

note: set rtn[7:5] =2

rtn [4:0]	TE(Hz)	rtn [4:0]	TE(Hz)
8'd00	77.5	8'd10	66.43
8'd01	76.7	8'd11	65.84
8'd02	75.92	8'd12	65.27
8'd03	75.15	8'd13	64.7
8'd04	74.4	8'd14	64.14
8'd05	73.67	8'd15	63.59
8'd06	72.95	8'd16	63.05
8'd07	72.24	8'd17	62.53
8'd08	71.54	8'd18	62
8'd09	70.86	8'd19	61.49
8'd0A	70.19	8'd1A	60.99
8'd0B	69.54	8'd1B	60.49
8'd0C	68.89	8'd1C	60
8'd0D	68.26	8'd1D	59.52
8'd0E	67.64	8'd1E	59.05
8'd0F	67.03	8'd1F	58.59

note: set rtn[7:5] =1

rtn [4:0]	TE(Hz)	rtn [4:0]	TE(Hz)	rtn [4:0]	TE(Hz)	rtn [4:0]	TE(Hz)
8'd12	90.74	8'd16	86.52	8'd1A	82.67	8'd1E	79.15
8'd13	89.64	8'd17	85.52	8'd1B	81.76	8'd1F	78.32
8'd14	88.58	8'd18	84.55	8'd1C	80.8		
8'd15	87.54	8'd19	83.6	8'd1D	80.01		

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

<table>
<tr><td rowspan="5">Default</td><td>Status</td><td colspan="3">Default Value</td></tr>
<tr><td></td><td>RTN[7:0]</td><td>RTN_IDLE[7:0]</td><td>RTN_DUMMY[7:0]</td></tr>
<tr><td>Power On Sequence</td><td>8’h5C</td><td>8’h40</td><td>8’h40</td></tr>
<tr><td>SW Reset</td><td>8’h5C</td><td>8’h40</td><td>8’h40</td></tr>
<tr><td>HW Reset</td><td>8’h5C</td><td>8’h40</td><td>8’h40</td></tr>
</table>

6.4.2. SPI 2DATA control(B4h)

B4h	SPI 2DATA control												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	0	1	0	0	B4h
1st Parameter	1	1	↑	XX	X	X	X	X	2data_en	2data_mdt[2:0]			00
Description	**2DATA_EN:** Set 2_data_line mode in 3-wire/4-wire SPI. **2DATA_MDT[2:0]** Set pixel data format in 2_data_line mode.												
Restriction	Inter command should be set high to enable this command												

2DATA_MDT[2:0]	Data Format
000	65K color 1pixle/transition
001	262K color 1pixle/transition
010	262K color 2/3pixle/transition
100	4M color 1pixle/transition
110	4M color 2/3pixle/transition

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
	2DATA_EN	2DATA_MDT[2:0]
Power On Sequence	1’b0	3’b000
SW Reset	1’b0	3’b000
HW Reset	1’b0	3’b000

6.4.3. DINV (B5h)

B5h	DINV												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	0	1	0	1	B5h
1st Parameter	1	1	↑	XX	dinv_idle [1:0]		dinv[1:0]		0				50

Description

dinv_idle[1:0]: Set display inversion mode at idle mode

dinv[1:0] : Set display inversion mode at normal mode

dinv_idle[1:0]/ dinv[1:0]	Inversion
0	column inversion
1	1 dot inversion
2	2 dot inversion
3	column inversion

Restriction

Inter_command should be set high to enable this command

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
	dinv_idle [1:0]	dinv[1:0]
Power On Sequence	2’h1	2’h1
SW Reset	2’h1	2’h1
HW Reset	2’h1	2’h1

6.4.4. Power Control 1 (67h)

67h	Power Control 2											
	D/C X	RD X	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	0	0	1	1	1	67h
1st Parameter	1	1	↑	XX	X	vbp_d[6:0]							27

Description

Set the voltage level value to output the VREG1A and VREG1B OUT level, which is a reference level for the grayscale voltage level.(Table is valid when vrh=0x18)

VREG1A=(vrh+vbp_d)*0.025+3.925

VREG1B=vbp_d*0.025+0.025

vbp_d[6:0]	VREG1A/V	VREG1B/V
7'h00	4.525	0.025
…	…	…
N	(N+24)*0.05+3.925	N*0.025+0.025
…	…	…
7'h3C	6.025	1.525
…	…	
7'h4A	6.375	1.875

Restriction

Inter_command should be set high to enable this command

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
	vbp_d[6:0]
Power On Sequence	7h27
SW Reset	7h27
HW Reset	7h27

6.4.5. Power Control 2 (68h)

68h	Power Control 3											
	D/C X	RD X	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	0	1	0	0	0	68h
1st Parameter	1	1	↑	XX	X	vbn_d[6:0]							27

Description

Set the voltage level value to output the VREG2A OUT level, which is a reference level for the grayscale voltage level(Table is valid when vrh=0x18)

VREG2A=(vbn_d-vrh)*0.025-3.875

VREG2B=vbn_d*0.025+0.025

vbn_d[6:0]	VREG2A/V	VREG2B/V
7'h00	-4.475	0.025
…	…	…
N	N*0.025-4.475	N*0.025+0.025
…	…	…
7'h3C	-2.975	1.525
…	…	…
7'h4A	-2.625	1.875

Restriction: Inter_command should be set high to enable this command

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
	vbn_d[6:0]
Power On Sequence	7'h27
SW Reset	7'h27
HW Reset	7'h27

6.4.6. Power Control 3 (66h)

66h	Power Control 4												
	D/C X	RD X	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	0	0	1	1	0	66h
1st Parameter	1	1	↑	XX	X	X	vrh[5:0]						18

Description

Set the voltage level value to output the VREG1A OUT level, which is a reference level for the grayscale voltage level. (Table is valid when vbp_d=0x27 and vbn_d=0x27)

VREG1A=(vrh+vbp_d)*0.025+3.925

VREG2A=(vbn_d-vrh)*0.025-3.875

vrh[5:0]	VREG1A/V	VREG2A/V
6'h00	4.9	-2.9
…	…	…
N	(N+39)*0.025+3.925	(39-N)*0.025-3.875
…	…	…
6'h28	5.9	-3.9
…	…	…
6'h3F	6.475	-4.475

Restriction: Inter_command should be set high to enable this command

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
	vrh[5:0]
Power On Sequence	6'h18
SW Reset	6'h18
HW Reset	6'h18

6.4.7. Power Control 4 (CAh)

CAh	Power Control 6												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	0	1	0	1	0	CAh
1st Parameter	1	1	↑	XX				VGH_AD[4:0]					09

Description

VGH_AD[4:0]: Set the voltage level value to output the VGH level

VGH_AD<4:0>	VGH(V)
0	7.5
1	8
2	8.5
3	9
4	9.5
5	10
6	10.5
7	11
8	11.5
9	12
A	12.5
B	13
C	13.5
D	14
E	14.5
F	15
10~16	Reserved

Restriction

Inter_command should be set high to enable this command

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default	<table><tr><th rowspan="2">Status</th><th>Default Value</th></tr><tr><th>VGH_AD[4:0]</th></tr><tr><td>Power On Sequence</td><td>5’h09</td></tr><tr><td>SW Reset</td><td>5’h09</td></tr><tr><td>HW Reset</td><td>5’h09</td></tr></table>

6.4.8. Power Control 5(CBh)

CBh	Power Control 7												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	0	1	0	1	1	CBh
1st Parameter	1	1	↑	XX	0	0	0	0	0	VGL_AD[2:0]			04

Description

VGL_AD[2:0]: Set the voltage level value to output the VGL level,

VGL_AD<2:0>	VGL(V)
0	-7
1	-8
2	-9
3	-9.5
4	-10
5	-10.5455
6	-11.5455
7	Reserved

Restriction

Inter_command should be set high to enable this command

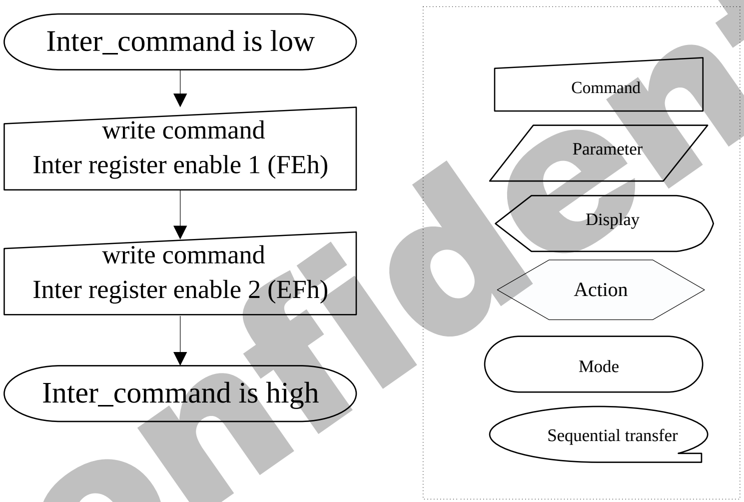
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
	VGL_AD[2:0]
Power On Sequence	3’b04
SW Reset	3’b04
HW Reset	3’b04

6.4.9. Inter Register Enable1(FEh)

FEh	Inter register enable 1												
	D/C X	RD X	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	1	1	1	1	0	FEh
Parameter	No Parameter												
Description	This command is used for Inter_command controlling. To set Inter_command high ,you should write Inter register enable 1 (FEh) and Inter register enable 2 (EFh) continuously. Once Inter_command is set high, only hardware or software reset can turn it to low. 												
Restriction													
Register Availability													
Default													

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

6.4.10. Inter Register Enable2(EFh)

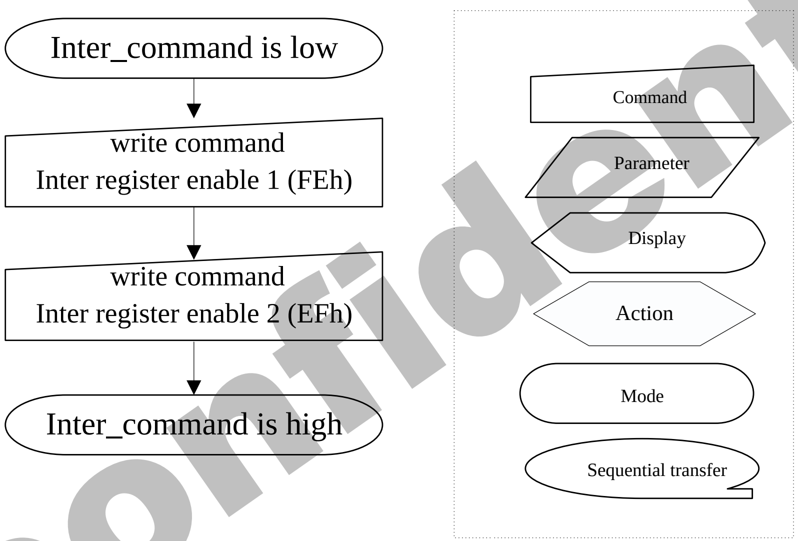
EFh	Inter register enable 2												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	0	1	1	1	1	EFh
Parameter	No Parameter												

Description

This command is used for Inter_command controlling.

To set Inter_command high ,you should write Inter register enable 1 (FEh) and Inter register enable 2 (EFh) continuously.

Once Inter_command is set high, only hardware or software reset can turn it to low.



Restriction	

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default	

6.4.11. SET_GAMMA1 (F0h)

F0h	SET_GAMMA1											
	D/CX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	1	1	1	1	0	0	0	0	F0
1st Parameter	1	1	↑	1	0	VR1_N[5:0]						86
2nd Parameter	1	1	↑	1	0	VR2_N[5:0]						85
3rd Parameter	1	1	↑	0	0	0	VR4_N[4:0]					05
4th Parameter	1	1	↑	0	0	0	VR6_N[4:0]					05
5th Parameter	1	1	↑	VR0_N[3:0]				VR13_N[3:0]				84
6th Parameter	1	1	↑		VR20_N[6:0]							07
7th Parameter	1	1	↑		VR43_N[6:0]							07
8th Parameter	1	1	↑	VR27_N[2:0]			VR57_N[4:0]					65
9th Parameter	1	1	↑	VR36_N[2:0]			VR59_N[4:0]					65
10th Parameter	1	1	↑			VR61_N[5:0]						06
11th Parameter	1	1	↑			VR62_N[5:0]						06
12th Parameter	1	1	↑	VR50_N[3:0]				VR63_N[3:0]				44
Description	Set the negetive voltage to adjust the gamma characteristics of the TFT panel.											
Restriction	Inter_command should be set high to enable this command											
Register Availability												

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

6.4.12. SET_GAMMA2 (F1h)

F1h	SET_GAMMA2											
	D/CX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	1	1	1	1	0	0	0	1	F1
1st Parameter	1	1	↑	1	0	VR1_P[5:0]						86
2nd Parameter	1	1	↑	1	0	VR2_P[5:0]						85
3rd Parameter	1	1	↑	0	0	0	VR4_P[4:0]					05
4th Parameter	1	1	↑	0	0	0	VR6_P[4:0]					05
5th Parameter	1	1	↑	VR0_P[3:0]				VR13_P[3:0]				84
6th Parameter	1	1	↑		VR20_P[6:0]							07
7th Parameter	1	1	↑		VR43_P[6:0]							07
8th Parameter	1	1	↑	VR27_P[2:0]			VR57_P[4:0]					65
9th Parameter	1	1	↑	VR36_P[2:0]			VR59_P[4:0]					65
10th Parameter	1	1	↑			VR61_P[5:0]						06
11th Parameter	1	1	↑			VR62_P[5:0]						06
12th Parameter	1	1	↑	VR50_P[3:0]				VR63_P[3:0]				44
Description	Set the positive voltage to adjust the gamma characteristics of the TFT panel.											
Restriction	Inter_command should be set high to enable this command											
Register Availability												

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

7. Electrical Characteristics

7.1. Absolute Maximum Ratings

The absolute maximum rating is listed on following table. When GC9309NA is used out of the absolute maximum ratings, GC9309NA may be permanently damaged. To use GC9309NA within the following electrical characteristics limitation is strongly recommended for normal operation. If these electrical characteristic conditions are exceeded during normal operation, GC9309NA will malfunction and cause poor reliability.

Table43.

Item	Symbol	Unit	Value
Supply voltage	VDDB	V	-0.3~+4.6
Supply voltage(Logic)	VDDI	V	-0.3~+4.6
Supply voltage(Digital)	DVDD	V	-0.3~+2.0
Driver supply voltage	VGH-VGL	V	-0.3~+27.0
Logic input voltage range	VIN	V	-0.3~VDDI+0.3
Logic output voltage range	VO	V	-0.3~VDDI+0.3
Operation temperature	Topr	℃	-30~+85
Storage temperature	Tstg	℃	-40~+110

Note: If the absolute maximum rating of even is one of the above parameters is exceeded even momentarily, the quality of the product may be degraded. Absolute maximum ratings, therefore specify the values exceeding which the product may be physically damaged. Be sure to use the product within the range of the absolute maximum ratings.

7.2. DC Characteristics

General DC Characteristics

Table44.

Item	Symbol	Unit	Condition	Min.	Typ.	Max.	Note
Power and Operation Voltage							
Analog Operating Voltage	VCI	V	Operating voltage	2.6	2.8	3.5	Note2
Logic Operating Voltage	IOVCC	V	I/O supply voltage	1.65	2.8	3.5	Note2
Digital Operating voltage	DVDD	V	Digital supply voltage	-	1.53	-	Note2
Gate Driver High Voltage	VGH	V	-	8.0	-	15.0	Note3
Gate Driver Low Voltage	VGL	V	-	-11.5	-	-7.0	Note3
Input and Output							
Logic High Level Input Voltage	VIH	V	-	0.7*IOVCC	-	IOVCC	Note1,2,3
Logic Low Level Input Voltage	VIL	V	-	VSSC	-	0.3*IOVCC	Note1,2,3
Logic High Level Output Voltage	VOH	V	IOL=-1.0mA	0.8*IOVCC	-	IOVCC	Note1,2,3
Logic Low Level Output Voltage	VOL	V	IOL=1.0mA	VSSC	-	0.2*IOVCC	Note1,2,3
Logic High Level Input Current	IIH	uA	-	-	-	1	Note1,2,3
Logic Low Level Input Current	IIL	uA	-	-1	-	-	Note1,2,3
Logic Input Leakage Current	ILEA	uA	VIN=IOVCC or VSSC	-0.1	-	+0.1	Note1,2,3
Source Driver							
Positive Source Output Range	Vsout	V	-	VREG1B	-	VREG1A	
Negative Source Output Range	Vsout	V	-	VREG2A	-	VREG2B	

Note 1: IOVCC=1.65 to 3.5V, VCI=2.6 to 3.5V, AGND=VSS=0V, Ta=-30 to 70 (to +85 no damage)℃

Note2: Please supply digital IOVCC voltage equal or less than analog VCI voltage.

Note3: CSX, RDX, WRX, D[17:0], D/CX, RESX, TE, DOTCLK, VSYNC, HSYNC, DE, SDA, SCL, IM3, IM2, IM1,IM0, and Test pins.

7.3. Power ON/OFF Sequence

IOVCC and VCI can be applied in any order.
VCI and IOVCC can be power down in any order.

During power off, if LCD is in the Sleep Out mode, VCI and IOVCC must be powered down minimum 120msec after RESX has been released.

During power off, if LCD is in the Sleep In mode, IOVCC or VCI can be powered down minimum 0msec after RESX has been released.

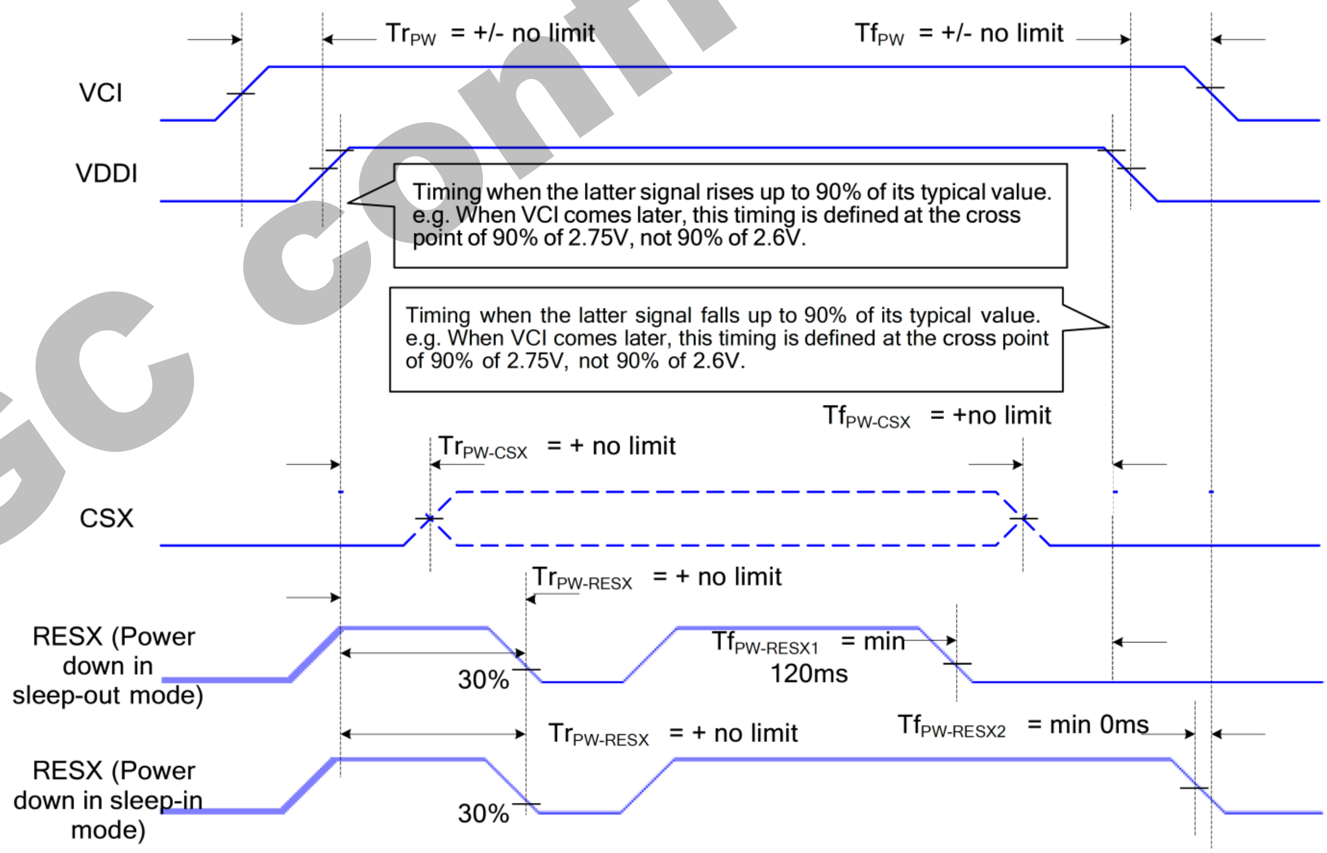
CSX can be applied at any timing or can be permanently grounded. RESX has priority over CSX.

Note 1: There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.

Note 2: There will be no abnormal visible effects on the display between end of Power On Sequence and before receiving Sleep Out command. Also between receiving Sleep In command and Power Off Sequence.

Note 3: If RESX line is not held stable by host during Power On Sequence as defined in the sequence below, then it will be necessary to apply a Hardware Reset (RESX) after Host Power On Sequence is complete to ensure correct operation. Otherwise function is not guaranteed.

The power on/off sequence is illustrated below



7.4. AC Characteristics

7.4.1. Display Parallel 18/16/9/8-bit Interface Timing Characteristics (8080-Ⅰ)

Figure90.

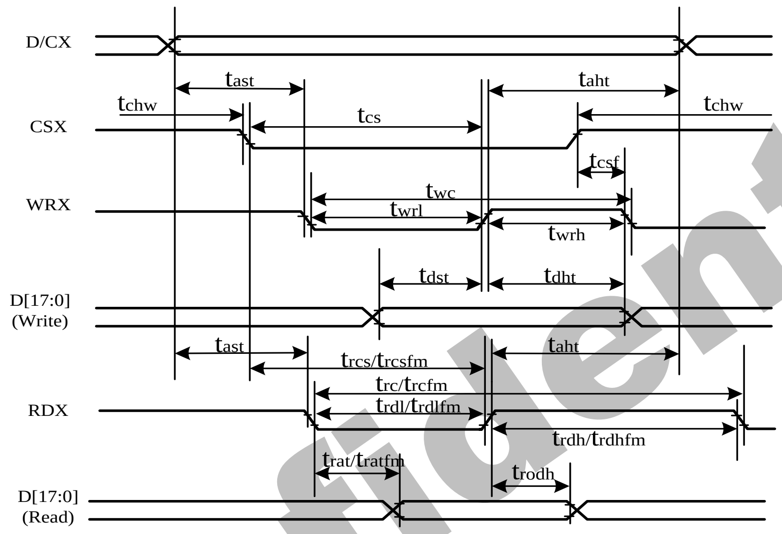


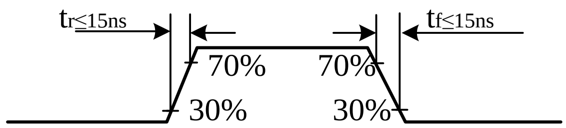
Table45.

Signal	Symbol	Parameter	min	max	Unit	Description
DCX	tast	Address setup time	0	-	ns	
	taht	Address hold time(Write/Read)	0	-	ns	
CSX	tchw	CSX "H" pulse width	0	-	ns	
	tcs	Chip Select setup time(Write)	15	-	ns	
	trcs	Chip Select setup time(Read ID)	45	-	ns	
	trcsfm	Chip Select setup time(Read FM)	355	-	ns	
	tcsf	Chip Select Wait time (Write/Read)	10	-	ns	
WRX	twc	Write Cycle	66	-	ns	
	twrh	Write Control pulse H duration	15	-	ns	
	twrl	Write Control pulse L duration	15	-	ns	
RDX(FM)	trcfm	Read Cycle (FM)	380	-	ns	
	trdhfm	Read Control H duration(FM)	180	-	ns	
	trdlfm	Read Control L duration(FM)	200	-	ns	
RDX(ID)	trc	Read Cycle (ID)	160	-	ns	
	trdh	Read Control H pulse duration	90	-	ns	
	trdl	Read Control L pulse duration	70	-	ns	
D[17:0], D[15:0], D[8:0],	tdst	Write data setup time	10	-	ns	For maximum CL=30pF For minimum
	tdht	Write data hold time	10	-	ns	
	trat	Read access time	-	40	ns	

D[7:0]	tratfm	Read access time	-	340	ns	CL=8pF
	trod	Read output disable time	20	80	ns	

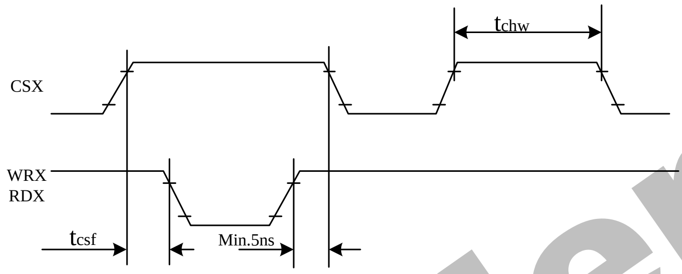
Note: Ta = -30 to 70 °C, IOVCC=1.65V to 3.3V, VCI=2.5V to 3.3V, VSS=0V

Figure91.



CSX timings :

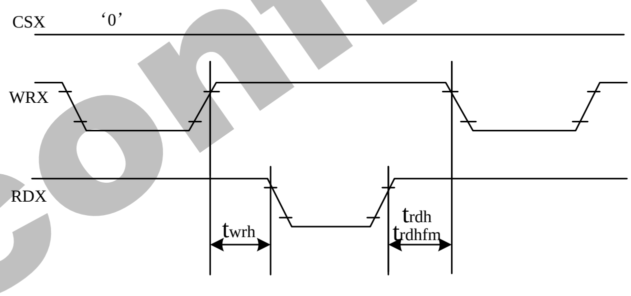
Figure92.



Note: Logic high and low levels are specified as 30% and 70% of IOVCC for Input signals.

Write to read or read to write timings:

Figure92.



Note: Logic high and low levels are specified as 30% and 70% of IOVCC for Input signals.

7.4.2. Display Parallel 18/16/9/8-bit Interface Timing Characteristics (8080-Ⅱ)

Figure93.

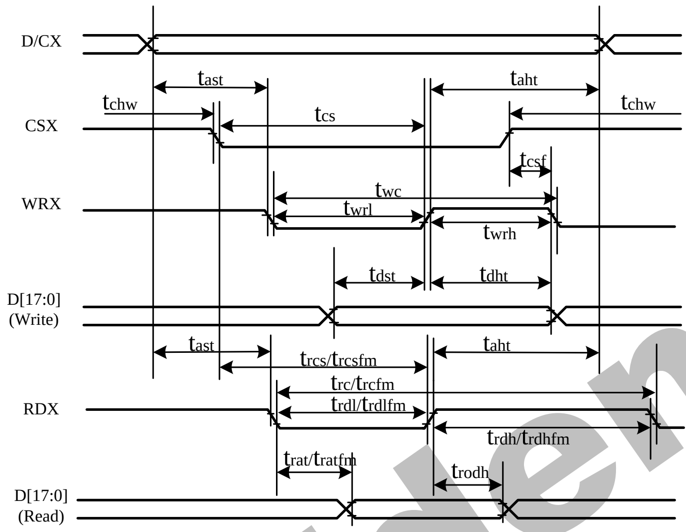
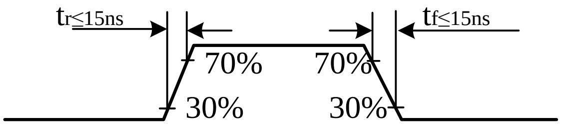


Table46.

Signal	Symbol	Parameter	min	max	Unit	Description
DCX	tast	Address setup time	0	-	ns	
	taht	Address hold time(Write/Read)	0	-	ns	
CSX	tchw	CSX "H" pulse width	0	-	ns	
	tcs	Chip Select setup time(Write)	15	-	ns	
	trcs	Chip Select setup time(Read ID)	45	-	ns	
	trcsfm	Chip Select setup time(Read FM)	355	-	ns	
	tcsf	Chip Select Wait time (Write/Read)	10	-	ns	
WRX	twc	Write Cycle	66	-	ns	
	twrh	Write Control pulse H duration	15	-	ns	
	twrl	Write Control pulse L duration	15	-	ns	
RDX(FM)	trcfm	Read Cycle (FM)	380	-	ns	
	trdhfm	Read Control H duration(FM)	180	-	ns	
	trdlfm	Read Control L duration(FM)	200	-	ns	
RDX(ID)	trc	Read Cycle (ID)	160	-	ns	
	trdh	Read Control pulse H duration	90	-	ns	
	trdl	Read Control pulse L duration	70	-	ns	
D[17:0], D[17:10] &D[8:1], D[17:10] ,D[17:9]	tdst	Write data setup time	10	-	ns	For maximum CL=30pF For minimum CL=8pF
	tdht	Write data hold time	10	-	ns	
	trat	Read access time	-	40	ns	
	tratfm	Read access time	-	340	ns	
	trod	Read output disable time	20	80	ns	

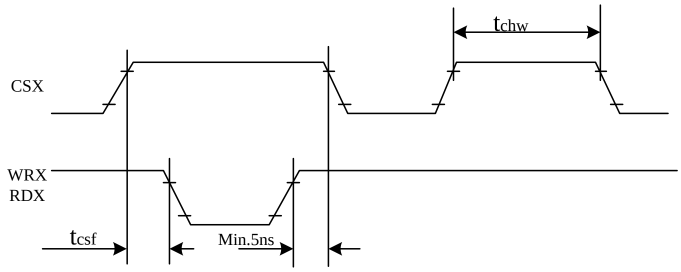
Note: Ta = -30 to 70 °C, IOVCC=1.65V to 3.3V, VCI=2.5V to 3.3V, VSS=0V.

Figure94.



CSX timings :

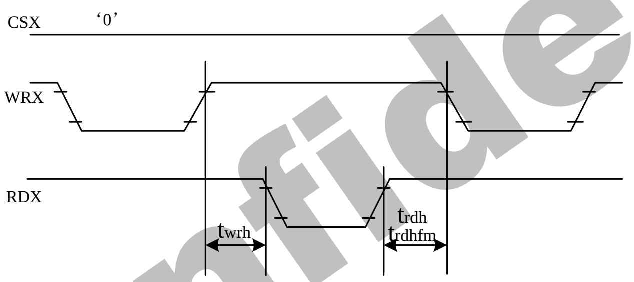
Figure95.



Note: Logic high and low levels are specified as 30% and 70% of IOVCC for Input signals.

Write to read or read to write timings:

Figure96.



Note: Logic high and low levels are specified as 30% and 70% of IOVCC for Input signals.

7.4.3. Display Serial Interface Timing Characteristics (3-line SPI system)

Figure97.

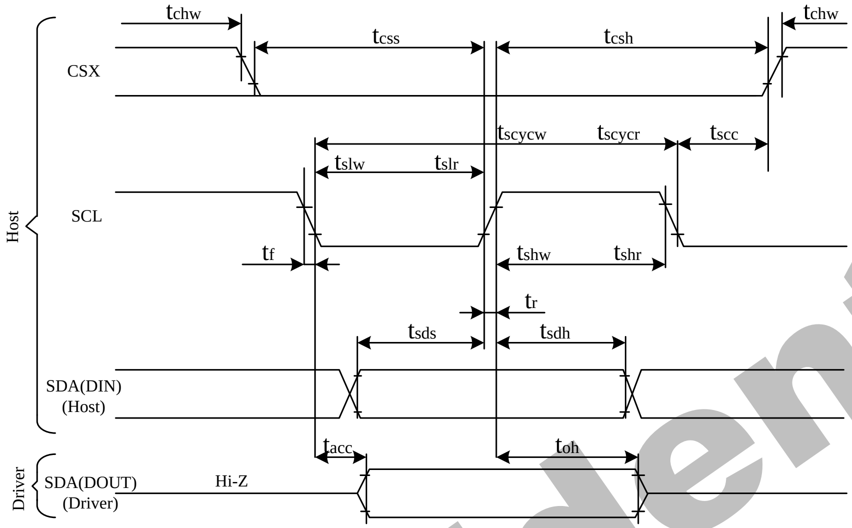
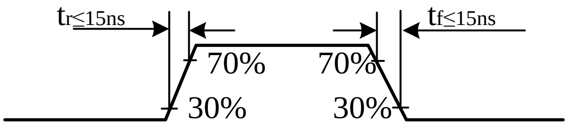


Table47.

Signal	Symbol	Parameter	min	max	Unit	Description
SCL	tscycw	Serial Clock Cycle (Write)	10	-	ns	
	tshw	SCL "H" Pulse Width (Write)	5	-	ns	
	tslw	SCL "L" Pulse Width (Write)	5	-	ns	
	tscycr	Serial Clock Cycle (Read)	150	-	ns	
	tshr	SCL "H" Pulse Width (Read)	60	-	ns	
	tslr	SCL "L" Pulse Width (Read)	60	-	ns	
SDA/SDI (Input)	tsds	Data setup time (Write)	5	-	ns	
	tsdh	Data hold time (Write)	5	-	ns	
SDA/SD0(Outp)	tacc	Access time (Read)	10	-	ns	
CSX	tscc	SCL-CSX	10	-	ns	
	tchw	CSX "H" Pulse Width	10	-	ns	
	tcss	CSX-SCL Time	20	-	ns	
	tcsh		40	-	ns	

Note: Ta = 25 °C, IOVCC=1.65V to 3.3V, VCI=2.5V to 3.3V, VSSA=VSSC=0V

Figure98.



7.4.4. Display Serial Interface Timing Characteristics (4-line SPI system)

Figure98.

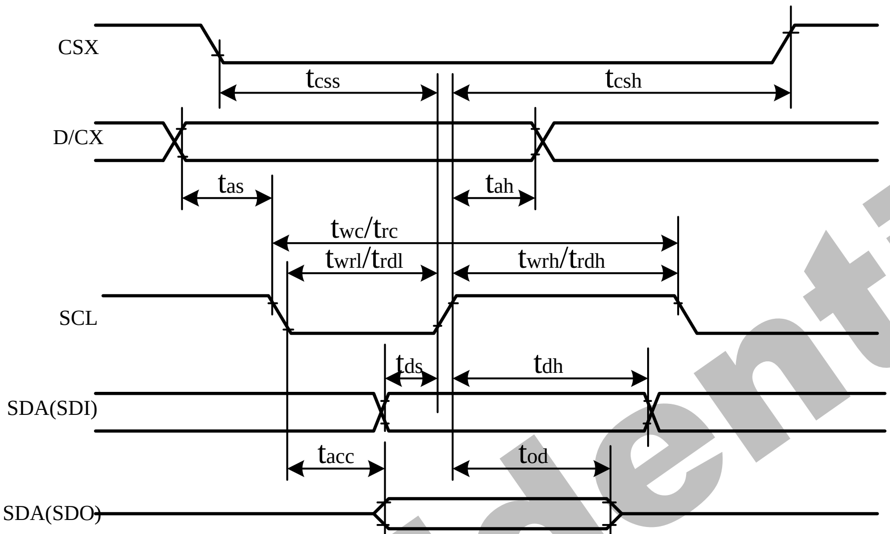
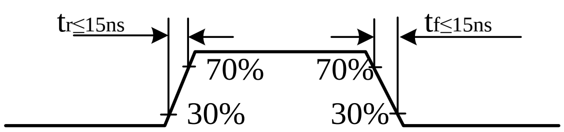


Table48.

Signal	Symbol	Parameter	min	max	Unit	Description
CSX	tcss	Chip select time (Write)	20	-	ns	
	tcsh	Chip select hold time (Read)	40	-	ns	
SCL	twc	Serial Clock Cycle (Write)	10	-	ns	
	twrh	SCL "H" Pulse Width (Write)	5	-	ns	
	twrl	SCL "L" Pulse Width (Write)	5	-	ns	
	trc	Serial Clock Cycle (Read)	150	-	ns	
	trdh	SCL "H" Pulse Width (Read)	60	-	ns	
	trdl	SCL "L" Pulse Width (Read)	60	-	ns	
D/CX	tas	D/CX setup time	10	-	ns	
	tah	D/CX hold time (Write/Read)	10	-	ns	
SDA/SDI (Input)	tds	Data setup time (Write)	5	-	ns	
	tdh	Data hold time (Write)	5	-	ns	
SDA/SD0 (Output)	tacc	Access time (Read)	10	-	ns	

Note: Ta = 25 °C, IOVCC=1.65V to 3.3V, VCI=2.5V to 3.3V, AGND=VSS=0V

Figure99.



7.4.5. Parallel 18/16/6-bit RGB Interface Timing Characteristics

Figure100.

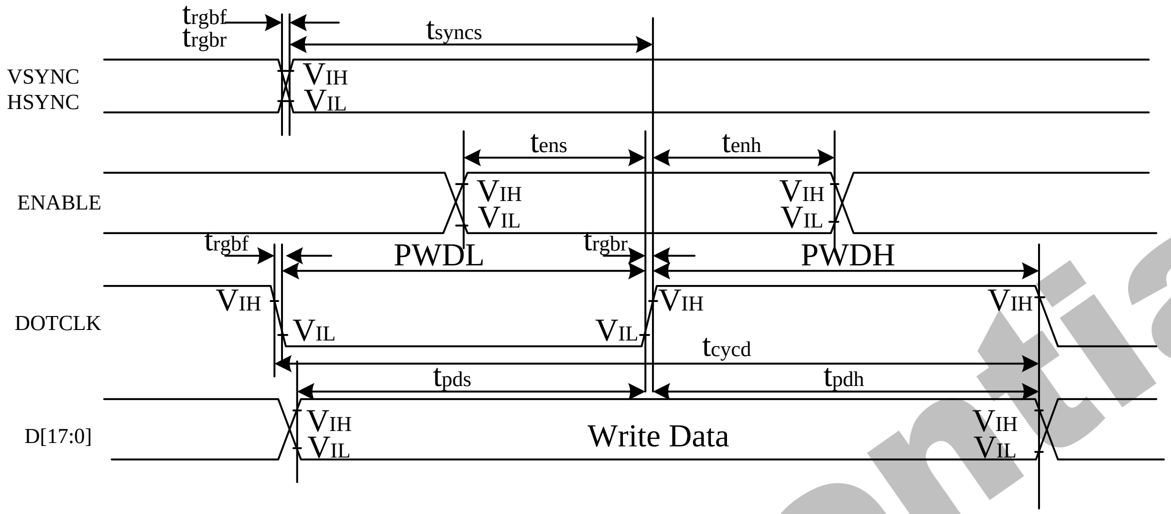


Table49.

Signal	Symbol	Parameter	min	max	Unit	Description
VSYNC/HSYNC	tsyncs	VSYNC/HSYNC setup time	15	-	ns	18/16-bit bus RGB interface mode
	tsynch	VSYNC/HSYNC hold time	15	-	ns	
DE	tens	DE setup time	15	-	ns	
	tenh	DE hold time	15	-	ns	
D[17:0]	tpos	Data setup time	15	-	ns	
	tpdh	Date hold time	15	-	ns	
DOTCLK	PWDH	DOTCLK high-level period	15	-	ns	
	PWDL	DOTCLK low-level period	15	-	ns	
	tcycd	DOTCLK cycle time	150	-	ns	
	trgbr,trgbf	DOTCLK,HSYNC,VSYNC rise/fall time	-	15	ns	
VSYNC/HSYNC	tsyncs	VSYNC/HSYNC setup time	15	-	ns	6-bit bus RGB interface mode
	tsynch	VSYNC/HSYNC hold time	15	-	ns	
DE	tens	DE setup time	15	-	ns	
	tenh	DE hold time	15	-	ns	
D[17:0]	tpos	Data setup time	15	-	ns	
	tpdh	Date hold time	15	-	ns	
DOTCLK	PWDH	DOTCLK high-level pulse period	15	-	ns	
	PWDL	DOTCLK low-level pulse period	15	-	ns	
	tcycd	DOTCLK cycle time	55	-	ns	
	trgbr,trgbf	DOTCLK,HSYNC,VSYNC rise/fall time	-	15	ns	

Note: Ta = -30 to 70 °C, IOVCC=1.65V to 3.3V, VCI=2.5V to 3.3V, AGND=VSS=0V

Figure101.